Studies In Riemann-Hilbert Correspondence



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I would also like to thank my parents, who supported me as this project grew to dangerous proportions and my sleeping hours dwindled.

Und die Stärke des Fadens liegt nicht darin, daß irgendeine Faser durch seine ganze Länge läuft, sondern darin, daß viele Fasern einander übergreifen.

The strength of a thread lies not in a single fibre running through its whole length, but in the overlapping of many fibres.

— Ludwig Wittgenstein, *Philosophische Untersuchungen*, §67

הַזֹּרְעִים בְּדִמְעָה בְּרִנָּה יִקְצֹרוּ׃
הָלוֹךְ יֵלֵךְ וּבָכֹה נֹשֵׂא מֶשֶׁךְ־הַזָּרַע
בֹּא־יָבוֹא בְרִנָּה נֹשֵׂא אֲלֻמֹּתָיו׃

Those who sow in tears shall reap with rejoicing. He goes forth weeping, bearing seed for sowing; he shall surely return with rejoicing, bearing his sheaves.

— Psalms 126:5–6

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Introduction

This work is an introduction to the Riemann–Hilbert correspondence, with an emphasis on clarity and on understanding each of the ideas needed to reach it. But before we plunge into the thick of it, we shall begin with the simplest case that illustrates the phenomenon the Riemann–Hilbert correspondence seeks to capture - our beloved linear algebra. For vector spaces V, W and a linear map $T : V \to W$ the equation$T(v) = w$ for $w \in W$, asks us to find a vector $v \in V$ that is mapped to w. If $w = 0$, the set of solutions is the kernel $\ker T$, and in general a solution exists when w lies in the image of T. If v_0 is one solution, then all solutions are of the form $v_0 + \ker T$. The kernel thus describes the freedom of choice among solutions, while the quotient space $W/(\operatorname{im} T)$ describes a complementary phenomenon. The class of w in this space vanishes precisely when w lies in the image of T, that is, precisely when the equation has a solution. A nonzero class indicates that there is no v for which $T(v) = w$, and in this sense one may say that it 'measures' the obstruction to solving the equation.

An almost immediate observation is that the same problem can be described in two ways. One may give the map T itself, for example by means of a matrix, and one may instead give, for every $w \in W$, the set of solutions $\{v \in V \mid T(v) = w\}$

The family of solution sets for all $w \in W$ determines T completely: the pair (v, w) belongs to the graph of T precisely when v is a solution of the equation $T(v) = w$. The homogeneous solution space is only the fibre over 0; it describes the kernel, but does not determine how T acts outside it. For this reason, the "solution side" will not refer below merely to the homogeneous solution space, but to its full structure: how solutions are determined by local data, how they vary and continue, and which obstructions may prevent them from existing.

A linear differential equation is a linear problem of the same kind, except that the unknown is a function and the linear map includes differentiation. An ordinary linear differential equation of order m is an equation of the form

$$a_{m(z)}u^{(m)}(z) + ... + a_1(z)u'(z) + a_0(z)u(z) = f(z),$$

where the functions $a_0, ..., a_m$ are given and the unknown is u. The operator

$$a_{m(z)}\frac{\mathrm{d}^m}{\mathrm{d}z^m} + ... + a_1(z)\frac{\mathrm{d}}{\mathrm{d}z} + a_0(z)$$

is a linear map between suitable spaces of functions, and so the equation may be written briefly as $Pu = f$. As before, the homogeneous solutions are the elements of $\ker P$, and the inability to solve $Pu = f$ is expressed by the fact that f does not lie in the image of P.

A linear system of ordinary differential equations is a system of the form

$$Y'(z) = A(z)Y(z) + b(z),$$

where $Y(z)$ is a vector of unknown functions, $A(z)$ is a given matrix and $b(z)$ is a given vector. The system is called homogeneous when $b = 0$. Every higher-order linear scalar equation can be written as such a system by means of the state vector

$$Y = \left(u, u', ..., u^{(m-1)}\right)$$

The passage from one equation to a system therefore makes it possible to formulate the problem by means of linear algebra that varies from point to point.

Near an ordinary point, the existence-and-uniqueness theorem states that every initial value $Y(z_0) = v$ determines a unique solution. The equation therefore determines linear transition maps that carry initial values from point to point, and may in turn be recovered from their infinitesimal behaviour. In this sense, the passage from an equation to its solutions is a direct development of the linear-algebraic picture: a single linear map is replaced by a linear rule that varies along the space, and the action of a matrix is replaced by the transport of vectors between points.

Here, however, phenomena appear that do not occur in linear algebra, which acts on fixed and complete spaces (in the sense that they are not fragmented). First, the solutions are local in a genuine sense. Let U be a domain on which A is holomorphic, and write

$$\mathrm{Sol}(U) = \{Y : U \to \mathbb{C}^n \mid Y' = AY\}$$

for the solution space of the homogeneous system on U. If $U' \subseteq U$, the restriction of a solution from U to U' is again a solution: the values of the function are already defined on the smaller domain, and differentiation commutes with restriction. In the opposite direction, new values must be found outside U', and this may be impossible if enlarging the domain reaches a singular point of the equation. For example, the equation with polynomial coefficients

$$(1 - z)u' - u = 0$$

has, on a small neighbourhood of 0, the solution$u(z) = \frac{1}{1-z}$.

This solution does not continue to a holomorphic function on a neighbourhood containing 1, where the coefficient of the derivative vanishes. If such an extension existed, it could of course be restricted back to the original solution; the problem is that the extension need not exist. Restriction is therefore automatic, whereas extension is not.

Solutions that agree on overlaps glue to a unique solution on their union, because holomorphicity and the equation $Y' = AY$ are both local conditions. Restriction and gluing are therefore not artificial additions to the equation, but a translation of the fact that a differential equation is checked point by point. A single space of global solutions does not retain this information, and so the solutions must be recorded on every local domain. As we shall see in Chapter 1, the structure that organizes such local information is called a sheaf.

Existence and uniqueness also allow a local solution to be continued along a path by repeatedly solving on overlapping discs; uniqueness makes the successive solutions agree on the overlaps. Continuation along a path is thus not a new global operation, but a finite repetition of solving local problems and gluing their results.

Second, a local solution may change when it is continued along a path. Even if every small neighbourhood has the same number of linearly independent solutions, continuation around a closed path may return a different linear combination of the solutions with which we began. This is the phenomenon of monodromy, which we shall discuss at length below. It is not visible in the solution space at a single point, but only when one remembers how the solution spaces at different points are identified along paths. On a domain without singularities one obtains in this way a sheaf that looks locally like a constant vector space, but may twist globally. Such a sheaf is called a local system.

It is now clear why the kernel of the operator as an abstract vector space is insufficient. Near every ordinary point, a system of size n produces an n-dimensional solution space. If we retained only the dimension or the isomorphism type of this space, almost all systems of size n would look identical. The information that distinguishes them lies instead in the way the solutions are transported and continued along the space.

Here an additional condition appears that does not occur in one variable: progress in different directions must be consistent. Computing the solution first

in one direction and then in another should agree, as one would expect, with computing it in the reverse order. When this compatibility condition holds, the system again describes linear transport along paths. In one variable there are no different directions to compare, and so this difficulty does not appear.

So far we have assumed that the system is nonsingular. If the matrix $A(z)$ is not defined at a certain point, or if the leading coefficient of a higher-order equation vanishes, a singularity may appear—a neighbourhood containing the problematic point in which the function behaves unusually. One may still solve the equation outside the singular set and continue solutions there along paths, but the information on the nonsingular domain does not necessarily determine what happens on the removed set. Two systems may produce the same local system away from the singularity, and nevertheless behave differently when one tries to continue them through it.

One example displays both phenomena at once. On a punctured disc around the origin, consider the equation

$$u' = \frac{\lambda}{z} u$$

On a small simply connected domain that does not contain the origin, one may choose a branch of $\log z$, and then every solution is of the form

$$u(z) = c \exp(\lambda \log z)$$

When the solution is continued once around the origin, the value of $\log z$ changes by $2\pi i$, and therefore the solution is multiplied by $e^{2\pi i \lambda}$. In every small disc the solution space is still one-dimensional, but the circuit remembers an additional linear datum that is not visible inside a single disc, and this is how continuation of local solutions produces monodromy.

Even when λ is an integer and the monodromy is trivial, restoring the origin remains a separate problem. For $\lambda = 0$ the solutions are constant and therefore continue holomorphically to the origin. For $\lambda = -1$ the solutions are $\frac{c}{z}$: on the punctured disc they form a one-dimensional local system isomorphic to that of the constant solutions, but no nonzero solution continues holomorphically to 0. Hence the solutions away from the singularity, even together with their monodromy, do not determine by themselves what is retained when the point is restored. One must also include information about what is allowed to occur

at the point itself and about the way this information is connected to the surrounding solutions.

The problem is not limited to isolated points. In a singular space, or in a system whose singularities extend along subspaces, different points may have different local geometry. We therefore divide the space into finitely many smooth pieces, called strata, so that along each stratum the local type remains uniform. The information is constructible when on every stratum it varies as a finite-dimensional local system and changes occur only in passing between strata. Monodromy may still occur along loops within a stratum; the requirement is only that no new local type appears at every point. We shall treat this notion at the end of Chapter 1.

A constructible object may also contain information supported on lower-dimensional strata and maps connecting it to the surrounding local systems. These are precisely the data that were missing when the origin was restored to the punctured disc.

Suppose the space is divided into an open part U, on which the system is nonsingular, and a closed part Z, on which the singularities are concentrated. The same local system on U may be extended to the whole space in different ways, distinguished by the information supported on Z and by the way it is connected to the surrounding solutions. This creates the need for operations that restrict information, extend it across the boundary, or extract the contribution supported on the closed part. The intermediate extension selects an extension that does not artificially add a subobject or quotient object supported only on the boundary. These operations will be developed in Chapter 3, and the intermediate extension will return in Chapter 4.

Even the sheaf of homogeneous solutions still does not retain all the necessary information. Consider on the complex line the two equations

$$u' = 0, \quad zu' = 0.$$

They have the same homogeneous holomorphic solutions—the constant functions. The behaviour of the inhomogeneous equations, however, is different. Every holomorphic function f has a local antiderivative, and therefore the equation $u' = f$ is locally solvable. By contrast, the equation

$$zu' = f$$

has no holomorphic solution near the origin when $f(0) \neq 0$. The homogeneous solution space does not distinguish the operators, but the obstruction to solving them distinguishes them immediately.

This is again the distinction from linear algebra between a kernel and the failure to lie in the image. To retain both together, one does not keep a single solution space, but records the operator as an arrow

$$\mathcal{O} \xrightarrow{P} \mathcal{O}$$

and retains both its kernel and the quotient of the space on the right by its image. In a system of several equations, relations among the equations may also appear, followed by relations among those relations. The information thus arranges itself into a chain of linear maps. Complexes and homological algebra are the language that makes it possible to treat such a chain as a single object. The derived category makes it possible to compare different presentations of the same system without erasing the solutions, obstructions and relations that appear in the different degrees.

The algebraic foundations of this construction will be given in Appendix A and Appendix B, and its sheaf-theoretic version will be developed in Chapter 2. We shall return to it in Chapter 5 for the solution complex of a differential system.

The passage to complexes is also important for understanding duality. In linear algebra, passage to the dual space reverses maps and exchanges kernels with appropriate quotient spaces. When the information is spread over a space, one must also distinguish between information visible near a point and information supported around it, and between arbitrary sections and sections with restricted support. Verdier duality extends linear duality while retaining location, support and local topology. It will be developed in Chapter 3.

Duality and constructibility also create a problem of degrees. On a smooth stratum of complex dimension d, a local system occupies its natural position after a shift by d, whereas information supported at a point requires no shift. The solution is to fold, or recenter, the scale of degrees according to the dimension of the stratum. The middle choice is arranged so that Verdier duality preserves the resulting degree zero. Its objects are called perverse sheaves: constructible complexes whose degrees have been adapted to the dimensions of the strata on which their information lies. They are also apparently not perverse, but that is a matter of personal taste. We shall discuss them in Chapter 4.

On the equation side too, a passage to a more intrinsic presentation is required. A list of operators is not an internal description of a system, since one may replace an equation by a combination of other equations, add an equation that follows from the existing ones, or differentiate a relation and obtain a new relation. All these differential consequences must be included in the system from the outset. To achieve this, the list of equations is replaced by a module over the algebra of differential operators. This is a $\mathcal{D}$-module. It retains the unknowns, the equations and all the relations that follow from them without depending on a particular list of generators. Chapter 5 will construct $\mathcal{D}$-modules from systems of equations, and in the same context introduce the derived solution complexes.

Not every $\mathcal{D}$-module produces finite and controlled information on the solution side. Holonomicity is the finiteness condition that ensures that its solutions, obstructions and singular information form a constructible complex. Regularity further excludes irregular singularities, near which solutions may grow or decay differently in different sectors around the same point. For regular holonomic systems, monodromy, contributions supported on the singular strata and the maps connecting them to the smooth information constitute a complete description. These conditions will be developed in Chapter 5.

We can now state the idea of the Riemann–Hilbert correspondence. On one side lies the system of equations, organized as a module of differential operators and including all its differential consequences and relations. On the other lies the solution data organized according to its position in the space: local systems on the smooth strata, monodromy along loops, contributions supported on the singular strata, local obstructions to solving, and the gluing maps between all the parts. For regular holonomic systems, the two descriptions determine one another.

The equivalence between the two descriptions, and the compatibility of the structures and operations built along the way, will be proved in Chapter 6.

The route to this correspondence determines the structure of the text. Sheaf theory carries this language to spaces and records local solutions, gluing and continuation. Constructibility imposes finite geometric control, while the operations developed in Chapter 3 describe restriction, extension, support and duality. Perverse sheaves (Chapter 4) organize information carried by strata of different dimensions. Finally, $\mathcal{D}$-modules, holonomicity and regularity provide the equation side on which the correspondence is defined. To make this introduction as

accessible and clear as possible, I have also included two substantial appendices containing the necessary background in homological algebra (Appendix A) and derived categories (Appendix B), so that readers unfamiliar with these subjects can follow the material without requiring prior knowledge of them. The examples woven through these appendices form the basis for those that appear later in the thesis. The appendices may therefore be regarded as Chapters -2 and -1 of the thesis, if the Introduction is taken to be Chapter 0. They are best read together as a single unit.

When an adapted finite cell decomposition is available, a constructible complex can be represented by finite-dimensional data on the cells and computed by linear algebra. Sections Section 1.5 and Section 2.4 develop this description, and Section 6.5 applies it to constructible de Rham and solution complexes.

1 Sheaves (and stratification)

1.1 Motivation

A topological space specifies which points occur together in open neighbourhoods, and the inclusions among those neighbourhoods record how local regions fit inside larger ones. Data that live on a space can therefore carry geometric information through their behaviour on open sets, even when their values at individual points look identical.

Continuous functions are the basic example. For every open set $U \subseteq X$, let

$$C^0_X(U) \coloneqq \{f : U \to \mathbb{R} \mid f \text{ is continuous}\}$$

A function on a larger open set determines one on every smaller open subset, and functions defined on an open cover determine one function on the union when they agree on every overlap. The point of sheaf theory is to isolate this pattern from the particular notion of function.

Passing to open sets allows us to study points from two perspectives: what happens as we shrink a neighbourhood of a point to arbitrarily small neighbourhoods, and what happens when we remove the point and compare the neighbourhood with its punctured counterpart. We can collect the information obtained by shrinking around each point and assemble these local pieces into a topological space in its own right. This geometric viewpoint gives a different meaning to information that becomes zero or disappears when the point is removed.

To complete this geometric view of information, we will examine the different ways in which information can be transported between spaces, as well as the ways in which it can be treated within a single space according to its relation to the underlying geometry. Finally, we will develop a useful tool for decomposing complicated geometric spaces into layers on which the information becomes simpler, without losing the ability to describe the original information. Throughout this chapter and the next, R denotes a commutative ring with unit. When constructibility is considered over R, Noetherianity will be imposed explicitly. The cellular calculations later in the chapter use a field k.

1.2 Basic Definitions

Begin with the weakest possible structure: for every open $U \subseteq X$, choose a set $\mathcal{F}(U)$. At this stage the sets attached to two nested neighbourhoods are unrelated.

Definition 1.1 (Section) *Let $\mathcal{F}$ assign a set $\mathcal{F}(U)$ to every open subset $U \subseteq X$. An element $s \in \mathcal{F}(U)$ is called a **section of $\mathcal{F}$ over U**. A section over X is a **global section**.*

To make these local descriptions coherent, data on a region must determine data on every smaller region. For $V \subseteq U$, we therefore require a restriction map from $\mathcal{F}(U)$ to $\mathcal{F}(V)$, and successive restrictions must agree with direct restriction.

Definition 1.2 (Presheaf of sets) *A **presheaf of sets** $\mathcal{F}$ on X assigns a set $\mathcal{F}(U)$ to every open $U \subseteq X$ and, for every inclusion $V \subseteq U$, a restriction map*

$$\rho_V^U : \mathcal{F}(U) \to \mathcal{F}(V)$$

such that

$$\rho_U^U = \mathrm{id}_{\mathcal{F}(U)}, \quad \rho_W^V \circ \rho_V^U = \rho_W^U$$

whenever $W \subseteq V \subseteq U$.

The two identities say that restriction is itself coherent. Categorically, they say that the assignment is a contravariant functor

$$\mathcal{F} : \mathrm{Op}(X)^{\mathrm{op}} \to \mathrm{Set}.$$

For a section $s \in \mathcal{F}(U)$, we write $s|_V$ for its image in $\mathcal{F}(V)$.

Restrictions alone still allow incompatible local descriptions. Suppose that $U = \bigcup_i U_i$ and sections $s_i \in \mathcal{F}(U_i)$ have been chosen. Two separate requirements are

needed. First, a global section is determined by its restrictions, so two sections on U that agree on every U_i must be equal. Second, whenever the s_i agree on every overlap $U_i \cap U_j$, they should arise from one section on U.

Definition 1.3 (Sheaf of sets) *A presheaf $\mathcal{F}$ of sets on X is a **sheaf** if every open cover $U = \bigcup_i U_i$ satisfies the following two conditions.*

Locality. *If $s, t \in \mathcal{F}(U)$ and*

$$s|_{U_i} = t|_{U_i}$$

for every i, then $s = t$.

Gluing. *If sections $s_i \in \mathcal{F}(U_i)$ satisfy*

$$s_i\big|_{U_i \cap U_j} = s_j\big|_{U_i \cap U_j}$$

for every i, j, then there is a section $s \in \mathcal{F}(U)$ with

$$s|_{U_i} = s_i$$

for every i.

By locality, the glued section is unique.

For C^0_X, locality is immediate because two functions that agree on a cover agree at every point. Gluing is also immediate: compatible functions on the U_i define a pointwise function on their union, and continuity can be checked locally. Thus C^0_X is a sheaf of $\mathbb{R}$-algebras. The same argument gives the sheaf

$$C^\infty_X(U) \coloneqq \{f : U \to \mathbb{R} \mid f \text{ is smooth}\}$$

on a smooth manifold and

$$\mathcal{O}_X(U) \coloneqq \{f : U \to \mathbb{C} \mid f \text{ is holomorphic}\}$$

on a complex manifold.

The two parts of the sheaf axiom are logically independent. The constant-presheaf example below satisfies locality but can fail gluing. Conversely, let $X = \mathbb{R}$, put $\mathcal{A}(X) = \{a, b\}$, and put $\mathcal{A}(U) = \{\bullet\}$ for every proper open subset $U \subseteq X$ with $U \neq X$, with the only possible restriction maps. Compatible local sections always have at least one global lift, but a cover of $\mathbb{R}$ by proper open sets cannot distinguish a from b. Thus gluing exists while locality fails. This artificial example (as examples often are) isolates why both clauses are required.

Definition 1.4 (Constant presheaf and constant sheaf) *Fix an R-module M. Define the* ***constant presheaf*** $\mathcal{M}_{\text{pre}}$ *by*

$$\mathcal{M}_{\text{pre}}(U) \coloneqq \begin{cases} MU \neq \varnothing \\ 0 \;\; U = \varnothing \end{cases}$$

with identity restrictions between nonempty open sets and the unique zero map to the empty set.

The corresponding ***constant sheaf*** $\underline{M}$ *is defined by*

$$\underline{M}(U) \coloneqq \{f : U \to M \mid f \;\text{ is locally constant}\}$$

where M has the discrete topology. The later sheafification construction will show that this sheaf is the canonical sheaf associated with $\mathcal{M}_{\text{pre}}$.

The difference appears as soon as an open set is disconnected. If $U = U_1 \sqcup U_2$ with both pieces nonempty, then $\mathcal{M}_{\text{pre}}(U) = M$: one element must supply the value on both components. In the constant sheaf, the two components can carry independent values, so $\underline{M}(U) \cong M \oplus M$. More generally, when U is locally connected,

$$\underline{M}(U) \cong \prod_{C \in \pi_0(U)} M$$

Thus the constant presheaf already has coherent restrictions, and it satisfies locality, but it fails the gluing requirement on a disconnected cover when different values are chosen on different components. On connected open sets the two constructions agree. Thus the term "constant sheaf" is some-what misleading, even abusive.

The elementwise definitions are the form used throughout the chapter. They also have a categorical formulation that will matter later, because our sections will usually form modules rather than bare sets.

Definition 1.5 (Category-valued presheaves and sheaves) *Let $\mathcal{C}$ be a category. A* ***$\mathcal{C}$-valued presheaf*** *on X is a contravariant functor*

$$\mathcal{F} : \text{Op}\ (X)^{\text{op}} \to \mathcal{C}$$

If $\mathcal{C}$ has the products and equalizers required by an open cover $U = \bigcup_i U_i$, the presheaf is a ***$\mathcal{C}$-valued sheaf*** *when the canonical morphism*

$$\mathcal{F}\,(U) \to \prod_i \mathcal{F}\,(U_i)$$

identifies $\mathcal{F}(U)$ with the equalizer of the two restriction morphisms

$$r_1, r_2 : \prod_i \mathcal{F}(U_i) \to \prod_{i,j} \mathcal{F}(U_i \cap U_j)$$

obtained from the two inclusions of each overlap into U_i and U_j. For $\mathcal{C} = \mathrm{Set}$ or $\mathcal{C} = R\text{-Mod}$, this is equivalent to the locality and gluing conditions above.

When $\mathcal{C} = R\text{-Mod}$, the elements of $\mathcal{F}(U)$ are again called sections, and all restriction maps are R-linear.

Kashiwara–Schapira give the presheaf definition in [1, Ch. II, Def. 2.1.1], the terminology of sections in [1, Ch. II, Terminology 2.1.2], and the ordinary sheaf axiom in [1, Ch. II, Def. 2.2.1].

The covering map

$$p : \mathbb{R} \to S^1, \quad p(t) = e^{2\pi i t}$$

provides the first example in which local sections exist everywhere while global compatibility fails. Over every sufficiently small open arc $U \subseteq S^1$ there is a continuous lift

$$s : U \to \mathbb{R}, \quad p \circ s = \mathrm{id}_U$$

If s is one lift, then $s + n$ is another for every locally constant integer-valued function n on U.

There is no continuous lift on the whole circle. If $s : S^1 \to \mathbb{R}$ were such a lift and $\gamma(u) = e^{2\pi i u}$, then $a(u) \coloneqq s(\gamma(u))$ would satisfy $a(u) - u \in \mathbb{Z}$. This is a continuous integer-valued function and hence constant, so $a(1) - a(0) = 1$. Since $\gamma(0) = \gamma(1)$, the same function s would give $a(0) = a(1)$.

Define

$$\mathcal{P}(U) \coloneqq \{s : U \to \mathbb{R} \mid s \ \text{ continuous with } \ p \circ s = \mathrm{id}_U\}.$$

Restriction makes $\mathcal{P}$ a sheaf of sets. The equality $\mathcal{P}(S^1) = \varnothing$ is compatible with the sheaf axiom: local lifts exist, while a family of local lifts on a cover need not agree on every overlap. The sheaf condition says that a compatible family glues once compatibility has actually been achieved.

Definition 1.6 (Morphisms of sheaves) *Let $\mathcal{F}$ and $\mathcal{G}$ be presheaves of sets or R-modules. A **morphism** $\varphi : \mathcal{F} \to \mathcal{G}$ is a family of maps*

$$\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$$

compatible with passage to smaller open sets. A morphism of sheaves is a morphism of their underlying presheaves. For sheaves of R-modules, every φ_U is R-linear.

We write

$$\mathrm{Sh}(X; R)$$

for the category of sheaves of R-modules on X. From this point onward this is the default coefficient category unless a set-valued example is explicitly named.

Definition 1.7 (Skyscraper sheaf) *Fix $x \in X$ and an R-module M. The **skyscraper sheaf** at x is*

$$\big(i_{x,*}M\big)(U) = \begin{cases} M \text{ if } & x \in U \\ 0 \quad \text{if } & x \notin U \end{cases}$$

where $i_x : \{x\} \to X$ is the inclusion.

1.2.1 Stalks, support, and costalks

So far, a sheaf has assigned information to open subsets of X, not directly to points. To recover what the sheaf says near a point x, we should forget every feature that disappears after shrinking the neighbourhood of x.

Consider $C^0_{\mathbb{R}}$ near 0. Let

$$f(t) = 0, \quad g(t) = \begin{cases} 0 & |t| < 1 \\ |t| - 1 |t| \geq 1 \end{cases}, \quad h(t) = t$$

The global sections f and g are different, but they become identical after restriction to $(-1, 1)$. The function h has the same value as f at the point 0, while it disagrees with f on every neighbourhood of 0. The local object attached to 0 should therefore identify f with g and distinguish h from both.

Definition 1.8 (Stalk) *Let $\mathcal{F}$ be a presheaf of sets or R-modules, let $x \in X$, and let $\mathcal{N}_x$ denote the set of open neighbourhoods of x. Consider pairs (U, s) with $U \in \mathcal{N}_x$ and $s \in \mathcal{F}(U)$. Declare $(U, s) \sim (V, t)$ if there is an open neighbourhood W satisfying $x \in W \subseteq U \cap V$ and $s|_W = t|_W$. The **stalk** of $\mathcal{F}$ at x, written $\mathcal{F}_x$, is the quotient*

$$\mathcal{F}_x := \left(\bigsqcup_{U \in \mathcal{N}_x} \mathcal{F}(U) \right) / \sim$$

For an R-module-valued presheaf, addition and scalar multiplication are defined after restricting representatives to a common neighbourhood.

Definition 1.9 (Germ) *If $s \in \mathcal{F}(U)$ and $x \in U$, the equivalence class of (U, s) in $\mathcal{F}_x$ is the **germ** of s at x, written s_x.*

Proposition 1.10 (Equality of germs) *For $s \in \mathcal{F}(U)$ and $t \in \mathcal{F}(V)$ with $x \in U \cap V$,*

$$s_x = t_x \iff s|_W = t|_W \quad \text{for some open } \; W \; \text{ with } \; x \in W \subseteq U \cap V$$

This gives the intended answer in the example:

$$f_0 = g_0, \quad h(0) = f(0) = 0, \quad h_0 \neq f_0.$$

The first equality forgets behaviour away from 0. The last inequality shows why the stalk is richer than evaluation at a point: it retains behaviour on arbitrarily small neighbourhoods.

The same object has a categorical description. The neighbourhoods of x form a directed system under restriction.

Proposition 1.11 (Stalk as a filtered colimit) *There is a natural isomorphism*

$$\mathcal{F}_x \cong \operatorname{colim}_{x \in U} \mathcal{F}(U)$$

where U runs through the open neighbourhoods of x and the transition maps are restrictions.

Proof. Every section $s \in \mathcal{F}(U)$ determines its germ s_x, and these maps are compatible with restriction. Conversely, two sections have the same image in the quotient of Definition 1.8 exactly when they become equal after restriction to some smaller neighbourhood. The quotient therefore satisfies the universal property of the filtered colimit. □

Kashiwara–Schapira use the filtered-colimit description in [1, Ch. II, Eq. (2.1.3)] and call the image of a section its germ immediately afterward. Goresky gives the same construction in [2, §1.2]. The concrete quotient in Definition 1.8 is the elementwise realization of that colimit.

The construction is also functorial. A morphism of sheaves sends representatives to representatives and respects the equivalence relation.

Definition 1.12 (Stalk functor) *For a fixed point $x \in X$, a morphism $\varphi : \mathcal{F} \to \mathcal{G}$ induces $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x, \quad s_x \mapsto (\varphi_U(s))_x$. The assignments $\mathcal{F} \mapsto \mathcal{F}_x, \quad \varphi \mapsto \varphi_x$ define the* ***stalk functor***

$$i_x^{-1} : \mathrm{Sh}(X; R) \to R\text{-Mod}, \quad i_x^{-1}\ \mathcal{F} \coloneqq \mathcal{F}_x$$

Once the abelian structure of $\mathrm{Sh}(X; R)$ has been established below, this will imply that the stalk functor is exact.

Definition 1.13 (Support of a section and of a sheaf) *Let $\mathcal{F}$ be a sheaf of R-modules on X, let $U \subseteq X$ be open, and let $s \in \mathcal{F}(U)$. The* ***support of s in U*** *is*

$$\mathrm{supp}(s) \coloneqq \{x \in U : s_x \neq 0\}$$

It is closed in U. The section has ***compact support*** *if* $\mathrm{supp}(s)$ *is compact.*

The ***closed support*** *of $\mathcal{F}$ is*

$$\mathrm{supp}(\mathcal{F}) \coloneqq \overline{\{x \in X : \mathcal{F}_x \neq 0\}}^X$$

The function $h(t) = t$ has $h(0) = 0$, while $0 \in \mathrm{supp}(h)$ because $h_0 \neq 0$. The tent function

$$b(t) = \begin{cases} 1 - |t| & |t| \leq 1 \\ 0 & |t| > 1 \end{cases}$$

has support $[-1, 1]$ and therefore compact support. The function e^{-t^2} tends rapidly to zero, yet it never vanishes, so its support is all of $\mathbb{R}$ and is not compact.

The constant sheaf and the skyscraper sheaf separate another local question. Both have nonzero stalk at 0. A nonzero section of $\underline{k}_{\mathbb{R}}$ on a connected neighbourhood of 0 remains nonzero on a smaller interval, so it cannot be supported only at 0. A section of the skyscraper $i_{0,*}M$ is supported there by definition. The stalk sees that both sheaves have local data at the point; it does not by itself distinguish data passing through the point from data concentrated there.

This motivates a second pointwise construction. For the next definition and the adjunction below, assume that $\{x\}$ is closed. This holds in the Hausdorff spaces used later and ensures that $U - \{x\}$ is open whenever U is an open neighbourhood of x.

Definition 1.14 (Ordinary costalk) *Define*

$$\Gamma_{\{x\}}(U, \mathcal{F}) \coloneqq \{s \in \mathcal{F}(U) : \mathrm{supp}(s) \subseteq \{x\}\}$$

Using the stalk's equivalent relation (Definition 1.8) the ***ordinary costalk*** *of* $\mathcal{F}$ *at* x *is*

$$i_x^! \ \mathcal{F} := \left(\bigsqcup_{U \in \mathcal{N}_x} \Gamma_{\{x\}}(U, \mathcal{F}) \right) / \sim$$

For a sheaf of R*-modules, addition and scalar multiplication are defined after restricting representatives to a common neighbourhood.*

Equivalently,

$$i_x^! \ \mathcal{F} \cong \operatorname{colim}_{U \in \mathcal{N}_x} \Gamma_{\{x\}}(U, \mathcal{F})$$

The inclusions $\Gamma_{\{x\}}(U, \mathcal{F}) \hookrightarrow \mathcal{F}(U)$ are compatible with restriction as U shrinks, and therefore induce a natural map $i_x^! \ \mathcal{F} \to \mathcal{F}_x$ This map forgets the support condition. The stalk retains every germ visible after shrinking around x, while the ordinary costalk retains those germs that can be represented by sections supported at x. After these constructions are derived, we will see that in some cases we can exchange the corresponding two local tests.

Proposition 1.15 (Stalk, skyscraper, and costalk) *For every* $x \in X$*, the stalk functor and the ordinary costalk are the left and right adjoints of the skyscraper functor:*

$$i_x^{-1} \dashv i_{x,*} \dashv i_x^!$$

Equivalently,

$$\operatorname{Hom}_R(\mathcal{F}_x, M) \cong \operatorname{Hom}_{\operatorname{Sh}(X;R)}\big(\mathcal{F}, i_{x,*}M\big),$$

$$\operatorname{Hom}_{\operatorname{Sh}(X;R)}\big(i_{x,*}M, \mathcal{F}\big) \cong \operatorname{Hom}_R\big(M, i_x^! \ \mathcal{F}\big)$$

The second adjunction requires $\{x\}$ to be closed.

Proof. A morphism $\mathcal{F} \to i_{x,*}M$ is determined on neighbourhoods of x. Compatibility with restriction gives a compatible family of maps $\mathcal{F}(U) \to M$, hence a unique map from the colimit $\mathcal{F}_x$ to M.

For the right adjunction, a morphism $i_{x,*}M \to \mathcal{F}$ sends every $m \in M$ to compatible sections that vanish on open sets avoiding x. Their germs at x are therefore represented by sections supported at x, giving a map $M \to i_x^! \ \mathcal{F}$.

Conversely, let a supported germ be represented by $s \in \mathcal{F}(W)$ with $\operatorname{supp}(s) \subseteq \{x\}$, and let U be any neighbourhood of x. On $U \cap W$ use $s|_{U \cap W}$. On $U - \{x\}$ use the zero section. These two sections agree on $(U \cap W) - \{x\}$: every germ there is

zero, so the locality axiom forces the restriction itself to vanish. The sheaf axiom therefore glues them uniquely to a section on U. Changing the representative of the germ gives the same glued section after shrinking once more. In this way every supported germ determines compatible sections on all neighbourhoods of x. Applying the construction elementwise to a homomorphism $M \to i_x^! \mathcal{F}$ gives a sheaf morphism $i_{x,*}M \to \mathcal{F}$. The two constructions are natural and inverse. □
For the supported-section construction attached to a closed subset, see [1, Ch. II, §2.3, Def. 2.3.8 and Prop. 2.3.9]. The point case above is all that is needed at this stage.

Example 1.16 (Stalk and costalk in elementary cases)

At $0 \in \mathbb{R}$*:*

1. *For* $C^0_{\mathbb{R}}$*, the stalk consists of germs of continuous functions, while* $i_0^! C^0_{\mathbb{R}} = 0$ *because a continuous function supported at one point must vanish.*
2. *For* $\underline{k}_{\mathbb{R}}$*, the stalk is* k *and the ordinary costalk is* 0*. A locally constant section is visible near* 0*, while no nonzero locally constant section can be supported only at* 0*.*
3. *For the skyscraper* $i_{0,*}M$*, both the stalk and the ordinary costalk are* M*.*
4. *For* $\mathcal{F} = \underline{k}_{\mathbb{R}} \oplus i_{0,*}M$*,*

$$\mathcal{F}_0 \cong k \oplus M, \quad i_0^! \mathcal{F} \cong M$$

In these examples the map $i_0^! \mathcal{F} \to \mathcal{F}_0$ *is* $0 \to k$ *for the constant sheaf and* $M \to M$ *for the skyscraper.*

1.2.2 Étale spaces and local systems

Stalks can be assembled into a topological space over X. In this presentation, sections of a sheaf become continuous sections of a local homeomorphism.

Definition 1.17 (Étale space over a topological space) *An* ***étale space over*** $\boldsymbol{X}$ *is a local homeomorphism* $\pi : E \to X$*. A morphism is a continuous map over* X*. The resulting category is denoted* $\text{Ét}\ (X)$*.*

Definition 1.18 (Étale space of a presheaf) *For a presheaf* $\mathcal{F}$ *of sets, define*

$$\text{Et}\ (\mathcal{F}) \coloneqq \bigsqcup_{x \in X} \mathcal{F}_x$$

with projection $\pi(s_x) = x$*. For* $U \subseteq X$ *open and* $s \in \mathcal{F}\,(U)$*, put* $[U, s] \coloneqq \{s_x : x \in U\}$*. These subsets form a basis for the* ***germ topology*** *on* $\text{Et}\ (\mathcal{F})$*.*

Proposition 1.19 (The germ topology is étale) *The projection* $\pi : \mathrm{Et}\ (\mathcal{F}) \to X$ *is a local homeomorphism.*

Proof. If $r_x \in [U, s] \cap [V, t]$, then $s_x = t_x$. By Proposition 1.10, the two sections agree on a smaller neighbourhood W of x, and $[W, s|_W]$ lies in the intersection. Hence the sets $[U, s]$ form a basis. The map

$$\pi|_{[U,s]} : [U, s] \to U$$

is a homeomorphism with inverse $x \mapsto s_x$. □

Definition 1.20 (Geometric section) *For* $\pi : E \to X$*, a* ***section over U*** *is a continuous map* $\sigma : U \to E$ *satisfying* $\pi \circ \sigma = \mathrm{id}_U$*. We write* $\Gamma(U, E)$ *for the set of such sections.*

Every $s \in \mathcal{F}\ (U)$ gives a section $x \mapsto s_x$ of $\mathrm{Et}\ (\mathcal{F})$. For a sheaf this accounts for every geometric section.

Proposition 1.21 (A sheaf is recovered from its germs) *If* $\mathcal{F}$ *is a sheaf of sets, the germ map* $\mathcal{F}\ (U) \to \Gamma(U, \mathrm{Et}\ (\mathcal{F})), \quad s \mapsto (x \mapsto s_x)$ *is a bijection for every open* U*.*

Proof. Injectivity is the locality axiom. A continuous section of the étale space is locally contained in basic opens $[U_i, s_i]$ and is therefore locally represented by sections s_i. Their germs agree on overlaps, so after refinement the sections agree and glue uniquely. □

Theorem 1.22 (Étale space and sheaf equivalence) *Taking the étale space and taking continuous sections define an equivalence of categories*

$$\mathrm{Sh}(X; \mathrm{Set}) \equiv \text{Ét}\ (X)$$

Proof.

Both constructions are functorial. By Proposition 1.21, for every sheaf $\mathcal{F}$ there are natural bijections $\mathcal{F}\ (U) \cong \Gamma(U, \mathrm{Et}\ (\mathcal{F}))$ for every open U, hence a natural isomorphism $\mathcal{F} \cong \Gamma(-, \mathrm{Et}\ (\mathcal{F}))$.

Conversely, for an étale space $\pi : E \to X$, evaluation defines

$$\varepsilon_E : \mathrm{Et}\ (\Gamma(-, E)) \to E, \quad \sigma_x \mapsto \sigma(x)$$

Every $e \in E$ has a neighbourhood on which π is a homeomorphism, whose inverse gives the unique germ of a local section mapped to e. On such a neighbourhood, ε_E is a homeomorphism. Thus ε_E is a natural isomorphism of étale spaces. □

This equivalence returns us to the motivate idea, of continuous functions, that opened the chapter.

The statement is set-valued. For the helix sheaf $\mathcal{P}$, the equivalence recovers the original covering $p : \mathbb{R} \to S^1$: a point $t \in \mathbb{R}$ determines the germ of the unique local inverse of p through t. For a sheaf $\mathcal{F}$ of R-modules Addition and scalar multiplication are defined locally by representatives $s_x + t_x := (s+t)_x, \quad rs_x := (rs)_x$.

Definition 1.23 (Local system) *A sheaf $\mathcal{L}$ of R-modules is a **local system** if every point has an open neighbourhood U such that*

$$\mathcal{L}|_U \cong \underline{M}|_U$$

*for some R-module M. It has **finite type** if the local fibres are finitely generated.*

The étale space of $\underline{M}$ is $X \times M_{\text{disc}}$. More generally, the étale space of a local system is a covering whose fibres carry compatible R-module structures.

Definition 1.24 (Parallel transport and monodromy) *Let $q : Y \to X$ be a covering map and $\gamma : [0,1] \to X$ a path. For every $y \in q^{-1}(\gamma(0))$, there is a unique lift η of γ beginning at y. Its endpoint defines a bijection*

$$T_\gamma : q^{-1}(\gamma(0)) \to q^{-1}(\gamma(1))$$

*For a loop based at x_0, these maps give the **monodromy representation***

$$\rho_q : \pi_1(X, x_0) \to \mathrm{Bij}(q^{-1}(x_0))$$

For a local system of R-modules, the image lies in $\mathrm{Aut}_R\big(\mathcal{L}_{x_0}\big)$.

For the helix, the positive loop around S^1 sends $n \in p^{-1}(1) = \mathbb{Z}$ to $n+1$. This is the monodromy of the sheaf of lifts $\mathcal{P}$. It is a permutation of the fibre, not an automorphism of the additive group with a chosen zero.

Example 1.25 (A linear local system on the circle) *Let V be an R-module, with discrete topology, and let $T \in \mathrm{Aut}_R(V)$. Form*

$$E_T := (\mathbb{R} \times V)/\sim$$

where $(t+1, v) \sim (t, Tv)$, and project $[t, v]$ to $e^{2\pi i t} \in S^1$. Over a sufficiently short arc, choosing one lift of the angular coordinate identifies the inverse image with $U \times V$. The locally constant sections therefore form a local system $\mathcal{L}_T$ with fibre V.

Follow $[0, v]$ *once around the positive generator of* $\pi_1(S^1)$*. The lifted path is* $t \mapsto [t, v]$*, and at its endpoint* $[1, v] = [0, Tv]$*. The constant-coefficient cellular calculation for the same circle appears in the homological-algebra appendix, in Example A.19. The present construction adds the return map* T*: after one circuit, the fibre is identified with itself through* T *rather than through the identity. Thus the fibre over one point does not determine the local system; its monodromy also records how the local data are glued around the circle. For* $V = k$ *and* $T(v) = \lambda v$ *with* $\lambda \in k^\times$*, this gives a rank-one local system whose monodromy is multiplication by* λ*. See [2, §1.7(d)].*

Example 1.26 (Riemann surface of the logarithm) *Begin with a germ of* $\log$ *at* $1 \in \mathbb{C}^\times$ *and continue it along paths. One counterclockwise turn around the origin adds* $2\pi i$*. The corresponding connected component of the étale space of* $\mathcal{O}$ *is the universal cover* $\exp : \mathbb{C} \to \mathbb{C}^\times$*. Analytic continuation is path lifting in this étale space. Over* 1*, the fibre is* $\exp^{-1}(1) = 2\pi i\mathbb{Z}$*, and a positive loop acts by the translation* $w \mapsto w + 2\pi i$*. As for the helix, this is a permutation of the fibre rather than the linear monodromy of Example 1.25.*

1.2.3 Sheafification and exactness

We now return to presheaves of R-modules. Their étale spaces are the étale spaces of the underlying set-valued presheaves, equipped with the fibrewise module operations described above. A presheaf may have suitable germs but fail to glue. Sheafification replaces it by the sheaf of continuous sections of this étale space.

Definition 1.27 (Sheafification) *The* ***sheafification*** *of a presheaf* $\mathcal{F}$ *is*

$$\mathcal{F}^+(U) := \Gamma(U, \mathrm{Et}\ (\mathcal{F}))$$

The natural map $\theta : \mathcal{F} \to \mathcal{F}^+$ *sends a section to its germs.*

Proposition 1.28 (Sheafification preserves stalks) *For every* $x \in X$*, the unit induces a natural isomorphism*

$$\mathcal{F}_x \cong \mathcal{F}^+_x$$

Proof. A germ of a continuous section of $\mathrm{Et}\ (\mathcal{F})$ is locally represented by a section of $\mathcal{F}$, and its value at x is the corresponding germ in $\mathcal{F}_x$. The two constructions are inverse. □

Theorem 1.29 (Universal property of sheafification) *For every sheaf* $\mathcal{G}$*, composition with* θ *induces a natural isomorphism*

$$\mathrm{Hom}_{\mathrm{Sh}(X;R)}(\mathcal{F}^+, \mathcal{G}) \cong \mathrm{Hom}_{\mathrm{PSh}(X;R)}(\mathcal{F}, \mathcal{G})$$

Equivalently, sheafification is left adjoint to the inclusion of sheaves into presheaves.

Proof. A section of $\mathcal{F}^+$ is locally represented by sections of $\mathcal{F}$. Applying a morphism $\mathcal{F} \to \mathcal{G}$ to these representatives gives compatible local sections of $\mathcal{G}$, which glue uniquely. This produces the required factorization, and local representability gives uniqueness. □

The constant presheaf $\mathcal{M}_{\mathrm{pre}}$ sheafifies to $\underline{M}$. Likewise, the presheaf of bounded continuous functions sheafifies to the sheaf of all continuous functions: boundedness is local near each point but need not be preserved by an infinite gluing.

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. The kernel presheaf is already a sheaf. The image and cokernel presheaves need not satisfy gluing, so their sheaf-theoretic versions are obtained by sheafification.

Proposition 1.30 (The category of sheaves is abelian) *The category* $\mathrm{Sh}(X;R)$ *is abelian. Kernels are computed sectionwise; images and cokernels are the sheafifications of the corresponding presheaves.*

Proof. The presheaf category is abelian because its kernels and cokernels are computed open set by open set in R-modules. Sheafification supplies cokernels in the full subcategory of sheaves. On every stalk, the canonical map from coimage to image is the ordinary module isomorphism. Since sheafification preserves stalks, the comparison is an isomorphism of sheaves. □

Proposition 1.31 (Stalkwise criterion) *A morphism of sheaves is a monomorphism, epimorphism, or isomorphism if and only if every stalk map has the corresponding property. A sequence of sheaves is exact if and only if it is exact on every stalk.*

Proof. Stalks are filtered colimits of R-modules and are therefore exact. Conversely, a sheaf with zero stalk at every point is zero: every section is locally zero and hence zero by locality. Apply this observation to kernels and cokernels. □

Proposition 1.32 (Sheaves have enough injectives) *The abelian category* $\mathrm{Sh}(X;R)$ *has enough injective objects.*

Proof. For each $x \in X$, choose an embedding $\mathcal{F}_x \hookrightarrow I_x$ into an injective R-module. The stalk functor is exact and left adjoint to the skyscraper functor, so $i_{x,*}I_x$ is injective. Products of injective objects are injective, hence $\prod_{x\in X} i_{x,*}I_x$ is injective. The chosen embeddings define a morphism $\mathcal{F} \to \prod_{x\in X} i_{x,*}I_x$. For every x, the composition of its stalk map with the projection to I_x is the

chosen embedding $\mathcal{F}_x \hookrightarrow I_x$. The morphism is therefore a monomorphism by Proposition 1.31. □

The constructions of Section B.7 and Section B.8 therefore apply to sheaves.

Example 1.33 (The exponential sequence) *Let C^0_R be the additive sheaf of continuous real-valued functions on S^1, and let $C^0_{S^1}$ be the sheaf of continuous circle-valued functions under pointwise multiplication. Exponentiation gives an exact sequence of sheaves of abelian groups*

$$0 \to \underline{\mathbb{Z}} \to C^0_R \xrightarrow{t \mapsto e^{2\pi i t}} C^0_{S^1} \to 0$$

The final map is an epimorphism because every circle-valued function has a continuous argument locally. It is not surjective on global sections: id_{S^1} *has no global lift. The obstruction is the one already visible in Example 1.26: a local branch exists everywhere, and one circuit changes it. Thus an epimorphism of sheaves need not be surjective after applying global sections.*

1.3 Basic sheaf operations

The point inclusion has already given two familiar transport constructions: taking a stalk and forming a skyscraper sheaf. Both are special cases of operations attached to an arbitrary continuous map. We define those general operations first. Proper direct image is introduced separately because the support of a section can matter when data are moved between spaces.

1.3.1 Direct and inverse image

Definition 1.34 (Direct image) *For a sheaf $\mathcal{F}$ on X, its **direct image** is the sheaf on Y defined by*

$$f_* \mathcal{F}(V) \coloneqq \mathcal{F}\left(f^{-1}(V)\right)$$

for every open $V \subseteq Y$.

The sheaf axiom is inherited from $\mathcal{F}$ because inverse images preserve unions and intersections.

For the map $a_X : X \to \{*\}$ sending every point to the unique point,

$$\left(a_{X,*}\,\mathcal{F}\right)(\{*\}) = \mathcal{F}(X)$$

Under the identification of sheaves on a point with R-modules, this says

$$a_{X,*}\,\mathcal{F} = \Gamma(X, \mathcal{F})$$

Similarly, direct image along $i_x : \{x\} \to X$ gives the skyscraper sheaf.

Inverse image is most transparent in the étale description. Let $\pi : \mathrm{Et}\ (\mathcal{G}) \to Y$ be the étale space of a sheaf $\mathcal{G}$. Form the pullback

$$E_f := X \times_Y \mathrm{Et}\ (\mathcal{G})$$

with its projection to X.

Definition 1.35 (Inverse image) *The **inverse image** $f^{-1}\ \mathcal{G}$ is the sheaf of continuous sections of the pullback étale space $E_f \to X$.*

Equivalently, it is the sheafification of the presheaf

$$U \mapsto \mathrm{colim}_{f(U) \subseteq V}\, \mathcal{G}\ (V),$$

where V ranges over the open subsets of Y containing $f(U)$. The set $f(U)$ need not be open.

Proposition 1.36 (Inverse image pulls back stalks) *For every $x \in X$ there is a natural isomorphism $f^{-1}\ \mathcal{G}_{\ x} \cong \mathcal{G}_{f(x)}$. Consequently, f^{-1} is exact.*

Proof. The fibre of $E_f \to X$ over x is the fibre of $\mathrm{Et}\ (\mathcal{G}) \to Y$ over $f(x)$, which is $\mathcal{G}_{f(x)}$. The stalk of the sheaf of sections of an étale space is its fibre. Exactness follows from the stalkwise criterion. □

For an inclusion $j : A \hookrightarrow X$, $j^{-1}\ \mathcal{F}$ is the restriction of $\mathcal{F}$ to A. For the point inclusion $i_x : \{x\} \to X$, inverse image recovers the stalk functor introduced in Definition 1.12:

$$i_x^{-1}\ \mathcal{F} \cong \mathcal{F}_x$$

If $\mathcal{L}$ is a local system on S^1, then $p^{-1}\ \mathcal{L}$ is constant on $\mathbb{R}$. Choose an isomorphism $p^{-1}\ \mathcal{L} \cong \underline{V}$. The deck transformation $\tau(t) = t + 1$ acts on V by the monodromy operator of $\mathcal{L}$. The pullback is constant, while the descent action retains the twisting on the circle.

Proposition 1.37 (Inverse and direct image are adjoint) *There is a natural isomorphism*

$$\mathrm{Hom}_{\mathrm{Sh}(X;R)}(f^{-1}\ \mathcal{G}, \mathcal{F}) \cong \mathrm{Hom}_{\mathrm{Sh}(Y;R)}(\mathcal{G}, f_*\ \mathcal{F})$$

Hence $f^{-1} \dashv f_*$. For $f = i_x$, this is the first adjunction of Proposition 1.15. Since f_* is a right adjoint, it is left exact.

Proof. Using the colimit description of inverse image, a morphism $f^{-1}\ \mathcal{G} \to \mathcal{F}$ is equivalent to compatible maps

$$\mathcal{G}(V) \to \mathcal{F}\left(f^{-1}(V)\right) = (f_*\,\mathcal{F})(V)$$

for the open subsets $V \subseteq Y$. These are exactly the components of a morphism $\mathcal{G} \to f_*\,\mathcal{F}$. □

1.3.2 Proper direct image

Assume in this subsection that the spaces are locally compact and Hausdorff.

Definition 1.38 (Proper map) *A continuous map $g : Z \to Y$ is **proper** if $g^{-1}(K)$ is compact for every compact subset $K \subseteq Y$.*

Definition 1.39 (Proper direct image) *For a sheaf $\mathcal{F}$ on X, define*

$$(f_!\,\mathcal{F})(V) \coloneqq \left\{s \in \mathcal{F}\left(f^{-1}(V)\right) : f|_{\operatorname{supp}(s)} : \operatorname{supp}(s) \to V \ \text{ is proper}\right\}$$

Equivalently, s belongs to $(f_!\,\mathcal{F})(V)$ when

$$\operatorname{supp}(s) \cap f^{-1}(K)$$

is compact for every compact $K \subseteq V$.

These assignments form a sheaf on Y, and the resulting functor $f_!$ is left exact. For the map to a point,

$$\Gamma_c(X, \mathcal{F}) \coloneqq a_{X,!}\,\mathcal{F}$$

is the module of global sections with compact support.

Definition 1.40 (Extension by zero) *Let $j : U \hookrightarrow X$ be an open embedding and let $\mathcal{F}$ be a sheaf on U. Then*

$$(j_!\,\mathcal{F})(V) \coloneqq \left\{s \in \mathcal{F}(V \cap U) : \overline{\operatorname{supp}(s)}^{V} \subseteq U\right\}$$

*This sheaf is the **extension by zero** of $\mathcal{F}$ from U to X.*

Proposition 1.41 (Extension by zero is exact) *For an open embedding $j : U \hookrightarrow X$, the functor*

$$j_! : \operatorname{Sh}(U; R) \to \operatorname{Sh}(X; R)$$

is exact. Its stalks are

$$(j_!\,\mathcal{F})_x \cong \begin{cases} \mathcal{F}_x & x \in U \\ 0 & x \in X - U \end{cases}$$

Proof. If $x \in U$, choose a neighbourhood contained in U; on such neighbourhoods extension by zero agrees with $\mathcal{F}$, so $(j_!\,\mathcal{F})_x \cong \mathcal{F}_x$. If $x \in X - U$, every germ allowed by the support condition vanishes after shrinking around x, so the stalk

is zero. A short exact sequence is therefore carried to a sequence that is exact at every stalk, and Proposition 1.31 gives exactness of $j_!$. □

Example 1.42 (Extension by zero at the boundary) *Let $j : U := (-1, 1) \hookrightarrow X := [-1, 1]$, let $K := [-1/2, 1/2]$, and choose $W := (-3/4, 3/4)$. Since the compact Hausdorff space X is normal, Urysohn's lemma gives a continuous function $f : X \to [0, 1]$ that equals 1 on K and vanishes on $X - W$. Thus $f|_U$ has support contained in $\overline{W}^X \subseteq U$ and defines a section of $j_!C^0_U$. In the étale-space description, its germ map agrees with the zero section near the endpoints, so extension by zero is continuous.*

The function $g(t) := 1 - t^2$ extends continuously to X by zero, but $\operatorname{supp}(g) = U$, so it does not define a section of $j_!C^0_U$. Similarly, for $\mathcal{F} := \underline{\mathbb{R}}_U$, a nonzero constant section near 1 defines a nonzero germ in $(j_* \mathcal{F})_1$, while its support reaches the boundary and hence $(j_! \mathcal{F})_1 = 0$. Since X is compact, $\Gamma(X, j_! \mathcal{F}) \cong \Gamma_c(U, \mathcal{F})$.

Let $j : U := B^n \hookrightarrow X := \overline{B^n}$ be the inclusion of the open unit ball in the closed unit ball. Given a compact subset $K \subseteq U$, choose an open set W such that $K \subseteq W \subseteq \overline{W}^X \subseteq U$. By Urysohn's lemma, there is a continuous function $f : X \to [0, 1]$ that equals 1 on K and vanishes on $X - W$. Thus $f|_U$ has support contained in $\overline{W}^X$ and defines a section of $j_!C^0_U$. Thus $f|_U$ has support contained in $\overline{W}^X$ and defines a section of $j_!C^0_U$. By Proposition 1.21, every section $s \in \mathcal{F}(U)$ of a sheaf is represented by the continuous map $x \mapsto s_x$ from U to $\operatorname{Et}(\mathcal{F})$. The function $f|_U$ is therefore a concrete instance of this: because its support stays inside U, its germ map agrees locally with the zero section near $X - U$ and extends continuously to X by following the zero section outside U.

The function $g(x_1, ..., x_n) := 1 - \sum_{i=1}^n x_i^2$ extends continuously to X by assigning the value 0 on the boundary. However, $\operatorname{supp}(g) = U$, so its germ map disagrees with the zero section on every neighbourhood of a boundary point and cannot be extended there by the zero section. Hence g does not define a section of $j_!C^0_U$.

Similarly, if $\mathcal{F} := \underline{R}_U$ and p is a boundary point, then $(j_* \mathcal{F})_p \cong R$, whereas $(j_! \mathcal{F})_p = 0$. Since X is compact, $\Gamma(X, j_! \mathcal{F}) \cong \Gamma_c(U, \mathcal{F})$.

For a point inclusion $i_x : \{x\} \to X$, proper and ordinary direct image agree:

$$i_{x,!} = i_{x,*}.$$

This is the skyscraper functor, whose right adjoint is the costalk $i_x^!$ of Definition 1.14. Thus i_x^{-1} reads the stalk at x, whereas $i_x^!$ reads information supported at x.

1.3.3 Tensor product and internal Hom

Let $\mathcal{F}, \mathcal{G}$ be sheaves of R-modules on X.

Definition 1.43 (Tensor product) *The tensor product is*

$$\mathcal{F} \otimes_R \mathcal{G} := U \mapsto \mathcal{F}(U) \otimes_R \mathcal{G}(U)^+$$

Its stalks satisfy

$$(\mathcal{F} \otimes_R \mathcal{G})_x \cong \mathcal{F}_x \otimes_R \mathcal{G}_x$$

Definition 1.44 (Internal sheaf Hom) *The internal Hom is the sheaf*

$$\mathcal{H}om_X(\mathcal{F}, \mathcal{G})(U) := \mathrm{Hom}_{\mathrm{Sh}(U;R)}(\mathcal{F}|_U, \mathcal{G}|_U)$$

If $U \subseteq V$, restricting a morphism over V gives a map

$$\mathcal{H}om_X(\mathcal{F}, \mathcal{G})(V) \to \mathcal{H}om_X(\mathcal{F}, \mathcal{G})(U)$$

Compatible morphisms glue, so these modules and restriction maps already form a sheaf. Taking the largest open set gives

$$\Gamma(X, \mathcal{H}om_X(\mathcal{F}, \mathcal{G})) \cong \mathrm{Hom}_{\mathrm{Sh}(X;R)}(\mathcal{F}, \mathcal{G})$$

Thus $\mathcal{H}om_X(\mathcal{F}, \mathcal{G})$ retains the morphisms available on each open subset together with their restrictions, while the ordinary Hom module is its value on the whole space.

There is a natural map

$$\mathcal{H}om_X(\mathcal{F}, \mathcal{G})_x \to \mathrm{Hom}_R(\mathcal{F}_x, \mathcal{G}_x)$$

which need not be an isomorphism for arbitrary sheaves. A morphism of stalks need not extend to a morphism on a neighbourhood.

If $\mathcal{L}_T$ and $\mathcal{L}_S$ are local systems on S^1 with fibres V, W and monodromies T, S, then their tensor product has fibre $V \otimes_R W$ and monodromy $T \otimes S$. Their internal Hom has fibre $\mathrm{Hom}_R(V, W)$ and monodromy

$$\varphi \mapsto S \circ \varphi \circ T^{-1}$$

Consequently,

$$\Gamma(S^1, \mathcal{H}om_{S^1}(\mathcal{L}_T, \mathcal{L}_S)) \cong \{\varphi \in \mathrm{Hom}_R(V, W) : S \circ \varphi = \varphi \circ T\}$$

Proposition 1.45 (Tensor and Hom adjunction) *There is a natural isomorphism*

$$\mathrm{Hom}(\mathcal{F} \otimes_R \mathcal{G}, \mathcal{H}) \cong \mathrm{Hom}(\mathcal{F}, \mathcal{H}om_X(\mathcal{G}, \mathcal{H}))$$

Thus $(- \otimes_R \mathcal{G}) \dashv \mathcal{H}om_X(\mathcal{G}, -)$.

Proof. The universal property of tensor product identifies a morphism from the tensor presheaf to $\mathcal{H}$ with a compatible family of bilinear maps, equivalently with a morphism $\mathcal{F} \to \mathcal{H}om_X(\mathcal{G}, \mathcal{H})$. The universal property of sheafification gives the stated bijection. □

The same constructions apply on a ringed space $(X, \mathcal{O}_X)$ with tensor product and internal Hom taken over $\mathcal{O}_X$. This is the coefficient setting used later for $\mathcal{D}$ -modules.

1.4 Stratification and constructibility

We now assume that X is a complex algebraic or analytic variety embedded in a smooth manifold M. As we saw, sheaves themselves are defined on an arbitrary topological space. However, on a singular space, different points may have different local geometry. If we want to describe a sheaf by finitely many kinds of local behaviour, we must first divide the space into areas on which the geometry is locally uniform. In this sense, we want each piece to recover a familiar feature of a connected smooth manifold, in which there is no distinguished point from the differential-topological point of view. This is what the following theorem proves.

Theorem 1.46 (Homogeneity of a connected smooth manifold) *Let M be a connected smooth manifold. For any $p, q \in M$, there is a diffeomorphism*

$$\varphi : M \to M$$

such that $\varphi(p) = q$.

Proof. Choose a smooth path from p to q and cover its image by finitely many coordinate balls. Inside each ball, a compactly supported vector field can be chosen whose local flow moves one prescribed point to the next and is the identity outside that ball. Composing the resulting diffeomorphisms sends p to q. □

The theorem concerns the geometry of the manifold. However, a sheaf may vary even on a connected smooth manifold. The stratification will specify the pieces on which constructibility separately requires local constancy.

This homogeneity fails at a singularity, as the origin stubbornly refuses to look like a point on one of the arms. In the cross $xy = 0$, for example, the origin cannot play the same geometric role as a point on one of the open arms. This suggests decomposing a singular space into smooth pieces. Smoothness of the pieces alone, however, is too weak. We must also prevent pieces from accumulating arbitrarily and specify consistently which pieces occur in the boundary of which others.

Recall that a family of subsets is **locally finite** if every point has a neighbourhood meeting only finitely many members of the family. A subset is **locally closed** if it is open in its closure, equivalently if it is the intersection of an open and a closed subset.

Definition 1.47 (Smooth stratification) *A **smooth stratification** of X is a locally finite decomposition*

$$X = \bigsqcup_{S \in \mathcal{S}} S$$

*into pairwise disjoint, connected, locally closed smooth submanifolds of M, called **strata**, such that the following frontier condition holds: if $S \cap \overline{T} \neq \varnothing$, then $S \subseteq \overline{T}$*

Here the closure is taken in X. If $S \subseteq \overline{T}$ and $S \neq T$, we write $S < T$ and require

$$\dim S < \dim T$$

Connectedness is only a convenient normalization: if a smooth piece has several connected components, we regard those components as separate strata. The frontier condition is more substantial. It says that incidence cannot change while one moves inside a stratum. If one point of S occurs in the boundary of T, then every point of S has the same incidence relation with T.

For the real cross, the origin is therefore a zero-dimensional stratum and the four open rays are four one-dimensional strata. Each ray has the origin in its closure.

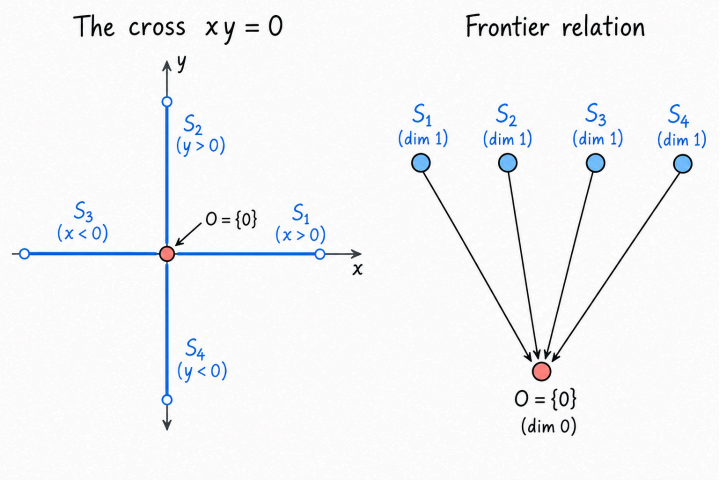


Figure 1.1: The cross $xy = 0$ and its frontier relation. Each open ray has the origin in its closure.

The frontier condition controls which strata approach one another. It still says almost nothing about how one stratum approaches another. This distinction becomes important as soon as the transverse geometry changes while we move along a lower-dimensional stratum.

Consider the real algebraic surface

$$X = \{(x, y, z) \in \mathbb{R}^3 : x^2 + z^3 - y^2 z^2 = 0\}$$

Its singular locus is the y-axis $\{x = z = 0\}$. If we fix a point with coordinate $y = t$ and cut transversely to that axis, the resulting plane curve is

$$x^2 + z^3 - t^2 z^2 = 0$$

For $t \neq 0$ and z sufficiently close to zero, the equation may be written $x = \pm z\sqrt{t^2 - z}$ so the slice consists of two smooth branches meeting at the origin. At $t = 0$, the slice becomes $x^2 + z^3 = 0$ which is a cusp. Thus the singular axis is itself smooth and satisfies the elementary frontier picture, while its transverse geometry changes at the origin. If a stratum is supposed to be a region on which the local geometry is uniform, the origin cannot remain in the same stratum as the rest of the singular axis.

We therefore need a condition on the manner of approach between incident smooth strata. Whitney condition (B) supplies exactly this control.

Let $S < T$ and let $y \in S$. Tangent spaces $T_x T$ for different points x live in different tangent spaces of the ambient manifold, so they cannot literally be compared without making an identification. Choose a smooth coordinate chart

around y. Inside this chart the tangent spaces may be regarded as subspaces of one fixed Euclidean space, and convergence is then convergence in the appropriate Grassmannian. The condition below is independent of the chosen chart.

Definition 1.48 (Whitney condition (B)) *Let $S < T$ be incident strata and let $y \in S$. Consider sequences $x_i \in T$, $y_i \in S$ with $x_i \neq y_i$ and $x_i \to y$, $y_i \to y$.*

Suppose that, in local coordinates around y,

- *the tangent spaces $T_{x_i}T$ converge in the Grassmannian to a subspace τ, and*
- *the secant lines through x_i and y_i converge in projective space to a line ℓ.*

The pair (S, T) satisfies ***Whitney condition (B) at y*** *if every such pair of convergent sequences satisfies $\ell \subseteq \tau$.*

It satisfies Whitney condition (B) if this holds at every $y \in S$.

The secant line records the direction from the lower stratum toward the nearby point of the higher stratum. The limiting tangent space records the directions that remain tangent to the higher stratum as the two points approach one another. The inclusion $\ell \subseteq \tau$ requires these two limiting descriptions to be compatible, an actual limiting direction of approach cannot disappear from the limiting tangent geometry.

For the surface above, suppose that the whole singular axis is one stratum S, and let T be the adjacent smooth stratum. Choose $t_i > 0$ with $t_i \to 0$ and set

$$a_i = (0, t_i, t_i^2) \in T, \quad b_i = (0, t_i, 0) \in S$$

The secant lines through a_i and b_i are the z-axis. For

$$f(x, y, z) = x^2 + z^3 - y^2 z^2$$

we have

$$\nabla f(a_i) = (0, -2t_i^5, t_i^4)$$

Hence the normal directions converge to the z-axis, so the tangent planes $T_{a_i}T$ converge to the xy-plane. The limiting secant line is not contained in this limiting tangent plane. Thus the pair (S, T) fails Whitney condition (B) at the origin, and the origin must be separated from the rest of the singular axis.

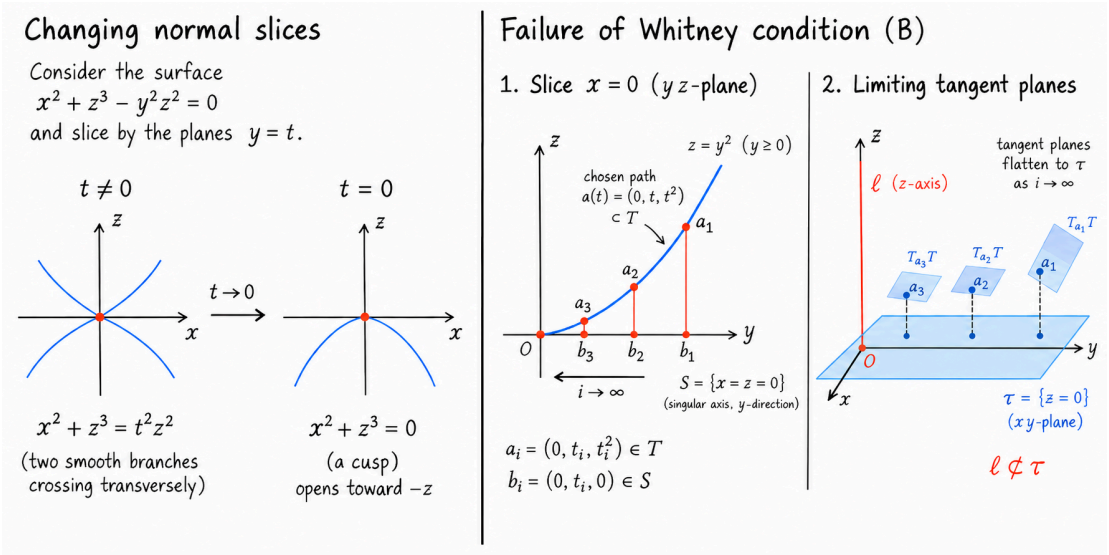


Figure 1.2: The Whitney cusp. Normal slices change at the origin. Whitney condition (B) compares limiting tangent spaces with limiting secant lines; failure requires a refinement of the lower stratum.

Definition 1.49 (Whitney stratification) *A **Whitney stratification** of X is a smooth stratification for which every incident pair $S < T$ satisfies Whitney condition (B).*

In the complex setting used later, we take the strata to be complex analytic submanifolds and apply Whitney condition (B) to their underlying real manifolds. Thus a stratum of complex dimension d has real dimension $2d$.

Complex algebraic varieties admit finite Whitney stratifications. Complex analytic spaces admit locally finite Whitney stratifications; for a noncompact analytic space, finiteness is an additional hypothesis. From this point onward we work with a fixed finite Whitney stratification.

Whitney (B) is a differential-geometric condition, whereas the later sheaf theory needs a topological conclusion: as one moves inside a stratum, the local stratified topology should remain unchanged. The passage between these two statements is supplied by the isotopy and control-data theorems for Whitney stratifications. To state the local model they produce, we first isolate the geometry transverse to a stratum.

1.4.1 Cones, slices, and links

Suppose for a moment that all transverse information around a point could be recovered from the set of directions leaving that point. Restoring only a radial

coordinate would then reconstruct the whole transverse neighbourhood. The appropriate construction is a cone.

Definition 1.50 (Open cone) *Let L be a compact topological space. Its **open cone** is*

$$c^{\circ}L := (L \times [0,1))/(L \times \{0\}),$$

*where all points of $L \times \{0\}$ are identified to a single point v, called the **vertex**. Away from the vertex there is a canonical identification*

$$c^{\circ}L - \{v\} \cong L \times (0,1)$$

By convention, $c^{\circ}\varnothing$ is a single point.

If L is stratified, then $c^{\circ}L$ carries the cone stratification: its strata are the vertex $\{v\}$ and the subsets

$$A \times (0,1)$$

where A ranges over the strata of L. We give $V \times c^{\circ}L$ the product stratification. A homeomorphism of stratified spaces is **stratified** if it carries every stratum onto a stratum.

A point away from the vertex consists of a direction in L together with a positive radial coordinate. Moving the radial coordinate to zero forgets the direction and approaches the vertex. The cone on one point is a half-line, $c^{\circ}S^0$ is a line, and $c^{\circ}S^{m-1}$ is homeomorphic to $\mathbb{R}^m$. The cone on four points is the local model of the four-armed cross.

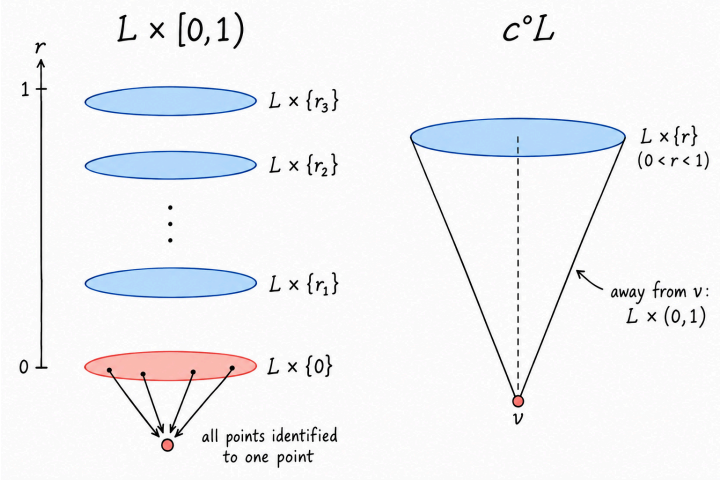


Figure 1.3: The open cone is obtained by collapsing $L \times \{0\}$ in $L \times [0,1)$ to one vertex.

We now need to extract the set L of transverse directions from a stratified space. If p belongs to an i-dimensional stratum S, a small ambient sphere around p is unsuitable: it contains directions tangent to S as well as directions normal to it. We first remove the tangent directions by taking a normal slice.

Definition 1.51 (Normal slice and link) *Let $p \in S$, where S is a stratum of a stratified subset $X \subseteq M$. A **normal slice to S at p** is a smooth submanifold $N \subseteq M$ through p such that $T_pM = T_pS \oplus T_pN$*

*For the link construction, choose such a slice N sufficiently small and transverse to every stratum that it meets near p. Choose a Riemannian metric on N and $\varepsilon > 0$ sufficiently small so that the corresponding sphere $\mathbb{S}_{\varepsilon(p)} \subseteq N$ is transverse to every stratum of $X \cap N$. The **link associated with N and ε** is $L_{p,N,\varepsilon} := X \cap N \cap \mathbb{S}_{\varepsilon(p)}$ Its strata are the connected components of the intersections $T \cap N \cap \mathbb{S}_{\varepsilon(p)}$ where T ranges over the strata of X.*

The normal slice removes motion along S. Intersecting it with a small sphere also removes the radial coordinate, leaving the directional part of the transverse geometry. The notation $L_{p,N,\varepsilon}$ still records the choices of slice, metric, and radius. The local conical theorem below shows that, once these choices are sufficiently small and transverse, its stratified homeomorphism type depends only on the point and its stratum.

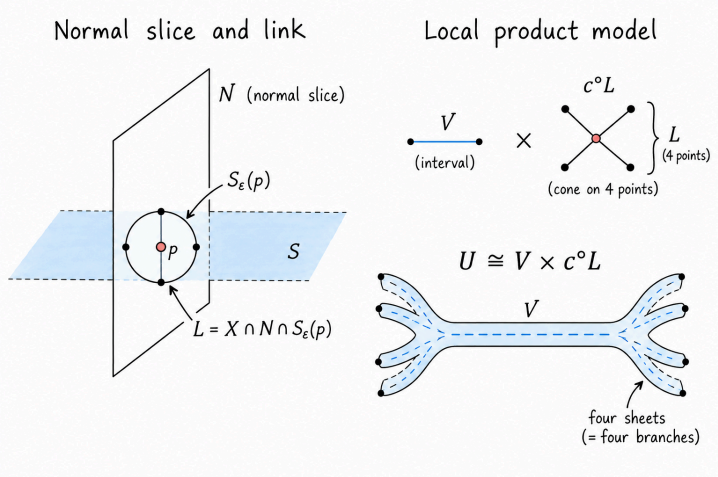


Figure 1.4: A normal slice removes the tangent directions to S. The resulting link gives the transverse conical factor in the local product model.

We can now state the topological consequence for which the Whitney condition was introduced.

Theorem 1.52 (Local conical structure) *Let X be a complex algebraic or analytic space equipped with a Whitney stratification, and let p lie in a stratum S of real dimension i. For all sufficiently small transverse choices in Definition 1.51, the stratified homeomorphism type of $L_{p,N,\varepsilon}$ is independent of the slice N, the metric, and the radius ε. We denote this type by L_p.*

After shrinking around p, there are neighbourhoods $V \subseteq S$ and $U \subseteq X$ of p, with V diffeomorphic to an open ball in $\mathbb{R}^i$, and a stratified homeomorphism $U \cong V \times c^\circ L_p$ which identifies $S \cap U$ with $V \times \{v\}$

At a point of a top-dimensional smooth stratum, the normal slice meets X only at p, so $L_p = \varnothing$ and the conical factor is a point. The local model therefore reduces to the smooth neighbourhood V. At the origin of the real cross, the link consists of four points, and its open cone recovers the four local arms.

We use this theorem as a black box; see [2, §8]. Its role is worth separating from its hypotheses. Whitney (B) controls limiting tangent geometry. The isotopy and control-data theory converts that differential control into local topological triviality. The resulting conical model is what we will actually use later.

The factor V in $V \times c^\circ L_p$ describes motion along the stratum, while the link and the radial coordinate describe the transverse geometry. If $T > S$, then each local component of $T \cap U$ corresponds to $V \times A \times (0,1)$ for a stratum A of L_p. A single stratum T may contribute more than one stratum of the link. Approaching S means sending the radial coordinate to zero.

Geometric input	**Local consequence**
Smooth strata	Each stratum is a smooth manifold.
Frontier condition	The incident strata are constant along a stratum.
Whitney condition (B)	Every limiting secant direction lies in the corresponding limiting tangent space.
Local conical structure	A neighbourhood of $p \in S$ has the stratified form $V \times c^\circ L_p$

Table 1.1: The geometric conditions and their local consequences.

The local conical theorem says that the stratified geometry is locally constant as one moves inside S. It imposes no condition on a sheaf. Constructibility adds such a condition by requiring the restriction to each stratum to be a finite-rank local system. This restriction may have monodromy around loops in the stratum even when the transverse links have no twisting. Variation within a stratum is

governed by local-system transport, while variation across strata is recorded by the attachment data between their restrictions.

1.4.2 Constructible sheaves

The conical model separates variation along a stratum from variation across strata. Constructibility requires local constancy in the first direction and finite algebraic data.

Definition 1.53 (Constructible sheaf) *Let $(X, \mathcal{S})$ be a finite Whitney-stratified space and let R be Noetherian. A sheaf $\mathcal{F} \in \mathrm{Sh}(X; R)$ is **$\mathcal{S}$-constructible** if $\mathcal{F}|_S$ is a local system of finitely generated R-modules for every stratum $S \in \mathcal{S}$.*

If $R = k$ is a field, finite generation means finite-dimensional stalks.

Proposition 1.54 (Constructible sheaves form an abelian subcategory) *For a fixed finite Whitney stratification $\mathcal{S}$ and a Noetherian coefficient ring R[1], the category $\mathrm{Sh}_{\mathcal{S}}(X; R)$ is a full abelian subcategory of $\mathrm{Sh}(X; R)$, and its inclusion is exact. It is also closed under extensions in $\mathrm{Sh}(X; R)$.*

Proof. Restriction to a stratum is exact. Given a morphism between two constructible sheaves, work on a sufficiently small connected open subset of a stratum on which both restrictions are constant. The morphism is then represented by a homomorphism of finitely generated R-modules. Its kernel and cokernel are constant sheaves associated with the corresponding module kernel and cokernel, which are finitely generated because R is Noetherian. Thus kernels and cokernels computed in the ambient sheaf category remain constructible.

For extension closure, consider an exact sequence $0 \to \mathcal{F} \to \mathcal{E} \to \mathcal{G} \to 0$ with constructible end terms. Restrict to a stratum and choose a point x and a neighbourhood on which the end terms are constant. The stalk $\mathcal{E}_x$ is finitely generated, hence finitely presented over R. Representatives of a finite set of generators, followed by shrinking to enforce their finitely many relations, define a morphism from the constant sheaf with fibre $\mathcal{E}_x$ to $\mathcal{E}$ near x. After shrinking again, this morphism commutes with the maps from the constant sheaf with fibre $\mathcal{F}_x$ and to the constant sheaf with fibre $\mathcal{G}_x$. Finite generation makes it enough to check these equalities on finitely many generators. Comparing the resulting short exact sequences on stalks shows that the middle morphism is an isomorphism. Hence $\mathcal{E}$ is locally constant on the stratum and has finitely generated stalks. This proves extension closure, and since finite direct sums,

[1] A ring R is **Noetherian** if every ideal of R is finitely generated; equivalently, every ascending chain of ideals eventually stabilizes.

kernels, and cokernels remain in the full subcategory and are computed in the ambient abelian category, the subcategory is abelian and its inclusion is exact. □

For a constructible sheaf $\mathcal{F}$, the nonzero-stalk locus is the union of the strata on which its restriction is nonzero. With the closed-support convention of Definition 1.13, this gives $\operatorname{supp}(\mathcal{F}) = \bigcup_{\mathcal{F}|_S \neq 0} \overline{S}$, where S ranges over $\mathcal{S}$ and closures are taken in X. Since the stratification is finite, this union of closures equals the closure of the nonzero-stalk locus. The frontier condition ensures that taking these closures adds only lower-dimensional strata, so the support dimension is the largest dimension of a stratum on which $\mathcal{F}$ is nonzero. We set $\dim \operatorname{supp}(0) = -\infty$.

Proposition 1.55 (Open and closed embeddings preserve constructibility) *Let $i : Z \to X$ be a closed union of strata and $j : U = X - Z \to X$ the complementary open union. With the induced stratifications, the exact functors*

$$j^{-1}, \quad j_!, \quad i^{-1}, \quad i_*$$

preserve constructible sheaves.

Proof. The assertion is checked on each stratum. Restriction retains the same local systems, extension by zero inserts the zero local system on the complementary strata, and closed pushforward changes only the ambient support. □

For the stratification

$$\mathbb{R} = (-\infty, 0) \sqcup \{0\} \sqcup (0, \infty)$$

a constructible sheaf is described by three finitely generated modules and two generization maps $V_- \longleftarrow V_0 \longrightarrow V_+$.

For $U = \mathbb{R} - \{0\}$,

1. $j_! \underline{R}_U$ has $V_0 = 0$ and $V_- = V_+ = R$;
2. $j_* \underline{R}_U$ has $V_0 = R \oplus R$, with the two projections as generization maps;
3. $i_* V$ has $V_0 = V$ and $V_- = V_+ = 0$.

Constructibility preserves the singular boundary data while restricting it to finitely many local systems and maps. A cellular model makes this description computational.

1.5 Cellular sheaves

The three-stratum description of the real line above is already a finite diagram, and now we isolate the setting in which such a diagram represents a sheaf. Let X

be a finite regular cell complex, meaning that the closure of each cell is a closed ball and its boundary is a union of lower-dimensional cells. We take coefficients in a field k.

Definition 1.56 (Cellular sheaf) *Let P_X be the face poset of X, with $\sigma \leq \tau$ when $\sigma \subseteq \overline{\tau}$. A **cellular sheaf** is a covariant functor $\mathcal{F} : P_X \to k\text{-Vect}$ whose values are finite-dimensional. Thus it assigns a vector space $\mathcal{F}(\sigma)$ to every cell and a map $\rho_{\sigma,\tau} : \mathcal{F}(\sigma) \to \mathcal{F}(\tau)$ to every face relation, compatibly with identities and composition. We write $\mathrm{Cell}(X; k)$ for the category of these diagrams.*

For $\sigma \in P_X$, write $P_{\geq\sigma} := \{\tau \in P_X : \sigma \leq \tau\}$ and let $\mathrm{st}(\sigma) \subseteq |X|$ be the union of the corresponding open cells. If $\sigma \leq \tau$, then $\mathrm{st}(\tau) \subseteq \mathrm{st}(\sigma)$. Restriction from $\mathrm{st}(\sigma)$ to $\mathrm{st}(\tau)$ therefore has the same direction as the cellular map $\mathcal{F}(\sigma) \to \mathcal{F}(\tau)$. This accounts for the covariance in Definition 1.56.

Theorem 1.57 (Cellular realization of cell-constructible sheaves) *Let X be a finite regular cell complex. Sheaves of k-vector spaces on $|X|$ that are constructible with respect to the stratification by open cells and have finite-dimensional stalks are equivalent to finite-dimensional cellular sheaves on the face poset P_X. Under this equivalence, the vector space attached to a cell σ is identified with the stalk on the open cell, and the face maps encode the corresponding generization maps.*

We denote the topological sheaf corresponding to $\mathcal{F}$ by $\mathrm{Real}_X(\mathcal{F})$. The derived form of this realization will be introduced in Section 2.4 after complexes of sheaves are available.

We keep the two categories notationally distinct and use Theorem 1.57 when passing between them. See [3, Rem. 4.9 and §6] for the cell-poset description and [4, Ch. 5, especially §5.2], for the derived comparison.

The theorem applies to the stratification by open cells. A sheaf constructible for a Whitney stratification enters this framework only after choosing an adapted finite regular cell decomposition, when one is available.

Example 1.58 (A cellular interval as a finite diagram) *Let $X = [0, 1]$ have vertices v_0, v_1 and one open edge e. Its face poset has the two relations $v_0 \leq e$, $v_1 \leq e$. A cellular sheaf is therefore the finite diagram*

$$V_0 \xrightarrow{\rho_0} V_e \xleftarrow{\rho_1} V_1$$

Under Theorem 1.57, the three vector spaces are the stalk values of $\mathrm{Real}_X(\mathcal{F})$ *on the open cells. The maps carry germs from the endpoints into the edge, where they can be compared. For the constant sheaf all three spaces are* k *and both maps are the identity.*

For constant coefficients this recovers the interval of Example A.18. There the endpoint values already belonged to one coefficient space. A cellular sheaf allows different spaces at the endpoints and compares them after applying the generization maps. Section Section 2.4 adds the incidence signs and turns this diagram into a cochain complex.

Let Δ be a small complex disc and set $\Delta^* = \Delta - \{0\}$. The decomposition $\Delta = \Delta^* \sqcup \{0\}$ is a Whitney stratification. Its open stratum is not an open cell, and a local system on Δ^* with fibre V may have nontrivial monodromy T. A cellular model therefore requires an adapted decomposition into contractible cells whose incidence diagram contains a cycle around the origin. In Section 2.4, the corresponding cellular complex reduces to $\left[V \xrightarrow{T-\mathrm{id}_V} V\right]$.

2 Sheaf Cohomology and Constructible Complexes

2.1 Motivation

The preceding chapter described a single sheaf and the information it carries on open sets, near points, and across a stratification. The language of complexes developed in Appendix A allows us to attach several stages of data to a single topological space and to each of its open subsets, with the differentials describing the relations among these stages. As we shall see, however, the global-section functors are not exact. Moreover, information affected by the removal of a point need not disappear, but may instead occur in higher degrees; in other words, the ordinary costalk functor is not exact either. The tools of Appendix B allow us to replace these operations with their derived versions and retain the cohomology they produce.

In the preceding chapter, we saw that, for the map from X to a point, direct image gives global sections, while direct image with a support condition gives compactly supported sections. For the inclusion of a point into X, the two corresponding inverse-image operations give the stalk and the costalk. We also discussed tensor product and Hom. Therefore we continue studying these operations, now from the perspective of exactness and derived functors.

2.2 Complexes of sheaves

Appendix A defined cochain complexes and their morphisms in an arbitrary abelian category. Since $\mathrm{Sh}(X;R)$ is abelian, no new algebraic definition is required here.

A **cohomological complex of sheaves** is a complex in $\mathrm{Sh}(X;R)$, written

$$\cdots \to K^{n-1} \xrightarrow{d^{n-1}} K^n \xrightarrow{d^n} K^{n+1} \to \cdots$$

The n-th **cohomology sheaf** of K is

$$\mathcal{H}^n(K) := \ker(d^n)/\operatorname{im}(d^{n-1})$$

where the kernel and image are taken in $\mathrm{Sh}(X;R)$, i.e - after sheafification.

Example 2.1 (Support created by a two-term complex) *On $\mathbb{C}$, put two copies of the holomorphic structure sheaf in degrees 0 and 1: $K_z = \left[\mathcal{O}_{\mathbb{C}} \xrightarrow{z} \mathcal{O}_{\mathbb{C}}\right]$. Multiplication by z is injective. Away from the origin it is invertible, while at the origin $\underset{\mathbb{C}}{\mathcal{O}}_0/z\underset{\mathbb{C}}{\mathcal{O}}_0 \cong \mathbb{C}$. Therefore*

$$\mathcal{H}^0(K_z) = 0, \quad \mathcal{H}^1(K_z) \cong {i_0}_*\mathbb{C}.$$

The cohomology is supported where multiplication by z fails to be invertible.

Taking stalks term by term gives a complex $K_x^\bullet$ of R-modules. Exactness of the stalk functor gives the basic comparison between the sheaf complex and these ordinary complexes.

Proposition 2.2 (Cohomology sheaves are computed on stalks) *For every complex K and every $x \in X$,*

$$\mathcal{H}^n(K)_x \cong H^n(K_x^\bullet).$$

Consequently, a morphism $K \to L$ is a quasi-isomorphism if and only if $K_x^\bullet \to L_x^\bullet$ is a quasi-isomorphism for every $x \in X$.

Proof. Exactness of the stalk functor identifies the stalks of kernels, images, and quotients with the corresponding kernels, images, and quotients of stalks. Hence $\mathcal{H}^n(K)_x \cong H^n(K_x^\bullet)$. A morphism of sheaves is an isomorphism exactly when it is an isomorphism on every stalk, so the same identity gives the stalkwise criterion for quasi-isomorphisms. □

We write

$$D^b(X;R) := D^b(\mathrm{Sh}(X;R))$$

for the bounded derived category of sheaves. The notation $D^+(X;R)$ will be used when bounded-below complexes are required. Exact functors on sheaves, such as f^{-1}, act term by term and preserve quasi-isomorphisms; left- or right-exact functors require the derived constructions of the preceding chapters.

The symbol $\mathcal{H}^q(K)$ therefore denotes a sheaf on X. The next section applies derived global sections and produces cohomology modules instead.

2.3 Sheaf cohomology

For a sheaf $\mathcal{F}$, the module of global sections is $\Gamma(X,\mathcal{F}) = \mathcal{F}(X)$. If X is locally compact and Hausdorff, the compactly supported global sections form the submodule $\Gamma_c(X,\mathcal{F}) \subseteq \Gamma(X,\mathcal{F})$. The distinction is already visible on the real line. For the constant sheaf,

$$\Gamma(\mathbb{R},\underline{k}) \cong k, \quad \Gamma_c(\mathbb{R},\underline{k}) = 0.$$

A global locally constant section is determined by one scalar. A nonzero such section has support all of $\mathbb{R}$, so none is compactly supported. For a skyscraper $i_{0,*}M$, every global section is supported at one point and therefore

$$\Gamma\big(\mathbb{R}, i_{0,*}M\big) \cong \Gamma_c\big(\mathbb{R}, i_{0,*}M\big) \cong M.$$

The next issue is exactness. Return to the exponential sequence of Example 1.33. The map of sheaves is locally surjective because a circle-valued function has a continuous argument locally. On global sections, the identity map of S^1 has no continuous lift to $\mathbb{R}$. Thus global sections do not preserve an epimorphism of sheaves.

Definition 2.3 (Sheaf cohomology) *For a sheaf $\mathcal{F}$ regarded as a complex concentrated in degree 0, define*

$$H^n(X,\mathcal{F}) := R^n\Gamma(X,\mathcal{F}), \quad H^n_c(X,\mathcal{F}) := R^n\Gamma_c(X,\mathcal{F}).$$

Equivalently,

$$H^n(X,\mathcal{F}) = H^n(R\Gamma(X,\mathcal{F})), \quad H^n_c(X,\mathcal{F}) = H^n(R\Gamma_c(X,\mathcal{F})).$$

In degree 0 these groups recover the ordinary section functors:

$$H^0(X,\mathcal{F}) = \Gamma(X,\mathcal{F}), \quad H^0_c(X,\mathcal{F}) = \Gamma_c(X,\mathcal{F}).$$

The higher groups record the failure of exactness.

Example 2.4 (The helix as a cohomology class) *Return to the sheaf of lifts from the helix. The corresponding short exact sequence of sheaves of abelian groups has the form*

$$0 \to \underline{\mathbb{Z}} \to C^0_R \to C^0_{S^1} \to 0,$$

where the middle map sends a real-valued function to its value modulo $\mathbb{Z}$*, viewed as an* S^1*-valued function. Locally every circle-valued function admits a lift, so the final map is an epimorphism of sheaves. Applying global sections produces part of the long exact sequence*

$$\Gamma\big(S^1, C^0_R\big) \to \Gamma\big(S^1, C^0_{S^1}\big) \xrightarrow{\delta} H^1(S^1, \underline{\mathbb{Z}}).$$

The identity map of S^1 *has no global lift to* $\mathbb{R}$*, exactly as in the helix calculation. Exactness therefore gives* $\delta(\mathrm{id}_{S^1}) \neq 0$*. This is the obstruction to a global lift. Differences between local lifts define an integer-valued cocycle whose value around the circle is* 1*, matching the geometric change of level in the helix.*

A complex of sheaves can carry several cohomology sheaves, so we also need to apply derived sections directly to a complex.

Definition 2.5 (Hypercohomology) *For* $K \in D^+(X; R)$*, define*

$$\mathbb{H}^n(X, K) := H^n(R\Gamma(X, K)), \quad \mathbb{H}^n_c(X, K) := H^n(R\Gamma_c(X, K)).$$

These are the ***hypercohomology modules*** *of* K*.*

D^+

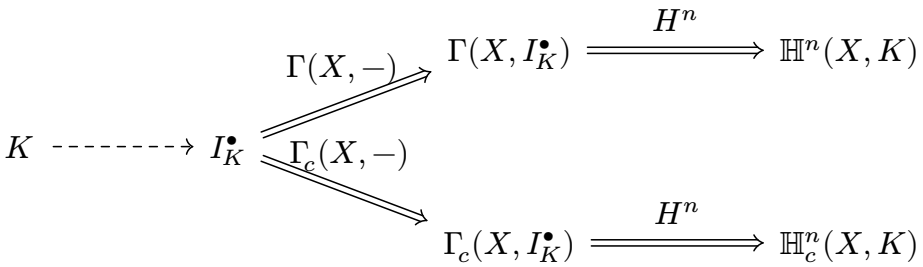
The notation separates the cohomology sheaves of K from the cohomology obtained after taking sections. To compute the latter, choose an injective replacement $K \to I^\bullet_K$. Then

$$\begin{array}{ccccc}
K^{q-1} & \longrightarrow & K^q & \longrightarrow & K^{q+1} \\
\downarrow{\scriptstyle \iota^{q-1}} & & \downarrow{\scriptstyle \iota^{q}} & & \downarrow{\scriptstyle \iota^{q+1}} \\
I_K^{q-1} & \longrightarrow & I_K^{q} & \longrightarrow & I_K^{q+1} \\
\downarrow & & \downarrow{\scriptstyle \Gamma(X,-)} & & \downarrow \\
\Gamma\big(X, I_K^{q-1}\big) & \longrightarrow & \Gamma(X, I_K^q) & \longrightarrow & \Gamma\big(X, I_K^{q+1}\big)
\end{array}$$

Thus

$$R\Gamma(X, K) \cong \Gamma(X, I^\bullet_K), \quad \mathbb{H}^n(X, K) = H^n(\Gamma(X, I^\bullet_K)).$$

The derived object $R\Gamma(X, K)$ contains the full complex of global sections. Passing to $\mathbb{H}^n(X, K)$ selects its n-th cohomological degree, as the derived functors of Appendix B are obtained from a derived object by taking cohomology.



The definitions above use injective replacements. For a sheaf represented by a finite cellular diagram, derived sections can instead be computed by a finite cochain complex. This gives explicit calculations of both ordinary and compactly supported cohomology.

2.4 Cellular calculations

The interval diagram of Example 1.58 records the local values and the maps between them. To obtain a global complex, we combine those maps with the incidence signs of the cell structure. Choose orientations of the cells. For cells $\sigma \le \tau$ with $\dim \tau = \dim \sigma + 1$, let $[\sigma : \tau]$ denote the coefficient of σ in the cellular boundary of τ.

Theorem 2.6 (Cellular computation of sheaf cohomology) *Let $\mathcal{F}_{\mathrm{cell}}$ be a cellular sheaf on a finite regular cell complex X, and let $\mathcal{F}_{\mathrm{top}}$ be its realization under Theorem 1.57. After choosing orientations of the cells, set*

$$C^p(X; \mathcal{F}_{\mathrm{cell}}) := \bigoplus_{\dim \sigma = p} \mathcal{F}_{\mathrm{cell}}(\sigma)$$

with differential

$$d^p|_{\mathcal{F}_{\mathrm{cell}}(\sigma)} := \sum_{\tau \supset \sigma,\, \dim \tau = p+1} [\sigma : \tau] \rho_{\sigma,\tau}$$

Then $d^{p+1} \circ d^p = 0$ and

$$C^\bullet(X; \mathcal{F}_{\mathrm{cell}}) \cong R\Gamma\big(|X|, \mathcal{F}_{\mathrm{top}}\big)$$

in the derived category of k-vector spaces.

We use the cellular comparison theorem from [3, Defs. 4.14–4.16 and §§5–6]; see also Shepard, Ch. 5, especially §5.2.

Example 2.7 (A cellular interval) *Decompose* $[-1,1]$ *into vertices* v_-, v_0, v_+ *and oriented edges* e_-, e_+*. For the constant cellular sheaf,*

$$C^0(X;k) = k^3, \quad C^1(X;k) = k^2,$$

$$d(a_-, a_0, a_+) = (a_0 - a_-, a_+ - a_0).$$

For constant coefficients, this is the same terminal-minus-initial differential computed in Example A.18. The new point is that it now arises from the cellular sheaf diagram of Example 1.58, and Theorem 2.6 identifies the resulting finite complex with derived global sections. Combining the generization maps with these signs turns the comparison of Example A.18 into the cellular differential. Hence

$$H^0(X;k) \cong k, \quad H^1(X;k) = 0.$$

The kernel consists of triples with one common value; the cokernel vanishes because every pair of edge differences can be realized.

Proposition 2.8 (Compact support and extension by zero) *Let* $j : U \hookrightarrow X$ *be an open embedding. For* $K \in D^b(U;R)$ *there is a natural isomorphism*

$$R\Gamma_c(U,K) \cong R\Gamma_c(X, j_!K)$$

If X *is compact, this becomes*

$$R\Gamma_c(U,K) \cong R\Gamma(X, j_!K)$$

Proof. Extension by zero identifies compactly supported sections on U with compactly supported sections on X. Since $j_!$ is exact by Proposition 1.41, the same identity holds after deriving. If X is compact, every closed support in X is compact. □

Example 2.9 (Compact support on the real line) *Consider the case* $n = 1$ *of Example 1.42. Thus* $U = (-1,1)$*,* $X = [-1,1]$*, and* $j : U \hookrightarrow X$*. Give* X *the regular cell structure with endpoint vertices* v_-, v_+ *and open edge* $e = U$*. For* $\mathcal{F} := \underline{k}_U$*, extension by zero has cellular values* $(j_!\,\mathcal{F})(v_-) = 0$*,* $(j_!\,\mathcal{F})(v_+) = 0$*, and* $(j_!\,\mathcal{F})(e) = k$*. Hence its cellular cochain complex is* $C^\bullet(X; j_!\,\mathcal{F}) = [0 \longrightarrow k]$*, with the terms in degrees* 0 *and* 1*.*

By Theorem 2.6, $H^0(X, j_!\,\mathcal{F}) = 0$ *and* $H^1(X, j_!\,\mathcal{F}) \cong k$*. By Proposition 2.8,*

$$R\Gamma_c(U,\mathcal{F}) \cong R\Gamma(X, j_!\,\mathcal{F}) \cong k[-1]$$

Identifying U *with* $\mathbb{R}$*, we obtain* $H^0_c(\mathbb{R};k) = 0$ *and* $H^1_c(\mathbb{R};k) \cong k$*.*

The same cellular calculation applies to the open ball B^n. Choose a finite regular cellulation of its boundary and adjoin the open n-cell B^n. The extension $j_!\underline{k}_{B^n}$ has value 0 on every boundary cell and value k on the open n-cell. The chosen boundary cellulation therefore contributes only zero terms, and

$$C^q\big(\overline{B^n}; j_!\underline{k}_{B^n}\big) = \begin{cases} k & q = n \\ 0 & q \neq n \end{cases}$$

Consequently, $R\Gamma_c(B^n, \underline{k}_{B^n}) \cong k[-n]$, and hence

$$H^q_c(B^n; k) \cong \begin{cases} k & q = n \\ 0 & q \neq n \end{cases}$$

By contrast, $R\Gamma(B^n, \underline{k}_{B^n}) \cong k$, independently of n. Ordinary cohomology sees that every ball is contractible; compactly supported cohomology records its dimension in the degree of the surviving class.

For $0 < r < R$, extension by zero along $B^n_r \hookrightarrow B^n_R$ induces

$$R\Gamma_c\big(B^n_r, \underline{k}_{B^n_r}\big) \to R\Gamma_c\big(B^n_R, \underline{k}_{B^n_R}\big).$$

With the orientations inherited from $\mathbb{R}^n$, this map identifies the copies of k in degree n. See [3, Def. 4.14 and §§5–6].

Example 2.10 (Monodromy as a cellular differential) *Let $\mathcal{L}_T$ be the local system of Example 1.25, with fibre V and monodromy $T \in \mathrm{Aut}_{k(V)}$. Use the regular two-vertex subdivision of Example A.24. There all four generization maps were identities; here three remain identities and the fourth, across the cut where the circuit closes, is T. Hence the cellular differential is*

$$D_T : V \oplus V \to V \oplus V, \quad D_T(a, b) = (b - a, Ta - b)$$

Set $(u, v) = (a, b - a)$. Then $D_{T(u,v)} = (v, (T - \mathrm{id}_V)u - v)$. The target coordinate change $(x, y) \mapsto (x + y, x)$ turns this into $((T - \mathrm{id}_V)u, v)$. Thus the cellular complex is the direct sum of a contractible identity complex and $\Big[V \overset{T-\mathrm{id}_V}{\longrightarrow} V\Big]$. Since the subdivision is regular, Theorem 2.6 gives

$$R\Gamma(S^1, \mathcal{L}_T) \cong \Big[V \overset{T-\mathrm{id}_V}{\longrightarrow} V\Big].$$

Consequently,

$$H^0(S^1, \mathcal{L}_T) = \ker(T - \mathrm{id}_V), \quad H^1(S^1, \mathcal{L}_T) = \mathrm{coker}(T - \mathrm{id}_V).$$

Example 2.11 (Circle and torus in the derived category) *The ordinary cellular cochain complexes of the circle and the torus were computed in Example A.19. We now reinterpret those calculations as models for derived global sections. Taking $T = \mathrm{id}_k$ in Example 2.10 gives $R\Gamma(S^1, \underline{k}) \cong \left[k \xrightarrow{0} k\right]$, recovering the circle complex of Example A.19.*

For the torus, the one-vertex, two-edge, one-face CW structure used in Example A.19 is not regular, so it does not directly satisfy the hypotheses of Theorem 2.6. Choose a finite regular cellulation K of T^2. Its cellular cochain complex represents $R\Gamma(T^2, \underline{k})$ and has the cohomology computed in Example A.19:

$$H^0(T^2; k) \cong k, \quad H^1(T^2; k) \cong k^2, \quad H^2(T^2; k) \cong k.$$

Over a field, a bounded complex is quasi-isomorphic to its cohomology with zero differential. Choose splittings $Z^n = B^n \oplus H^n$, $C^n = Z^n \oplus L^n$ so that $L^n \to B^{n+1}$ is an isomorphism. These summands form contractible pairs. Hence

$$R\Gamma(T^2, \underline{k}) \cong \left[k \xrightarrow{0} k^2 \xrightarrow{0} k\right]$$

in degrees $0, 1, 2$.

The same construction applies degreewise to a bounded complex of cellular sheaves. A **bounded cellular complex** is an object $B^\bullet \in \mathrm{Ch}^b(\mathrm{Cell}(X; k))$. Equivalently, it assigns a bounded complex $B^\bullet_\sigma$ to every cell and a cochain map $r_{\sigma,\tau} : B^\bullet_\sigma \to B^\bullet_\tau$ to every face relation. Shepard's derived comparison associates to it a constructible complex $\mathrm{Real}_X(B^\bullet)$.

For chosen orientations, set $C^{p,q}(X; B) := \bigoplus_{\dim \sigma = p} B^q_\sigma$. For $c = (c_\sigma)_\sigma$ and $\dim \tau = p + 1$, define $(\delta c)_\tau := \sum_{\sigma < \tau,\, \dim \sigma = p} [\sigma : \tau] r_{\sigma,\tau}(c_\sigma)$. Put $C^m(X; B) := \bigoplus_{p+q=m} C^{p,q}(X; B)$ and let $D := \delta + (-1)^p d_B$ on bidegree (p, q).

Proposition 2.12 (Cellular hypercohomology) *The operator D satisfies $D^2 = 0$, and the total complex satisfies*

$$C^\bullet(X; B) \cong R\Gamma(|X|, \mathrm{Real}_X(B^\bullet))$$

in the derived category of k-vector spaces.

Proof. The incidence identity gives $\delta^2 = 0$. The maps $r_{\sigma,\tau}$ are cochain maps, so $\delta d_B = d_B \delta$. On bidegree (p, q), the mixed terms in D^2 have signs $(-1)^p$ and $(-1)^{p+1}$ and therefore cancel. Together with $d_B^2 = 0$, this proves $D^2 = 0$. Applying the preceding cellular comparison in every internal degree and total-

ising gives the stated identification. Boundedness ensures that only finite direct sums occur in each total degree. □

2.5 Local cohomology and derived costalks

Return to the constant sheaf on $\mathbb{R}$. Its ordinary costalk at 0 is zero, because a nonzero locally constant section cannot be supported on one point. The previous calculation shows that supported information may still appear after derivation. The correct local comparison uses a neighbourhood together with the same neighbourhood after the point has been removed.

However, its higher right-derived functors do not vanish. For a neighbourhood U of x, the derived costalk is computed locally from the restriction map

$$R\Gamma(U, K) \to R\Gamma(U - \{x\}, K)$$

Definition 2.13 (Derived costalk) *For $x \in X$, let $i_x : \{x\} \hookrightarrow X$ be the inclusion. The **derived costalk** of a bounded-below complex K is*

$$Ri_x^! K$$

This is the right-derived functor of the ordinary costalk $i_x^!$ from Definition 1.14.

Proposition 2.14 (Local cone model for the derived costalk) *Let U be an open neighbourhood of x and put $U^\times := U - \{x\}$. Write $R\Gamma_{\{x\}}(U, K)$ for the right-derived complex of sections on U supported at x. It fits into a distinguished triangle*

$$R\Gamma_{\{x\}}(U, K) \to R\Gamma(U, K) \to R\Gamma(U^\times, K) \to R\Gamma_{\{x\}}(U, K)[1].$$

As U shrinks to x, these supported complexes compute the derived costalk $Ri_x^! K$. In the locally conical situations used below, a sufficiently small neighbourhood gives the stable local model, and then

$$Ri_x^! K \cong \mathrm{Cone}(R\Gamma(U, K) \to R\Gamma(U^\times, K))[-1].$$

Proof. Choose an injective resolution $K \to I^\bullet$. For each degree q, sections supported at x are the kernel of

$$\Gamma(U, I^q) \to \Gamma(U^\times, I^q).$$

Injective sheaves are flabby, so this restriction map is surjective; see [1, Ch. II, Prop. 2.4.6(vii)]. We therefore obtain a short exact sequence of complexes

$$0 \to \Gamma_{\{x\}}(U, I^\bullet) \to \Gamma(U, I^\bullet) \to \Gamma(U^\times, I^\bullet) \to 0.$$

The associated distinguished triangle gives the first statement. Passing to arbitrarily small neighbourhoods is exactly the local supported-section construction underlying $Ri_x^!$. When the local stratified type stabilizes on a sufficiently small neighbourhood, one such U computes the resulting costalk. The underived supported-section functor is defined in [1, Ch. II, Def. 2.3.8], and its derived cohomology is developed in [1, Ch. II, §2.6]. □

Example 2.15 (Stalk and derived costalk on the real line) *Let $I = (-\varepsilon, \varepsilon)$ and $I^\times = I - \{0\}$. For the constant sheaf,*

$$R\Gamma(I, \underline{k}) \cong k, \quad R\Gamma(I^\times, \underline{k}) \cong k \oplus k$$

in degree 0. Restriction is the diagonal map $k \to k \oplus k, \quad a \mapsto (a, a)$. Its kernel is zero and its cokernel is k. Hence the cone has one copy of k in degree 0, and shifting by -1 places that copy in degree 1. Therefore $Ri_0^! \underline{k}_{\mathbb{R}} \cong k[-1]$. The stalk is $(\underline{k}_{\mathbb{R}})_0 \cong k$ in degree 0. The stalk records the local value that persists through the point, and the derived costalk records the relative contribution detected by comparing the neighbourhood with the punctured one. The ordinary costalk vanished as calculated in Example 1.16, but the derived costalk does not, since comparing the interval with the punctured interval recovers the missing datum as $k[-1]$.

Example 2.16 (A skyscraper at the point) *For $K = i_{0,*}M$, the punctured neighbourhood contains no part of the sheaf. Hence*

$$Ri_0^! K \cong M$$

in degree 0. This is the row of Example 1.16 on which stalk and costalk already agreed and hence deriving makes no changes.

Derived sections and derived costalks take sheaf complexes to complexes of R-modules. Verdier duality also requires a sheaf-valued complex of local morphisms. Internal derived Hom provides that object, while derived global sections recover the ordinary derived Hom complex.

2.6 Internal derived Hom and Ext

The internal sheaf Hom of Definition 1.44 is left exact in its second variable. Deriving that variable for complexes means combining an injective replacement with the Hom complex of Definition A.22.

Let $K^\bullet$ be a bounded complex, let $L \in D^+(X;R)$, and let $L \to I_L^\bullet$ be a bounded-below injective replacement. Applying internal Hom degreewise produces the double complex

$$C^{p,q} := \mathcal{H}om_X(K^p, I_L^q).$$

Apply the Hom-complex construction of Definition A.22 with internal sheaf Hom in place of ordinary Hom.

Definition 2.17 (Internal derived Hom) *Let $K \in D^b(X;R)$ and $L \in D^+(X;R)$. Choose a bounded representative $K^\bullet$ and a bounded-below injective replacement $L \to I_L^\bullet$. The **internal derived Hom** is the object of $D^+(X;R)$ represented by*

$$R\mathcal{H}\mathrm{om}_X(K,L) := \mathcal{H}om_X^\bullet(K^\bullet, I_L^\bullet).$$

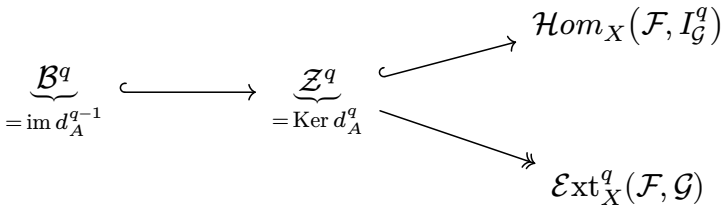
As in Theorem B.40, the construction is independent of the chosen bounded representative and injective replacement. The only additional input is that $\mathcal{H}om_X(-,I)$ is exact for injective I. Indeed, for an open inclusion $j : U \hookrightarrow X$, exactness of $j_!$ and the adjunction $j_! \dashv j^{-1}$ imply that $I|_U$ is injective, so $\mathcal{H}om_X(-,I)(U)$ is exact.

Take now ordinary sheaves $\mathcal{F}$ and $\mathcal{G}$, placed in degree 0. The double complex has a single nonzero row, the totalization collapses to $A^\bullet := \mathcal{H}om_X(\mathcal{F}, I_\mathcal{G}^\bullet)$, $d_A^q(\varphi) = d_I^q \circ \varphi$ For every open $U \subseteq X$, the sections of $A^\bullet$ form the Hom complex (Definition A.22)

$$A^\bullet(U) = \mathrm{Hom}_{\mathrm{Sh}(U;R)}\Big(\mathcal{F}|_U, I_\mathcal{G}^\bullet\big|_U\Big).$$

The **local Ext sheaves** are its cohomology sheaves,

$$\mathcal{E}\mathrm{xt}_X^q(\mathcal{F},\mathcal{G}) := \mathcal{H}^q(R\mathcal{H}\mathrm{om}_X(\mathcal{F},\mathcal{G})) = \mathrm{Ker}\big(d_A^q\big)/\,\mathrm{im}\Big(d_A^{q-1}\Big).$$



An epimorphism of sheaves need not be surjective on sections over an open set. Thus a section of $\mathcal{E}xt_X^{q(\mathcal{F},\mathcal{G})}$ lifts locally to a cocycle, but a global lift need not exist. The following lemmas relate internal derived Hom to the derived Hom complex of global morphisms by applying derived sections.

Lemma 2.18 (Restriction of internal derived Hom) *For every open $U \subseteq X$,*

$$R\mathcal{H}\mathrm{om}_X(K, L)|_U \cong R\mathcal{H}\mathrm{om}_U(K|_U, L|_U).$$

Proof. Restriction preserves injectives, as noted above, so $I_L^\bullet|_U$ is a bounded-below injective replacement of $L|_U$, and internal Hom restricts degreewise to the corresponding double complex on U. Each total degree involves finitely many factors, $K^\bullet$ being bounded, so totalization commutes with restriction. □

Lemma 2.19 (Internal and external derived Hom) *For every open $U \subseteq X$,*

$$R\Gamma\big(U, R\mathcal{H}\mathrm{om}_X(K, L)|_U\big) \cong \mathrm{RHom}_{\mathrm{Sh}(U;R)}(K|_U, L|_U).$$

In particular,

$$R\Gamma(X, R\mathcal{H}\mathrm{om}_X(K, L)) \cong \mathrm{RHom}_{\mathrm{Sh}(X;R)}(K, L).$$

Proof. By Lemma 2.18 it suffices to argue on U. Suppose I is injective and $\mathcal{F}$ is a sheaf on U; we claim $\mathcal{H}om_U(\mathcal{F}, I)$ is flabby. Let $j : V \hookrightarrow U$ be an open inclusion. A morphism $\mathcal{F}|_V \to I|_V$ corresponds under $j_! \dashv j^{-1}$ to a morphism $j_!\mathcal{F}|_V \to I$, and the canonical map $j_!\mathcal{F}|_V \to \mathcal{F}$ is monic, so injectivity of I extends the latter morphism to $\mathcal{F} \to I$. The restriction map from U to V is thus surjective.

Every total degree of $\mathcal{H}om_U^\bullet\big(K|_U, I_L|_U\big)$ is a finite product of flabby sheaves and so is acyclic for global sections. The complex is bounded below, since $K^\bullet$ is bounded and $I_L^\bullet$ is bounded below, which is what allows derived global sections to be computed term by term. The identification

$$\Gamma(U, \mathcal{H}om_U(\mathcal{F}, \mathcal{G})) \cong \mathrm{Hom}_{\mathrm{Sh}(U;R)}(\mathcal{F}, \mathcal{G})$$

then matches the result with the external Hom complex. Compare [1, §II.2.6]. □

Corollary 2.20 (Ext as hypercohomology of internal derived Hom) *For ordinary sheaves $\mathcal{F}, \mathcal{G}$ and every $n \geq 0$,*

$$\mathbb{H}^n(X, R\mathcal{H}\mathrm{om}_X(\mathcal{F}, \mathcal{G})) \cong \mathrm{Ext}^n_{\mathrm{Sh}(X;R)}(\mathcal{F}, \mathcal{G}).$$

Proof. Apply H^n to the second isomorphism of Lemma 2.19 and use Corollary B.41. □

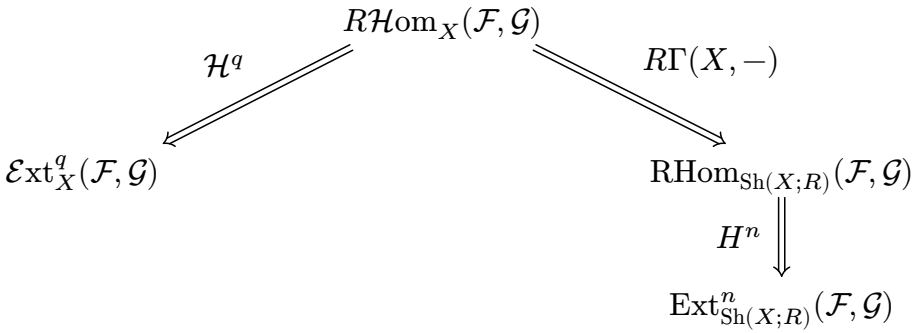


Figure 2.3: Cohomology sheaves give local Ext on X; derived global sections give the external Hom complex, whose cohomology gives global Ext.

Over a point there is no restriction datum to record, $\mathrm{Sh}(\{*\};R)$ is the category of R-modules, and the construction returns ordinary derived Hom:

$$R\mathcal{H}\mathrm{om}_{\{*\}}(V,W) \cong \mathrm{RHom}_R(V,W).$$

The bifunctor $R\mathcal{H}\mathrm{om}_X$ is contravariant and triangulated in its first variable, and covariant and triangulated in its second. A distinguished triangle in either variable therefore gives a distinguished triangle of internal derived Homs.

Example 2.21 (Derived morphisms between local systems on the circle) *Let $\mathcal{L}_T, \mathcal{L}_S$ be the local systems of Example 1.25, with fibres V, W and monodromies T, S. Their internal Hom has fibre $\mathrm{Hom}_{k(V,W)}$ and monodromy $A(\varphi) := S \circ \varphi \circ T^{-1}$.*

Since $\mathcal{L}_T$ is locally free of finite rank,

$$R\mathcal{H}\mathrm{om}_{S^1}(\mathcal{L}_T,\mathcal{L}_S) \cong \mathcal{H}om_{S^1}(\mathcal{L}_T,\mathcal{L}_S)$$

in degree 0*. Applying Example 2.10 gives*

$$\mathrm{RHom}_{\mathrm{Sh}(S^1;k)}(\mathcal{L}_T,\mathcal{L}_S) \cong \left[\mathrm{Hom}_{k(V,W)} \xrightarrow{A-\mathrm{id}} \mathrm{Hom}_{k(V,W)}\right]$$

And hence, $\mathrm{Ext}^0 \cong \ker(A-\mathrm{id})$, $\mathrm{Ext}^1 \cong \mathrm{coker}(A-\mathrm{id})$, *and the higher Ext groups vanish. Degree* 0 *consists of the intertwiners satisfying $S \circ \varphi = \varphi \circ T$.*

2.7 Constructible complexes

Constructibility was defined in the preceding chapter for a single sheaf. A derived object can have several cohomology sheaves, so the same finite geometric control must hold in every degree.

Definition 2.22 (Constructible derived categories) *Assume that R is Noetherian. An object $K \in D^b(X;R)$ is **$\mathcal{S}$-constructible** if every cohomology sheaf $\mathcal{H}^n(K)$ is $\mathcal{S}$-constructible. We write*

$$D^b_{\mathcal{S}}(X;R)$$

for the full subcategory of such objects.

The category $D^b_c(X;R)$ is the full subcategory of objects constructible with respect to some allowed finite Whitney stratification. We assume that any two allowed stratifications admit a common Whitney refinement.

Proposition 2.23 (Stability of constructible complexes) *For fixed $\mathcal{S}$, $D^b_{\mathcal{S}}(X;R)$ is a full triangulated subcategory of $D^b(X;R)$, stable under canonical truncations. Its standard heart is $\mathrm{Sh}_{\mathcal{S}}(X;R)$. Under the common-refinement assumption, the same statements hold for $D^b_c(X;R)$.*

Proof. A distinguished triangle $A \to B \to C \to A[1]$ gives, for every n, a short exact sequence $0 \to \operatorname{coker}\big(\mathcal{H}^{n(A)} \to \mathcal{H}^{n(B)}\big) \to \mathcal{H}^{n(C)} \to \ker\big(\mathcal{H}^{n+1}(A) \to \mathcal{H}^{n+1}(B)\big) \to 0$. If A and B are constructible, the two outer terms are constructible by the kernel and cokernel statements of Proposition 1.55, and the middle term is constructible by extension closure. Rotation handles the other choices of two constructible vertices. Thus the subcategory is closed under distinguished triangles and shifts.

For canonical truncations,

$$\mathcal{H}^m(\tau_{\leq n}K) \cong \begin{cases} \mathcal{H}^m(K) & m \leq n \\ 0 & m > n \end{cases}$$

and analogously for $\tau_{\geq n}K$. Hence both truncation functors preserve constructibility. The heart consists precisely of objects whose only nonzero cohomology sheaf is in degree 0, and therefore identifies with $\mathrm{Sh}_{\mathcal{S}}(X;R)$.

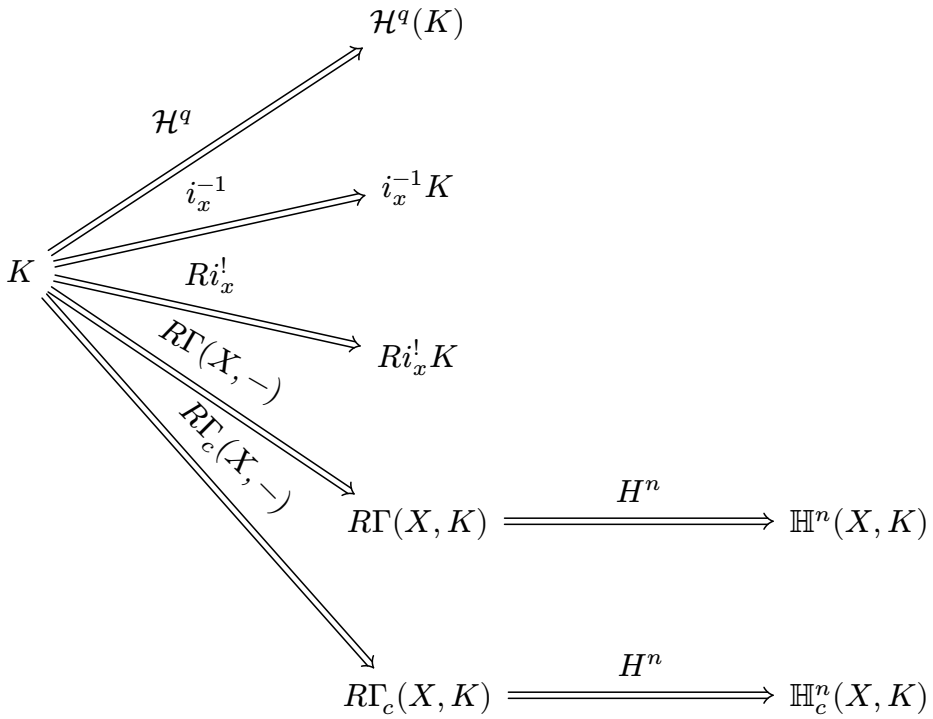
For objects constructible with respect to different allowed stratifications, pass to a common Whitney refinement and apply the same argument. □

The notation

$$D^b(\mathrm{Sh}_{\mathcal{S}}(X;R)) \quad \text{and} \quad D^b_{\mathcal{S}}(X;R)$$

should not be identified by definition. The first category is derived inside the abelian category of constructible sheaves; the second is the full subcategory of the ambient derived category selected by its cohomology sheaves. Their equivalence, when used, is an additional theorem.

The category $D_c^b(X;R)$ is the derived category used from this point onward. The constructions of this chapter separate sheaf-valued layers, pointwise tests, and global tests:



Constructibility is a condition on the first branch: every $\mathcal{H}^q(K)$ must be locally constant of finite rank along the strata. Verdier duality will compare the two pointwise branches after dualization and will relate ordinary global sections to compactly supported sections of the dual complex.

3 Verdier Duality

3.1 Motivation

Duality's story begins with a simple observation - a finite-dimensional vector space can be described through its vectors or through the linear tests applied to them. By a linear test we mean a linear functional $\ell : V \to k$, which assigns a scalar to every vector. The collection of all such tests is the dual space

$$V^\vee := \mathrm{Hom}_{k(V,k)}$$

A vector $v \in V$, in turn, defines a linear test on the space of tests: it sends each $\ell \in V^\vee$ to $\ell(v)$. This gives the canonical evaluation map

$$V \to V^{\vee\vee}, \quad v \mapsto (\ell \mapsto \ell(v))$$

which evaluates each functional ℓ at the vector v. Finite-dimensionality makes this map an isomorphism, so the answers to all linear tests determine the original vector. There is generally no canonical identification $V \cong V^\vee$ since the original space consists of the objects being tested, while the dual space consists of the tests themselves.

Sheaves of vector spaces, as we have seen, organise algebraic data over a geometric space while retaining much of the algebraic structure of their values. We ask whether linear duality can also be made local and sensitive to the geometry. In oter words, for every open subset $U \subseteq X$, we would like to test the information carried by $K|_U$ through its derived sections and organise these tests

so that passing to a smaller open subset gives the restriction maps of a sheaf. But if $U \subseteq V$ are open, ordinary derived sections give a restriction map

$$R\Gamma(V, K|_V) \to R\Gamma(U, K|_U)$$

Linear duality reverses arrows, so dualising this map would give

$$\mathrm{RHom}_k(R\Gamma(U, K|_U), k) \to \mathrm{RHom}_k(R\Gamma(V, K|_V), k)$$

which points from the smaller open set to the larger one, opposite to the direction of sheaf restriction.

Compact support resolves this problem. Compactly supported information on a smaller open set can be extended by zero to a larger one. Concretely, a continuous function with compact support inside an open ball vanishes on a neighbourhood of its boundary, so extending it by zero to a larger ball remains continuous. This is the mechanism encoded by extension by zero in Definition 1.40 and used in Example 2.9. Thus, if $U \subseteq V$, $R\Gamma_c(U, K|_U) \to R\Gamma_c(V, K|_V)$.

The derived linear dual $\mathrm{RHom}_k(-, k)$ reverses both the maps and the cohomological degrees of a complex. Dualising the preceding maps therefore gives

$$\mathrm{RHom}_k(R\Gamma_c(V, K|_V), k) \to \mathrm{RHom}_k(R\Gamma_c(U, K|_U), k)$$

which now points from the larger open set to the smaller one. In this chapter we will organise the test complexes $\mathrm{RHom}_k(R\Gamma_c(U, K|_U), k)$ as a single derived object on X, compatibly with restriction as U varies.

Compact support also retains local dimension. As computed in Example 2.9, the compactly supported cohomology of a small real d-ball is concentrated in degree d, although its ordinary cohomology is concentrated in degree 0. Verdier duality therefore assigns different cohomological shifts to data supported in different dimensions.

A second boundary problem appears when a space is decomposed as $X = U \cup Z$, with U open and Z closed. The restrictions of K to U and Z do not determine how the two pieces are attached. We therefore need operations that distinguish extension from U with or without boundary contributions, and that isolate the part supported on Z. Verdier duality exchanges the ordinary operations with their support-sensitive counterparts. These open–closed constructions will also be used in the next chapter to describe how constructible complexes pass between strata.

Throughout, k is a field; in the Riemann–Hilbert setting $k = \mathbb{C}$. Spaces are locally compact Hausdorff and countable at infinity. Whenever an exceptional inverse image is used, we impose the corresponding finite-cohomological-dimension hypothesis explicitly. When algebraic constructibility is invoked, the spaces and maps are complex algebraic varieties and algebraic morphisms. Analytic constructibility is used only under the hypotheses stated there. We write $\mathrm{Sh}(X;k)$ for sheaves of k-vector spaces and $D^b_c(X)$ for the bounded constructible derived category in the relevant setting. Dimension means complex dimension unless real dimension is stated.

3.2 Verdier duality

We seek an object whose derived sections on every open $U \subseteq X$ are $\mathrm{RHom}_k(R\Gamma_c(U, K|_U), k)$. Internal derived Hom is compatible with restriction and converts sheaf-valued Hom into the external derived Hom complex:

$$R\mathcal{H}\mathrm{om}_X(K,L)|_U \cong R\mathcal{H}\mathrm{om}_U(K|_U, L|_U)$$

$$R\Gamma\Big(U, R\mathcal{H}\mathrm{om}_X(K,L)|_U\Big) \cong \mathrm{RHom}_{\mathrm{Sh}(U;k)}(K|_U, L|_U)$$

(Lemma 2.18, Lemma 2.19). It remains to choose an object L for which the latter complex is the k-linear dual of compactly supported sections.

Let $a_X : X \to \{*\}$ be the structure map. Since $Ra_{X,!}(-) = R\Gamma_c(X, -)$, the required coefficient object is obtained from the right adjoint of $Ra_{X,!}$. Under the finite-cohomological-dimension hypothesis of Theorem 3.16 (stated below), this right adjoint exists:

$$a_X^! : D^+(\mathbf{Vect}_k) \to D^+(\mathrm{Sh}(X;k)), \quad Ra_{X,!} \dashv a_X^!$$

Definition 3.1 (Dualizing complex) *The **dualizing complex** of X is*

$$\omega_X^\bullet := a_X^! k$$

For a point, $a_{\{*\}}$ is the identity and hence $\omega^\bullet_{\{*\}} \cong k$. Thus the zero-dimensional case is the ordinary coefficient object used in linear duality.

Proposition 3.2 (The defining property) *For every $K \in D^b(\mathrm{Sh}(X;k))$ there is a natural isomorphism*

$$\mathrm{RHom}_k(R\Gamma_c(X,K), k) \cong \mathrm{RHom}_{\mathrm{Sh}(X;k)}(K, \omega_X^\bullet)$$

Proof sketch. The derived adjunction gives, for every n,

$$\mathrm{Hom}_{D^+(\mathrm{Sh}(X;k))}\big(K, a_X^! k[n]\big) \cong \mathrm{Hom}_{D^+(\mathbf{Vect}_k)}\big(Ra_{X,!}K, k[n]\big)$$

These groups are the degree-n cohomology groups of the two derived Hom complexes. In the standard resolution models for derived Hom, the adjunction maps assemble to the natural quasi-isomorphism

$$\mathrm{RHom}_{\mathrm{Sh}(X;k)}\big(K, a_X^! k\big) \cong \mathrm{RHom}_k\big(Ra_{X,!}K, k\big)$$

Substituting $\omega_X^\bullet = a_X^! k$ and $Ra_{X,!}K = R\Gamma_c(X, K)$ gives the formula. □

The representing object is compatible with restriction to open subsets.

Proposition 3.3 (Restriction of the dualizing complex) *For every open embedding $j : U \hookrightarrow X$,*

$$\omega_X^\bullet|_U \cong \omega_U^\bullet$$

Proof. Let $j : U \hookrightarrow X$ be open. Extension by zero $j_!$ is exact for an open embedding, as is checked on stalks. For $L \in D^b(\mathrm{Sh}(U;k))$, the adjunction $j_! \dashv j^{-1}$ therefore gives

$$\mathrm{RHom}_{\mathrm{Sh}(U;k)}\big(L, \omega_X^\bullet|_U\big) \cong \mathrm{RHom}_{\mathrm{Sh}(X;k)}(j_! L, \omega_X^\bullet) \cong \mathrm{RHom}_k(R\Gamma_c(U, L), k)$$

Thus $\omega_X^\bullet|_U$ has the same representing property as $\omega_U^\bullet$. Uniqueness of representing objects finishes the proof. □

The Verdier dual is the internal derived Hom into the dualizing complex.

Definition 3.4 (Verdier dual) *For $K \in D^b(\mathrm{Sh}(X;k))$, the **Verdier dual** is*

$$\mathbb{D}_X(K) \coloneqq R\mathcal{H}\mathrm{om}_X(K, \omega_X^\bullet)$$

It sends $D^b(\mathrm{Sh}(X;k))^{\mathsf{op}}$ to $D^+(\mathrm{Sh}(X;k))$.

For a point this becomes ordinary derived linear duality $\mathbb{D}_{\{*\}}(V) \cong \mathrm{RHom}_k(V, k)$, and therfore the point case recovers the opening construction.

By Proposition 3.3 and restriction of internal derived Hom (Lemma 2.18), $\mathbb{D}_X(K)|_U \cong \mathbb{D}_U(K|_U)$ and therfore the Verdier dual is therefore local in the sense that it commutes with restriction to open subsets.

Proposition 3.5 (Local tests from supported information) *For every open $U \subseteq X$ and $K \in D^b(\mathrm{Sh}(X;k))$,*

$$R\Gamma\big(U, \mathbb{D}_X(K)|_U\big) \cong \mathrm{RHom}_k(R\Gamma_c(U, K|_U), k)$$

Proof. Restriction identifies the left-hand side with

$$R\Gamma(U, R\mathcal{H}\mathrm{om}_U(K|_U, \omega_U^\bullet))$$

by Lemma 2.18 and Proposition 3.3. The internal–external comparison (Lemma 2.19) turns this into $\mathrm{RHom}_{\mathrm{Sh}(U;k)}(K|_U, \omega_U^\bullet)$, and the defining property Proposition 3.2 identifies that complex with the dual of $R\Gamma_c(U, K|_U)$. □

This is the object sought in the motivation. One first collects the information on U that is compactly supported, then takes all derived linear tests on it, and Verdier duality organises those test complexes compatibly as U varies.

Proposition 3.6 (Global Verdier duality) *For every $K \in D^b(\mathrm{Sh}(X;k))$,*

$$R\Gamma(X, \mathbb{D}_X(K)) \cong \mathrm{RHom}_k(R\Gamma_c(X, K), k).$$

Consequently,

$$\mathbb{H}^m(X, \mathbb{D}_X(K)) \cong (\mathbb{H}_c^{-m}(X, K))^\vee.$$

Proof. Take $U = X$ in Proposition 3.5. Over a field, linear duality is exact, so $H^m(\mathrm{RHom}_k(C, k)) \cong H^{-m}(C)^\vee$ for a complex C; this gives the second formula. □

3.3 Computing Verdier duals

First let us examine the compactly supported cochain complexes on sufficiently small neighbourhoods. Their derived duals determine the local models of $\omega_X^\bullet$; their cohomology gives the stalks described in Theorem 3.7, while the restriction maps, and in several cohomological degrees also the extension data, determine how those local pieces assemble as a derived object. Once $\omega_X^\bullet$ is known, compute

$$\mathbb{D}_X(K) = R\mathcal{H}\mathrm{om}_X(K, \omega_X^\bullet).$$

For a constructible complex the same procedure may be carried out stratum by stratum or, after choosing an adapted cell decomposition, through the derived cellular realization of Section 2.4. The local-test formula provides an independent check on the resulting object.

Theorem 3.7 (Stalks of the dualizing complex) *For every $x \in X$ and $m \in \mathbb{Z}$,*

$$H^m(\omega_X^\bullet)_x \cong \mathrm{colim}_{U \ni x} \, (H_c^{-m}(U, k))^\vee,$$

where U runs through open neighbourhoods of x ordered by reverse inclusion.

The theorem tests the defining property against extensions by zero from small neighbourhoods; see [5, Thm. 3.2.9(i)] and [6, Prop. V.2.4]. It gives a concrete calculation - compute compactly supported cohomology near the point, dualise it, and then analyzing how those local groups vary.

For $X = \{*\}$, as we previously $\omega^\bullet_{\{*\}} \cong k$, $\mathbb{D}_{\{*\}}(V) \cong \mathrm{RHom}_k(V, k)$. Nothing geometric has yet been added.

Example 3.8 (The real line revisited) *Fix $x \in \mathbb{R}$ and choose a sufficiently small open interval I around it. In Example 2.9 we computed*

$$R\Gamma_c(I, \underline{k}_I) \cong k[-1],$$

so $H^q_c(I, k) = 0$ for $q \neq 1$ and $H^1_c(I, k) \cong k$. Substituting these groups into the stalk formula gives

$$H^{-1}(\omega^\bullet_{\mathbb{R}})_x \cong k, \quad H^m(\omega^\bullet_{\mathbb{R}})_x = 0 \ \text{ for } \ m \neq -1.$$

The canonical orientation of $\mathbb{R}$ identifies these one-dimensional local spaces compatibly as x varies, so the surviving cohomology sheaf is constant. Hence $\omega^\bullet_{\mathbb{R}} \cong \underline{k}_{\mathbb{R}}[1]$. The same one-dimensional local class appeared as $H^1_c(\mathbb{R}; k)$ in Example 2.9 and, through the punctured-neighbourhood comparison, in the derived costalk of Example 2.15. Here its dual gives the local generator of $\omega^\bullet_{\mathbb{R}}$ and accounts for the shift.

On a real d-manifold, every small coordinate ball has compactly supported cohomology k in degree d and zero in the other degrees. Dimension therefore still produces only one shift. The local generators may nevertheless acquire nontrivial monodromy as they are transported around X.

Definition 3.9 (Orientation local system) *If X is a real d-manifold, the* ***orientation local system*** or_X *is the rank-one local system of local orientations. Equivalently, its stalk at x is*

$$(\mathrm{or}_X)_x := \mathrm{colim}_{U \ni x} \big(H^d_c(U, k)^\vee\big),$$

where U ranges over coordinate balls; for nested balls, the transition maps are obtained by dualising extension by zero on compactly supported cohomology.

Proposition 3.10 (Dualizing complex of a manifold) *Let X be a real d-manifold without boundary. Then*

$$\omega^\bullet_X \cong \mathrm{or}_X[d].$$

If X is k-orientable,

$$\omega_X^\bullet \cong \underline{k}_X[d].$$

In particular, for a complex manifold of complex dimension n,

$$\omega_X^\bullet \cong \underline{k}_X[2n].$$

Proof. A coordinate ball has compactly supported cohomology k in degree d and zero in the other degrees. By Theorem 3.7, the dualizing complex therefore has one cohomology sheaf, in degree $-d$. The transition maps between its local generators are precisely those defining the orientation local system, giving $\omega_X^\bullet \cong \mathrm{or}_X[d]$. If the orientation local system is trivial, this becomes the shifted constant sheaf. Complex coordinates provide a canonical real orientation, so a complex n-manifold gives the shift by real dimension $2n$. □

The open Möbius band changes only the gluing. Every small disc still contributes one class in degree 2, but transport around the core reverses its generator. Over a field of characteristic different from 2, the orientation local system has monodromy -1, and

$$\omega_M^\bullet \cong \mathrm{or}_M[2]$$

The dimension has not changed; the new datum is the sign acquired when local fundamental classes are transported around a nonorientable loop.

Example 3.11 (The complex torus) *Return to the torus whose derived global sections were computed in Example 2.11. There we obtained $R\Gamma(T^2, \underline{k}) \cong \left[k \xrightarrow{0} k^2 \xrightarrow{0} k\right]$ in degrees $0, 1, 2$.*

Choose $\omega_1, \omega_2 \in \mathbb{C}$ linearly independent over $\mathbb{R}$ and set

$$\Lambda := \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2, \quad E := \mathbb{C}/\Lambda$$

The one-dimensional compact complex torus E, with the class of 0 as origin, is the analytic space underlying an elliptic curve. Its underlying topological space is a torus, so the preceding calculation applies to $\underline{k}_E$. As a complex manifold of dimension 1, it is a canonically oriented real manifold of dimension 2. Hence Proposition 3.10 gives

$$\omega_E^\bullet \cong \underline{k}_E[2]$$

Because E is compact, $R\Gamma_c(E, -) \cong R\Gamma(E, -)$. Applying the defining property of the dualizing complex (Proposition 3.2) to the constant sheaf gives

$$\mathrm{RHom}_k(R\Gamma(E, \underline{k}_E), k) \cong \mathrm{RHom}_{\mathrm{Sh}(E;k)}(\underline{k}_E, \underline{k}_E[2])$$

Taking cohomology recovers

$$H^m(E,k) \cong H^{2-m}(E,k)^\vee$$

The earlier calculation gave dimensions $1,2,1$ *in degrees* $0,1,2$*. Verdier duality explains this - the shift by the real dimension* 2 *pairs degrees* m *and* $2-m$*. Its local origin is the compactly supported cohomology class of a small real* 2*-ball in degree* 2*, while compactness of* E *identifies ordinary and compactly supported cohomology.*

Smooth manifolds contribute local compactly supported cohomology in one degree, but a singularity can remove that simplification.

Example 3.12 (A singular point produces several local degrees) *Let*

$$X \coloneqq \{(x,y) \in \mathbb{C}^2 \mid xy = 0\}$$

Near the origin, the local conical structure theorem (Theorem 1.52) identifies a sufficiently small neighbourhood with the open cone (Definition 1.50) on the link

$$L = S^1 \sqcup S^1$$

Since the link L *is compact, the one-point compactification of* $C(L)$ *is the suspension* ΣL*. The suspension is obtained by collapsing the two ends of* $L \times [0,1]$ *to two cone points. We write* $\tilde{H}^\bullet$ *for reduced cohomology; in positive degrees it agrees with ordinary cohomology, while* $\tilde{H}^0$ *records connected components modulo the constant class. Compactly supported cohomology of the open cone is therefore reduced cohomology of its one-point compactification, so*

$$H_c^m(C(L),k) \cong \tilde{H}^m(\Sigma L,k) \cong \tilde{H}^{m-1}(L,k).$$

The link has two connected components and one loop in each component, so

$$\tilde{H}^0(L,k) \cong k, \quad \tilde{H}^1(L,k) \cong k^2$$

Hence

$$H_c^1(C(L),k) \cong k, \quad H_c^2(C(L),k) \cong k^2$$

By the stalk formula (Theorem 3.7),

$$H^{-1}(\omega_X^\bullet)_0 \cong k, \quad H^{-2}(\omega_X^\bullet)_0 \cong k^2$$

Away from the origin the space is a smooth complex curve, so only degree -2 *remains. At the origin an additional local degree* -1 *appears alongside the smooth*

degree -2. No single shifted sheaf can record both groups, so a genuine derived complex is required.

Example 3.13 (The cellular Verdier dual on the interval) *Let $X = [0,1]$, with vertices v_0, v_1 and open edge e, oriented from v_0 to v_1. Write the cellular sheaf of Example 1.58 as*

$$A \xrightarrow{\alpha} C \xleftarrow{\beta} B$$

where the coefficient spaces are finite-dimensional over k.

By the derived realization of Section 2.4, the assignment

$$\sigma \mapsto C_c^\bullet(\mathrm{st}(\sigma); \mathcal{F})^\vee$$

forms a bounded cellular complex. Curry's cellular duality identifies its realization with the Verdier dual of $\mathrm{Real}_X(\mathcal{F})$; see [3, Def. 7.1 and Prop. 7.13]. With the chosen orientation, the three compactly supported complexes are

$$C_c^\bullet(\mathrm{st}(v_0)); \mathcal{F}) = \left[A \xrightarrow{-\alpha} C\right]$$

$$C_c^\bullet(\mathrm{st}(v_1); \mathcal{F}) = \left[B \xrightarrow{\beta} C\right]$$

$$C_c^\bullet(\mathrm{st}(e); \mathcal{F}) = C[-1]$$

The vertex complexes are in degrees $0, 1$, and the edge complex is concentrated in degree 1.

Their linear duals are

$$D_0 = \left[C^\vee \xrightarrow{-\alpha^\vee} A^\vee\right], \quad D_1 = \left[C^\vee \xrightarrow{\beta^\vee} B^\vee\right], \quad D_e = C^\vee[1]$$

with the vertex complexes now in degrees $-1, 0$.

The cellular maps $D_0 \to D_e$ and $D_1 \to D_e$ are the identity on $C^\vee$ in degree -1 and zero in degree 0. These maps complete the description of the dual as a complex of cellular sheaves.

For the constant sheaf, $A = B = C = k$ and $\alpha = \beta = \mathrm{id}$. Both vertex complexes are acyclic, while the edge complex is $k[1]$. The resulting realized object is therefore $\mathbb{D}_X(\underline{k}_X) \cong j_!\big(\underline{k}_{(0,1)}[1]\big)$, where $j : (0,1) \hookrightarrow X$.

The dependence on the open star is essential. If $A = C = k$ but $\alpha = 0$, the left vertex complex instead has nonzero cohomology in both degrees -1 and 0.

Keeping the same vertex stalk while changing its map to the edge changes the Verdier dual at that vertex.

The manifold, singular and cellular calculations describe the local mechanism of Verdier duality. We still need to show that these local duals remain constructible, glue across a finite stratification and reconstruct the original complex after dualising twice.

Theorem 3.14 (Verdier biduality) *Let X be a complex algebraic variety, or a complex analytic space in a constructible setting for which Verdier duality preserves $D_c^b(X)$. Then $\mathbb{D}_X$ is a contravariant triangulated equivalence on $D_c^b(X)$. For every $K \in D_c^b(X)$, the canonical evaluation morphism*

$$K \to \mathbb{D}_X(\mathbb{D}_X(K))$$

is an isomorphism.

Proof sketch. We use Verdier biduality and preservation of constructibility as external theorems; see [5, Thm. 3.3.6 and §4.1], [1, Prop. 3.4.3], and [6, §VII.5]. The preceding examples explain the local calculations involved. On a smooth open stratum, Verdier duality combines finite-dimensional linear duality with the dimension shift. At a boundary point, the dual stalk is determined by compactly supported local cohomology, as in Theorem 3.7 and Example 3.12. These calculations illustrate the theorem, but do not by themselves prove constructibility of the dual or global biduality. □

3.4 Boundary operations and recollement

We now study how sheaves are transported along a map $f : X \to Y$, with particular attention to an open subset and its closed complement. The direct and inverse image functors associated with these inclusions give the adjunctions and distinguished triangles used to construct perverse sheaves. Additional operations will be introduced in Chapter 6 alongside their differential-module counterparts.

3.4.1 Direct and proper direct image

The ordinary direct image f_* of Definition 1.34 collects all sections over inverse images of opens. The proper direct image $f_!$ of Definition 1.39 keeps only sections whose support is proper over the target. Both are left exact in the present setting, so their derived versions retain the higher obstruction data.

Definition 3.15 (Derived direct images) *Let $f : X \to Y$ be continuous. The right derived functors of f_* and $f_!$ are denoted Rf_* and $Rf_!$. For $K \in$*

$D^+(\mathrm{Sh}(X;k))$, *they may be computed by choosing an injective resolution* $K \to I^\bullet$ *and applying the ordinary functors termwise:*

$$Rf_*(K) \cong f_* I^\bullet, \quad Rf_!(K) \cong f_! I^\bullet$$

For $a_X : X \to \{*\}$, ordinary pushforward is the global-section functor $\Gamma(X, -)$, while proper pushforward is the compactly supported section functor $\Gamma_{c(X,-)}$ of the proper push forward (Definition 1.39). Their derived functors satisfy

$$Ra_{X_*}(K) = R\Gamma(X, K), \quad Ra_{X!}(K) = R\Gamma_c(X, K)$$

Because the target is a point, the distinction between the two pushforwards is exactly the distinction between ordinary and compactly supported global information.

Let $j : U = \mathbb{R} \setminus \{0\} \hookrightarrow \mathbb{R}$. Since extension by zero along an open inclusion is exact, $Rj_! k_U = j_! k_U$, and its stalk at the origin is zero in every degree. To compute the stalk of $Rj_* k_U$, take $V_\varepsilon = (-\varepsilon, \varepsilon)$. The punctured interval $V_\varepsilon \cap U = (-\varepsilon, 0) \cup (0, \varepsilon)$ consists of two contractible components, so $H^0(V_\varepsilon \cap U; k) \approx k \oplus k$ and $H^{q(V_\varepsilon \cap U; k)} = 0$ for $q > 0$. Restriction to smaller punctured intervals preserves the two component values, so passing to the stalk gives $\mathcal{H}^0(Rj_* k_U)_0 \approx k \oplus k$ and $\mathcal{H}_0^{q(Rj_* k_U)} = 0$ for $q \neq 0$. Thus $(Rj_* k_U)_0 \approx (k \oplus k)[0]$, whereas $(j_! k_U)_0 = 0$. The two copies of k record the independent locally constant values on the two sides of the missing point.

3.4.2 Inverse and exceptional inverse image

Ordinary inverse image reads information on the source from information on the target. It preserves stalks and is exact for sheaves of vector spaces (Proposition 1.36), so the sheaf-level adjunction Proposition 1.37 passes directly to the derived category:

$$f^{-1} \dashv Rf_*$$

No derived replacement of f^{-1} is needed.

The support-sensitive pushforward $Rf_!$ has a different right adjoint. We already met its special case for a_X when constructing the dualizing complex. The general theorem says that the same supported reading exists relative to a map.

Theorem 3.16 (Existence of the exceptional inverse image) *Let* $f : X \to Y$ *be continuous, with* X *and* Y *locally compact Hausdorff and countable at infinity. Assume that there is an integer* N *such that, for every sheaf* $\mathcal{F}$ *on* X *and every* $q > N$,

$$\mathcal{H}^q(Rf_!(\mathcal{F})) = 0$$

Then there is a functor

$$f^! : D^+(\mathrm{Sh}(Y;k)) \to D^+(\mathrm{Sh}(X;k))$$

characterised by natural isomorphisms

$$\mathrm{Hom}_{D^+(\mathrm{Sh}(Y;k))}(Rf_!(K), L) \cong \mathrm{Hom}_{D^+(\mathrm{Sh}(X;k))}\big(K, f^!(L)\big)$$

Thus

$$Rf_! \dashv f^!.$$

The hypothesis gives a uniform upper bound on the nonzero derived functors of $f_!$ on sheaves. The cited existence theorem uses this finite cohomological dimension to construct the right adjoint on D^+; preservation of boundedness alone would not establish its existence. For $f = a_X$, this is the adjoint used in Definition 3.1. We take the existence proof from [5, Thm. 3.2.3 and Cor. 3.2.4], [1, Thm. 3.1.5], and [6, §VII.3].

For a point inclusion, exceptional inverse image is the derived costalk already constructed in the Chapter 2.

Proposition 3.17 (The point case of exceptional inverse image) *Let $i_x : \{x\} \hookrightarrow X$. The exceptional inverse image $i_x^!(K)$ is canonically identified with the derived costalk of Definition 2.13. Equivalently, for every $V \in D^+(\mathbf{Vect}_k)$,*

$$\mathrm{Hom}_{D^+(\mathrm{Sh}(X;k))}\big(i_{x,*}V, K\big) \cong \mathrm{Hom}_{D^+(\mathbf{Vect}_k)}\big(V, i_x^!(K)\big).$$

Proof. The inclusion of a point is closed, so proper and ordinary direct image agree: $Ri_{x!} = i_{x,*}$. The derived costalk is the right adjoint to this point-supported direct image, while Theorem 3.16 characterises $i_x^!$ by the same adjunction. Uniqueness of right adjoints gives the identification. □

The left-hand side of the displayed adjunction asks for maps into K from an object supported entirely at x; the right-hand side says that every such map is encoded by $i_x^!K$. The relative-neighbourhood cone formula (Proposition 2.14) computes this same object by comparing a small neighbourhood of x with the neighbourhood after removing x.

On the line at the origin, i^{-1} simply reads the stalk k of the constant sheaf. The exceptional inverse image instead computes the relative local complex comparing a neighbourhood of the origin with the punctured neighbourhood, and the calculation of Example 2.15 gives

$$i^!(\underline{k}_{\mathbb{R}}) \cong k[-1]$$

The two inverse images therefore answer different local questions: i^{-1} asks what remains visible at the point, while $i^!$ asks what contribution is supported there relative to deleting it.

3.4.3 Dictionary of basic operations

The preceding examples can be compressed into a small dictionary of basic maps.

Proposition 3.18 (Dictionary of basic maps) *Let $K \in D^b(\mathrm{Sh}(X;k))$.*

1. *For $a_X : X \to \{*\}$,*
$$Ra_{X*}(K) = R\Gamma(X,K), \quad Ra_{X!}(K) = R\Gamma_c(X,K).$$
2. *For $i_x : \{x\} \hookrightarrow X$,*
$$i_x^{-1}K = K_x$$
is the stalk and $i_x^!(K)$ is the derived costalk.
3. *For an open embedding $j : U \hookrightarrow X$,*
$$j_! \dashv j^{-1} \dashv Rj_*, \quad j^! = j^{-1}.$$
4. *For a closed embedding $i : Z \hookrightarrow X$,*
$$i^{-1} \dashv i_* \dashv i^!, \quad Ri_! = Ri_* = i_*.$$

The first line turns geometry over a point into ordinary versus compactly supported cohomology. The second turns a point inclusion into stalk versus costalk. The last two compare the two ways of crossing an open–closed boundary. Each row was realised above on the line at the origin: ordinary against compactly supported sections, stalk against costalk, and the two extensions across the puncture.

Theorem 3.19 (Preservation of algebraic constructibility) *Let $f : X \to Y$ be a morphism of complex algebraic varieties. The functors*
$$Rf_*, \quad Rf_!, \quad f^{-1}-, \quad f^!$$
preserve bounded complexes with algebraically constructible cohomology sheaves.

This preservation theorem allows the geometric direct- and inverse-image operations introduced above to be performed inside the constructible derived categories. We use it as an external result; see [5, §§2.3 and 4.1] and [1, §§8.4–8.5].

Verdier duality interacts with these operations by exchanging ordinary and support-sensitive transport.

Theorem 3.20 (Verdier duality and geometric operations) *Let $f : X \to Y$ be an algebraic morphism and assume the displayed operations preserve the bounded constructible categories. For $K \in D^b_c(X)$ and $L \in D^b_c(Y)$ there are natural isomorphisms*

$$\mathbb{D}_Y(Rf_*(K)) \cong Rf_!(\mathbb{D}_X(K)),$$

and

$$\mathbb{D}_X\big(f^{-1}L\big) \cong f^!(\mathbb{D}_Y(L)).$$

Equivalently, using biduality,

$$\mathbb{D}_Y(Rf_!(K)) \cong Rf_*(\mathbb{D}_X(K)),$$

and

$$\mathbb{D}_X\big(f^!(L)\big) \cong f^{-1}\mathbb{D}_Y(L).$$

The same identities hold for analytic maps whenever the corresponding operations preserve D^b_c.

The first identity says that dualising after ordinary pushforward agrees with properly supported pushforward of the dual. The second is the relative form of the stalk–costalk exchange; for a point inclusion it specialises to the local exchange displayed in the recollement discussion below. The proof rests on a sheaf-valued relative form of the adjunction $Rf_! \dashv f^!$ together with biduality, and the constructible finiteness needed for the passage to D^b_c is part of the external theorem; the formal mechanism is developed with the common calculus in the Riemann–Hilbert chapter. See [5, Thm. 3.3.6 and §4.1], [1, Prop. 3.4.3], and [6, §VII.5].

3.4.4 Recollement

A closed subset and its open complement provide the smallest nontrivial example of gluing. Let $i : Z \hookrightarrow X$ be closed and let $j : U = X - Z \hookrightarrow X$ be the complementary open embedding. We work in a constructible setting in which the six operations in this decomposition preserve D^b_c; this holds in the algebraic setting of Theorem 3.19, and in the analytic setting whenever the corresponding preservation theorem applies. Assume moreover that Z and U are constructible for a common finite stratification, for example that Z is a closed union of strata.

The line at the origin already showed the local difference between $j_!$ and Rj_* and between i^{-1} and $i^!$. The corresponding global statement is that an object on X can be reconstructed from its open and closed pieces in two dual ways.

Theorem 3.21 (The two open–closed triangles) *For every $K \in D^b_c(X)$ there are natural distinguished triangles*

$$j_!(j^{-1}K) \to K \to i_*(i^{-1}K) \to j_!(j^{-1}K)[1],$$

and

$$i_*\big(i^!(K)\big) \to K \to Rj_*(j^{-1}K) \to i_*\big(i^!(K)\big)[1].$$

Proof. For an ordinary sheaf F, the sequence

$$0 \to j_!j^{-1}F \to F \to i_*i^{-1}F \to 0$$

is exact on stalks: on U it is $0 \to F_x \to F_x \to 0$, while on Z it is $0 \to 0 \to F_x \to F_x \to 0$. Applying this degreewise to a representative of K gives the first distinguished triangle. Apply the first triangle to $\mathbb{D}_X(K)$ and then apply $\mathbb{D}_X$. Biduality Theorem 3.14 and the exchange identities Theorem 3.20 transform the three terms into $i_*i^!K$, K and $Rj_*j^{-1}K$, giving the second triangle. □

The first triangle reconstructs K from its restriction to U and the information ordinarily visible on Z. The second reconstructs it from the contribution supported on Z and the open information retained by Rj_*. Applying cohomology to either triangle produces the corresponding long exact sequence, while the final morphism of the triangle packages the entire family of connecting maps. This is the same passage from exact sequences to distinguished triangles developed in Appendix B, now applied to the attachment of open and closed geometric pieces.

Proposition 3.22 (Open–closed recollement) *Under the constructibility hypothesis above, the six functors*

$$i^{-1}, i_*, i^!, \quad j_!, j^{-1}, Rj_*$$

preserve the corresponding constructible derived categories, and together with the adjoint triples $i^{-1} \dashv i_* \dashv i^!$ and $j_! \dashv j^{-1} \dashv Rj_*$ (Proposition 3.18) and the two triangles of Theorem 3.21 they constitute a ***recollement*** *of $D^b_c(X)$ by $D^b_c(Z)$ and $D^b_c(U)$. In particular the insertion functors i_*, $j_!$ and Rj_* are fully faithful, the essential image of i_* consists exactly of the objects killed by j^{-1}, and the incompatible readings vanish:*

$$i^{-1}j_! = 0, \quad j^{-1}i_* = 0, \quad i^!Rj_* = 0.$$

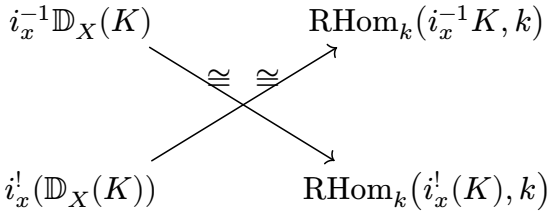
The second statement says that the closed category is exactly the part of $D^b_c(X)$ invisible on the open piece. Preservation of constructibility holds in the algebraic setting by Theorem 3.19 and is part of the standing hypothesis in the analytic one; the remaining assertions are formal consequences of the adjunctions and the two triangles. For the axiomatic formulation of recollement see [7, §1.4].

Example 3.23 (The two reconstructions across the origin) *Take $X = \mathbb{R}$, $Z = \{0\}$, $U = \mathbb{R} - \{0\}$ and $K = \underline{k}_{\mathbb{R}}$. The object $j_! j^{-1} K$ has zero stalk at the origin, while $Rj_* j^{-1} K$ has stalk $k \oplus k$, one copy from each side. The first triangle restores the missing ordinary stalk k. The second uses the derived costalk $k[-1]$ computed in Example 2.15 to correct the two boundary values retained by Rj_*. Both triangles reconstruct the same constant sheaf, but they express the attaching data from the two dual local directions.*

Verdier duality exchanges the two open–closed reconstructions through

$$\mathbb{D}_X j_! \cong Rj_* \mathbb{D}_U, \quad \mathbb{D}_Z i^{-1} \cong i^! \mathbb{D}_X.$$

For a point inclusion $i_x : \{x\} \hookrightarrow X$, the corresponding local identities can be displayed together as



The crossed arrows display the exchange between the two local readings. Ordinary information of the dual is determined by the linear dual of the supported information of K, while supported information of the dual is determined by the linear dual of the ordinary information of K. The relative-neighbourhood cone formula of Proposition 2.14 gives a model for the supported term by comparing a neighbourhood of x with the punctured neighbourhood, and this is the local contribution that appears in the second recollement triangle.

A finite stratification allows recollement to be applied inductively. One separates an open stratum from its closed complement, studies the object on each part, and then reconstructs the original object from this data. Repeating this process, which called **dévissage**, reduces questions about constructible complexes on the whole space to corresponding questions on the individual strata and hence very useful to our purposes.

3.5 Support dimension

Let E be the smooth complex curve of Example 3.11, and let $i : \{p\} \hookrightarrow E$ be the inclusion of a point. Verdier duality acts on the following two objects by shifts determined by the dimensions of their supports.

Example 3.24 (Different supports require different shifts) *The curve-supported side was computed in Example 3.11:* $\mathbb{D}_E(\underline{k}_E) \cong \underline{k}_E[2]$*, so* $\underline{k}_E[1]$ *is self-dual. The point-supported side requires no shift*

$$\mathbb{D}_E(i_*k) \cong i_*\Big(\mathbb{D}_{\{p\}}(k)\Big) \cong i_*k$$

The second equality is the point case, $\omega^\bullet_{\{p\}} \cong k$*, so finite-dimensional duality does not create a dimensional shift. The difference between the two objects comes from the dimensions of their supports.*

The curve-supported object is centred by [1], whereas the point-supported object is centred in degree 0. Thus there is no single global shift that makes both types of objects self-dual in the same degree. The appropriate degree bounds must instead depend on the dimension of the support and must exchange stalk and costalk conditions under Verdier duality. - as we will see in the next chapter.

4 Perverse Sheaves

4.1 Motivation

The calculation of Example 3.24 shows that Verdier duality does not preserve the standard t-structure. Thus, if we want a t-structure compatible with Verdier duality, the allowed cohomological degrees must depend on the dimension of the stratum on which an object is supported. As we saw, Verdier duality exchanges stalks and costalks. Since these encode complementary kinds of local information, the desired conditions should likewise come in a dual pair.

In this chapter, k is a field. We fix a finite Whitney stratification $\mathcal{S}$ of X by connected complex analytic strata, and every object is constructible with respect to $\mathcal{S}$. For a stratum S, write $i_S : S \hookrightarrow X$ for its inclusion. Dimension means complex dimension throughout this chapter.

4.2 Dimension-dependent conditions

On the torus E, a local system is centred as $\mathcal{L}$ [1], while a skyscraper supported at a point is centred without a shift. More generally, let S be a stratum of complex dimension d. We seek a degree bound $p(S)$ for the ordinary local information along S:

$$H^m\big(i_S^{-1}K\big) = 0 \quad \text{for} \quad m > p(S)$$

The example (Example 3.24) suggests that $p(S)$ should depend on d and the compatibility with Verdier duality determines it

Since S is a complex manifold of complex dimension d, its dualizing complex is $\underline{k}_S[2d]$ (Proposition 3.10). The stalk–costalk exchange gives

$$i_S^! \mathbb{D}_X K \cong \mathbb{D}_S i_S^{-1} K$$

and therefore

$$H^m\big(i_S^! \mathbb{D}_X K\big) \cong \big(H^{-m-2d}\big(i_S^{-1} K\big)\big)^\vee$$

If the stalk condition vanishes above $p(S)$, the formula shows that the corresponding costalk condition for the Verdier dual vanishes below

$$-p(S) - 2d.$$

For Verdier duality to exchange the two halves of the same degree convention, this lower cutoff must again be $p(S)$. Hence $p(S) = -p(S) - 2d$, so $p(S) = -d$.

Thus $-d$ is the unique stratumwise cutoff fixed by the transformation

$$p \mapsto -p - 2d$$

induced by Verdier duality.

The calculation gives the upper bound on ordinary local information,

$$H^m\big(i_S^{-1} K\big) = 0 \quad \text{for} \quad m > -\dim S$$

Verdier duality forces the complementary lower bound on supported local information:

$$H^m\big(i_S^! K\big) = 0 \quad \text{for} \quad m < -\dim S$$

Definition 4.1 (Perverse support and costalk conditions) *Define full subcategories of $D_c^b(X;k)$ by*

$$^{\mathrm{p}}\mathrm{D}^{\leq 0}(X) := \big\{K \mid H^m\big(i_S^{-1} K\big) = 0 \ \text{ for every } \ S \in \mathcal{S} \ \text{ and } \ m > -\dim S\big\},$$

$$^{\mathrm{p}}\mathrm{D}^{\geq 0}(X) := \big\{K \mid H^m\big(i_S^! K\big) = 0 \ \text{ for every } \ S \in \mathcal{S} \ \text{ and } \ m < -\dim S\big\}.$$

For $n \in \mathbb{Z}$, we use the shifted notation

$$^{\mathrm{p}}\mathrm{D}^{\leq n}(X) := {}^{\mathrm{p}}\mathrm{D}^{\leq 0}(X)[-n], \quad {}^{\mathrm{p}}\mathrm{D}^{\geq n}(X) := {}^{\mathrm{p}}\mathrm{D}^{\geq 0}(X)[-n]$$

The stratumwise formulation is convenient for the definition. For calculations at a point, the corresponding bounds take a slightly different form because the point costalk includes the real-dimensional shift of the stratum.

Remark (Stratum conditions and pointwise bounds) *The definition is stated for the restriction and exceptional restriction along an entire stratum. If $x \in S$, write $i_x : \{x\} \hookrightarrow X$ and let $\dim S = d$. The equivalent pointwise bounds are*

$$H^m\big(i_x^{-1}K\big) = 0 \ \text{ for } \ m > -d, \quad H^m\big(i_x^{!}K\big) = 0 \ \text{ for } \ m < d.$$

The two tests behave differently under this passage:

	Along the stratum S	***At a point $x \in S$***
Ordinary local test	$H^m = 0$ *for* $m > -d$	$H^m = 0$ *for* $m > -d$
Supported local test	$H^m = 0$ *for* $m < -d$	$H^m = 0$ *for* $m < d$

Only the supported test changes, by the real-dimensional shift $2d$ of the stratum. To see the shift, factor the point inclusion as

$$\{x\} \overset{i_{x,S}}{\to} S \overset{i_S}{\to} X$$

For closed embeddings, ordinary and proper direct image agree, so their right adjoints compose -

$$i_x^{!}K \cong i_{x,S}^{!}\big(i_S^{!}K\big).$$

Put $M := i_S^{!}K$. Constructibility makes the cohomology sheaves of M locally constant on S. Since S is a complex manifold of real dimension $2d$ with its canonical orientation, the manifold calculation of Proposition 3.10 gives

$$i_{x,S}^{!}(M) \cong i_{x,S}^{-1}M[-2d]$$

Consequently,

$$H^m\big(i_x^{!}K\big) \cong H^{m-2d}\Big(i_{x,S}^{-1}i_S^{!}K\Big)$$

The stratumwise condition makes the right-hand side vanish when $m - 2d < -d$, or equivalently when $m < d$. This is the pointwise costalk bound in the table. Geometrically, we work inside the smooth stratum S and choose a sufficiently small ball B around x. Its real dimension is $2d$, and the local cohomology of the pair formed by B and the punctured ball $B - \{x\}$ is one-dimensional in degree $2d$. Hence

$$i_{x,S}^{!}\underline{k}_S \cong k[-2d]$$

For a local system $\mathcal{L}$ on S, the same calculation is performed fibrewise. After the shift by d it gives

$$i_{x,S}^{-1}\,\mathcal{L}\,[d] \cong \mathcal{L}_x[d] \quad i_{x,S}^{!}\,\mathcal{L}\,[d] \cong \mathcal{L}_x[-d]$$

Thus the point stalk is concentrated in ordinary degree $-d$, while the point costalk is concentrated in degree $+d$. The normalization by the complex dimension places these two local readings symmetrically around degree zero, and the real-dimensional shift $2d$ contributed by the ball is divided equally between them.

Example 4.2 (The torus and a point) *Let $i : \{p\} \hookrightarrow E$ be the inclusion of a point in the torus. The shifted local system $\underline{k}_E[1]$ and the skyscraper i_*V both satisfy the perverse conditions. At p,*

$$i^{-1}\underline{k}_E[1] \cong k[1], \quad i^{!}\underline{k}_E[1] \cong k[-1],$$

so the ordinary stalk and costalk degrees are -1 and $+1$. For i_*V both are V in degree 0. The dimension-dependent bounds therefore place both objects in perverse degree zero.

The local conditions can also be read as a bound on the dimensions of the supports of the ordinary cohomology sheaves.

Proposition 4.3 (Support formulation and duality) *For $K \in D_c^b(X;k)$,*

$$K \in {}^{\mathrm{p}}\mathrm{D}^{\leq 0}(X) \iff \dim \operatorname{supp} H^m(K) \leq -m \ \text{ for every } \ m,$$

and

$$K \in {}^{\mathrm{p}}\mathrm{D}^{\geq 0}(X) \iff \mathbb{D}_X K \in {}^{\mathrm{p}}\mathrm{D}^{\leq 0}(X) \iff \dim \operatorname{supp} H^m(\mathbb{D}_X K) \leq -m \ \text{ for every } \ m.$$

Proof sketch. On a stratum S, exactness of restriction gives

$$H^m(K)|_S \cong H^m\big(i_S^{-1}K\big).$$

The first perverse condition says that $H^m(K)$ vanishes on every stratum of dimension greater than $-m$. Because the support of a constructible sheaf is a union of closures of strata, this is equivalent to the displayed support bound. The second equivalence follows from the stalk–costalk exchange under Verdier duality (Theorem 3.14) and exactness of finite-dimensional linear duality. See [7, §2.2] and [5, Prop. 5.1.16]. □

The support half can be read as a staircase relating ordinary degree to support dimension. For a complex threefold:

Ordinary degree m	**Where $H^m(K)$ may be supported**
-3	Anywhere in X; support dimension may be 3
-2	On subsets of complex dimension at most 2
-1	On curves and points
0	On points
$m > 0$	Nowhere

On a complex curve the two local conditions admit a sheaf-theoretic dictionary.

Proposition 4.4 (The perverse conditions on a curve) *Let X be a complex curve stratified by a smooth open stratum and finitely many point strata. A constructible complex K satisfies both perverse conditions if and only if*

1. $H^m(K) = 0$ *for* $m \neq -1, 0$*;*
2. $H^0(K)$ *is supported on the point strata;*
3. $H^{-1}(K)$ *has no nonzero subsheaf supported on the point strata.*

Proof. Let $j : U \hookrightarrow X$ be the smooth open part and $i : Z \hookrightarrow X$ the finite set of point strata. Suppose first that K satisfies the perverse conditions. On U, the two conditions force $j^{-1}K$ to be a local system placed in degree -1. The support condition gives $\mathcal{H}^{m(K)} = 0$ for $m > 0$ and confines $\mathcal{H}^0(K)$ to Z.

Apply i^{-1} to the second recollement triangle to obtain $i^!K \to i^{-1}K \to i^{-1}Rj_*j^{-1}K \to \big(i^!K\big)[1]$. Since $j^{-1}K$ is concentrated in degree -1 and Rj_* is a right derived functor of a left-exact functor, the third term has no cohomology below degree -1. The long exact sequence therefore gives $H^{m(i^!K)} \cong H^{m(i^{-1}K)}$ for $m \leftarrow 1$. The point-costalk condition makes the left-hand side zero. Together with the calculation on U and exactness of stalks, this proves $\mathcal{H}^{m(K)} = 0$ for $m \leftarrow 1$.

Put $\mathcal{F} = \mathcal{H}^{-1}(K)$ and $\mathcal{G} = \mathcal{H}^0(K)$. The canonical truncation triangle is $\mathcal{F}[1] \to K \to \mathcal{G} \to \mathcal{F}[2]$. Exceptional inverse image is the right derived functor of the ordinary left-exact supported-section functor, so $i^!\mathcal{F}$ and $i^!\mathcal{G}$ have no negative cohomology. Applying $i^!$ to this triangle gives $H^{-1}\big(i^!K\big) \cong H^0\big(i^!\mathcal{F}\big)$. The sheaf $i_*H^0\big(i^!\mathcal{F}\big)$ is the largest subsheaf of $\mathcal{F}$ supported on Z. The point-costalk condition therefore excludes every nonzero such subsheaf.

Conversely, suppose the three stated conditions hold. On U, they make $j^{-1}K$ a local system in degree -1, so both perverse conditions hold there. At Z, the

stalk condition follows from the absence of positive cohomology. Applying $i^!$ to the same truncation triangle shows that its cohomology vanishes below degree -1, and that its degree--1 cohomology is $H^0\big(i^!\mathcal{F}\big)$. The third assumption makes this group zero. Thus the costalk condition holds at every point stratum as well.

□

4.3 The perverse t-structure

To obtain kernels, cokernels and images, the two classes of Definition 4.1 must form a t-structure. On a single smooth stratum of dimension d they are the standard t-structure shifted by d; on a stratified space these shifted structures are glued across open–closed decompositions.

Theorem 4.5 (Gluing a t-structure) *Let $i : Z \hookrightarrow X$ be a closed embedding with open complement $j : U \hookrightarrow X$. Suppose bounded t-structures*

$$\big(\mathcal{T}_Z^{\leqslant 0}, \mathcal{T}_Z^{\geqslant 0}\big) \quad \text{and} \quad \big(\mathcal{T}_U^{\leqslant 0}, \mathcal{T}_U^{\geqslant 0}\big)$$

are given on $D_c^b(Z;k)$ and $D_c^b(U;k)$. Then the conditions

$$j^{-1}K \in \mathcal{T}_U^{\leqslant 0}, \quad i^{-1}K \in \mathcal{T}_Z^{\leqslant 0}$$

and

$$j^{-1}K \in \mathcal{T}_U^{\geqslant 0}, \quad i^{!}K \in \mathcal{T}_Z^{\geqslant 0}$$

define a bounded t-structure on $D_c^b(X;k)$.

Proof sketch. Only the mechanism of the construction is sketched here. Orthogonality follows from the first open–closed triangle

$$j_! j^{-1}A \to A \to i_* i^{-1}A \to,$$

together with the adjunctions $j_! \dashv j^{-1}$ and $i_* \dashv i^!$ from the open–closed recollement (Proposition 3.22). For truncations, first truncate the restriction of an object to U and lift the resulting triangle to X. The remaining term is supported on Z, where it can be truncated using the given t-structure on Z. The octahedral axiom (Definition B.21) splices the two corrections into a truncation triangle on X.

The rest of the proof verifies functoriality of this construction and the remaining t-structure axioms after the two truncations are spliced. Those checks use the same adjunction and octahedral diagrams. See [7, §1.4] and [5, Thm. 5.2.4]. □

Theorem 4.6 (The perverse t-structure) *The pair* $({}^{\mathrm{p}}\mathrm{D}^{\leq 0}(X), {}^{\mathrm{p}}\mathrm{D}^{\geq 0}(X))$ *is a bounded t-structure on* $D_c^b(X;k)$.

Proof. Induct on the number of strata. With a single stratum the perverse conditions are the shifted standard t-structure, as observed above. For the inductive step, remove an open stratum $j : U \hookrightarrow X$ and let $i : Z \hookrightarrow X$ be its closed complement. The perverse conditions (Definition 4.1) split along this decomposition: the upper half is tested by $j^{-1}-$ and $i^{-1}-$, the lower half by $j^{-1}-$ and $i^{!}$. These are exactly the hypotheses of the gluing theorem (Theorem 4.5), applied to the t-structures already constructed on U and on Z by the induction. □

Changing the t-structure does not change $D_c^b(X;k)$ or its morphisms. It changes which objects count as degree zero, and with them the meanings of subobject, quotient, kernel, cokernel and short exact sequence. The heart of this new t-structure is the category of perverse sheaves.

Definition 4.7 (Perverse sheaf) *For a fixed adapted stratification $\mathcal{S}$, a* ***perverse sheaf*** *on X is an object of the heart*

$$\mathrm{Perv}_{\mathcal{S}}(X) := {}^{\mathrm{p}}\mathrm{D}^{\leq 0}(X) \cap {}^{\mathrm{p}}\mathrm{D}^{\geq 0}(X).$$

Note that a perverse sheaf can have ordinary cohomology sheaves in several degrees; "degree zero" here refers to the perverse t-structure. Also, perversity of a constructible object is independent of the chosen adapted stratification in our allowed class (Remark 4.9), so we write $\mathrm{Perv}(X)$ when working in $D_c^b(X;k)$.

On a point, $\mathrm{Perv}(\{*\})$ is the category of finite-dimensional k-vector spaces. On a smooth connected complex manifold of dimension d with one stratum, the assignment $\mathcal{L} \mapsto \mathcal{L}\,[d]$ identifies finite-rank local systems with perverse sheaves. On a mixed-dimensional stratification, the restriction of a general perverse sheaf to every stratum need not be a single shifted local system. In other words, the heart allows objects supported in different dimensions to be treated in one abelian category and assembled by extensions.

We denote the perverse truncation functors by ${}^{\mathrm{p}}\tau_{\leq n}$ and ${}^{\mathrm{p}}\tau_{\geq n}$ and the corresponding cohomology functors by ${}^{\mathrm{p}}\mathrm{H}^n$ (Perverse cohomology uses the dimension-dependent grading defined above). Thus ${}^{\mathrm{p}}\mathrm{H}^0(K)$ is the degree-zero cohomology object of K for the perverse t-structure.

Corollary 4.8 (The perverse heart) $\mathrm{Perv}(X)$ *is an abelian category. If a morphism $P \to Q$ in the heart is completed to a triangle*

$$P \to Q \to C \to P[1],$$

then

$$\ker_{\mathrm{Perv}(X)}(P \to Q) \cong {}^{\mathrm{p}}\mathrm{H}^{-1}(C), \quad \operatorname{coker}_{\mathrm{Perv}(X)}(P \to Q) \cong {}^{\mathrm{p}}\mathrm{H}^{0}(C).$$

Proof. The abelian-heart theorem (Theorem B.48) says that the heart of every t-structure is abelian. Applying that result to the perverse t-structure gives the first statement. The displayed kernel and cokernel formulas are the usual heart construction applied to the cone triangle of $P \to Q$. □

Corollary 4.9 (Verdier duality preserves the perverse heart) *Verdier duality restricts to an exact contravariant equivalence*

$$\mathbb{D}_X : \mathrm{Perv}\,(X)^{\mathrm{op}} \to \mathrm{Perv}(X),$$

and biduality remains valid on the heart.

Proof. The support formulation and its dual half (Proposition 4.3) show that $\mathbb{D}_X$ exchanges ${}^{\mathrm{p}}\mathrm{D}^{\leq 0}(X)$ and ${}^{\mathrm{p}}\mathrm{D}^{\geq 0}(X)$. Hence it preserves their intersection. Exactness follows because a contravariant functor exchanging the two halves of a t-structure reverses short exact sequences in the heart. □

Remark (Middle perversity) *It is worth noting that perverse sheaves can be defined around different folding points (i.e - dimension-dependent cutoffs), giving different perversity functions. The choice used here is the middle perversity. It is distinguished by the property just proved, as Verdier duality exchanges the two halves of the resulting t-structure and therefore preserves its heart. In other words, among perversities, the middle one has unusually good manners under duality. See [5, §5.1] and [2, §20.5].*

Remark (Refining the stratification) *A* ***common refinement*** *of two stratifications $\mathcal{S}$ and $\mathcal{T}$ is a third stratification $\mathcal{R}$ whose every stratum is contained in a stratum of $\mathcal{S}$ and in a stratum of $\mathcal{T}$. Geometrically, $\mathcal{R}$ cuts the two decompositions into smaller pieces until both are simultaneously respected. In the constructible derived category we already assumed that the allowed finite Whitney stratifications admit such common refinements (Definition 2.22).*

If K is constructible with respect to two adapted stratifications, passing to a common refinement does not change the support dimensions of its cohomology sheaves. The support formulation (Proposition 4.3) therefore gives the same perverse condition after refinement. However, the statement concerns a fixed constructible object, while a finer stratification may, of course, make additional objects constructible.

4.4 Extensions across a boundary

The first use of the new heart is the boundary-extension problem, whose basic phenomenon already appears on a curve.

Example 4.10 (The two extreme extensions on a punctured curve) *Let Δ be a small complex disc, let $U = \Delta - \{0\}$, and write*

$$j : U \hookrightarrow \Delta \quad i : \{0\} \hookrightarrow \Delta$$

Let $\mathcal{L}_T$ be a finite-rank local system on U with fibre V and monodromy T, and put $P := \mathcal{L}_T[1]$.

The cellular calculation of Example 2.10 gives

$$C_T^\bullet := R\Gamma(U, \mathcal{L}_T) \cong \left[V \overset{T-\mathrm{id}_V}{\longrightarrow} V \right]$$

with

$$H^0(C_T^\bullet) \cong V^T := \ker(T - \mathrm{id}_V) \quad H^1(C_T^\bullet) \cong V_T := \mathrm{coker}(T - \mathrm{id}_V)$$

Consider the two derived extensions $P_! := j_!(P) \quad P_* := Rj_*(P)$. The stalk of P_* at the origin is computed by derived sections on sufficiently small punctured discs, so $i^{-1}P_* \cong C_T^\bullet[1]$. The incompatible costalk reading vanishes by the open-closed (Proposition 3.22) $i^!(P_*) = 0$.

For extension by zero, $i^{-1}P_! = 0$. Applying $i^{-1}-$ to the second triangle of Theorem 3.21 gives

$$i^!(P_!) \cong i^{-1}P_*[-1] \cong C_T^\bullet$$

Object	***Stalk at the origin***	***Costalk at the origin***
P_*	$C_T^\bullet[1]$ *in degrees* -1 *and* 0	0
$P_!$	0	$C_T^\bullet$ *in degrees* 0 *and* 1

On U, both extensions restrict to P, which is concentrated in degree -1. At the point stratum, the table gives no positive stalk cohomology and no negative costalk cohomology. The pointwise bounds of Remark 4.1 therefore show that both $P_!$ and P_* are perverse. This is special to the one-dimensional local model; Example 4.11 shows that the conclusion fails already in complex dimension 2.

When $T = \mathrm{id}_V$, put

$$P_0 := \underline{V}_\Delta[1]$$

Rotating the two open–closed triangles gives short exact sequences in $\mathrm{Perv}(\Delta)$*:*

$$0 \to i_* V \to P_! \to P_0 \to 0$$

$$0 \to P_0 \to P_* \to i_* V \to 0$$

Thus the two extreme extensions differ from the middle extension by a boundary subobject on the $j_!$ *side and a boundary quotient on the* Rj_* *side.*

Suppose a perverse sheaf is known on an open union of strata U. Extending it to X is not automatic: restriction to U forgets every constituent supported on the closed boundary. The punctured-curve sequences above (Example 4.10) already show that several perverse extensions can have the same restriction. The problem is to determine which boundary information is forced by the object on U and which information could be added independently.

Let $i : Z \hookrightarrow X$ be the inclusion of a closed union of strata, and let $U = X - Z$ with inclusion $j : U \hookrightarrow X$. The dictionary of basic maps already supplied the two extreme derived extensions, $j_!$ and Rj_* (Section 3.4.3), while open–closed recollement describes their relation to the boundary (Theorem 3.21). The new question is how these extensions sit relative to the perverse heart.

First, neither extreme derived extension needs to lie in the heart, even for the simplest coefficients.

Example 4.11 (Why perverse truncation is needed) *Let* $X = \mathbb{C}^2$*,* $U = X - \{0\}$*, and* $P = \underline{k}_U[2]$*. For the inclusion* $j : U \hookrightarrow X$ *and the origin* $i_0 : \{0\} \hookrightarrow X$*, a punctured ball retracts onto* S^3*, hence*

$$i_0^{-1} Rj_* P \cong R\Gamma(S^3, k)[2]$$

In Example 3.12, disconnectedness of the link produced an additional low degree. Here the link is connected, but its top class in $H^3(S^3; k)$ *appears after the shift in ordinary degree* 1*. Thus the stalk complex has cohomology in ordinary degrees* -2 *and* 1*. The point stratum has complex dimension* 0*, so the perverse support condition permits stalk cohomology there only in degrees at most* 0*; the class in ordinary degree* 1 *therefore places* Rj_*P *outside the perverse heart. Applying* ${}^{\mathrm{p}}\mathrm{H}^0$ *removes that contribution and leaves the degree-zero extension.*

From this point on, perverse truncation is the operation that returns a derived extension to the dimension-sensitive degree zero before abelian constructions are applied.

Theorem 4.12 (One-sided perverse exactness) *For the perverse t-structures on Z, X, and U:*

1. *$j^{-1} = j^!$ and i_* are t-exact;*
2. *$j_!$ and i^{-1} carry the nonpositive half to the nonpositive half;*
3. *Rj_* and $i^!$ carry the nonnegative half to the nonnegative half.*

Proof sketch. The first line follows directly from the gluing description of the perverse t-structure (Theorem 4.5). For $A \in {}^{\mathrm{p}}\mathrm{D}^{\leq 0}(U)$ and $B \in {}^{\mathrm{p}}\mathrm{D}^{\geq 1}(X)$,

$$\mathrm{Hom}(j_!A, B) \cong \mathrm{Hom}(A, j^{-1}B) = 0,$$

so orthogonality places $j_!A$ in ${}^{\mathrm{p}}\mathrm{D}^{\leq 0}(X)$. Verdier duality gives the corresponding assertion for Rj_*. The adjunctions $i^{-1} \dashv i_*$ and $i_* \dashv i^!$ give the two statements for the closed inclusion. See [7, §1.4.16]. □

The one-sided exactness statements visualised in the following table.

Functor	**Perverse exactness**	**Geometric role**
$j^{-1} = j^!$	t-exact	Restriction back to the open part
i_*	t-exact	Boundary data regarded as an object on X
$j_!$	Right t-exact	Adds no ordinary sections on the boundary
i^{-1}	Right t-exact	Tests the stalk-side boundary contribution
Rj_*	Left t-exact	Retains cohomology of punctured neighbourhoods
$i^!$	Left t-exact	Tests the supported or costalk-side contribution

For $P \in \mathrm{Perv}(U)$, extension by zero satisfies $j_!P \in {}^{\mathrm{p}}\mathrm{D}^{\leq 0}(X)$, while derived direct image satisfies $Rj_*P \in {}^{\mathrm{p}}\mathrm{D}^{\geq 0}(X)$. Either object can still have perverse cohomology away from degree 0. Its degree-zero part gives an extension that lies in the heart.

Definition 4.13 (The two perverse boundary extensions) *For $P \in \mathrm{Perv}(U)$, define*

$$j_!^p P := {}^{\mathrm{p}}\mathrm{H}^0(j_!P), \quad j_*^p P := {}^{\mathrm{p}}\mathrm{H}^0(Rj_*P).$$

These are called the standard and costandard ***perverse extensions****.*

The identity of P, transported through the adjunctions $j_! \dashv j^{-1} \dashv Rj_*$, induces a canonical morphism

$$j_!^p P \to j_*^p P.$$

The relation among the four objects is

$$j_! P \to j_!^p P \to j_*^p P \to Rj_* P,$$

where the two middle terms lie in $\mathrm{Perv}(X)$. The left extension may contribute a boundary subobject on the supported side and the right extension may contribute a boundary quotient on the ordinary side.

The desired extension keeps only the boundary information forced by P on U. It should introduce neither an independent subobject nor an independent quotient supported on Z. Its construction is therefore an image, which is why the abelian heart is needed to begin with.

Definition 4.14 (Intermediate extension) *For $P \in \mathrm{Perv}(U)$, define*

$$j_{!*} P := \mathrm{im}_{\mathrm{Perv}(X)}\big(j_!^p P \to j_*^p P\big).$$

The operations $j_!$ and Rj_* already exist in the derived category, and $j_!^p$, j_*^p and $j_{!*}$ become available after the perverse heart has been chosen. In particular, the definition of $j_{!*}$ uses the image of a morphism in the abelian category $\mathrm{Perv}(X)$.

Theorem 4.15 (Boundary characterization of the intermediate extension) *The exact functor i_* identifies $\mathrm{Perv}(Z)$ with the full subcategory of $\mathrm{Perv}(X)$ consisting of objects Q for which $j^{-1}Q = 0$. Moreover, for $Q \in \mathrm{Perv}(X)$:*

1. *Q has no nonzero quotient supported on Z exactly when ${}^{\mathrm{p}}\mathrm{H}^0\big(i^{-1}Q\big) = 0$;*
2. *Q has no nonzero subobject supported on Z exactly when ${}^{\mathrm{p}}\mathrm{H}^0\big(i^! Q\big) = 0$.*

Consequently, $j_{!*}P$ is, up to unique isomorphism, the unique perverse sheaf Q satisfying

1. *$j^{-1}Q \cong P$;*
2. *Q has no nonzero subobject supported on Z;*
3. *Q has no nonzero quotient supported on Z.*

Equivalently,

$$i^{-1}Q \in {}^{\mathrm{p}}\mathrm{D}^{\leq -1}(Z), \quad i^! Q \in {}^{\mathrm{p}}\mathrm{D}^{\geq 1}(Z).$$

Proof. If $j^{-1}Q = 0$, the open–closed triangles (Theorem 3.21) identify Q with $i_*i^{-1}Q$ and with $i_*i^!Q$. One-sided perverse exactness (Theorem 4.12) then places the corresponding object on Z in its perverse heart. Thus i_* identifies $\mathrm{Perv}(Z)$ with the perverse sheaves supported on Z.

Since $i^{-1}Q \in {}^{\mathrm{p}}\mathrm{D}^{\leq 0}(Z)$, adjunction gives, for $R \in \mathrm{Perv}(Z)$,

$$\mathrm{Hom}(Q, i_*R) \cong \mathrm{Hom}\big({}^{\mathrm{p}}\mathrm{H}^0\big(i^{-1}Q\big), R\big).$$

A nonzero quotient of Q supported on Z therefore exists when ${}^{\mathrm{p}}\mathrm{H}^0\big(i^{-1}Q\big)$ is nonzero. Verdier duality gives the corresponding statement for subobjects and ${}^{\mathrm{p}}\mathrm{H}^0\big(i^!Q\big)$.

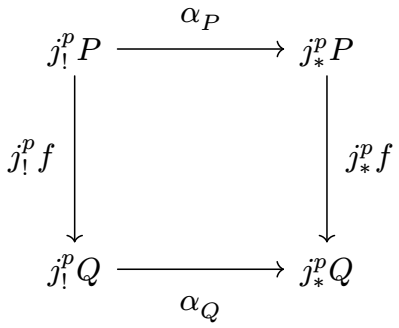
The perverse direct-image extension j^p_*P has no nonzero subobject supported on Z, and the perverse extension by zero $j^p_!P$ has no nonzero quotient supported there. Their image inherits both properties. Conversely, an extension Q with neither kind of boundary constituent receives a morphism from $j^p_!P$ and maps to j^p_*P. The cokernel of the first map and the kernel of the second are supported on Z, so both vanish. Hence Q is the image of the canonical morphism. The final degree conditions restate the two vanishings in the perverse t-structure. □
In short, the functor i^{-1} detects boundary information that remains ordinarily visible and can produce a quotient. The exceptional inverse image $i^!$ detects contributions supported on the boundary and can produce a subobject. Intermediate extension excludes both independent boundary contributions - and Verdier duality exchanges the two conditions.

Proposition 4.16 (Formal properties of intermediate extension) *The functor $j_{!*} : \mathrm{Perv}(U) \to \mathrm{Perv}(X)$ is fully faithful and commutes with Verdier duality:*

$$\mathbb{D}_X(j_{!*}P) \cong j_{!*}(\mathbb{D}_U P).$$

Proof. A morphism between two intermediate extensions that vanishes on U has image supported on Z. That image is simultaneously a quotient of the source and a subobject of the target, so the boundary characterization of intermediate extension (Theorem 4.15) forces it to vanish. This proves faithfulness.

For fullness, let $f : P \to Q$ be a morphism in $\mathrm{Perv}(U)$, and write $\alpha_P : j^p_!P \to j^p_*P$ and $\alpha_Q : j^p_!Q \to j^p_*Q$ for the canonical maps. Functoriality gives the commu-



tative square and therefore a morphism between the two images. This proves fullness. Verdier duality exchanges $j^p_!$ with j^p_*, so it preserves the image of the canonical morphism. □

The punctured-curve model shows what boundary information survives this image construction.

On a curve the intermediate extension can moreover be reached by an ordinary truncation of Rj_*, as the calculation below shows. That formula is a special feature of the one-dimensional dictionary (Proposition 4.4), where only ordinary degrees -1 and 0 occur; in general $j_{!*}$ is defined only as the image in the perverse heart.

Example 4.17 (Intermediate extension on a punctured curve) *Retain the notation of Example 4.10. Since $P_!$ and P_* are already perverse,*

$$j^p_! P \cong P_! \quad j^p_* P \cong P_*$$

The ordinary degree-0 cohomology of P_* is supported at the origin and has stalk V_T. The boundary characterization of Theorem 4.15 therefore gives

$$j_{!*}(P) \cong \tau_{\leq -1}(P_*)$$

The truncation removes the degree-0 boundary quotient and retains the invariant term:

$$i^{-1} j_{!*}(P) \cong V^T[1]$$

For the dual local system,

$$\ker(T^\vee - \mathrm{id}_{V^\vee}) \cong (\mathrm{coker}(T - \mathrm{id}_V))^\vee \cong V_T^\vee$$

Compatibility with Verdier duality from Proposition 4.16 then gives

$$i^! (j_{!*}(P)) \cong V_T[-1]$$

Thus the same map $T - \mathrm{id}_V$ controls both boundary readings: its kernel occurs in stalk degree -1, while its cokernel occurs in costalk degree 1.

The same formula gives three boundary tests -

Monodromy	V^T	V_T	**Boundary behaviour of $j_{!*}\mathcal{L}_T[1]$**
$T = \mathrm{id}_V$	V	V	The local system extends across the puncture
$T - \mathrm{id}_V$ invertible	0	0	No point stalk or costalk contribution remains
$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	k	k	One invariant and one coinvariant class survive

4.5 Intersection complexes and cohomology

The intermediate extension now gives a canonical coefficient object on a singular space. Let X be pure of complex dimension n, and let $j : X_{\mathrm{reg}} \hookrightarrow X$ be its smooth locus. On X_{reg}, a local system $\mathcal{L}$ has the natural perverse normalization $\mathcal{L}\,[n]$. The question is how to extend this generic top-dimensional data across the singular locus without introducing additional subobjects or quotients supported only there.

The boundary characterization of $j_{!*}$ gives such an extension. Thus we define the intersection complex by extending $\mathcal{L}\,[n]$ from the smooth locus using $j_{!*}$. Since, as we saw, intermediate extension also commutes with Verdier duality, this construction preserves the duality enjoyed by the normalized local system on the smooth part.

However, let $X = \mathbb{P}^1 \sqcup_p \mathbb{P}^1$, be two projective lines meeting at one point. Topologically, X is a wedge of two 2-spheres, so

$$\mathbb{H}^0(X, k) \cong k, \quad \mathbb{H}^1(X, k) = 0, \quad \mathbb{H}^2(X, k) \cong k^2$$

Thus ordinary cohomology does not have matching dimensions in degrees 0 and 2. Hence, we need a different coefficient object.

Definition 4.18 (Intersection complex) *Let S be a stratum of complex dimension d, let $j : S \hookrightarrow \overline{S}$ be its inclusion, and let $a_S : \overline{S} \hookrightarrow X$ be the closed embedding. For a finite-rank local system $\mathcal{L}$ on S, define*

$$\mathrm{IC}_{\overline{S}}(\mathcal{L}) := {a_S}_* j_{!*}(\mathcal{L}\,[d]).$$

A complex space is **pure n-dimensional** when all of its irreducible components have dimension n. If X is irreducible and pure n-dimensional, with smooth locus $j : X_{\mathrm{reg}} \hookrightarrow X$, we write

$$\mathrm{IC}_X(\mathcal{L}) := j_{!*}(\mathcal{L}\,[n]), \quad \mathrm{IC}_X := \mathrm{IC}_X\Big(\underline{k}_{X_{\mathrm{reg}}}\Big).$$

For a reducible pure-dimensional space, IC_X is the direct sum of the corresponding intersection complexes of its irreducible components.

Example 4.19 (The elliptic curve: dualizing and perverse shifts) *Return to the elliptic curve $E = \mathbb{C}/\Lambda$ from the opening of the chapter. Since E is smooth of complex dimension 1, its regular locus is all of E, and the definition gives $\mathrm{IC}_E \cong \underline{k}_E[1]$. Verdier duality gives $\mathbb{D}_E(\underline{k}_E) \cong \underline{k}_E[2]$, and therefore $\mathbb{D}_E(\mathrm{IC}_E) \cong \mathrm{IC}_E$. The shift by 2 comes from the real dimension of the dualizing complex. The shift by 1 is the complex-dimensional normalization that places the constant local system in the perverse heart.*

Let X be a reduced pure complex curve, let $\nu : \tilde{X} \to X$ be its normalization, and let $j : X_{\mathrm{reg}} \hookrightarrow X$ be the smooth-locus inclusion. Put $P := R\nu_*\underline{k}_{\tilde{X}}[1]$. The normalization is finite, so ν_* is exact and $R\nu_* = \nu_*$. Let $x \in X$ and write $r_x := \mathrm{card}(\nu^{-1}(x))$ for the number of local branches at x. For a sufficiently small neighbourhood B of x, the inverse image $\nu^{-1}(B)$ is a disjoint union of r_x discs. The stalk of P is therefore the direct sum of their constant branch values. Its costalk is computed by the corresponding disjoint union of local pairs $(D, D - \{0\})$; each pair contributes k in real degree 2, which appears in ordinary degree $+1$ after the shift by [1]. Hence

$$i_x^{-1}P \cong k^{r_x}[1], \quad i_x^{!}P \cong k^{r_x}[-1].$$

The restriction to the smooth locus and these pointwise stalk and costalk bounds show that P is perverse. Both displayed complexes have vanishing ordinary cohomology in degree 0, so the boundary characterization of intermediate extension (Theorem 4.15) shows that P has no nonzero subobject or quotient supported at a singular point. Since ν is an isomorphism over X_{reg}, it follows that

$$\mathrm{IC}_X \cong R\nu_*\underline{k}_{\tilde{X}}[1].$$

This is the one-dimensional normalization formula used in the curve examples below; see [8, §8.2.2].

Definition 4.20 (Intersection cohomology) *For a pure n-dimensional space, define*

$$I\mathbb{H}^m(X,\mathcal{L}) := \mathbb{H}^{m-n}(X, \mathrm{IC}_X(\mathcal{L})).$$

For the constant local system, write $I\mathbb{H}^m(X)$.

The shift $\mathcal{L}\,[n]$ is the perverse normalization determined by the support and costalk conditions (Definition 4.1). Hypercohomology of the normalized object is shifted by n relative to the usual topological indexing, and the term $m-n$ restores degrees 0 through $2n$ on a compact n-dimensional space.

Proposition 4.21 (Duality of intersection complexes) *For the fibrewise dual local system* $\mathcal{L}^\vee$,

$$\mathbb{D}_X \mathrm{IC}_{\overline{S}}(\mathcal{L}) \cong \mathrm{IC}_{\overline{S}}(\mathcal{L}^\vee).$$

In particular, the intersection complex with constant coefficients is Verdier self-dual.

Proof. On the smooth stratum, complex orientation gives

$$\mathbb{D}_S\ \mathcal{L}\ [d] \cong \mathcal{L}^\vee[d].$$

Compatibility of intermediate extension with Verdier duality (Proposition 4.16) carries this isomorphism across the boundary. Proper pushforward along the closed embedding a_S then gives the displayed formula. □

Global Verdier duality turns the self-duality of the intersection complex into a pairing between complementary intersection-cohomology degrees.

Theorem 4.22 (Poincaré duality for intersection cohomology) *Let* X *be compact, irreducible, and pure of complex dimension* n. *Then*

$$I\mathbb{H}^m(X,\mathcal{L}) \cong \left(I\mathbb{H}^{2n-m}(X,\mathcal{L}^\vee)\right)^\vee.$$

For constant coefficients the same conclusion holds componentwise on a reducible pure-dimensional space.

Proof. Apply global Verdier duality (Proposition 3.6) to $\mathrm{IC}_{X(\mathcal{L})}$. Compactness identifies ordinary and compactly supported global sections. Duality of intersection complexes (Proposition 4.21) identifies the Verdier dual with $\mathrm{IC}_{X(\mathcal{L}^\vee)}$. Taking cohomology and translating by the indexing in the definition of intersection cohomology (Definition 4.20) gives the displayed degrees. □

For the reducible curve above, the coefficient change can be seen through normalization.

Example 4.23 (Two projective lines meeting at a point) *For* $X = \mathbb{P}^1 \sqcup_p \mathbb{P}^1$, *the germ at* p *is the two-branch singularity of Example 3.12. The*

intersection cohomology treats it globally as follows. Let $\nu : \mathbb{P}^1 \sqcup \mathbb{P}^1 \to X$ be the normalization map. The constant sheaf on X has one value at p, so sections on the two branches must agree there. The normalization has two preimages of p, and the intersection complex keeps the corresponding branch data separate -

$$\mathrm{IC}_X \cong R\nu_* \underline{k}_{\mathbb{P}^1 \sqcup \mathbb{P}^1}[1].$$

And therefore

$$I\mathbb{H}^0(X) \cong k^2, \quad I\mathbb{H}^1(X) = 0, \quad I\mathbb{H}^2(X) \cong k^2.$$

The space X is connected, so ordinary degree-zero cohomology is $\mathbb{H}^0(X, k) \cong k$. The group $I\mathbb{H}^0(X) \cong k^2$ retains the two top-dimensional branches supplied by the intersection complex. In particular, $I\mathbb{H}^0$ should not be read as the number of connected components of the underlying topological space.

	$m = 0$	$m = 1$	$m = 2$
$\mathbb{H}^{m(X,k)}$	k	0	k^2
$I\mathbb{H}^{m(X)}$	k^2	0	k^2

The reducible node above compared ordinary and intersection cohomology globally. The irreducible nodal curve below has the same two-branch local germ as Example 3.12, there the germ will exhibit an intersection-complex factor and a point-supported factor inside one object of the perverse heart.

4.6 Simple perverse sheaves

The intersection complex gives a canonical perverse object from an irreducible local system on one stratum. To use these objects as building blocks, we need to know which perverse sheaves are simple and whether arbitrary perverse sheaves can be assembled from finitely many such pieces.

Definition 4.24 (Simple objects and irreducible local systems) *A nonzero object P of an abelian category is **simple** if its only subobjects are* 0 *and P. A finite-rank local system is **irreducible** if it has no nonzero proper sub-local system.*

Irreducibility controls subobjects on the stratum itself. The boundary characterization of intermediate extension (Theorem 4.15) prevents new subobjects or quotients from appearing only on the complement. Together these two facts identify the simple objects of the perverse heart.

Theorem 4.25 (Simple objects and finite composition) *Every object* $P \in \mathrm{Perv}(X)$ *admits a finite filtration*

$$0 = P_0 \subset P_1 \subset \ldots \subset P_r = P$$

whose successive quotients are simple. The simple objects are exactly

$$\mathrm{IC}_{\overline{S}}(\mathcal{L}),$$

where S *is a stratum and* $\mathcal{L}$ *is an irreducible finite-rank local system on* S*.*

Proof sketch. We first prove the part that displays the mechanism. If $P \in \mathrm{Perv}(U)$ is simple, any nonzero subobject of $j_{!*}P$ restricts either to 0 or to all of P. In the first case it is supported on the boundary and vanishes by the boundary characterization of intermediate extension (Theorem 4.15). In the second case the quotient is supported on the boundary and vanishes for the same reason. Hence intermediate extension preserves simplicity. In particular, $\mathrm{IC}_{\overline{S}}(\mathcal{L})$ is simple whenever $\mathcal{L}$ is irreducible.

Let P be simple and choose a stratum S that is open in $\operatorname{supp} P$. Its restriction has the form $\mathcal{M}\,[\dim S]$ for a finite-rank local system $\mathcal{M}$. A proper nonzero sub-local system of $\mathcal{M}$ would produce, by adjunction and image in the perverse heart, a proper nonzero subobject of P; thus $\mathcal{M}$ is irreducible. The canonical maps from the two perverse boundary extensions to and from P then force P to be their image, hence

$$P \cong \mathrm{IC}_{\overline{S}}(\mathcal{M}).$$

For finite length, one inducts on the number of strata, taking a finite composition series on an open stratum and applying the induction hypothesis to the closed complement. The detailed induction is given in [7, Thm. 4.3.1] and [5, Thm. 5.2.12]. □

A finite composition series separates two kinds of information. Its factors specify which irreducible local systems occur on which supports, and the successive extension classes in the chosen filtration record how those factors are assembled. The filtration need not split, so the composition factors alone do not determine the perverse sheaf.

Example 4.26 (A perverse sheaf with two composition factors) *Let* X *be an irreducible curve with one ordinary node* p*. Let* $\nu : \tilde{X} \to X$ *be the normalization, which replaces* p *by two points, and let* $i : \{p\} \hookrightarrow X$ *be the singular-point*

inclusion. Away from p, the object $\underline{k}_X[1]$ restricts to the perverse-normalized constant local system on a smooth curve. It remains to test the point stratum.

Choose a sufficiently small neighbourhood B of the node and put $B^\times = B - \{p\}$. The neighbourhood B is contractible, while $B^\times$ has two connected components, each homotopy equivalent to S^1. Hence

$$H^0(B;k) = k, \quad H^0(B^\times;k) = k^2, \quad H^1(B^\times;k) = k^2,$$

and the map $H^0(B;k) \to H^0(B^\times;k)$ is the diagonal inclusion. The long exact sequence of the pair $(B, B^\times)$ gives

$$H^1(B, B^\times;k) \cong k, \quad H^2(B, B^\times;k) \cong k^2,$$

with no relative cohomology in lower degrees. Thus

$$i^{-1}\underline{k}_X[1] \cong k[1],$$

while the costalk, represented by the shifted local pair, has cohomology

$$H^0\big(i^!\underline{k}_X[1]\big) \cong k, \quad H^1\big(i^!\underline{k}_X[1]\big) \cong k^2,$$

and vanishes in negative degrees. These are the pointwise perverse bounds (Remark 4.1), so $\underline{k}_X[1]$ is perverse.

The normalization has two local values over the node, so there is an exact sequence of ordinary sheaves

$$0 \to \underline{k}_X \to \nu_*\underline{k}_{\tilde{X}} \to i_*k \to 0$$

where the first map is diagonal at the node and the quotient measures the difference between the two branch values. After shifting by 1 this becomes a distinguished triangle

$$\underline{k}_X[1] \to R\nu_*\underline{k}_{\tilde{X}}[1] \to i_*k[1]$$

The middle term is the intersection complex,

$$\mathrm{IC}_X \cong R\nu_*\underline{k}_{\tilde{X}}[1]$$

and i_*k is perverse in degree 0. Taking perverse cohomology gives the short exact sequence

$$0 \to i_*k \to \underline{k}_X[1] \to \mathrm{IC}_X \to 0$$

The normalization calculation is therefore read differently in the two hearts:

Heart	***Short exact sequence***
Ordinary sheaves	$0 \to \underline{k}_X \to \nu_* \underline{k}_{\tilde{X}} \to i_* k \to 0$
Perverse sheaves	$0 \to i_* k \to \underline{k}_X[1] \to \mathrm{IC}_X \to 0$

The second row is not obtained by simply shifting the first row as a short exact sequence. One shifts the associated distinguished triangle and then takes cohomology in the perverse heart. The derived object has not changed; the change of heart changes which triangles become short exact sequences and therefore which subobjects and quotients are visible.

The two simple composition factors are therefore the point-supported skyscraper and the intersection complex. The example separates the two issues controlled by the theory: the perverse bounds decide which degrees are allowed, while the abelian heart keeps track of subobjects and extensions among objects satisfying those bounds.

4.7 Toward Riemann–Hilbert

The perverse t-structure places a local system on a smooth d-dimensional stratum in degree zero after the shift $[d]$. Recollement glues these shifted local systems across strata, while intermediate extension removes subobjects and quotients supported only on the boundary. The simple objects of $\mathrm{Perv}(X)$ are therefore the intersection complexes $\mathrm{IC}_{\overline{S}}(\mathcal{L})$ attached to irreducible local systems, and arbitrary perverse sheaves are assembled from them by extensions.

Riemann–Hilbert supplies the differential counterparts of these data. Regular connections produce the local systems on smooth strata, Kashiwara's equivalence treats objects supported on closed strata, and the compatibility of the solution functor with the six operations and Verdier duality carries the gluing and extension data between the two categories.

5 Differential Equations and Their Solution

At the introduction we described how local solutions restrict and glue, continuation around a deleted singular point may change a branch, and restoring the point asks what information must be kept there. This chapter translates (formally) that picture into the sheaf language developed above.

We move between two descriptions of the same differential problem. On the analytic side, solutions form sheaves and derived complexes of sheaves. On the algebraic side, equations become modules over a sheaf of differential operators. Whenever the relation between the two descriptions follows directly from the definitions, we prove it here, but the full identification will come from the Riemann–Hilbert correspondence of the next chapter.

Throughout this chapter, the equation side is algebraic on a smooth complex algebraic variety X, while holomorphic solutions live on the associated analytic manifold X^{an}. The analytification step connecting these settings is made explicit below.

5.1 Local solution data

Let $Y' = A(z)Y$ be a linear system on an open set $U \subseteq \mathbb{C}$. Its local solution sheaf is defined as follows.

Definition 5.1 (The local solution sheaf of a linear system) *For an open subset $W \subseteq U$, set*

$$\mathcal{S}ol_A(W) \coloneqq \{Y \in \mathcal{O}_U(W)^r \mid Y' = AY\}$$

where $\mathcal{O}_U(W)^r$ consists of holomorphic maps $W \to \mathbb{C}^r$. Restriction is componentwise. We write $\mathcal{S}ol$ for the ordinary solution sheaf and Sol *for the derived solution functor introduced later.*

Solutions agreeing on overlaps glue componentwise to a holomorphic map. The equation holds locally, so it also holds for the glued map; hence $\mathcal{S}ol_A$ is a sheaf. At an ordinary point, where A is holomorphic, the initial-value theorem gives the stronger local-system property.

Theorem 5.2 (Nonsingular linear systems produce local systems) *If $A : U \to \mathrm{Mat}_r(\mathbb{C})$ is holomorphic, then the local solution sheaf $\mathcal{S}ol_A$ (Definition 5.1) is a local system (Definition 1.23) of rank r. Here the rank means the complex dimension of a fibre, or equivalently of any stalk. For every $z_0 \in U$, evaluation at z_0 identifies its stalk with $\mathbb{C}^r$.*

Proof. Fix $z_0 \in U$. For every $v \in \mathbb{C}^r$, local existence and uniqueness give a unique solution Y_v near z_0 with $Y_v(z_0) = v$. Hence evaluation identifies the stalk with $\mathbb{C}^r$.

Choose the solutions $Y_1, ..., Y_r$ whose initial values are the standard basis of $\mathbb{C}^r$, and put them into the columns of a matrix $\Phi = (Y_1|...|Y_r)$. Then $\Phi(z_0) = I_r$. After shrinking the neighbourhood, $\det \Phi$ remains nonzero. Every local solution can therefore be written uniquely as

$$Y = \Phi c$$

for a holomorphic vector-valued function c. The equation forces this coefficient vector to be constant. Indeed, $c = \Phi^{-1}Y$, and differentiating gives

$$c' = \left(\Phi^{-1}Y\right)' = -\Phi^{-1}\Phi'\Phi^{-1}Y + \Phi^{-1}Y' = 0$$

because $\Phi' = A\Phi$ and $Y' = AY$. Thus $c \in \mathbb{C}^r$ is constant. On this neighbourhood the solution sheaf is isomorphic to the constant sheaf with fibre $\mathbb{C}^r$, so it is a local system of rank r. □

If U is connected, continuation along paths gives the monodromy representation

$$\rho : \pi_1(U, z_0) \to \mathrm{GL}\Big(\left(\mathcal{S}ol_A\right)_{z_0}\Big)$$

Evaluation at z_0 identifies global solutions with the vectors fixed by every monodromy operator.

Example 5.3 (The Euler equation and monodromy) *For*

$$zu'(z) = \alpha u(z)$$

on $\mathbb{C}^\times$*, choose a branch* $\mathrm{Log}_W z$ *on a simply connected open set* W*. The local solutions are*

$$u(z) = ce^{\alpha \mathrm{Log}_W z}$$

After one counterclockwise circuit around the origin, $\mathrm{Log}\, z$ *changes by* $2\pi i$*, so the solution is multiplied by*

$$\lambda := e^{2\pi i \alpha}.$$

Thus the solution sheaf is a rank-one local system $\mathcal{L}_\lambda$ *with monodromy* λ*. Monodromy, it seems, has a poor memory for integers, as it remembers* α *only modulo* $\mathbb{Z}$*, since replacing* α *by* $\alpha + m$ *with* $m \in \mathbb{Z}$ *leaves* $e^{2\pi i \alpha}$ *unchanged.*

For a local system $\mathcal{L}_T$ on $\mathbb{C}^\times$ with fibre V and monodromy T, the cellular calculation (Example 2.10) gives $R\Gamma(\mathbb{C}^\times, \mathcal{L}_T) \cong \left[V \overset{T-\mathrm{id}_V}{\longrightarrow} V \right]$ in degrees 0 and 1. Thus

$$H^0(\mathbb{C}^\times, \mathcal{L}_T) \cong \ker(T - \mathrm{id}_V), \quad H^1(\mathbb{C}^\times, \mathcal{L}_T) \cong \frac{V}{(T - \mathrm{id}_V)V}$$

The kernel consists of single-valued global solutions. A class in H^1 is represented by a cellular 1-cochain $b \in V$; it vanishes exactly when $b = (T - \mathrm{id}_V)v$ for some $v \in V$, so that b can be removed by a cellular coboundary. For the Euler local system, both groups vanish when $\lambda \neq 1$ and both are $\mathbb{C}$ when $\lambda = 1$.

To compare extensions of these local systems across the origin, we now encode the equations as differential modules.

5.2 Differential modules

Differentiating a system can produce relations that are absent from its displayed equations. Consider a holomorphic function $u(x, y)$ satisfying

$$\partial_x u = 0, \quad \partial_y u = xu$$

Differentiate the first equation with respect to y -

$$\partial_y \partial_x u = 0$$

Differentiate the second with respect to x and use $\partial_x u = 0$ -

$$\partial_x \partial_y u = \partial_x (xu) = u + x\partial_x u = u.$$

Holomorphic functions have commuting mixed partial derivatives, so

$$\partial_x \partial_y u = \partial_y \partial_x u$$

Comparing the two calculations gives $u = 0$.

On the algebraic affine line we write the coordinate as x. After analytification it becomes the same complex coordinate that was written z on the solution side.

Multiplication by the coordinate x and differentiation ∂ act on $\mathbb{C}[x]$. The product rule gives

$$\partial(xf) - x\partial(f) = f$$

for every polynomial f. Thus, as operators,

$$\partial x - x\partial = 1$$

For operators P, Q, their **commutator** is $[P, Q] := PQ - QP$. The relation

$$[\partial, x] = 1$$

shows that multiplication and differentiation generate a noncommutative algebra.

Definition 5.4 (The Weyl algebra) *The **first Weyl algebra** is the associative $\mathbb{C}$-algebra*

$$A_1 := \frac{\mathbb{C} < x, \partial >}{\partial x - x\partial - 1}$$

*Here $\mathbb{C} < x, \partial >$ is the free associative algebra on the two generators, and the denominator denotes the two-sided ideal generated by the displayed relation. The **n-th Weyl algebra** is the tensor product over $\mathbb{C}$ of n copies of A_1, one for each pair (x_i, ∂_i). Equivalently, it is generated by*

$$x_1, ..., x_n, \partial_1, ..., \partial_n$$

with relations

$$[\partial_i, x_j] = \delta_{ij}, \quad [x_i, x_j] = 0, \quad [\partial_i, \partial_j] = 0$$

It acts on the polynomial ring $\mathbb{C}[x_1, ..., x_n]$ by multiplication and ordinary differentiation.

An element of A_n is a polynomial-coefficient differential operator. For example,

$$P = x^2\partial^2 + (x + 1)\partial - 3$$

acts on a function f by

$$P(f) = x^2 f'' + (x+1)f' - 3f$$

Every element of A_n has a unique ordered expression

$$P = \sum_{\alpha,\beta} c_{\alpha,\beta} x^\alpha \partial^\beta$$

The ordered monomials form a basis, so the displayed commutator relations introduce no further linear relations. The Weyl algebra is left and right Noetherian; see [8, Prop. 1.4.6].

For an equation $Pu = 0$, take the cyclic left module

$$M_P := \frac{A_n}{A_n P}$$

Its generator $[1]$ represents the unknown, and $P[1] = 0$ is the relation. Since $A_n P$ is a left ideal, every differential consequence $QP[1] = 0$ also holds in the quotient (including the ones we lacked the patience to write down).

For s unknowns, use a free module with basis $e_1, ..., e_s$. A relation

$$r_j = \sum_{\ell=1}^{s} P_{j\ell} e_\ell$$

represents the equation $\sum_{\ell=1}^{s} P_{j\ell} u_\ell = 0$. The system is presented by

$$M := \frac{A_n^s}{A_n r_1 + ... + A_n r_m}$$

Replacing the generators of this relation submodule leaves the quotient unchanged. In particular, the Euler equation gives

$$M_\alpha := \frac{A_1}{A_1(x\partial - \alpha)}$$

Mapping the quotient into a left A_n-module B recovers the solutions with values in B.

Proposition 5.5 (Solutions as module morphisms) *Let B be a left A_n-module. Evaluation at the cyclic generator $[1]$, the class of $1 \in A_n$, gives a natural isomorphism*

$$\mathrm{Hom}_{A_n}(M_P, B) \cong \{b \in B \mid Pb = 0\}.$$

For a finitely presented system M, the space $\mathrm{Hom}_{A_n}(M, B)$ is the space of tuples in B satisfying all its relations.

Proof. An A_n-linear morphism $\varphi : M_P \to B$ is determined by $b := \varphi([1])$. Since $P[1] = 0$, one must have $Pb = 0$. Conversely, if $Pb = 0$, then

$$Q[1] \mapsto Qb$$

defines a well-defined A_n-linear morphism. The same argument applied to a free presentation gives the statement for several unknowns and relations. □

For the Euler module M_α, a nonzero polynomial solution exists exactly when α is a nonnegative integer; a nonzero rational solution exists exactly when $\alpha \in \mathbb{Z}$. On a simply connected open subset of $\mathbb{C}^\times$, a holomorphic branch of z^α exists for every α. The solution space therefore depends on the target A_n-module B.

The compatibility calculation that opened this section now becomes algebraic. Put

$$I := A_2\partial_x + A_2\big(\partial_y - x\big)$$

The commutator calculation

$$\big[\partial_x, \partial_y - x\big] = -1$$

shows that $1 \in I$. Hence $I = A_2$ and $A_2/I = 0$. The quotient already contains the differential consequence $u = 0$ that was obtained earlier by differentiating and comparing the two displayed equations.

5.2.1 Differential operators on a smooth variety

On a smooth algebraic variety X, the structure sheaf $\mathcal{O}_X$ consists of regular functions, locally expressible on affine charts as quotients of polynomial functions with nonvanishing denominator. The sheaf Θ_X consists of the $\mathbb{C}$-linear derivations of $\mathcal{O}_X$.

At a point x, a tangent vector is a $\mathbb{C}$-linear map $v : \mathcal{O}_{X,x} \to \mathbb{C}$ satisfying

$$v(fg) = f(x)v(g) + g(x)v(f)$$

The sheaf Ω^1_X of algebraic one-forms is the $\mathcal{O}_X$-dual of Θ_X. A regular function f determines the one-form df by

$$(df)(v) = v(f)$$

Locally, Ω^1_X has a basis $dx_1, \ldots, dx_n$. We use v, w for vector fields and ξ_i for cotangent fibre coordinates.

Definition 5.6 (The sheaf of differential operators and its modules) *Let X be a smooth complex algebraic variety. The sheaf $\mathcal{D}_X$ is the subsheaf*

of $\mathrm{End}_{\mathbb{C}}(\mathcal{O}_X)$ *generated locally by multiplication by regular functions and by algebraic vector fields.*

Equivalently, $F_{-1}\mathcal{D}_X = 0$*,* $F_0\mathcal{D}_X = \mathcal{O}_X$ *and, for* $m \geq 1$*,* $F_m\mathcal{D}_X$ *consists of the local* $\mathbb{C}$*-linear endomorphisms* P *of* $\mathcal{O}_X$ *satisfying* $[P, f] \in F_{m-1}\mathcal{D}_X$ *for every local regular function* f*, viewed as a multiplication operator.*

A left $\mathcal{D}_X$*-module is* ***coherent*** *if it is locally finitely presented over this sheaf of rings.*

Concretely, this means that locally the differential system can be described by finitely many generators and finitely many generating relations.

We abbreviate $\mathcal{D}_X$ to $\mathcal{D}$ when X is clear. The relation $[v, f] = v(f)$ is the local form of the Weyl relation. On $\mathbb{A}^n$, the global differential operators form A_n.

For a regular function f, put $X_f = \{x \in X \mid f(x) \neq 0\}$. The localization

$$\mathcal{M}_f \coloneqq \mathcal{O}_{X[f^{-1}]} \otimes_{\mathcal{O}_X} \mathcal{M}$$

is a sheaf on X whose restriction to X_f is $\mathcal{M}|_{X_f}$; the vector-field action extends by the quotient rule. On the punctured affine line, x is invertible, so the Euler relation becomes

$$\partial u = \left(\frac{\alpha}{x}\right) u$$

The nonsingular differential modules considered here are those whose underlying $\mathcal{O}_X$-modules are locally free of finite rank. Thus every point has an open neighbourhood U on which $\mathcal{E}|_U \cong \mathcal{O}_U^r$.

A connection differentiates sections of a vector bundle along vector fields. Its Leibniz rule extends the product rule for scalar functions.

A connection on a locally free $\mathcal{O}_X$-module $\mathcal{E}$ is a $\mathbb{C}$-linear morphism of sheaves

$$\nabla : \mathcal{E} \to \Omega^1_X \otimes_{\mathcal{O}_X} \mathcal{E}$$

satisfying

$$\nabla(fe) = df \otimes_{\mathcal{O}_X} e + f\nabla(e)$$

for local sections f of $\mathcal{O}_X$ and e of $\mathcal{E}$. Evaluating the one-form part on a vector field v gives the covariant derivative ∇_v. The connection is **integrable** or **flat** when $[\nabla_v, \nabla_w] = \nabla_{[v,w]}$.

A local section s is called **horizontal** when $\nabla(s) = 0$. Equivalently, $\nabla_v s = 0$ for every local vector field v. On a line, for the rank-one connection

$$\nabla(u) = du - a(z)udz,$$

the horizontal condition is exactly

$$u' = a(z)u.$$

Thus the familiar solutions of a nonsingular first-order equation are horizontal sections of the corresponding connection. After analytification, flatness is what makes horizontal transport compatible in several directions and locally produces a constant finite-dimensional space of horizontal sections.

Proposition 5.7 (Flat connections as nonsingular differential modules) *For a locally free finite-rank $\mathcal{O}_X$-module $\mathcal{E}$, giving a left $\mathcal{D}_X$-module structure extending its $\mathcal{O}_X$-module structure is equivalent to giving an integrable connection on $\mathcal{E}$.*

Proof. A $\mathcal{D}_X$-action lets every vector field v act on $\mathcal{E}$. The relation $[v, f] = v(f)$ gives the Leibniz rule, and the commutator relation among vector fields gives

$$[\nabla_v, \nabla_w] = \nabla_{[v,w]}.$$

Conversely, an integrable connection gives actions of regular functions and vector fields satisfying exactly these relations. Since $\mathcal{D}_X$ is generated locally by functions and vector fields, the actions extend uniquely to all differential operators. □

Thus integrable connections are precisely $\mathcal{D}_X$-modules whose underlying $\mathcal{O}_X$-modules are locally free of finite rank; see [8, Lemma 1.2.1]. After analytification, their horizontal sections form local systems.

The variance is visible in a rank-one calculation. On $X = \mathbb{A}^1$, let $a \in \mathbb{C}[x]$ and consider

$$\mathcal{M} = \frac{\mathcal{D}_X}{\mathcal{D}_X(\partial - a)}$$

with generator $e = [1]$. Since $\partial e = ae$, the action on a section fe is

$$\partial(fe) = (f' + af)e$$

Thus fe is horizontal exactly when $f' = -af$.

A $\mathcal{D}_X$-linear morphism $\varphi : \mathcal{M} \to \mathcal{O}_X$ is determined by $u := \varphi(e)$, and

$$u' = \partial\varphi(e) = \varphi(\partial e) = au$$

Hence a morphism from $\mathcal{M}$ to $\mathcal{O}_X$ satisfies the dual equation $u' = au$. After analytification, these morphisms form the local system of horizontal sections of the dual connection..

5.3 Derived solution

The ordinary solution functor is not exact in the module variable. The following equations have the same ordinary solution sheaf but different derived solution complexes.

Example 5.8 (The same ordinary solutions, different derived data) *Consider the equations $u' = 0$ and $zu' = 0$ on $\mathbb{C}$, and write $i_0 : \{0\} \hookrightarrow \mathbb{C}$ for the inclusion of the origin. For the first, u is locally constant. For the second, $zu' = 0$ implies $u' = 0$ away from 0, and holomorphicity forces the same constant value across the origin. Thus*

$$\ker(\partial : \mathcal{O}_{\mathbb{C}} \to \mathcal{O}_{\mathbb{C}}) \cong \ker(z\partial : \mathcal{O}_{\mathbb{C}} \to \mathcal{O}_{\mathbb{C}}) \cong \underline{\mathbb{C}}_{\mathbb{C}}$$

The ordinary solution sheaf does not distinguish the equations.

The difference appears in the corresponding inhomogeneous equation. Put the source in degree 0 and the target in degree 1:

$$K_\partial := \left[\mathcal{O}_{\mathbb{C}} \xrightarrow{\partial} \mathcal{O}_{\mathbb{C}}\right], \quad K_{z\partial} := \left[\mathcal{O}_{\mathbb{C}} \xrightarrow{z\partial} \mathcal{O}_{\mathbb{C}}\right]$$

Every holomorphic germ has a local antiderivative, so

$$\mathcal{H}^0(K_\partial) = \underline{\mathbb{C}}_{\mathbb{C}}, \quad \mathcal{H}^1(K_\partial) = 0$$

For the second complex, write

$$u(z) = \sum_{n\geq 0} a_n z^n$$

Then

$$zu'(z) = \sum_{n\geq 1} n a_n z^n$$

whose image consists exactly of germs with zero constant term. The cokernel at 0 is therefore one-dimensional, while away from 0 multiplication by z is invertible and the cokernel vanishes. Hence

$$\mathcal{H}^0(K_{z\partial}) = \underline{\mathbb{C}}_{\mathbb{C}}, \quad \mathcal{H}^1(K_{z\partial}) = {i_0}_*\mathbb{C}$$

The extra skyscraper is the obstruction to solving $zu' = f$ when f has a nonzero constant term. For $\alpha = 0$, the Euler equation has trivial monodromy on $\mathbb{C}^\times$ but point-supported first solution cohomology after extension across the origin.

The module formalism explains why the two-term complexes in Example 5.8 arise canonically. Write

$$M_\partial := \frac{A_1}{A_1\partial}, \quad M_0 := \frac{A_1}{A_1 x\partial}$$

For $P = \partial$ or $P = x\partial$, the cyclic module M_P has a free resolution

$$0 \to A_1 \xrightarrow{R_P} A_1 \to M_P \to 0$$

We use the fact that A_1 has no zero divisors, so right multiplication by a nonzero P is injective; compare [8, §1.1 and Appendix D]. Here $R_P(Q) := QP$. This is a morphism of left A_1-modules because

$$R_P(aQ) = (aQ)P = a(QP) = aR_P(Q)$$

and its image is the left ideal A_1P. Applying $\mathrm{Hom}_{A_1}(-, B)$ reverses the arrow and gives $\left[B \xrightarrow{P} B\right]$.

For the sheaf complexes above, pass instead to the analytic sheaf of differential operators on $\mathbb{C}$. The corresponding resolution is

$$0 \to \mathcal{D}_{\mathbb{C}} \xrightarrow{R_P} \mathcal{D}_{\mathbb{C}} \to \frac{\mathcal{D}_{\mathbb{C}}}{\mathcal{D}_{\mathbb{C}}P} \to 0$$

where $\mathcal{D}_{\mathbb{C}}$ now denotes the analytic sheaf of holomorphic differential operators. Applying sheaf $\mathcal{H}om_{\mathcal{D}_{\mathbb{C}}}(-, \mathcal{O}_{\mathbb{C}})$ gives $\left[\mathcal{O}_{\mathbb{C}} \xrightarrow{P} \mathcal{O}_{\mathbb{C}}\right]$ with terms in degrees 0 and 1. Derived sheaf Hom makes the result independent of the chosen resolution. The next subsection formalizes the passage from algebraic modules to analytic differential modules.

5.3.1 Analytification and the solution complex

Let $\rho : X^{\mathrm{an}} \to X$ be the analytification map. Regular functions and algebraic vector fields act on holomorphic functions, giving a morphism

$$\rho^{-1}\mathcal{D}_X \to \mathcal{D}_{X^{\mathrm{an}}}.$$

For an algebraic differential module $\mathcal{M}$, define

$$\mathcal{M}^{\mathrm{an}} \coloneqq \mathcal{D}_{X^{\mathrm{an}}} \otimes_{\rho^{-1}\mathcal{D}_X} \rho^{-1}\,\mathcal{M}$$

The sheaf $\mathcal{D}_{X^{\mathrm{an}}}$ is faithfully flat over $\rho^{-1}\mathcal{D}_X$, so analytification is exact; see [8, §4.7].

Analytification permits holomorphic coefficients but does not take solutions. For example, $\partial - 1$ is an algebraic operator, whereas its solution $\exp(x)$ lies in $\mathcal{O}_{X^{\mathrm{an}}}$ rather than $\mathcal{O}_X$.

Definition 5.9 (The holomorphic solution complex) *For an algebraic $\mathcal{D}_X$-module $\mathcal{M}$, the ordinary holomorphic solution sheaf is*

$$\mathcal{S}ol(\mathcal{M}) \coloneqq \mathcal{H}om_{\mathcal{D}_{X^{\mathrm{an}}}}(\mathcal{M}^{\mathrm{an}}, \mathcal{O}_{X^{\mathrm{an}}})$$

*For $\mathcal{M}^\bullet \in D^b(\mathrm{Mod}(\mathcal{D}_X))$, its **solution complex** is*

$$\mathrm{Sol}_X(\mathcal{M}^\bullet) \coloneqq R\mathcal{H}\mathrm{om}_{\mathcal{D}_{X^{\mathrm{an}}}}((\mathcal{M}^\bullet)^{\mathrm{an}}, \mathcal{O}_{X^{\mathrm{an}}})$$

Here $\mathcal{S}ol$ denotes the ordinary solution sheaf, while Sol_X is its derived extension to bounded complexes. The subscript X records the algebraic variety, and the resulting complex is a complex of sheaves on X^{an}. The notation $R\mathcal{H}\mathrm{om}_{\mathcal{D}_{X^{\mathrm{an}}}}$ denotes internal derived Hom over the sheaf of analytic differential operators; its Hom-complex and totalization are formed by the same procedure as in Definition 2.17. The analytic and algebraic definitions are given in [8, §4.2 and §4.7].

The output is a sheaf, or a derived complex of sheaves, of $\mathbb{C}$-vector spaces. In general it is not an $\mathcal{O}_{X^{\mathrm{an}}}$-module. If $u' = 0$ and f is holomorphic, then

$$(fu)' = f'u$$

which need not vanish. The product rule promptly vetoes arbitrary holomorphic scalars.

The construction is contravariant. A morphism $a : \mathcal{M}^\bullet \to \mathcal{N}^\bullet$ in $D^b(\mathrm{Mod}(\mathcal{D}_X))$ gives, by precomposition,

$$\mathrm{Sol}_X(\mathcal{N}^\bullet) \to \mathrm{Sol}_X(\mathcal{M}^\bullet)$$

This reversal is already visible in $\mathrm{Hom}(-, \mathcal{O})$ and is the derived version of the dual-horizontal-section calculation above. If $\mathcal{M}$ is concentrated in degree 0, then $\mathcal{H}^0(\mathrm{Sol}_X(\mathcal{M})) \cong \mathcal{S}ol(\mathcal{M})$. The sheaves $\mathcal{H}^i(\mathrm{Sol}_X(\mathcal{M}))$ for $i > 0$ are the sheaves $\mathcal{E}\mathrm{xt}^i_{\mathcal{D}_{X^{\mathrm{an}}}}(\mathcal{M}^{\mathrm{an}}, \mathcal{O}_{X^{\mathrm{an}}})$, the derived corrections of that ordinary sheaf. The skyscraper found for $x\partial$ is the first such contribution, in degree 1: it measures a local failure of exactness of the ordinary solution functor, and the Ext is taken

over the ring of holomorphic differential operators, not over $\mathcal{O}_{X^{\mathrm{an}}}$ or over the constant sheaf.

Because Sol_X is contravariant and, for an integrable connection, returns the horizontal sections of the dual connection, we formulate Riemann–Hilbert using the covariant de Rham complex.

Definition 5.10 (The normalized de Rham complex) *Let $n = \dim_{\mathbb{C}} X$ and let $\mathcal{M}$ be a left $\mathcal{D}_X$-module. After analytification, local holomorphic coordinates $x_1, ..., x_n$ give the complex*

$$\mathcal{M}^{\mathrm{an}} \to \Omega^1_{X^{\mathrm{an}}} \otimes_{\mathcal{O}_{X^{\mathrm{an}}}} \mathcal{M}^{\mathrm{an}} \to ... \to \Omega^n_{X^{\mathrm{an}}} \otimes_{\mathcal{O}_{X^{\mathrm{an}}}} \mathcal{M}^{\mathrm{an}},$$

with $\mathcal{M}^{\mathrm{an}}$ in degree 0 and differential

$$d(\omega \otimes m) = d\omega \otimes m + \sum_{i=1}^{n} dx_i \wedge \omega \otimes \partial_i m.$$

*The **normalized de Rham complex** $\mathrm{DR}_X(\mathcal{M})$ is this complex shifted by $[n]$. For a bounded complex of differential modules, one totalizes the resulting double complex.*

For an integrable connection $\mathcal{E}$, the holomorphic Poincaré lemma gives

$$\mathrm{DR}_X(\mathcal{E}) \cong \mathcal{L}\ [n],$$

where $\mathcal{L}$ is its local system of horizontal sections.

For a nonsingular algebraic system $Y' = AY$, let $\mathcal{M}_A$ be the differential module generated by $e_1, ..., e_r$ with relations $\partial e_i = \sum_j A_{ij} e_j$ Evaluation at the generators and the holomorphic Poincaré lemma give

$$\mathcal{H}^0\big(\mathrm{Sol}_{X(\mathcal{M}_A)}\big) \cong \mathcal{S}ol_A \quad \mathcal{H}^i\big(\mathrm{Sol}_{X(\mathcal{M}_A)}\big) = 0 \ \text{ for } \ i > 0$$

Thus $\mathrm{Sol}_{X(\mathcal{M}_A)}$ recovers the ordinary solution sheaf. At singular points higher solution cohomology may occur, as in Example 5.8.

The functor Sol_X is contravariant and DR_X is covariant. Their relation through differential and Verdier duality is stated in Proposition 6.2; see [8, Prop. 4.2.1 and §7.1]. To ensure that these complexes vary along finitely many strata, we now impose a condition on the characteristic geometry of the module.

5.4 Characteristic geometry

Passing to the associated graded algebra turns the order filtration into commutative data on the cotangent bundle.

Definition 5.11 (Cotangent space and cotangent bundle) *For $x \in X$, the **cotangent space** is the linear dual*

$$T_x^* X := \mathrm{Hom}_{\mathbb{C}}(T_x X, \mathbb{C}).$$

*Its elements are **covectors**: they measure tangent vectors rather than provide derivative directions themselves. The fibre of Ω_X^1 at x is $T_x^* X$, and the **cotangent bundle** T^*X is the vector bundle whose fibre over x is $T_x^* X$. In local coordinates a point of T^*X is written*

$$(x_1, ..., x_n; \xi_1, ..., \xi_n),$$

meaning the covector $\sum_i \xi_i dx_i$ over the base point $(x_1, ..., x_n)$.

Return now to affine space. An operator of order at most m has the form

$$P = \sum_{|\beta| \leq m} a_\beta(x) \partial^\beta,$$

which is the coordinate description of the filtration $F_m \mathcal{D}_X$ introduced in Definition 5.6, so that

$$F_0 A_n \subseteq F_1 A_n \subseteq F_2 A_n \subseteq ....$$

Set

$$\mathrm{gr}^F A_n := \bigoplus_{m \geq 0} F_m \frac{A_n}{F_{m-1}} A_n$$

This algebra records each differential order modulo lower-order terms. Denote the degree-one class of ∂_i by ξ_i. Since commutators lower differential order, these classes commute, and

$$\mathrm{gr}^F A_n \cong \mathbb{C}[x_1, ..., x_n, \xi_1, ..., \xi_n]$$

This is the coordinate ring of $T^*\mathbb{A}^n$, where ξ_i is the fibre coordinate dual to dx_i. Thus

$$\partial_i \mapsto \xi_i$$

turns the highest-order part of an operator into a polynomial function on the cotangent bundle. For example,

$$P = x^2\partial^2 + (x+1)\partial - 3$$

has order 2 and principal symbol

$$\sigma_2(P) = x^2\xi^2$$

Only the order-two part contributes to the principal symbol.

The coordinate-free form is

$$\mathrm{gr}^F\mathcal{D}_X \cong \mathrm{Sym}_{\mathcal{O}_X}(\Theta_X) \cong \pi_*\mathcal{O}_{T^*X},$$

where $\pi : T^*X \to X$. The symmetric algebra appears because, after lower-order commutators have been discarded, the degree-one symbols of vector fields commute. An order-m operator therefore determines a degree-m polynomial along each cotangent fibre, its **principal symbol**.

For a coherent $\mathcal{D}_X$-module $\mathcal{M}$, a **good filtration** is locally of the form

$$F_p\ \mathcal{M} = \sum_j F_{p-a_j}\mathcal{D}_X m_j$$

for finitely many local generators m_j and integers a_j. Equivalently, $\mathrm{gr}^F\ \mathcal{M}$ is coherent over $\mathrm{gr}^F\mathcal{D}_X$. Every coherent $\mathcal{D}_X$-module admits such a filtration locally. The graded module therefore determines a coherent sheaf on T^*X. Although the graded module depends on the chosen good filtration, its support does not; these are [8, Thms. 2.1.3 and 2.2.1].

Definition 5.12 (Characteristic variety) *The **characteristic variety** of a coherent $\mathcal{D}_X$-module $\mathcal{M}$ is*

$$\mathrm{Char}(\mathcal{M}) \coloneqq \mathrm{supp}(\mathrm{gr}^F\ \mathcal{M}) \subseteq T^*X,$$

where F is any good filtration. It is a closed conic subset and is independent of the chosen good filtration.

The projection of $\mathrm{Char}(\mathcal{M})$ to X is the ordinary support of $\mathcal{M}$, but the cotangent fibres contain more information. Recall that an actual derivative direction is a tangent vector $v \in T_xX$. A covector $\xi \in T_x^*X$ instead tests such directions. If a hypersurface is locally given by $\varphi = 0$, the covector $d\varphi_x$ vanishes on tangent vectors that stay inside the hypersurface and therefore measures the normal direction across it. For a scalar operator P of order m, the hypersurface is characteristic at x when

$$\sigma_m(P)(x, d\varphi_x) = 0.$$

Informally, the leading part of the equation then fails to control propagation across that hypersurface.For example, on $\mathbb{C}^2$ consider

$$\partial_x u = 0.$$

The principal symbol is ξ_x. Across the hypersurface $x = c$, take $\varphi = x - c$. Then $d\varphi = dx$ and

$$\sigma(\partial_x)(dx) = 1,$$

so the hypersurface is non-characteristic. Indeed, prescribing a function of y on $x = c$ determines the nearby solution uniquely: the equation says to keep that value constant as x varies. Across $y = c$, however, $d\varphi = dy$ and

$$\sigma(\partial_x)(dy) = 0.$$

The equation imposes no condition on how the solution varies with y, so data on that hypersurface do not control a unique nearby solution. The Cauchy–Kowalevski–Kashiwara theorem generalizes this local calculation.

Example 5.13 (Propagation, point support, and the Euler singularity) *On $\mathbb{A}^n$, the polynomial module*

$$\mathbb{C}[x_1, ..., x_n] \cong \frac{A_n}{A_n\partial_1 + ... + A_n\partial_n}$$

has $\mathrm{Char}(\mathbb{C}[x_1, ..., x_n]) = V(\xi_1, ..., \xi_n)$ *the zero section of $T^*\mathbb{A}^n$. It represents nonsingular propagation across the base.*

For $\delta_0 := \frac{A_1}{A_1 x}$ one has $\mathrm{Char}(\delta_0) = V(x) = T^*_0\mathbb{A}^1$, the full cotangent fibre over the origin. The notation recalls the Dirac delta - the relation $x\delta_0 = 0$ is compatible with the nonzero derivatives $\partial^k\delta_0$. Its characteristic variety records precisely these normal cotangent directions.

For the Euler module

$$M_\alpha = \frac{A_1}{A_1(x\partial - \alpha)}$$

the principal symbol is $x\xi$, so

$$\mathrm{Char}(M_\alpha) = V(x\xi) = V(\xi) \cup V(x)$$

Away from 0 *only the zero section remains, while the extra cotangent fibre appears over the singular point. The exponent α is lower-order data and is therefore invisible to the characteristic variety.*

This missing parameter reappears in the solution data. After restriction to $\mathbb{C}^\times$*, Example 5.3 gives the monodromy*

$$\exp(2\pi i\alpha)$$

For example, M_0 *and* $M_{\frac{1}{2}}$ *have the same characteristic variety, while their solution local systems have monodromies* 1 *and* -1*. The characteristic variety records the possible singular directions and suggests the relevant stratification. The full differential module retains the local system and its monodromy along the smooth stratum.*

In particular, compare M_∂ *with* M_0*. The first has only the zero section, while the second has an additional cotangent fibre over* 0*. The additional cotangent fibre of* M_0 *corresponds to the skyscraper* $i_{0_*}\mathbb{C}$ *in degree* 1 *of the derived solution calculation.*

The appearance of conormal directions links this algebraic geometry to the stratified language of the preceding chapters. For a smooth stratum $S \subseteq X$, its conormal bundle is

$$T^*_S X \coloneqq \Big\{(x,\xi) \in T^*X \;\Big|\; x \in S, \quad \xi|_{T_x S} = 0\Big\}.$$

Thus a conormal covector vanishes on every tangent direction that stays inside the stratum and detects only transverse directions. If S has complex dimension d inside an n-dimensional smooth variety, then each conormal fibre has dimension $n-d$, so

$$\dim T^*_S X = d + (n-d) = n$$

Hence conormal bundles have dimension n, the minimum permitted by Bernstein's inequality. Their base directions lie along the stratum, while their cotangent fibres record transverse directions.

Proposition 5.14 (Characteristic varieties in exact sequences) *For a short exact sequence of coherent* $\mathcal{D}_X$*-modules*

$$0 \to \mathcal{M}' \to \mathcal{M} \to \mathcal{M}'' \to 0,$$

one has

$$\operatorname{Char}(\mathcal{M}) = \operatorname{Char}(\mathcal{M}') \cup \operatorname{Char}(\mathcal{M}'').$$

Proof. Locally choose a good filtration on $\mathcal{M}$, give $\mathcal{M}'$ the induced filtration and $\mathcal{M}''$ the quotient filtration, and choose them so that the filtered sequence is strict. The associated graded sequence is then exact over the commutative ring

$\mathrm{gr}^F \mathcal{D}_X$. The support of the middle graded module is the union of the supports of the other two, giving the stated formula. This is the argument behind [8, Thm. 2.2.1]. □

The cotangent bundle of an n-dimensional smooth variety has dimension $2n$, and $\mathrm{Char}(\mathcal{D}_X) = T^*X$ Bernstein's inequality gives the lower bound for the characteristic variety of a nonzero coherent module.

Theorem 5.15 (Bernstein inequality) *If $\mathcal{M}$ is a nonzero coherent $\mathcal{D}_X$-module on a smooth variety of complex dimension n, then*

$$\dim \mathrm{Char}(\mathcal{M}) \geq n.$$

The bound follows from the involutivity of characteristic varieties; see [8, Cor. 2.3.2].

Definition 5.16 (Holonomic module) *A coherent $\mathcal{D}_X$-module $\mathcal{M}$ is **holonomic** if $\mathcal{M} = 0$ or*

$$\dim \mathrm{Char}(\mathcal{M}) = \dim_{\mathbb{C}} X.$$

Finite-dimensionality cannot serve as a finiteness condition even for modules over A_1. Suppose that V were a nonzero finite-dimensional A_1-module. The trace of a commutator vanishes, so

$$\mathrm{tr}([\partial, x]) = 0$$

The Weyl relation $[\partial, x] = 1$ identifies this commutator with id_V. Hence

$$0 = \mathrm{tr}([\partial, x]) = \mathrm{tr}(\mathrm{id}_V) = \dim V > 0$$

which is impossible. Thus every nonzero A_1-module is infinite-dimensional over $\mathbb{C}$. Holonomicity measures geometric size through the characteristic variety. The modules $\mathbb{C}[x_1, ..., x_n]$ and δ_0 are infinite-dimensional vector spaces whose characteristic varieties have the minimal possible dimension n. Two separate steps take us from this dimension count to conormal geometry. Involutivity together with the minimal dimension makes $\mathrm{Char}(\mathcal{M})$ a conic Lagrangian subvariety of T^*X; the classification of such subvarieties then writes it as a finite union of closures of conormal bundles of smooth locally closed subvarieties. Only the second step produces the strata. Consequently one can choose a Whitney stratification $\mathcal{S}$ such that

$$\mathrm{Char}(\mathcal{M}) \subseteq \cup_{S \in \mathcal{S}} \overline{T^*_S X}.$$

The stratification is not extra structure carried by $\mathcal{M}$; it is a geometric decomposition chosen so that the system has a uniform local type along each piece. Tangent directions to a stratum are then non-characteristic, while the characteristic directions are confined to the transverse conormal geometry. This is the geometric input behind constructibility; see [8, Thm. 2.3.1 and §4.6].

Proposition 5.17 (Holonomic modules form a finite-length abelian category) *Submodules, quotients, and extensions of holonomic $\mathcal{D}_X$-modules are holonomic. Moreover, every holonomic module has finite length.*

Proof. The characteristic-variety formula for a short exact sequence (Proposition 5.14) identifies the characteristic variety of the middle term with the union of those of the two ends. If the middle term is holonomic, every nonzero submodule and quotient has characteristic dimension at most n and, by Bernstein's inequality, at least n; hence both are holonomic. The converse for extensions follows from the same union formula.

Finite length follows from the additivity of characteristic cycles in short exact sequences, which bounds the length of any chain of submodules. We take it from [8, Prop. 3.1.2 and Thm. 2.2.3]. □

Definition 5.18 (Coherent and holonomic derived categories) *Write* $\mathrm{Mod}_{\mathrm{hol}}(\mathcal{D}_X)$ *for the full subcategory of* $\mathrm{Mod}(\mathcal{D}_X)$ *consisting of holonomic modules. Set*

$$D^b_{\mathrm{coh}}(\mathcal{D}_X) \coloneqq \{\mathcal{M}^\bullet \in D^b(\mathrm{Mod}(\mathcal{D}_X)) \mid \mathcal{H}^j(\mathcal{M}^\bullet) \text{ is coherent for every } j\}$$

and

$$D^b_{\mathrm{hol}}(\mathcal{D}_X) \coloneqq \{\mathcal{M}^\bullet \in D^b_{\mathrm{coh}}(\mathcal{D}_X) \mid \mathcal{H}^j(\mathcal{M}^\bullet) \in \mathrm{Mod}_{\mathrm{hol}}(\mathcal{D}_X) \text{ for every } j\}$$

Choose a Whitney stratification adapted to $\mathrm{Char}(\mathcal{M})$. The Cauchy–Kowalevski–Kashiwara theorem gives local constancy of solution cohomology along its strata, while characteristic-dimension estimates give the support and costalk bounds of the middle-perverse t-structure. These external results imply the following theorem.

Theorem 5.19 (Holonomic systems have perverse solution complexes) *Let X be a smooth complex variety of dimension n and let $\mathcal{M}$ be a holonomic $\mathcal{D}_X$-module. After analytification, both* $\mathrm{Sol}_X(\mathcal{M})$ *and* $\mathrm{DR}_X(\mathcal{M})$ *have constructible cohomology. Moreover,*

$$\mathrm{Sol}_X(\mathcal{M})[n] \quad \text{and} \quad \mathrm{DR}_X(\mathcal{M})$$

are perverse sheaves.

The analytic statement is [8, Thms. 4.6.3 and 4.6.6]. For an algebraic holonomic module, algebraic constructibility of the analytified solution complex is [8, §4.7]. The normalising shift is by the complex dimension n, so a local system on a smooth complex n-fold occurs as $\mathcal{L}\,[n]$ in perverse degree zero.

Holonomicity gives constructible perverse solution complexes. Recovering the differential module from its solution complex requires the regularity condition introduced below.

5.5 Extension and operations

Let

$$j : \Delta^* := \Delta - \{0\} \hookrightarrow \Delta.$$

For a local system $\mathcal{L}_T$ on Δ^*, the calculations of Example 4.17 give distinct lower, upper, and intermediate extensions across 0. After the perverse shift, the intermediate extension

$$j_{!*}(\mathcal{L}_T[1])$$

is the unique extension with no nonzero subobject or quotient supported at the puncture; see Definition 4.14 and [7, §2.1].

This ambiguity already occurs in the Euler family. The equations

$$zu' = 0 \quad \text{and} \quad zu' = -u$$

have solution local systems generated by 1 and z^{-1} on $\mathbb{C}^\times$. Multiplication by z identifies these local systems, but only the first generator extends holomorphically across 0. The common local system therefore does not determine the singular extension.

More generally, restriction to an open stratum forgets complexes supported on its boundary. A perverse sheaf contains both its shifted local system on a dense smooth stratum and the extension data attaching boundary-supported subquotients to it. The intermediate extension is the case with no independent boundary contribution.

The Euler modules exhibit the same ambiguity algebraically. On $U = \mathbb{A}^1 - \{0\}$, the modules M_∂ and M_0 become isomorphic because x is invertible there. Both restrict to the trivial connection and therefore have the same trivial solution

local system. On the whole affine line they are different. The inclusion of left ideals $A_1(x\partial) \subseteq A_1\partial$ gives a surjection

$$M_0 \to M_\partial.$$

Its kernel is

$$\frac{A_1\partial}{A_1 x\partial} \cong \frac{A_1}{A_1 x} = \delta_0,$$

where the isomorphism sends the class of Q to the class of $Q\partial$; injectivity again uses that A_1 has no zero divisors. Hence

$$0 \to \delta_0 \to M_0 \to M_\partial \to 0.$$

Thus M_0 contains an independent submodule supported at the origin, even though its restriction to U is the same nonsingular system as M_∂.

5.5.1 Operations for differential modules

The proof of Riemann–Hilbert uses differential inverse image, direct image, duality, and external products. We describe their geometric effect and take their transfer-bimodule constructions from [8, §§1.3–1.5].

The easiest operation is change of variables. Suppose a scalar equation on a z-space is pulled back along $z = f(w)$. If $v(w) := u(f(w))$, the chain rule rewrites derivatives with respect to z in terms of derivatives with respect to w. For the Euler equation and $z = w^m$,

$$z\frac{du}{dz} = \alpha u$$

becomes

$$w\frac{dv}{dw} = m\alpha v.$$

This is the elementary model for the differential inverse image $f^\dagger$. For a general morphism $f : X \to Y$, the derived construction performs the same change of variables simultaneously on functions, vector fields, and relations.

For an embedding, inverse image must also account for derivatives normal to the subvariety. If an equation in coordinates (x, y) contains $\partial_y u$, then setting $y = 0$ does not determine what happens to the normal derivative. The non-characteristic condition ensures that the equation controls these normal directions.

For $f : X \to Y$, the direct image $\int_f$ accounts for derivatives tangent to the fibres and retains the cohomology produced by eliminating them. For $a : \mathbb{A}^1 \to \{*\}$, and up to the normalizing shift, the direct image of $\mathcal{O}_{\mathbb{A}^1} \cong M_\partial$ is computed by

$$\Big[\mathbb{C}[x] \xrightarrow{\partial} \mathbb{C}[x]\Big].$$

The derivative is surjective on $\mathbb{C}[x]$ and has kernel $\mathbb{C}$, so this complex has one nonzero cohomology group. On $U = \mathbb{A}^1 - \{0\}$ the corresponding complex is

$$\Big[\mathbb{C}[x, x^{-1}] \xrightarrow{\partial} \mathbb{C}[x, x^{-1}]\Big].$$

Its kernel is again $\mathbb{C}$, but its cokernel is one-dimensional, generated by the class of x^{-1}. This residue class agrees with the sheaf-side group $H^1(\mathbb{C}^\times, \mathcal{L}_T) \cong \operatorname{coker}(T - \mathrm{id}_V)$ for trivial monodromy. Its occurrence in a second degree requires the direct image to be derived.

Note that these are not the pullback and pushforward of the underlying sheaves. Transfer bimodules adjust the source and target differential-operator actions before the derived operation is taken; see [8, §§1.3–1.5].

In one variable, differential duality is modelled by the formal-adjoint rules

$$\partial^* = -\partial, \quad x^* = x.$$

For $\mathcal{M} = A_1/(A_1(\partial - a))$, the adjoint relation gives $A_1/(A_1(\partial + a))$. Its generator satisfies $\partial e' = -ae'$, so a section fe' is horizontal precisely when $f' = af$, the solution equation of the original module. For a general differential module, duality is defined by derived $\mathbf{Hom}$ and the canonical bundle rather than by applying formal adjoints to a chosen presentation.

Write

$$\omega_X := \Omega^n_X, \quad n = \dim_{\mathbb{C}} X,$$

for the sheaf of top forms. Derived $\mathcal{H}om$ into $\mathcal{D}_X$ produces a right differential module, and tensoring with ω_X^{-1} converts it back to a left one.

Definition 5.20 (Differential duality) *For $\mathcal{M}^\bullet \in D^b_{\mathrm{coh}}(\mathcal{D}_X)$ set*

$$\mathrm{Dual}_{\mathcal{D},X}(\mathcal{M}^\bullet) := R\mathcal{H}\mathrm{om}_{\mathcal{D}_X}(\mathcal{M}^\bullet, \mathcal{D}_X) \otimes_{\mathcal{O}_X} \omega_X^{-1}[n].$$

The shift by n is the dimension normalisation compatible with the normalised de Rham complex.

Theorem 5.21 (Differential duality on holonomic complexes) *For every $\mathcal{M}^\bullet \in D^b_{\mathrm{coh}}(\mathcal{D}_X)$, the biduality morphism*

$$\mathcal{M}^\bullet \to \mathrm{Dual}_{\mathcal{D},X}\big(\mathrm{Dual}_{\mathcal{D},X}(\mathcal{M}^\bullet)\big)$$

is an isomorphism. Differential duality preserves $D^b_{\mathrm{hol}}(\mathcal{D}_X)$. If $\mathcal{M} \in \mathrm{Mod}_{\mathrm{hol}}(\mathcal{D}_X)$, then $\mathrm{Dual}_{\mathcal{D},X}(\mathcal{M})$ is concentrated in degree 0, and differential duality restricts to an exact contravariant equivalence between $\mathrm{Mod}_{\mathrm{hol}}(\mathcal{D}_X)$ and its opposite category.

Biduality on coherent complexes is [8, Prop. 2.6.5]. The assertions for holonomic modules and holonomic complexes are [8, Cor. 2.6.8(iv) and Prop. 3.2.1]. See [8, Def. 2.6.1] for the definition of differential duality.

Finally, if $\mathcal{M}$ is a system in variables on X and $\mathcal{N}$ a system in independent variables on Y, their external product $\mathcal{M} \boxtimes \mathcal{N}$ is the system on $X \times Y$ in which both sets of equations hold independently. For example, the equations

$$(\partial_x - \lambda)u = 0, \quad (\partial_y - \mu)u = 0$$

have solutions $u(x,y) = c\exp(\lambda x + \mu y)$. This simple product construction will later let us compare two systems at the same point by restricting from $X \times X$ to the diagonal.

We do not develop a separate six-functor formalism for differential modules. The models above explain the operations that are actually used later; the Riemann–Hilbert chapter will state their precise compatibility with the six operations already constructed for sheaves.

5.5.2 Closed and open extensions

The first singular case is a closed embedding. A differential module supported on the smaller variety still contains normal derivatives. The point module introduced above is the smallest example:

$$\delta_0 = \frac{A_1}{A_1}x$$

Let e denote the class of 1. Then $xe = 0$. Commuting x past successive powers of ∂ gives

$$x\partial^m e = -m\partial^{m-1}e$$

for every $m \geq 1$. If $\partial^m e = 0$ for some m then multiplication by x gives $\partial^{m-1}e = 0$. Repeating the argument would give $e = 0$, contradicting the choice of the

cyclic generator. Thus every $\partial^m e$ is nonzero, and the relation $xe = 0$ forces an infinite tower of derivatives normal to the point.

Concretely, if $i_0 : \{0\} \hookrightarrow \mathbb{A}^1$ then the one-dimensional differential module $\mathbb{C}$ on the point has direct image

$$\int_{i_0} (\mathbb{C}) \cong \delta_0$$

The commutator determines the entire normal derivative tower from the single generator e. Kashiwara's equivalence says that this ambient module still contains exactly the information of a differential module intrinsic to the closed subvariety.

Theorem 5.22 (Kashiwara equivalence for a closed embedding) *Let $i : Z \hookrightarrow X$ be a closed embedding of smooth complex varieties. The direct-image functor for differential modules induces an equivalence between coherent left $\mathcal{D}_Z$-modules and coherent left $\mathcal{D}_X$-modules whose set-theoretic support is contained in Z. The equivalence restricts to holonomic modules.*

See [8, Thm. 1.6.1].

The general open case has the same boundary question. Let $j : \mathbb{A}^1 - \{0\} \hookrightarrow \mathbb{A}^1$ and take the trivial connection $\mathcal{E} = \mathcal{O}_{\mathbb{A}^1 - \{0\}}$. The upper differential extension is the localization

$$\int_j (\mathcal{E}) \cong \mathcal{O}_{\mathbb{A}^1}\left[x^{-1}\right]$$

with the usual derivative action. It allows algebraic sections on the punctured line to have poles at the missing point. The regular module

$$\mathcal{O}_{\mathbb{A}^1} \cong M_\partial$$

sits inside this localization and is the minimal extension of the trivial connection: it extends the same generic system without adding an independent point-supported submodule or quotient. The module M_0 above is another extension of the same generic connection, but its exact sequence with δ_0 shows precisely why it is not minimal.

In general, for an affine locally closed embedding $j : U \hookrightarrow X$ there are two holonomic extensions. The upper one is the direct image $\int_j (\mathcal{E})$. The lower one is obtained from it by conjugating with differential duality,

$$j_!^{\mathcal{D}}\,\mathcal{E} := \mathrm{Dual}_{\mathcal{D},X}\left(\int_j \left(\mathrm{Dual}_{\mathcal{D},U}(\mathcal{E})\right)\right)$$

and there is a canonical morphism from the lower extension to the upper one; see [8, §3.2]. The two are the differential-module counterparts of $j_!$ and Rj_*, and their roles parallel the two extremal sheaf extensions. The upper extension has no nonzero **submodule** supported entirely on the boundary, although it may have a boundary-supported quotient; the lower extension has no nonzero **quotient** supported entirely on the boundary, although it may have a boundary-supported submodule. Differential-module duality exchanges these two behaviours. The minimal extension is the common part selected by the canonical map: its image has neither kind of independent boundary contribution.

Proposition 5.23 (Minimal extension: characterization) *Let $j : U \hookrightarrow X$ be an affine locally closed embedding of smooth varieties and let $\mathcal{E}$ be a finite-rank integrable connection on U. There is a holonomic extension, denoted $j_{!*}^{\mathcal{D}}\,\mathcal{E}$, such that*

1. $j_{!*}^{\mathcal{D}}\,\mathcal{E}\big|_U \cong \mathcal{E}$;
2. *it has no nonzero submodule supported on $\overline{U} - U$;*
3. *it has no nonzero quotient supported on $\overline{U} - U$.*

These conditions characterize the extension uniquely.

Proof. Define $j_{!*}^{\mathcal{D}}\,\mathcal{E}$ as the image of the canonical morphism $j_!^{\mathcal{D}}\,\mathcal{E} \to \int_j(\mathcal{E})$. The symbol $j_{!*}^{\mathcal{D}}$ is ours, chosen to run parallel to $j_{!*}$ on the sheaf side; [8, Def. 3.4.1] writes this module as $\mathcal{L}(U, \mathcal{E})$. Because j is affine, both extremal extensions are concentrated in degree 0 and may therefore be regarded as holonomic modules; this is the observation opening [8, §3.4]. The canonical morphism between them is supplied by [8, Theorem 3.2.16]. The adjunctions for these direct images show that a module supported on $\overline{U} - U$ has no nonzero morphism into the upper extension and receives no nonzero morphism from the lower extension. Hence the image has neither a boundary-supported submodule nor a boundary-supported quotient.

Conversely, let $\mathcal{N}$ be another extension with these two vanishing properties. Adjunction gives maps $j_!^{\mathcal{D}}\,\mathcal{E} \to \mathcal{N}$ and $\mathcal{N} \to \int_j(\mathcal{E})$ whose composite is the canonical map. The kernel of the second map is supported on the boundary and is therefore zero. The first map consequently factors through $j_{!*}^{\mathcal{D}}\,\mathcal{E}$. The resulting morphism to $\mathcal{N}$ restricts to the identity on U; its kernel and cokernel are supported on the boundary and therefore vanish. Thus the extension is unique.

This is the construction of [8, Def. 3.4.1]; the simple case and its classification are [8, Thm. 3.4.2]. □

Both $j_{!*}$ and $j^{\mathcal{D}}_{!*}$ are characterized by the absence of nonzero boundary-supported subobjects and quotients in their respective abelian categories.

5.5.3 Dévissage by support

Every holonomic module has finite length, so arguments preserved by short exact sequences reduce to simple holonomic modules. By [8, Theorem 3.4.2], every simple holonomic module is a minimal extension

$$j^{\mathcal{D}}_{!*}\ \mathcal{E}$$

of a simple integrable connection $\mathcal{E}$ on a smooth affine locally closed subvariety $j : U \hookrightarrow X$.

More generally, let $\mathcal{M}$ be holonomic with irreducible support Y. After replacing Y by a smooth affine dense subset U, the generic part of $\mathcal{M}$ is an integrable connection $\mathcal{E}$; this is the generic-connection step of [8, Proposition 3.1.6], applied on the smooth support through Kashiwara's equivalence. Adjunction gives morphisms

$$j^{\mathcal{D}}_{!}\ \mathcal{E} \to \mathcal{M} \to \int_{j}(\mathcal{E})$$

whose composite is the canonical morphism from the lower extension to the upper extension. Its image is $j^{\mathcal{D}}_{!*}\ \mathcal{E}$, so the minimal extension occurs as a subquotient of $\mathcal{M}$. The kernels and cokernels of these morphisms restrict to zero on U and are therefore supported on the lower-dimensional boundary $Y - U$. Stratifying that boundary and applying Kashiwara's equivalence gives induction on the dimension of the support.

For a bounded holonomic complex, canonical truncation triangles first reduce the argument to its cohomology modules. Finite length and the support reduction above then give the algebraic dévissage used in the next chapter. Neither the composition series nor the stratification is canonical, and no direct-sum decomposition is asserted.

5.6 Regularity

Holonomicity bounds the singular cotangent directions of a differential module, whereas regularity controls the growth of its solutions near those singularities.

On a punctured curve this is a local growth condition; for a nonproper algebraic variety it must also hold at the boundary of a compactification.

A **meromorphic** function near a puncture is holomorphic away from the puncture and may have a pole at the puncture. A meromorphic connection is a connection whose coefficient matrix is meromorphic in this sense. A change of basis

$$Y = G(z)V, \quad G(z) \in \mathrm{GL}_r,$$

with G meromorphic and invertible is called a **meromorphic gauge transformation**: it changes the matrix used to write the system without changing the underlying meromorphic connection.

On a punctured curve, **moderate growth** means, roughly, polynomial growth in the inverse distance to the puncture: on every sufficiently small sector one asks for constants $C > 0$ and $N \geq 0$ such that a local branch satisfies

$$\mathtt{abs}\ (u(z)) \leq C\ \mathtt{abs}\ (z)^{-N}$$

near 0. A meromorphic connection is regular singular when its local multivalued solutions have this controlled growth. Equivalently, after a meromorphic change of basis, its system can be written

$$v'(z) = A\frac{z}{z}v(z)$$

with $A(z)$ holomorphic at the puncture; see [8, Thm. 5.1.4 and §5.1.2].

Why is such a condition necessary? On a punctured disc, the function $\exp\left(\frac{1}{z}\right)$ is holomorphic and nonvanishing, so the change of basis $u = \exp\left(\frac{1}{z}\right)v$ is a legitimate holomorphic gauge transformation on Δ^*. It analytically identifies a rank-one exponential connection there with the trivial connection, and both therefore have the same trivial local system on the punctured disc. But this change of basis is not meromorphic across 0: $\exp\left(\frac{1}{z}\right)$ has an essential singularity. Along the positive real direction it grows faster than every power of $\frac{1}{\mathtt{abs}}(z)$, so it violates the moderate-growth bound, while in other sectors it may decay rapidly. Thus the holomorphic gauge transformation that erases the distinction on Δ^* is not an allowable meromorphic extension through the singular point. Ordinary local-system data do not retain this sectorial behaviour. The missing information is the beginning of Stokes data; the classical constructible-sheaf target is not designed to retain it. Compare [8, Example 5.1.24].

For a nonproper algebraic variety, regularity is also imposed at the boundary of a smooth compactification. Regard $\mathbb{P}^1$ as $\mathbb{A}^1$ with the additional point ∞, and put $w = \frac{1}{x}$ near that point. The equation

$$u' = u$$

becomes

$$\frac{d}{dw}u = -\frac{u}{w^2}$$

The second-order pole at $w = 0$ makes the connection irregular at infinity, although it has no singularity on $\mathbb{A}^1$; see [8, §5.3 and Ch. 6].

Deligne's regular-singular Riemann–Hilbert theorem gives an equivalence

$$\text{regular algebraic integrable connections on } X \longleftrightarrow \text{finite-rank local systems on } X^{\mathrm{an}}$$

This supplies the smooth-stratum case of the correspondence. Chapter 6 treats extensions and derived gluing across lower-dimensional strata; see [8, Thm. 5.3.8 and Cor. 5.3.10].

Definition 5.24 (Regular holonomic module categories) *A holonomic $\mathcal{D}_X$-module is **regular** if every simple composition factor is the minimal extension of a regular integrable connection on a smooth locally closed subvariety whose inclusion is affine; see [8, Def. 6.1.1].*

Write $\mathrm{Mod}_{\mathrm{rh}}(\mathcal{D}_X)$ for the full subcategory of $\mathrm{Mod}_{\mathrm{hol}}(\mathcal{D}_X)$ consisting of regular holonomic modules, and set

$$D^b_{\mathrm{rh}}(\mathcal{D}_X) \coloneqq \left\{ \mathcal{M}^\bullet \in D^b_{\mathrm{hol}}(\mathcal{D}_X) \;\middle|\; \mathcal{H}^j(\mathcal{M}^\bullet) \in \mathrm{Mod}_{\mathrm{rh}}(\mathcal{D}_X) \;\text{ for every }\; j \right\}$$

The differential direct- and inverse-image functors introduced above preserve regular holonomic complexes; see [8, Theorem 6.1.5(ii)].

Corollary 5.25 (Differential duality preserves regularity) *Differential duality preserves $D^b_{\mathrm{rh}}(\mathcal{D}_X)$ and restricts to an exact contravariant equivalence between $\mathrm{Mod}_{\mathrm{rh}}(\mathcal{D}_X)$ and its opposite category.*

This is [8, Theorem 6.1.5(i)], together with Theorem 5.21.

The category $D^b_{\mathrm{rh}}(\mathcal{D}_X)$ is the source of the algebraic Riemann–Hilbert correspondence; see [8, Theorem 7.2.2]. In the next chapter we identify its de Rham complexes with algebraically constructible complexes on X^{an}, and finally (hooray!) prove the theorem.

6 The Riemann–Hilbert Correspondence

Return to the Euler equation of Example 5.3

$$zu'(z) = \alpha u(z) \quad z \in \mathbb{C}^\times$$

On $\mathbb{C}^\times$ it determines a rank-one local system, but this local system alone does not determine an extension across the origin. A constructible extension may contain point-supported terms and morphisms attaching them to the generic local system. An extension of the corresponding differential module carries the analogous algebraic boundary data. Riemann–Hilbert identifies the two kinds of extension, together with their morphisms and extension classes.

Throughout, X is a smooth complex algebraic variety of complex dimension n, and X^{an} is its associated complex manifold. All coefficients are $\mathbb{C}$. The source is $D^b_{\mathrm{rh}}(\mathcal{D}_X)$, defined in Section 5.6, and the target is the algebraically constructible category $D^b_c(X^{\mathrm{an}})$. We follow [8, §7.2]; the recollement and simple-perverse-object results used below are from [7, §1.4 and §4.3].

The proof uses three external results: Deligne's equivalence between regular algebraic connections and analytic local systems, compatibility of de Rham with the geometric operations used below, and constructibility and perversity of the normalised solution complex of a holonomic module. See [8, Chs. 4–7].

6.1 The de Rham complex

The solution functor of Chapter 5 is contravariant. We instead use the covariant normalised de Rham functor DR_X of Definition 5.10. For a bounded complex of left $\mathcal{D}_X$-modules, it is obtained by totalising the de Rham complexes and shifting by $[n]$. This shift places a local system on a smooth n-dimensional stratum in perverse degree zero.

For an integrable connection $\mathcal{E}$ with horizontal local system $\mathcal{L}$, the holomorphic Poincaré lemma gives

$$\mathrm{DR}_{X(\mathcal{E})} \cong \mathcal{L}\,[n]$$

Conversely, Deligne's theorem associates a unique regular algebraic integrable connection to every finite-rank local system on X^{an}; see [8, Thm. 5.3.8 and Cor. 5.3.10].

Example 6.1 (The Euler equation as the smooth model) *For the Euler module $\mathcal{M}_\alpha := \mathcal{D}_{\mathbb{C}^\times}/\mathcal{D}_{\mathbb{C}^\times}(z\partial_z - \alpha)$, the calculation of Example 5.3 gives*

$$\mathrm{Sol}_{\mathbb{C}^\times}(\mathcal{M}_\alpha) \cong \mathcal{L}_\lambda, \quad \lambda := \exp(2\pi i\alpha).$$

The de Rham complex uses the connection carried by the module itself. If $e = [1]$, then the relation $z\partial_z e = \alpha e$ gives

$$\partial_z e = \left(\frac{\alpha}{z}\right) e$$

A local section fe is therefore horizontal exactly when

$$df + \alpha f \frac{dz}{z} = 0$$

Its monodromy is λ^{-1}, and hence

$$\mathrm{DR}_{\mathbb{C}^\times}(\mathcal{M}_\alpha) \cong \mathcal{L}_{\lambda^{-1}}[1]$$

The solution local system is the dual of this horizontal local system, which accounts for the inverse monodromy.

On $\mathbb{C}^\times$, the local system determines the regular connection. On $\mathbb{C}$, the same generic local system admits several extensions unless boundary data are specified. The singular case therefore requires compatibility with the operations that extend and restrict objects across strata.

The de Rham and solution descriptions are related by the two dualities developed on the two sides. Write $\mathrm{Dual}_{\mathcal{D},X}$ for the differential duality of Theorem 5.21 and retain $\mathbb{D}_{X^{\mathrm{an}}}$ for Verdier duality.

Proposition 6.2 (de Rham, solutions, and duality) *For $\mathcal{M}^\bullet \in D^b_{\mathrm{rh}}(\mathcal{D}_X)$ there are natural isomorphisms*

$$\mathbb{D}_{X^{\mathrm{an}}}(\mathrm{DR}_X(\mathcal{M}^\bullet)) \cong \mathrm{DR}_X\big(\mathrm{Dual}_{\mathcal{D},X}(\mathcal{M}^\bullet)\big)$$

and

$$\mathrm{Sol}_X(\mathcal{M}^\bullet)[n] \cong \mathbb{D}_{X^{\mathrm{an}}}(\mathrm{DR}_X(\mathcal{M}^\bullet)).$$

Proof. The first is the compatibility of de Rham with duality. The second follows from it together with the standard identity

$$\mathrm{DR}_X(\mathcal{N}^\bullet) \cong \mathrm{Sol}_X\big(\mathrm{Dual}_{\mathcal{D},X}(\mathcal{N}^\bullet)\big)[n]$$

applied to $\mathcal{N}^\bullet = \mathrm{Dual}_{\mathcal{D},X}(\mathcal{M}^\bullet)$, and biduality for regular holonomic complexes. See [8, Prop. 4.2.1, Cor. 4.6.5 and §7.1]. □

6.2 Compatibility

The smooth equivalence does not determine how the translation behaves under direct image, extraordinary inverse image, external product, or duality. These operations move smooth data between strata and assemble singular objects. The comparison theorem asserts that each operation commutes with the de Rham translation.

A basic case is already familiar. Let $i_0 : \{0\} \hookrightarrow \mathbb{C}$ be the origin and let V be a finite-dimensional vector space regarded as a differential module on the point. Differential direct image produces a module supported at the origin; for $V = \mathbb{C}$ this is the module δ_0 of Section 5.2. On the sheaf side, direct image of the local system V on the point is the skyscraper sheaf $i_{0_*}V$. Compatibility predicts

$$\mathrm{DR}_{\mathbb{C}}\left(\int_{i_0}(V)\right) \cong i_{0_*}V.$$

Nothing propagates away from the origin on either side. The theorem below says that this agreement persists for general morphisms and derived objects, including situations where the fibres and the singular boundary have nontrivial cohomology.

We write $\int_f$ for the derived direct image of differential modules and $f^\dagger$ for the extraordinary inverse image; their transfer-bimodule constructions are not developed here. We also write $\mathcal{M} \boxtimes \mathcal{N}$ for the external product.

6.2.1 The common geometric calculus

Definition 6.3 (Derived tensor product) *For $K, L \in D^b(\mathrm{Sh}(X;k))$, totalise the degreewise tensor product by*

$$(K \otimes^{\boldsymbol{L}} L)^n := \bigoplus_{p+q=n} K^p \otimes_k L^q,$$

with differential

$$d(a \otimes b) = d_K(a) \otimes b + (-1)^p a \otimes d_L(b)$$

for $a \in K^p$ and $b \in L^q$. This defines the derived tensor product $K \otimes^{\boldsymbol{L}} L$.

Because the coefficients form a field, tensor product of sheaves of k-vector spaces is exact, so no flat replacement is required here.

Lemma 6.4 (Derived tensor–Hom adjunction) *For $K, L \in D^b(\mathrm{Sh}(X;k))$ and $M \in D^+(\mathrm{Sh}(X;k))$ there is a natural isomorphism*

$$\mathrm{Hom}_{D^+(\mathrm{Sh}(X;k))}(K \otimes^{\boldsymbol{L}} L, M) \cong \mathrm{Hom}_{D^+(\mathrm{Sh}(X;k))}(K, R\mathcal{H}\mathrm{om}_X(L, M)).$$

The degreewise tensor–Hom adjunction respects the differentials of the two total complexes and gives the displayed derived adjunction; see [1, Prop. 2.6.3].

Definition 6.5 (External product) *Let $p_X : X \times Y \to X$ and $p_Y : X \times Y \to Y$ be the projections. For $K \in D^b(\mathrm{Sh}(X;k))$ and $L \in D^b(\mathrm{Sh}(Y;k))$, define*

$$K \boxtimes L := p_X^{-1} K \otimes^{\boldsymbol{L}} p_Y^{-1} L.$$

The two inverse images place the objects on $X \times Y$, where the tensor product combines them while retaining the two geometric variables.

Proposition 6.6 (Internal exceptional adjunction) *Let $f : X \to Y$ satisfy the hypotheses of Theorem 3.16. For $K \in D^b(\mathrm{Sh}(X;k))$ and $L \in D^+(\mathrm{Sh}(Y;k))$ there is a natural isomorphism*

$$Rf_*\big(R\mathcal{H}\mathrm{om}_X\big(K, f^!(L)\big)\big) \cong R\mathcal{H}\mathrm{om}_Y(Rf_!(K), L).$$

Taking derived global sections recovers $Rf_! \dashv f^!$. The sheaf-valued identity is the relative adjunction used to convert a supported pullback along the diagonal into an internal Hom; see [1, Prop. 3.1.10], [5, Thm. 3.2.3], and [6, §VII.5].

Geometric task	**$\mathcal{D}$-modules**	**Sheaves**	**Use below**
Move from X to Y	$\int_f$	Rf_*	Insert a stratum and push to a point
Read from Y on X with support	$f^\dagger$	$f^!$	Restrict the comparison to the diagonal
Keep two variables separate	$\boxtimes$	$\boxtimes$	Place the source test and target on $X \times X$
Reverse the first input	$\mathrm{Dual}_{\mathcal{D},X}$	$\mathbb{D}_{X^{\mathrm{an}}}$	Turn the source into the dual input of RHom

Theorem 6.7 (Compatibility of de Rham with the operations used below) *Let $f : X \to Y$ be a morphism of smooth complex algebraic varieties. On regular holonomic complexes there are natural isomorphisms*

$$\mathrm{DR}_Y\left(\int_f (\mathcal{M}^\bullet)\right) \cong Rf_*(\mathrm{DR}_X(\mathcal{M}^\bullet)),$$

$$\mathrm{DR}_X\big(f^\dagger(\mathcal{N}^\bullet)\big) \cong f^!(\mathrm{DR}_Y(\mathcal{N}^\bullet)),$$

$$\mathrm{DR}_{X\times Y}(\mathcal{M}^\bullet \boxtimes \mathcal{N}^\bullet) \cong \mathrm{DR}_X(\mathcal{M}^\bullet) \boxtimes \mathrm{DR}_Y(\mathcal{N}^\bullet),$$

and

$$\mathrm{DR}_X\big(\mathrm{Dual}_{\mathcal{D},X}(\mathcal{M}^\bullet)\big) \cong \mathbb{D}_{X^{\mathrm{an}}}(\mathrm{DR}_X(\mathcal{M}^\bullet)).$$

This is [8, Thm. 7.1.1], together with the external-product compatibility of §4.7. Regularity is essential at singular boundaries. For an irregular module, direct image across an open embedding need not commute with de Rham; see [8, Rem. 7.1.2]. The exponential connection of Section 5.5.1 exhibits the missing sectorial Stokes data.

6.3 Full faithfulness

Full faithfulness asks whether every map between differential systems can be recovered from the corresponding sheaf objects, and whether every sheaf-theoretic map comes from a unique differential one. Before introducing the diagonal, consider the degenerate case $X = \{*\}$. Then $\mathcal{D}_X = \mathbb{C}$ and DR_X is the identity. For finite-dimensional vector spaces,

$$\mathrm{Hom}_{\mathbb{C}}(V,W) \cong V^{\vee} \otimes W$$

and for bounded complexes the corresponding statement is

$$\mathrm{RHom}_{\mathbb{C}}(V^{\bullet},W^{\bullet}) \cong \mathrm{RHom}_{\mathbb{C}}\big(\mathbb{C},(V^{\bullet})^{\vee} \otimes W^{\bullet}\big)$$

The dual of the first variable provides linear tests; tensoring with the second variable evaluates those tests against possible targets.

On a space X the two variables may lie at different points, so the analogue first lives on $X \times X$. The external product

$$\mathbb{D}_{X^{\mathrm{an}}}(K) \boxtimes L$$

contains a dual test for K at the first point and a value of L at the second. A morphism of sheaves must compare data over the same point, and the diagonal

$$\Delta : X \hookrightarrow X \times X$$

imposes that equality. Applying $\Delta^{!}$ keeps the supported part of the comparison along the diagonal, and the map $p : X \to \{*\}$ then collects the local comparisons into the global derived Hom complex. The formula below is therefore the sheaf version of the elementary expression $V^{\vee} \otimes W$.

Let $p : X \to \{*\}$ be the map to a point. There are two parallel diagonal formulas. On the differential side,

$$\mathrm{RHom}_{\mathcal{D}_X}(\mathcal{M}^{\bullet},\mathcal{N}^{\bullet}) \cong \int_{p} \big(\Delta^{\dagger}\big(\mathrm{Dual}_{\mathcal{D},X}(\mathcal{M}^{\bullet}) \boxtimes \mathcal{N}^{\bullet}\big)\big).$$

On the sheaf side, for $K, L \in D^{b}_{c}(X^{\mathrm{an}})$,

$$\mathrm{RHom}_{\mathrm{Sh}(X^{\mathrm{an}};k)}(K,L) \cong Rp_{*}\big(\Delta^{!}(\mathbb{D}_{X^{\mathrm{an}}}(K) \boxtimes L)\big).$$

The first is the differential tensor–Hom identity written on the diagonal; the second is its Verdier-dual analogue. Compare [8, Cor. 2.6.15 and equations (7.2.3)–(7.2.4)].

Proposition 6.8 (Derived Hom is preserved by de Rham) *For $\mathcal{M}^{\bullet},\mathcal{N}^{\bullet} \in D^{b}_{\mathrm{rh}}(\mathcal{D}_X)$ there is a natural isomorphism*

$$\mathrm{RHom}_{\mathcal{D}_X}(\mathcal{M}^{\bullet},\mathcal{N}^{\bullet}) \cong \mathrm{RHom}_{\mathrm{Sh}(X^{\mathrm{an}};k)}(\mathrm{DR}_X(\mathcal{M}^{\bullet}),\mathrm{DR}_X(\mathcal{N}^{\bullet})).$$

Consequently DR_X is fully faithful.

Proof sketch. Start with the sheaf-side diagonal formula for $K = \mathrm{DR}_X(\mathcal{M}^{\bullet})$ and $L = \mathrm{DR}_X(\mathcal{N}^{\bullet})$. By Theorem 6.7, Verdier duality of the first factor becomes the

de Rham complex of $\mathrm{Dual}_{\mathcal{D},X}(\mathcal{M}^\bullet)$, and the external product becomes the de Rham complex of the external product of the two differential objects. Compatibility with $\Delta^!$ then gives the de Rham complex of $\Delta^\dagger$, and compatibility with Rp_* gives the de Rham complex of $\int_p$. Since the de Rham functor on a point is the identity, the resulting complex is

$$\int_p \big(\Delta^\dagger\big(\mathrm{Dual}_{\mathcal{D},X}(\mathcal{M}^\bullet) \boxtimes \mathcal{N}^\bullet\big)\big),$$

which is the differential-side diagonal formula for $\mathrm{RHom}_{\mathcal{D}_X}(\mathcal{M}^\bullet, \mathcal{N}^\bullet)$.

Taking degree-zero cohomology gives

$$\mathrm{Hom}_{D^b_{\mathrm{rh}}(\mathcal{D}_X)}(\mathcal{M}^\bullet, \mathcal{N}^\bullet) \cong \mathrm{Hom}_{D^b_c(X^{\mathrm{an}})}(\mathrm{DR}_X(\mathcal{M}^\bullet), \mathrm{DR}_X(\mathcal{N}^\bullet)),$$

so DR_X is fully faithful. □

The diagonal calculation identifies the two derived Hom complexes. We take from the proof of [8, Thm. 7.2.2] the coherence of the comparison maps, and hence the identification of this isomorphism with the morphism induced by DR_X; see also [8, Rem. 7.2.3].

6.4 Essential surjectivity

Example 6.9 (The two-stratum model on the line) *Let $X = \mathbb{C}$, let $U = \mathbb{C}^\times$ and $Z = \{0\}$, and write $j : U \hookrightarrow X$ and $i : Z \hookrightarrow X$ for the inclusions. Let K be constructible for this two-stratum decomposition and put*

$$A := i^!(K) \quad B := j^{-1}K$$

The open–closed triangle of Theorem 3.21 is

$$Ri_*(A) \longrightarrow K \longrightarrow Rj_*(B) \xrightarrow{\delta} Ri_*(A)[1]$$

Thus A, B, and the connecting morphism

$$\delta : Rj_*(B) \to Ri_*(A)[1]$$

determine K. In terms of the mapping cone,

$$K \cong \mathrm{Cone}(\delta)[-1]$$

The open and closed pieces alone do not determine K. If $\delta = 0$ then

$$K \cong Ri_*(A) \oplus Rj_*(B)$$

and the open–closed triangle splits. A nonzero connecting morphism may produce a non-split object with the same pieces A and B. Thus the connecting morphism records how the two strata are attached. Essential surjectivity must consequently lift δ in addition to lifting the open and closed pieces. This is the role of full faithfulness in the construction below.

We first lift B. Its cohomology sheaves are finite-rank local systems on U. For each m, Deligne's theorem supplies a regular connection $\mathcal{E}_m$ whose horizontal local system is $\mathcal{H}^m(B)$; if this cohomology sheaf vanishes, take $\mathcal{E}_m = 0$. Since $\dim_{\mathbb{C}} U = 1$,

$$\mathrm{DR}_U(\mathcal{E}_m) \cong \mathcal{H}^m(B)[1]$$

Therefore

$$\mathcal{N}_m := \mathcal{E}_m[-m-1]$$

satisfies

$$\mathrm{DR}_U(\mathcal{N}_m) \cong \mathcal{H}^m(B)[-m]$$

Let a be the lowest degree with nonzero cohomology. Then

$$\tau_{\le a} B \cong \mathcal{H}^a(B)[-a]$$

so set $\mathcal{M}_a := \mathcal{N}_a$. Suppose inductively that

$$\mathrm{DR}_U(\mathcal{M}_{m-1}) \cong \tau_{\le m-1} B$$

The next truncation triangle is

$$\tau_{\le m-1} B \longrightarrow \tau_{\le m} B \longrightarrow \mathcal{H}^m(B)[-m] \xrightarrow{\theta_m} \tau_{\le m-1} B[1]$$

By Proposition 6.8, the connecting morphism lifts uniquely to

$$\tilde{\theta}_m : \mathcal{N}_m \to \mathcal{M}_{m-1}[1]$$

Set

$$\mathcal{M}_m := \mathrm{Cone}\big(\tilde{\theta}_m\big)[-1]$$

Applying DR_U gives the truncation triangle above and hence

$$\mathrm{DR}_U(\mathcal{M}_m) \cong \tau_{\le m} B$$

Since B is bounded, the induction terminates and produces

$$\mathcal{M}_U \in D^b_{\mathrm{rh}}(\mathcal{D}_U) \quad \text{with} \quad \mathrm{DR}_U(\mathcal{M}_U) \cong B$$

The closed stratum is a point, so $\mathcal{D}_Z = \mathbb{C}$ *and* DR_Z *is the identity. Regard* A *as an object*

$$\mathcal{M}_Z \in D^b_{\mathrm{rh}}(\mathcal{D}_Z) \quad \text{with} \quad \mathrm{DR}_Z(\mathcal{M}_Z) \cong A$$

Set

$$\mathcal{N}_U := \int_j (\mathcal{M}_U) \quad \mathcal{N}_Z := \int_i (\mathcal{M}_Z)$$

By Theorem 6.7,

$$\mathrm{DR}_X(\mathcal{N}_U) \cong Rj_*(B) \quad \mathrm{DR}_X(\mathcal{N}_Z) \cong Ri_*(A)$$

Under these identifications, Proposition 6.8 lifts δ *uniquely to*

$$\tilde{\delta} : \mathcal{N}_U \to \mathcal{N}_Z[1]$$

Define

$$\mathcal{M} := \mathrm{Cone}\big(\tilde{\delta}\big)[-1]$$

Since DR_X *preserves distinguished triangles,*

$$\mathrm{DR}_X(\mathcal{M}) \cong \mathrm{Cone}(\delta)[-1] \cong K$$

Thus the open and closed lifts, together with the lifted connecting morphism, recover K *up to isomorphism.*

Iterating the two-stratum construction over a finite algebraic stratification gives the following generation statement.

Lemma 6.10 (Stratumwise local systems generate the constructible derived category) *Let* $K \in D^b_c(X^{\mathrm{an}})$*. After refining an algebraic Whitney stratification,* K *belongs to the triangulated subcategory generated by objects*

$$Ri_*(\mathcal{L})[m]$$

where $i : S \hookrightarrow X$ *is the inclusion of a smooth affine locally closed stratum,* $\mathcal{L}$ *is a finite-rank local system on* S^{an}*, and* $m \in \mathbb{Z}$*.*

Proof. Choose a finite algebraic Whitney stratification adapted to K, refined so that the strata and their inclusions have the affine properties in the statement, as in [8, proof of Thm. 7.2.2]. We induct on the total number of strata. Let $j : U \hookrightarrow X$ be the union of the open strata and let $i : Z \hookrightarrow X$ be its closed complement. The frontier condition ensures that Z is a closed union of fewer

strata. Recollement gives the distinguished triangle $i_*i^!K \to K \to Rj_*j^{-1}K \to \big(i_*i^!K\big)[1]$.

On each component S of U, the cohomology sheaves of $j^{-1}K$ are finite-rank local systems. The canonical truncation triangles therefore build $j^{-1}K$ from shifts of local systems on these components. Applying Rj_* expresses $Rj_*j^{-1}K$ in terms of the required generators associated with the open strata. By induction, $i^!K$ is generated on Z by the corresponding objects for its strata. Applying i_* gives generators on X, since direct images compose and i_* is exact. The displayed triangle then places K in the same triangulated subcategory. The case of an empty closed complement starts the induction. □

This is an induction using the open–closed recollement of Section 3.4.4. The gluing theorem (Theorem 4.5) uses the same decomposition to glue t-structures, whereas the present argument uses it to generate objects of the triangulated category. See [7, §1.4].

Proposition 6.11 (Every constructible complex is a de Rham complex) *For every $K \in D^b_c(X^{\mathrm{an}})$ there exists $\mathcal{M}^\bullet \in D^b_{\mathrm{rh}}(\mathcal{D}_X)$ with*

$$\mathrm{DR}_X(\mathcal{M}^\bullet) \cong K.$$

Proof. By Proposition 6.8 the essential image of DR_X is a full triangulated subcategory of $D^b_c(X^{\mathrm{an}})$. By Lemma 6.10 it is therefore enough to lift one generator $Ri_*(\mathcal{L})[m]$.

Let $d = \dim_{\mathbb{C}} S$. Deligne's theorem supplies a regular algebraic integrable connection $\mathcal{E}$ on S whose horizontal local system is $\mathcal{L}$, and by Section 6.1

$$\mathrm{DR}_S(\mathcal{E}) \cong \mathcal{L}\,[d]$$

Set

$$\mathcal{M}^\bullet := \int_i (\mathcal{E})[m-d]$$

Regular holonomicity is preserved by this direct image, and Theorem 6.7 gives

$$\mathrm{DR}_X(\mathcal{M}^\bullet) \cong Ri_*(\mathrm{DR}_S(\mathcal{E}))[m-d] \cong Ri_*(\mathcal{L})[m].$$

Thus every generator lies in the essential image, and since that image is triangulated, so does every constructible complex. □

Theorem 6.12 (Riemann–Hilbert correspondence) *Let X be a smooth complex algebraic variety. The normalised de Rham functor is an equivalence of triangulated categories*

$$\mathrm{DR}_X : D^b_{\mathrm{rh}}(\mathcal{D}_X) \to D^b_c(X^{\mathrm{an}})$$

Equivalently, the solution functor gives a contravariant equivalence

$$\mathrm{Sol}_X : D^b_{\mathrm{rh}}(\mathcal{D}_X) \to D^b_c(X^{\mathrm{an}})^{\mathrm{op}}$$

Proof. Full faithfulness is Proposition 6.8 and essential surjectivity is Proposition 6.11. The statement for Sol_X follows from Proposition 6.2 and Verdier biduality on constructible complexes, Theorem 3.14. □

This is [8, Thm. 7.2.2 and Cor. 7.2.4].

Remark (Analytic and algebraic forms) *For a complex manifold X, the analytic theorem gives*

$$\mathrm{DR}_X : D^b_{\mathrm{rh}}(\mathcal{D}_X) \to D^b_c(X)$$

as an equivalence with the analytically constructible derived category; see [8, Thm. 7.2.1]. For algebraic X, the theorem above uses the smaller category of complexes constructible with respect to a finite algebraic stratification. Deligne's regular-singular correspondence supplies the algebraisation of the local systems on the strata, and [8, Thm. 4.7.7] shows that algebraic holonomic modules land in this smaller target.

6.5 Finite cellular models of constructible complexes

The construction below represents an algebraically constructible complex by finite cellular data. It applies in particular to the de Rham and solution complexes of regular holonomic complexes. Its input is the constructible complex, an adapted triangulation, and choices of local cohomology representatives and primitives. A presentation of the differential module alone does not supply these choices.

Recall the Euler module $M_0 := \frac{A_1}{A_1(x\partial)}$ from Example 5.8. After analytification, write z for the complex coordinate. Its unshifted solution complex is $F_0^\bullet := \mathrm{Sol}_{\mathbb{C}}(M_0) \cong \left[\mathcal{O}_{\mathbb{C}} \xrightarrow{z\partial_z} \mathcal{O}_{\mathbb{C}}\right]$, with terms in degrees 0 and 1. The germ of 1 in the degree-one term represents the generator of $\mathcal{H}^1(F_0^\bullet)_0$. On a simply connected punctured sector it becomes a boundary because $z\partial_z(\log z) = 1$. Branches of $\log z$ on overlapping sectors differ by constants, and their continuation around the origin produces the jump $2\pi i$. A cellular model records the finite-dimensional complexes on the cells together with the restriction maps containing these transition constants.

Let $F^\bullet \in D^b_c(X^{\mathrm{an}})$ be constructible with respect to a finite algebraic stratification. Choose an algebraic compactification $X \hookrightarrow \overline{X}$, put $Y := \overline{X}^{\mathrm{an}}$, and choose a finite simplicial pair (K, L) compatible with the stratification and the boundary such that $Y \cong |K|$ and $X^{\mathrm{an}} \cong |K| - |L|$. To obtain the pair, regard the boundary and the strata as subanalytic subsets of a real analytic ambient space. The compatible triangulation supplied by [1, Prop. 8.2.5] is finite because Y is compact. Write $j : X^{\mathrm{an}} \hookrightarrow Y$ for the open inclusion and put $E^\bullet := j_!(F^\bullet)$. Then $E^\bullet$ is constructible for the cells of K and has zero stalks on the cells of L. If X^{an} is compact, take $Y = X^{\mathrm{an}}$ and L empty.

Definition 6.13 (Finite cellular model) *A **finite cellular model** of $F^\bullet$ relative to (K, L) is a bounded cellular complex $B^\bullet$ on K with finite-dimensional values, satisfying $B^\bullet_\sigma = 0$ for every $\sigma \in L$, together with an isomorphism*

$$\mathrm{Real}_K(B^\bullet) \cong j_!(F^\bullet)$$

in the constructible derived category on Y.

For a cell σ of K, let $\mathrm{st}(\sigma)$ be its topological open star as in Section 1.5. If $\sigma \leq \tau$, then $\mathrm{st}(\tau) \subseteq \mathrm{st}(\sigma)$. Choose a bounded-below injective resolution $E^\bullet \to I^\bullet$ and set $A^\bullet_\sigma := \Gamma(\mathrm{st}(\sigma), I^\bullet)$. Restriction gives cochain maps $r_{\sigma,\tau} : A^\bullet_\sigma \to A^\bullet_\tau$ satisfying $r_{\tau,\upsilon} \circ r_{\sigma,\tau} = r_{\sigma,\upsilon}$ and $d_\tau \circ r_{\sigma,\tau} = r_{\sigma,\tau} \circ d_\sigma$.

For $x \in \sigma$, evaluation gives $\Gamma(\mathrm{st}(\sigma), I^\bullet) \to I^\bullet_x$, which is a quasi-isomorphism by [4, Lemma 5.2.2]. Exactness of stalks also makes $E^\bullet_x \to I^\bullet_x$ a quasi-isomorphism. Thus $A^\bullet_\sigma$ represents the stalk complex of $E^\bullet$ on σ, and every $H^q(A^\bullet_\sigma)$ is finite-dimensional. Shepard's comparison identifies the sheaf complex reconstructed from this open-star diagram with $E^\bullet$; see [8, Theorem 5.2.3]. The vector spaces in the complexes $A^\bullet_\sigma$ need not themselves be finite-dimensional.

Lemma 6.14 (Finite subdiagrams) *Let P be a finite poset and let $A^\bullet$ be a diagram of complexes of $\mathbb{C}$-vector spaces supported in a common finite range of degrees $[a, b]$. Assume that every $H^q(A^\bullet_\sigma)$ is finite-dimensional. There is a subdiagram $B^\bullet \subseteq A^\bullet$, finite-dimensional in every degree at every element of P, such that each inclusion $B^\bullet_\sigma \to A^\bullet_\sigma$ is a quasi-isomorphism.*

Proof. For every σ and q, choose cycles representing a basis of $H^q(A^\bullet_\sigma)$. Let $B^\bullet$ initially be their span, closed under the diagram maps. Finiteness of P makes every B^q_σ finite-dimensional. The chosen vectors and their images are cycles, and compatibility with composition makes $B^\bullet$ a subdiagram. The maps $H^q(B^\bullet_\sigma) \to H^q(A^\bullet_\sigma)$ are surjective.

Proceed downwards through $q = b, b-1, ..., a+1$. For every σ, choose cycles $z_{\sigma,1}, ..., z_{\sigma,m_\sigma}$ representing a basis of $\ker(H^q(B^\bullet_\sigma) \to H^q(A^\bullet_\sigma))$. Each $z_{\sigma,i}$ is a boundary in $A^\bullet_\sigma$, so choose $y_{\sigma,i} \in A^{q-1}_\sigma$ with $d_\sigma(y_{\sigma,i}) = z_{\sigma,i}$. Adjoin these elements and all their images under the diagram maps to B^{q-1}.

The enlarged spaces form a subcomplex because $d_\tau(r_{\sigma,\tau}(y_{\sigma,i})) = r_{\sigma,\tau}(z_{\sigma,i})$, and the right-hand side belongs to B^q_τ. This step kills the kernel in degree q and changes no terms in degrees at least q. Later steps handle the lower-degree kernels. Surjectivity persists because a cycle representing a nonzero class in $A^\bullet_\sigma$ cannot become a boundary in its subcomplex.

After the step $q = a+1$, the map on degree-a cohomology is injective: a cycle in B^a_σ that is a boundary in $A^\bullet_\sigma$ is the differential of an element of $A^{a-1}_\sigma = 0$. The resulting maps are isomorphisms on cohomology in every degree. Every step adds finitely many vectors and their images under the maps of P. □

Proposition 6.15 (Finite cellular model) *With the chosen simplicial pair (K, L), the complex $F^\bullet$ admits a finite cellular model in the sense of Definition 6.13. In particular, two regular holonomic complexes whose de Rham complexes have isomorphic realised cellular models are isomorphic.*

Proof. Use the diagram $A^\bullet_\sigma := \Gamma(U_\sigma, I^\bullet)$ constructed above. It is bounded below. Since K has finitely many cells and every $A^\bullet_\sigma$ has bounded cohomology, there is a common upper bound b for their cohomology. Replace each $A^\bullet_\sigma$ by its good truncation in degrees at most b: replace degree b by $\ker(d^b_\sigma)$ and set the higher degrees to zero. The restriction maps preserve these kernels, so the truncations form a bounded subdiagram, and their inclusions are quasi-isomorphisms at every cell.

Apply the preceding lemma. The inclusions $B^\bullet_\sigma \to A^\bullet_\sigma$ are objectwise quasi-isomorphisms, and Shepard's comparison therefore identifies $\mathrm{Real}_K(B^\bullet)$ with $E^\bullet = j_!(F^\bullet)$. The construction can be made with $B^\bullet_\sigma = 0$ on L: the stalk complexes of $E^\bullet$ on L are acyclic, and closure under the maps $r_{\sigma,\tau}$ introduces no vectors on L.

If $F^\bullet = \mathrm{DR}_X(\mathcal{M}^\bullet)$, the final assertion follows from the full faithfulness of Theorem 6.12. □

The choices are noncanonical, but the objectwise quasi-isomorphism with the open-star diagram fixes the realised object up to isomorphism in the constructible derived category. If $F^\bullet = \mathrm{DR}_X(\mathcal{M}^\bullet)$, derived full faithfulness

determines the isomorphism class of $\mathcal{M}^\bullet$ from this realised object. This determination is categorical and does not reconstruct a differential-module presentation.

Since a finite cellular model vanishes on L, its total cellular complex computes sections of the extension by zero on Y. We write $C_c^\bullet(K;B) \coloneqq C^\bullet(K;B)$ to record this compact-support interpretation.

Proposition 6.16 (Compactly supported cohomology from a finite model) *Let $B^\bullet$ be a finite cellular model of $F^\bullet$ relative to (K,L). Then*

$$C_c^\bullet(K;B) \cong R\Gamma_c(X^{\mathrm{an}}, F^\bullet)$$

If X^{an} is compact, the same total complex computes $R\Gamma(X^{\mathrm{an}}, F^\bullet)$.

Proof. By Proposition 2.12 and Definition 6.13,

$$C_c^\bullet(K;B) \cong R\Gamma(Y, \mathrm{Real}_K(B^\bullet)) \cong R\Gamma(Y, j_!(F^\bullet))$$

Since Y is compact, the last object is $R\Gamma_c(X^{\mathrm{an}}, F^\bullet)$. If X^{an} is compact, take $Y = X^{\mathrm{an}}$ and L empty. □

6.5.1 The Euler module

Apply the construction to the unshifted solution complex $F_0^\bullet$. Let Δ be a small open disc centred at 0 and set $Y \coloneqq \overline{\Delta}$, $X \coloneqq \Delta = \mathrm{int}(Y)$ and $L \coloneqq \partial Y$. Write $j : \Delta \hookrightarrow Y$ for the open inclusion. Choose a finite triangulation K of Y with central vertex v, three boundary vertices, three radial edges e_0, e_1, e_2, three boundary edges and three triangular two-cells t_0, t_1, t_2. Assign the zero complex to the cells of L.

The free resolution of M_0 gives $F_0^\bullet|_\Delta \cong \left[\mathcal{O}_\Delta \xrightarrow{z\partial_z} \mathcal{O}_\Delta\right]$ in degrees 0 and 1, as in Example 5.8. The open star of v is Δ. Every other retained open star is a simply connected punctured sector. These sets are Stein, so their complexes of holomorphic sections compute the required derived sections.

At the central vertex take $B_v^\bullet \coloneqq \left[\mathbb{C} \xrightarrow{0} \mathbb{C}\right]$ using the constant functions in both degrees. At every other retained cell σ, choose a branch ℓ_σ of $\log z$ on U_σ and take $B_\sigma^\bullet \coloneqq \left[\mathrm{span}_{\mathbb{C}}\{1, \ell_\sigma\} \xrightarrow{z\partial_z} \mathrm{span}_{\mathbb{C}}\{1\}\right]$. In the displayed bases its differential is $d_\sigma = \begin{pmatrix} 0 & 1 \end{pmatrix} : \mathbb{C}^2 \to \mathbb{C}$.

For a face relation between noncentral cells, write $\ell_\sigma|_{U_\tau} = \ell_\tau + c_{\sigma,\tau}$. Restriction is then represented by $r^0_{\sigma,\tau} = \begin{pmatrix} 1 & c_{\sigma,\tau} \\ 0 & 1 \end{pmatrix}$ and $r^1_{\sigma,\tau} = (1)$. The maps from the centre are $r^0_{v,\sigma} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $r^1_{v,\sigma} = (1)$.

Proposition 6.17 (Euler model) *These complexes and maps form a finite cellular model of $F_0^\bullet|_\Delta$. The branches may be chosen so that every c_{e_i,t_j} is zero except $c_{e_0,t_2} = -2\pi i$. With the radial edges oriented away from v and the two-cells oriented counterclockwise, Gaussian elimination reduces the total compactly supported cellular complex to $\left[\mathbb{C} \xrightarrow{2\pi i} \mathbb{C}\right]$ with terms in total degrees 1 and 2, so the total complex is acyclic.*

Proof. On Δ, the operator sends $\sum_{m\geq 0} a_m z^m \mapsto \sum_{m\geq 1} m a_m z^m$. Its kernel consists of constants, and its image consists exactly of the holomorphic functions with zero constant term. Thus the constants in degrees 0 and 1 represent bases of the two cohomology groups, so $B_v^\bullet$ is a quasi-isomorphic subcomplex.

On a noncentral star U_σ, a holomorphic function g has a preimage under $z\partial_z$ because $g\frac{z}{z}$ has a holomorphic primitive on the simply connected domain U_σ. The kernel again consists of constants. Since $z\partial_z(\ell_\sigma) = 1$, the chosen finite subcomplex has the same kernel and zero cokernel. Its inclusion is therefore a quasi-isomorphism.

The displayed matrices are the restrictions of the chosen holomorphic functions, so they commute with the differentials and satisfy the composition identities. For example, $\begin{pmatrix}0 & 1\end{pmatrix}\begin{pmatrix}1 & c_{\sigma,\tau}\\ 0 & 1\end{pmatrix} = \begin{pmatrix}0 & 1\end{pmatrix}$, and the map from the central degree-zero term lands in the kernel of $\begin{pmatrix}0 & 1\end{pmatrix}$.

Choose the arguments on t_0, t_1, t_2 in $\left(0, 2\frac{\pi}{3}\right)$, $\left(2\frac{\pi}{3}, 4\frac{\pi}{3}\right)$ and $\left(4\frac{\pi}{3}, 2\pi\right)$, respectively. Choose the argument on U_{e_0} in $\left(-2\frac{\pi}{3}, 2\frac{\pi}{3}\right)$, and those on U_{e_1} and U_{e_2} in $\left(0, 4\frac{\pi}{3}\right)$ and $\left(2\frac{\pi}{3}, 2\pi\right)$, respectively. All restrictions agree except from U_{e_0} to U_{t_2}, where the arguments differ by -2π. This gives the stated constant.

In every noncentral cell, the internal differential maps the logarithmic basis vector to the constant basis vector by the identity. Gaussian elimination cancels these pairs and the incidence identity maps along a spanning tree of the circular link. The central degree-one class in total degree 1 and the compactly supported two-cell class in total degree 2 remain. Their differential is the logarithmic jump $2\pi i$ across the omitted edge, up to the sign fixed by the cell orientations. Its nonzero value makes the total complex acyclic. □

On the punctured disc, degree-one hypercohomology comes from the monodromy of the degree-zero local system. The cochain 1 at the centre represents a class in $\mathbb{H}^1(\Delta, F_0^\bullet)$. On every simply connected punctured star, its image is the boundary of the chosen logarithm. The transition constants around the puncture sum to $2\pi i$. On $\Delta^\times := \Delta - \{0\}$, multiplication by $\frac{dz}{z}$ in degree 1 gives

an isomorphism of complexes $\left[\mathcal{O} \xrightarrow{z\partial_z} \mathcal{O}\right] \to \left[\mathcal{O} \xrightarrow{d} \Omega^1\right]$. The class of 1 is sent to $\frac{dz}{z}$, whose integral around a positively oriented circle is $2\pi i$. It therefore gives a nonzero class in $\mathbb{H}^1\left(\Delta^\times, F_0^\bullet|_{\Delta^\times}\right)$. This period is the final differential in the compactly supported total complex.

Remark (Analytic input) *The Euler calculation gives the required local representatives and primitives explicitly. For a general differential module, finding them is an analytic problem. Constructibility makes the local cohomology finite-dimensional; after the analytic choices, reduction and totalisation are finite linear-algebraic procedures.*

For an adapted finite triangulation, Proposition 6.15 represents every algebraically constructible de Rham complex by finitely many complexes and restriction maps. The transition constants in the Euler example record the gluing of local logarithms, and Proposition 6.16 computes compactly supported cohomology from the resulting finite diagram. It remains to recognise which such complexes correspond to differential modules concentrated in degree 0; this is the role of the perverse heart.

6.6 The perverse heart

For a regular holonomic module $\mathcal{M}$, Theorem 5.19 and Proposition 6.2 give that $\mathrm{DR}_X(\mathcal{M})$ is perverse. A short exact sequence of regular holonomic modules gives a distinguished triangle whose three de Rham complexes all lie in the perverse heart, and a distinguished triangle with all three vertices in the heart of a t -structure is a short exact sequence there. The restriction of DR_X to regular holonomic modules is therefore exact as a functor into $\mathrm{Perv}(X^{\mathrm{an}})$.

The two-stratum line makes the target heart concrete. By Theorem 4.25, with $X = \mathbb{C}$, $U = \mathbb{C}^\times$ and $Z = \{0\}$, the simple objects are the skyscraper at 0 and the intermediate extensions $j_{!*}(\mathcal{L}\,[1])$ of irreducible local systems on $\mathbb{C}^\times$. Their differential counterparts are δ_0 and, by Deligne, the differential minimal extensions of the corresponding irreducible regular connections. The next proposition identifies the two extension procedures.

Proposition 6.18 (Minimal extension is preserved) *Let $i : S \hookrightarrow X$ be an affine locally closed embedding of a smooth stratum of dimension d, let $\mathcal{E}$ be a regular integrable connection on S, and let $\mathcal{L}$ be its horizontal local system. Then*

$$\mathrm{DR}_X\big(i^{\mathcal{D}}_{!*}\,\mathcal{E}\big) \cong \mathrm{IC}_{\overline{S}}(\mathcal{L}).$$

Proof. Factor i as $S \xrightarrow{j} \overline{S} \xrightarrow{a} X$ with j open and a closed. For affine i the two differential extensions of Proposition 5.23 are regular holonomic modules, and $i^{\mathcal{D}}_{!*}\,\mathcal{E}$ is the image of their canonical morphism. De Rham is exact on regular holonomic modules, and by Theorem 6.7 it carries the upper differential extension to the corresponding derived direct image on sheaves; the lower extension follows from the same statement and duality. Because the differential extensions are modules, their de Rham images lie in the perverse heart, so they are the standard and costandard perverse extensions of $\mathcal{L}\,[d]$ in the sense of Definition 4.13.

Taking the image of their canonical morphism in the perverse heart gives $a_*j_{!*}(\mathcal{L}\,[d])$, which by Definition 4.14 and Definition 4.18 is $\mathrm{IC}_{\overline{S}}(\mathcal{L})$. Equivalently, Theorem 4.15 identifies it as the unique perverse extension with no nonzero subobject and no nonzero quotient supported on the boundary. This is the correspondence recorded in [8, Rem. 7.2.10]; on the perverse side it is the construction organised by [7, §1.4 and §4.3]. □

Theorem 6.19 (Abelian Riemann–Hilbert correspondence) *The de Rham functor restricts to an equivalence of abelian categories*

$$\mathrm{DR}_X : \mathrm{Mod}_{\mathrm{rh}}(\mathcal{D}_X) \to \mathrm{Perv}(X^{\mathrm{an}}).$$

Proof. We first show that de Rham is t-exact for the standard t-structure on the differential side and the perverse t-structure on the sheaf side. A regular holonomic module in degree 0 is sent to the perverse heart. Every bounded complex $\mathcal{M}^\bullet$ is built by its canonical truncation triangles from the layers $\mathcal{H}^q(\mathcal{M}^\bullet)[-q]$. Applying de Rham turns these into shifts by $[-q]$ of perverse sheaves. If $\mathcal{M}^\bullet$ has no positive cohomology, only $q \le 0$ occur; these shifts lie in the nonpositive perverse half, which is closed under extensions. If $\mathcal{M}^\bullet$ has no negative cohomology, only $q \ge 0$ occur; these shifts lie in the nonnegative perverse half, which is likewise closed under extensions. This proves both required inclusions.

Hence DR_X is t-exact, and for every i there is a natural isomorphism

$$^{\mathrm{p}}\mathrm{H}^i(\mathrm{DR}_X(\mathcal{M}^\bullet)) \cong \mathrm{DR}_X(\mathcal{H}^i(\mathcal{M}^\bullet))$$

Full faithfulness on the hearts follows by restricting Proposition 6.8 to objects concentrated in degree 0.

For essential surjectivity, let $P \in \mathrm{Perv}(X^{\mathrm{an}})$. By Theorem 6.12 there is $\mathcal{M}^\bullet \in D^b_{\mathrm{rh}}(\mathcal{D}_X)$ with $\mathrm{DR}_X(\mathcal{M}^\bullet) \cong P$. For $i \neq 0$, t-exactness gives

$$\mathrm{DR}_X(\mathcal{H}^i(\mathcal{M}^\bullet)) \cong {}^{\mathrm{p}}\mathrm{H}^i(P) = 0.$$

A fully faithful functor detects the zero object: it carries the identity of $\mathcal{H}^i(\mathcal{M}^\bullet)$ to the identity of the zero object, so that identity vanishes. Hence $\mathcal{H}^i(\mathcal{M}^\bullet) = 0$ for every $i \neq 0$, the complex $\mathcal{M}^\bullet$ is isomorphic to the regular holonomic module $\mathcal{H}^0(\mathcal{M}^\bullet)$ placed in degree 0, and DR_X carries that module to P. See [8, Thm. 7.2.5]. □

Remark (Perverse objects in a finite model) *Let $B^\bullet$ be a finite cellular model of an algebraically constructible object $F \in D^b_c(X^{\mathrm{an}})$. By the Riemann–Hilbert equivalence there is a regular holonomic complex $\mathcal{M}^\bullet$ with $\mathrm{DR}_X(\mathcal{M}^\bullet) \cong F$. The complex $\mathcal{M}^\bullet$ is isomorphic to a regular holonomic module concentrated in degree 0 when $F \in \mathrm{Perv}(X^{\mathrm{an}})$.*

For a cell σ contained in a complex stratum, the stalk of F is represented by $B^\bullet_\sigma$. After a local subdivision making x a vertex, its costalk is represented by the cone of the restriction

$$C^\bullet(\mathrm{st}(x); B) \to C^\bullet(\mathrm{st}(x) - \{x\}; B)$$

shifted by $[-1]$. The pointwise bounds of Remark 4.1 therefore decide by finite linear algebra whether the model corresponds to a regular holonomic module concentrated in degree 0.

The two basic objects on the line summarise the dimension shift. The trivial connection has

$$\mathrm{DR}_{\mathbb{C}}(\mathcal{O}_{\mathbb{C}}) \cong \underline{\mathbb{C}}_{\mathbb{C}}[1]$$

while the point-supported module satisfies

$$\mathrm{DR}_{\mathbb{C}}(\delta_0) \cong {i_0}_*\mathbb{C}$$

Both $\mathcal{O}_{\mathbb{C}}$ and δ_0 are ordinary regular holonomic modules in degree zero. Their de Rham images occupy different ordinary sheaf degrees because their supports have different dimensions, yet both lie in perverse degree zero. The perverse heart is therefore the degree convention in which the differential module category appears without artificial shifts from one support dimension to another.

The Euler example shows why monodromy on the nonsingular locus is insufficient. On $\mathbb{C}^\times$, a regular connection is determined by its local system; after the origin is restored, the constructible complex also records boundary-supported terms and their attaching morphisms. The correspondence sends closed support to closed support, differential minimal extension to perverse intermediate

extension, and differential duality to Verdier duality. For an adapted finite triangulation, the cellular model packages this constructible object into finite-dimensional complexes and restriction maps without discarding its derived gluing data.

Holonomicity alone still gives constructible, perversely normalized solution data. An irregular meromorphic connection may also carry Stokes data, which ordinary constructible sheaves do not retain. Regularity is therefore the condition under which the full constructible de Rham complex determines the differential system.

6.7 Further directions

The construction of Section 6.5 does not yet provide an intrinsic category of differential systems on a stratified cellulation or an effective procedure for reconstructing differential operators from a finite cellular diagram. One further direction is therefore to define a category $\mathrm{CellD}(K, \mathcal{S})$ that records local propagation on strata, extension maps, compatibility relations and monodromy, together with a realisation functor to $D^b_{\mathrm{rh}}(\mathcal{D}_X)$. The problem is to determine which additional analytic or algebraic data make such a presentation intrinsic. Such a category would extend the constructible-side diagrams of Section 6.5 by the analytic or algebraic data needed to recover differential operators.

A further possible application of such a cellular model appears in the theory of contextuality. Abramsky, Brandenburger, and their collaborators formulate ″contextuality″ as a failure to pass from local data to global data [9]. In their setting, a measurement is a question about a particular quantity in a physical system, whose outcome is one of several possible answers, and a context is a collection of measurements that can be performed jointly. A measurement scenario is represented by a finite cover of possible contexts and a sheaf of local assignments, and a model is contextual when the local assignments agree on overlaps but cannot be glued to a global assignment. The same structure also appears in logical paradoxes - for example, in the familiar liar cycle, every local relation between two adjacent statements can be realised, while no assignment of truth values satisfies the entire cycle. This gluing problem can be represented by a finite complex, for example the Čech complex of the context cover, in which a nontrivial cohomology class may witness contextuality. If the cellular model is realised as a constructible complex, the Riemann–Hilbert correspondence relates

it to a complex of regular holonomic $\mathcal{D}$-modules. It therefore seems possible to begin with this finite model, translate a finite logical obstruction into a system of differential equations whose solution complex recovers the cellular data, and then study it using monodromy, singularities, and nearby and vanishing cycles.

A Basic Homological Algebra (Briefly)

A.1 Motivation

This chapter is a fast and motivating tour through (some of) the main stops of homological algebra. The basic idea of homological algebra is to distil from the theory of R-modules the categorical structure that allows us to discuss kernels, images, quotients, and the relations between them without relying on elements. This allows us to recognise phenomena from module theory, such as the First Isomorphism Theorem, in other mathematical settings that satisfy the conditions required for this interface to apply. The same distillation also allows us to pass from a single map to a sequence of consecutive maps, to examine what happens when the image of one map does not exhaust the kernel of the next, and to identify the nature of this failure. This chapter is fundamentally introductory, we generally have refrained from giving full proofs and have preferred, whenever possible, to explain matters relatively concretely using R-modules. The standard references for the material of this chapter are [10] and [11]; for the purely categorical background we follow [12] and [13]. I also relied extensively on Yuhang Wei's notes, both as a guide to these references and as a source from which I learned much of the material. The notes are based on and substantially expand Kobi Kremnitzer's Oxford C2.2 Homological Algebra lectures; see [14].

Let us start with the setup of a group, a subgroup, and its quotient. Take the group of integers $\mathbb{Z}$ and its subgroup of even integers, represented by the multiplication-by-2 map from $\mathbb{Z}$ to $\mathbb{Z}$. The quotient is $\mathbb{Z}/2\mathbb{Z}$. We can display the three groups and the two maps as

$$0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$$

The zero on the left records that multiplication by 2 has kernel 0. The map $\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ is onto, while its kernel is the subgroup $2\mathbb{Z}$, exactly the image of the preceding multiplication-by-2 map. Thus the sequence records in one diagram the subgroup $2\mathbb{Z} \subseteq \mathbb{Z}$ and the quotient $\mathbb{Z}/2\mathbb{Z}$ obtained from it.

Now tensor the sequence with $\mathbb{Z}/2\mathbb{Z}$. Each copy of $\mathbb{Z}$ becomes $\mathbb{Z}/2\mathbb{Z}$, and multiplication by 2 becomes the zero map because $2 = 0$ in $\mathbb{Z}/2\mathbb{Z}$. The resulting sequence begins

$$0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z}$$

Before tensoring, the first map had kernel 0; afterwards its kernel is the whole module $\mathbb{Z}/2\mathbb{Z}$ and it is no longer injective, so tensoring has created elements in the kernel that did not come from the kernel of the original map. The rest of the chapter develops a systematic way to describe and record this kind of failure.

This example allows us to survey the main stages of our journey. First, we need a language in which to discuss kernels, cokernels, and images. Second, we need to discuss consecutive maps and describe what happens when a functor applied to them fails to preserve all their properties. In particular, we need a way to measure maps for which not everything that goes out is what came in: "leaks" and excesses. This will lead us to distinguish between different kinds of objects—those on which the functors behave properly—and to develop a strategy that uses them to measure these gaps precisely and degree by degree, ultimately producing "derived functors". We conclude the chapter with two explicit examples of such functors, which will be used below.

Throughout this chapter, R denotes a ring (with 1, not necessarily commutative), R- $\mathbf{Mod}$ the category of left R-modules, and $\mathbf{Mod}$ -R the category of right R-modules. When R is a field k these are vector spaces; when $R = \mathbb{Z}$ they are abelian groups, so $\mathbb{Z}$- $\mathbf{Mod} = \mathbf{Ab}$.

A.2 Abelian Categories

Which categorical structure retains the kernel, quotient and exact-sequence arguments familiar from modules?

Definition A.1 (Abelian Category) *An **abelian category** is a category $\mathcal{A}$ such that:*

(A1) each $\mathrm{Hom}_{\mathcal{A}}(X, Y)$ *is an abelian group and composition is bilinear (*we can add and subtract parallel maps*);*

(A2) there is a zero object, and any two objects have a biproduct $X \oplus Y$ *that is simultaneously their product and coproduct (*finite direct sums exist*);*

*(A3) every morphism has a kernel and a cokernel (*we can solve $f(x) = 0$ and pass to quotients*);*

(A4) the canonical map

$$\mathrm{coim}(f) := \mathrm{Coker}(\ker f) \to \mathrm{Ker}(\mathrm{coker}\, f) =: \mathrm{im}(f)$$

*is an isomorphism for every f (*the first isomorphism theorem holds*).*

A category satisfying only the first axiom is called an Ab-**enriched category**; a category that also satisfies the second axiom is called an **additive category**.

The non-trivial axiom is the fourth. In short, in any setting with kernels and cokernels a map $f : X \to Y$ has two a priori different "images": the coimage $\mathrm{Coker}(\ker f)$, a quotient of X, and the image $\mathrm{Ker}(\mathrm{coker}\, f)$, a subobject of Y. In R- **Mod** these are $X/\operatorname{Ker} f$ and $\{f(x)\} \subseteq Y$, and the comparison map between them is exactly the isomorphism $X/\operatorname{Ker} f \cong \operatorname{im} f$ of the First Isomorphism Theorem. So (A4) is nothing more than an arrow-only version of that theorem.

In other words, in an abelian category every morphism factors canonically as an epimorphism onto its image followed by a monomorphism, $X \twoheadrightarrow \mathrm{im}(f) \hookrightarrow Y$.

The following theorem guarantees that, in some cases, we can understand this abstract definition in terms of R-Mod.

Theorem A.2 (Freyd-Mitchell Embedding Theorem) *Every small abelian category $\mathcal{A}$ admits an exact, fully faithful functor $\mathcal{A} \to R$- **Mod** into the modules over some ring R, embedding $\mathcal{A}$ as a full subcategory.*

A proof is given in [10, p. 25].

Modules therefore provide a concrete model for the constructions below, in finite diagram arguments.

The relevant structure-preserving functors are the following:

Definition A.3 (Additive Functor) *A functor $F : \mathcal{C} \to \mathcal{D}$ is an* ***additive functor*** *if each induced map* $\mathrm{Hom}(X, Y) \to \mathrm{Hom}(FX, FY)$ *is a group homomorphism, such that*

$$F(f + g) = F(f) + F(g)$$

and $F(0) = 0$.

In an **Ab**-enriched category with a zero object, every finite product or finite coproduct, when it exists, has the canonical structure of a biproduct. In particular, every abelian category has finite biproducts, written $X \oplus Y$. As shown in [13, Proposition 1.2.4] Additive functors preserve finite biproducts.

A.3 Exact Sequences

The First Isomorphism Theorem gives the model to keep in mind. For R-modules, a map records both a submodule through its image and a quotient through its kernel. Exact sequences isolate the point at which these two descriptions meet.

Definition A.4 (Exact Sequence) *A sequence $A \to (f)B \to (g)C$ is* ***exact at B*** *if* $\operatorname{im} f = \operatorname{Ker} g$.

In R- **Mod**, this has a direct meaning: the elements of B killed by g are exactly those that arrive from A. Thus $g \circ f = 0$, and if $g(b) = 0$, then $b = f(a)$ for some $a \in A$.

Definition A.5 (Short Exact Sequence) *A sequence $0 \to A \to (i)B \to (q)C \to 0$ is a* ***short exact sequence*** *if it is exact at A, B, and C.*

For modules, this is precisely the familiar submodule–quotient situation. The map i identifies A with the submodule $i(A) \subseteq B$, exactness gives $i(A) = \operatorname{Ker} q$, and q is surjective. The First Isomorphism Theorem therefore gives $C \cong B/i(A)$. The sequence from the introduction, $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$, records the inclusion $2\mathbb{Z} \subseteq \mathbb{Z}$ together with the corresponding quotient.

There is still a question about how the two end modules are assembled inside B. The simplest case is when they separate completely.

Definition A.6 (Split short exact sequence) *A short exact sequence $0 \to A \xrightarrow{i} B \xrightarrow{q} C \to 0$ is* ***split*** *if $B \cong A \oplus C$ compatibly with i and q.*

A splitting can be recognized without first constructing the direct-sum isomorphism. A **retraction** is a map $r : B \to A$ with $r \circ i = \mathrm{id}_A$; a **section** is a map

$s : C \to B$ with $q \circ s = \mathrm{id}_C$. In module language, a section chooses one representative in B for every class in the quotient C, and does so R-linearly.

Lemma A.7 (Splitting Lemma) *For a short exact sequence* $0 \to A \xrightarrow{i} B \xrightarrow{q} C \to 0$*, the following are equivalent:*

1. *the sequence is split;*
2. *there is a retraction* $r : B \to A$ *with* $r \circ i = \mathrm{id}_A$*;*
3. *there is a section* $s : C \to B$ *with* $q \circ s = \mathrm{id}_C$*.*

Proof. Suppose a section s is given. For every $b \in B$, the difference $b - s(q(b))$ lies in $\operatorname{Ker} q = i(A)$, so there is a unique $a \in A$ with $b = i(a) + s(q(b))$. This decomposition is unique, and therefore $(a, c) \mapsto i(a) + s(c)$ gives an isomorphism $A \oplus C \cong B$. A split sequence clearly supplies both a section and a retraction, and the argument from a retraction is dual. See [15] and [11, Corollary 2.24]. □

Example A.8 (A nonsplit extension) *Consider the short exact sequence* $0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{\iota} \mathbb{Z}/4\mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \to 0$*, where* $\iota([a]) = [2a]$ *and* $\pi([b]) = [b] \bmod 2$*.*

The sequence does not split. Indeed, a section $s : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ *would have to send* $[1]$ *to an element mapping to* $[1]$ *modulo* 2 *and annihilated by multiplication by* 2*, which does not exist.*

Thus the two end modules do not determine the middle module: the split extension has middle term $(\mathbb{Z}/2\mathbb{Z}) \oplus (\mathbb{Z}/2\mathbb{Z})$*, whereas this extension has middle term* $\mathbb{Z}/4\mathbb{Z}$*.*

We now ask what happens to a short exact sequence when an algebraic functor is applied to it.

A.4 Exact Functors

Split sequences cause no difficulty: an additive functor preserves finite biproducts, so it carries $B \cong A \oplus C$ to $F(B) \cong F(A) \oplus F(C)$. The interesting case is therefore a general short exact sequence $0 \to A \xrightarrow{i} B \xrightarrow{q} C \to 0$.

For $\mathrm{Hom}_R(M, -)$, the kernel part survives. A map $M \to B$ whose composite with q is zero takes its values in $\operatorname{Ker} q = i(A)$ and therefore factors through A. What may fail is the final surjectivity: a map $M \to C$ need not admit a **lift** $M \to B$.

For $\mathrm{Hom}_R(-, M)$, the arrows reverse. A map $B \to M$ that vanishes on $i(A)$ depends only on the coset in $B/i(A) \cong C$, so it factors through C. What may fail is that a map $A \to M$ need not extend to B.

Tensor product behaves in the opposite direction. Since $C \cong B/i(A)$, tensoring preserves the quotient and gives an exact sequence $A \otimes_R N \to B \otimes_R N \to C \otimes_R N \to 0$. The map at the left need not remain injective.

Definition A.9 (Exact functor) *An additive functor F is **right exact** if $A \to B \to C \to 0$ exact implies $F(A) \to F(B) \to F(C) \to 0$ exact. It is **left exact** if $0 \to A \to B \to C$ exact implies $0 \to F(A) \to F(B) \to F(C)$ exact. It is **exact** if it is both.*

Lemma A.10 (One-sided exactness of Hom and tensor product) *For every object M of an abelian category, $\mathrm{Hom}(M, -)$ is left exact and $\mathrm{Hom}(-, M)$ is contravariantly left exact. If N is a left R-module, $- \otimes_R N : \mathbf{Mod}\text{-}R \to \mathbf{Ab}$ is right exact; dually, if M is a right R-module, $M \otimes_R - : R\text{-}\mathbf{Mod} \to \mathbf{Ab}$ is right exact.*

There is a structural reason for the opposite behaviour of tensor product and Hom. For this paragraph, suppose that R is commutative. A homomorphism $F : M \otimes_R N \to P$ can be read by fixing $m \in M$: it gives the homomorphism $n \mapsto F(m \otimes n)$ from N to P. Conversely, a map $G : M \to \mathrm{Hom}_R(N, P)$ defines $F(m \otimes n) \coloneqq G(m)(n)$. Thus $\mathrm{Hom}_R(M \otimes_R N, P) \cong \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$.

Definition A.11 (Adjoint functors) *Functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ are **adjoint**, written $F \dashv G$, if there are natural isomorphisms $\mathrm{Hom}_{\mathcal{D}}(F(A), B) \cong \mathrm{Hom}_{\mathcal{C}}(A, G(B))$.*

The tensor–Hom correspondence is therefore an adjunction in the sense of the preceding definition:

$$- \otimes_R N \dashv \mathrm{Hom}_R(N, -)$$

In module language, a homomorphism $M \otimes_R N \to P$ is equivalent to a homomorphism $M \to \mathrm{Hom}_R(N, P)$, obtained by fixing the first input.[2]

This adjunction explains the one-sided exactness established above. Left adjoints preserve cokernels, while right adjoints preserve kernels. Hence tensor product preserves the quotient $B/i(A)$, whereas $\mathrm{Hom}_R(N, -)$ preserves the kernel of $B \to C$. The opening sequence shows that the opposite halves need not be preserved. See [14] for the module-theoretic presentation and [12, Chapter 9] for the categorical preservation properties of adjoints.

[2]This adjunction is the commutative-ring form. For a noncommutative ring the same construction requires explicit right-, left-, and bimodule hypotheses, so we will not use the unqualified formula in that setting.

Return to $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$. Applying $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, -)$ fails at the final term because $\mathrm{id}_{\mathbb{Z}/2\mathbb{Z}}$ has no lift to $\mathbb{Z}$. Tensoring with $\mathbb{Z}/2\mathbb{Z}$ turns multiplication by 2 into the zero map and creates a new kernel.

A.5 Chain Complexes and Homology

We ended the last section with a problem - while short exact sequences appear to be a crucial part in the grammar / interface that abelian categories describe, our very useful functors break it. However, consecutive maps still compose to zero, and hence, instead of $\operatorname{im} d_{n+1} = \operatorname{Ker} d_n$, we get $\operatorname{im} d_{n+1} \subseteq \operatorname{Ker} d_n$; some information is killed without coming from the incoming map.

Before we can construct a new exact sequence that incorporates this defect, we first need to formally measure the failure of exactness. An obvious solution, which fits well the cokernel-kernel attributes of Abelian categories, is to simply use the quotient $\operatorname{Ker} d_n / \operatorname{im} d_{n+1}$, to which we will give a fancy name.

Definition A.12 (Chain Complex) *A **chain complex** $C_\bullet$ in an abelian category $\mathcal{A}$ is a family of objects $\{C_n\}_{n\in\mathbb{Z}}$ with morphisms (**differentials**) $d_n : C_n \to C_{n-1}$ such that consecutive differentials compose to zero:*

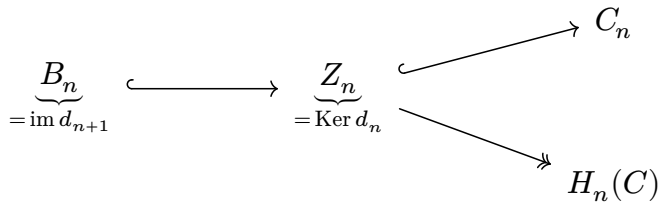
$$d_{n-1} \circ d_n = 0, \qquad \text{written} \qquad d^2 = 0.$$

*The **n-cycles** are $Z_n = \operatorname{Ker} d_n$; the **n-boundaries** are $B_n = \operatorname{im} d_{n+1}$. The condition $d^2 = 0$ says exactly $B_n \subseteq Z_n$.*

This lays the foundations to our measurment tool:

Definition A.13 (Homology) *The **n-th homology** is the quotient $H_n(C) = Z_n/B_n = \frac{\operatorname{Ker} d_n}{\operatorname{im} d_{n+1}}$*

At degree n, the two differentials determine $B_n \subseteq Z_n \subseteq C_n$, and homology takes the quotient of the cycles by the boundaries.



Two cycles determine the same class in $H_n(C)$ exactly when their difference lies in B_n. Consequently, $H_n(C) = 0$ precisely when the complex is exact at C_n. This gives the measurement of the difference between an incoming image and the following kernel.

Homology arose in topology from the same construction. For an oriented simplicial complex, let C_n be the free abelian group generated by its oriented n-simplices. The differential sends a simplex to the alternating sum of its faces. For an oriented triangle, $d_2[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]$, while $d_1[v_i, v_j] = v_j - v_i$. Applying d_1 once more gives $(v_2 - v_1) - (v_2 - v_0) + (v_1 - v_0) = 0$. Each vertex occurs twice with opposite signs, which gives the geometric model for $d^2 = 0$. The construction of simplicial and singular homology is developed in [16, Chapter 2].

In this setting, an n-cycle is a chain with zero boundary, while an n-boundary already occurs as the boundary of a chain one degree higher. The quotient Z_n/B_n therefore retains the classes of cycles that remain after boundaries have been identified with zero.

Example A.14 (The multiplication-by-two map as a differential) *A cellular calculation for $\mathbb{R}P^2$ gives the chain complex $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0$, with the three copies of $\mathbb{Z}$ in degrees 2, 1, and 0; See [16, Chapter 2].*

At degree 1, we have $Z_1 = \mathrm{Ker}(0) = \mathbb{Z}$ and $B_1 = \mathrm{im}(\cdot 2) = 2\mathbb{Z}$. Hence $H_1(\mathbb{R}P^2; \mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}$.

The multiplication by 2 map from the opening example has now appeared as a differential. If this complex is tensored with $\mathbb{Z}/2\mathbb{Z}$, that differential becomes zero, exactly as in the opening calculation.

Definition A.15 (Chain Map) *A **chain map** $f : C_\bullet \to D_\bullet$ between two chain complexes is a family of morphisms $f_n : C_n \to D_n$ that commute with the differentials. That is, for every n:*

$$d_n^D \circ f_n = f_{n-1} \circ d_n^C \quad (\text{or simply } df = fd).$$

Visually, this means the following ladder diagram commutes:

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & C_{n+1} & \xrightarrow{d_{n+1}^C} & C_n & \xrightarrow{d_n^C} & C_{n-1} & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle f_{n+1}} & & \downarrow{\scriptstyle f_n} & & \downarrow{\scriptstyle f_{n-1}} & & \\
\cdots & \longrightarrow & D_{n+1} & \xrightarrow{d_{n+1}^D} & D_n & \xrightarrow{d_n^D} & D_{n-1} & \longrightarrow & \cdots
\end{array}$$

These objects and chain maps form the category $C(\mathcal{A})$, which is abelian whenever $\mathcal{A}$ is abelian; see [11, Proposition 5.100].

Because a chain map respects the differentials, it carries cycles to cycles and boundaries to boundaries. Consequently, every chain map $f : C_\bullet \to D_\bullet$ induces a morphism $H_n(f) : H_n(C) \to H_n(D)$. Identities and compositions are preserved, so for every n the construction $C_\bullet \mapsto H_n(C)$ defines a functor from the category of chain complexes to $\mathcal{A}$.

Definition A.16 (chain homotopic) *Chain maps $f, g : C_\bullet \to D_\bullet$ are **chain homotopic**, written $f \simeq g$, if there exist maps $s_n : C_n \to D_{n+1}$ with*

$$f - g = d \circ s + s \circ d.$$

Chain homotopy is an equivalence relation on the chain maps $C_\bullet \to D_\bullet$, and it is compatible with composition; see [10, Section 1.4]. Thus, quotienting the morphisms in $C(\mathcal{A})$ by chain homotopy produces the homotopy category $K(\mathcal{A})$, which we will construct later, in Section B.4.

The importance of this notion is that homology cannot distinguish between chain-homotopic maps.

Lemma A.17 (Chain homotopy and homology) *If $f, g : C_\bullet \to D_\bullet$ are chain homotopic, then they induce the same map on homology:*

$$H_n(f) = H_n(g)$$

for every $n \in \mathbb{Z}$.

Proof. If c is a cycle, then $(f - g)(c) = d(s(c))$ is a boundary. Hence f and g determine the same class on every homology object. See [10, Lemma 1.4.5] □

A.5.1 Cochain complexes and cohomology

The cohomological convention reverses the grading of a chain complex. A **cochain complex** $A^\bullet$ is a family of objects with differentials $d_A^n : A^n \to A^{n+1}$ satisfying $d_A^{n+1} \circ d_A^n = 0$. Write $B^n := \operatorname{im} d_A^{n-1}$ and $Z^n := \operatorname{Ker} d_A^n$. Then $B^n \subseteq Z^n \subseteq A^n$, and the ***n*-th cohomology** object is $H^n(A^\bullet) := Z^n / B^n = \operatorname{Ker} d_A^n / \operatorname{im} d_A^{n-1}$. Thus cohomology measures the failure of the complex to be exact at A^n.

When a cochain complex has only two nonzero terms in degrees 0 and 1, we abbreviate $0 \to A \xrightarrow{d} B \to 0$ by $\left[A \xrightarrow{d} B\right]$. If the terms occur in different degrees, we state their placement explicitly.

A morphism of cochain complexes $f : A^\bullet \to B^\bullet$ is a family $f^n : A^n \to B^n$ satisfying $d_B^n \circ f^n = f^{n+1} \circ d_A^n$. It induces morphisms $H^n(f) : H^n(A^\bullet) \to H^n(B^\bullet)$, so each H^n is a functor from cochain complexes to $\mathcal{A}$. Two cochain maps $f, g : A^\bullet \to$

$B^\bullet$ are **cochain-homotopic** if there are morphisms $h^n : A^n \to B^{n-1}$ such that $f - g = d_B \circ h + h \circ d_A$. Cochain-homotopic maps induce the same morphism on cohomology, by the same calculation as for chain homotopy.

For complexes of vector spaces, cochains have a concrete interpretation as linear tests on chains. A cocycle is a test that vanishes on boundaries, while two cocycles differing by a coboundary define the same cohomology class. The standard construction from singular cochains is described in [16, Chapter 3]. The smallest cellular example already shows how the cochain differential is formed.

Example A.18 (The oriented interval and the cochain differential) *Let an interval have vertices v_0, v_1 and one edge e oriented from v_0 to v_1. Over a field k, its cellular chain groups are $C_1 = ke$ and $C_0 = kv_0 \oplus kv_1$, with boundary $d_1(e) = v_1 - v_0$. Applying $\mathrm{Hom}_k(-, k)$ gives $C^0 := \mathrm{Hom}_k(C_0, k)$ and $C^1 := \mathrm{Hom}_k(C_1, k)$, with $d^0(\varphi) := \varphi \circ d_1$. If $\varphi(v_0) = a_0$ and $\varphi(v_1) = a_1$, then $d^0(a_0, a_1) = a_1 - a_0$. The cochain differential therefore reads the terminal value minus the initial value.*

$$C_1 = ke \xrightarrow{d_0} c_0 = kv_0 \oplus kv_1$$

$$C^1 = \mathrm{Hom}_k(C_1, k) \xleftarrow{d_0} C^0 = \mathrm{Hom}_k(C_0, k)$$

Example A.19 (Cellular cohomology of the circle and the torus) *Let k be a field. Give S^1 its standard cell decomposition with one vertex v and one edge e, whose two endpoints are attached to v. The terminal-minus-initial calculation of Example A.18 becomes $a - a = 0$ after the endpoints are identified. Hence the cellular cochain complex is $0 \to k \xrightarrow{0} k \to 0$ in degrees 0 and 1, and*

$$H^0(S^1; k) \cong k, \quad H^1(S^1; k) \cong k.$$

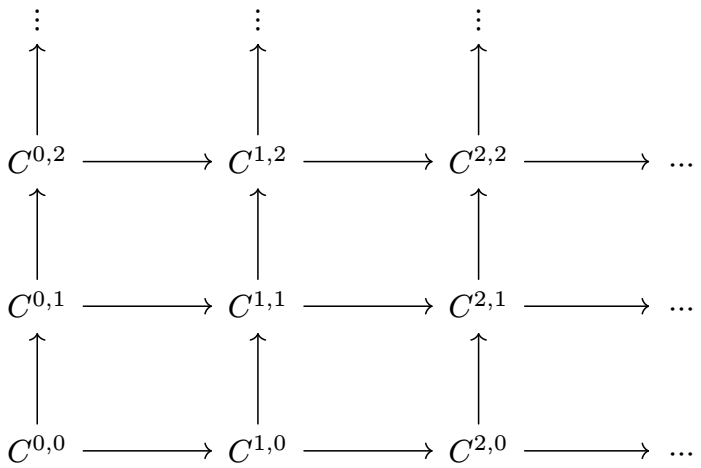
Thus $H^0(S^1; k) \cong k$ reflects connectedness, while $H^1(S^1; k) \cong k$ gives one independent degree-one cohomology class.

Give the torus T^2 its standard cell decomposition with one vertex, two edges a, b, and one two-cell. Its cellular cochain complex is $0 \to k \xrightarrow{0} k^2 \xrightarrow{0} k \to 0$ in degrees $0, 1, 2$. Both differentials vanish: the edges begin and end at the same vertex, and the boundary word $aba^{-1}b^{-1}$ contains each edge with opposite signs. Hence $H^0(T^2; k) \cong k$, $H^1(T^2; k) \cong k^2$, and $H^2(T^2; k) \cong k$.

A.5.2 Double complexes and Hom complexes

The examples above have one cohomological index. A second index appears when a construction depends simultaneously on two complexes.

Definition A.20 (Total complex) *A **first-quadrant double cochain complex** is a family $C^{p,q}$, indexed by $p, q \geq 0$, with differentials $d_h : C^{p,q} \to C^{p+1,q}$ and $d_v : C^{p,q} \to C^{p,q+1}$ satisfying $d_h^2 = 0$, $d_v^2 = 0$, and $d_h \circ d_v = d_v \circ d_h$. Its **total complex** is defined by $\mathrm{Tot}^n(C) := \bigoplus_{p+q=n} C^{p,q}$, with $d_{\mathrm{Tot}}(x) := d_h(x) + (-1)^p d_v(x)$ for $x \in C^{p,q}$.*



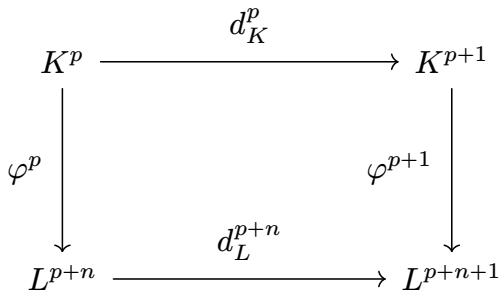
The horizontal differential moves one column to the right and the vertical differential moves one row upward. The terms with $p + q = n$ lie on one diagonal of this grid, and $\mathrm{Tot}^n(C)$ collects exactly that diagonal.

Proposition A.21 (The total differential squares to zero) *The differential of the total complex (Definition A.20) satisfies $d_{\mathrm{Tot}} \circ d_{\mathrm{Tot}} = 0$.*

Proof. For $x \in C^{p,q}$, $d_{\mathrm{Tot}}^2(x) = d_h^2(x) + (-1)^p d_h d_v(x) + (-1)^{p+1} d_v d_h(x) + d_v^2(x) = 0$. The first and last terms vanish because the rows and columns are complexes, while the middle terms cancel because the two differentials commute. □

Definition A.22 (Hom complex) *For cochain complexes $K^\bullet, L^\bullet$ in an abelian category $\mathcal{A}$, define $\mathrm{Hom}^n_{\mathcal{A}}(K^\bullet, L^\bullet) := \prod_p \mathrm{Hom}_{\mathcal{A}}(K^p, L^{p+n})$. For a homogeneous family $\varphi = (\varphi^p)_p$ of degree n, set $d(\varphi)^p := d_L^{p+n} \circ \varphi^p - (-1)^n \varphi^{p+1} \circ d_K^p$.*

The two terms are the two routes from K^p to L^{p+n+1}.



When the first argument is an object A concentrated in degree 0, the definition reduces to $\mathrm{Hom}^\bullet_{\mathcal{A}}(A, L^\bullet)$, whose degree-$q$ term is $\mathrm{Hom}_{\mathcal{A}}(A, L^q)$ and whose differential sends φ to $d_L^q \circ \varphi$. This is the form used below to compute Ext.

Definition A.23 (Acyclic complexes and quasi-isomorphisms) *A chain or cochain complex is* ***acyclic*** *if all its homology or cohomology objects vanish.*

A morphism of chain or cochain complexes is a ***quasi-isomorphism*** *if it induces isomorphisms on all homology or cohomology objects.*

A quasi-isomorphism means that, from the point of view of homology, $C_\bullet$ and $D_\bullet$ share the same information. This provides our strategy - we will replace difficult objects with quasi-isomorphic complexes made entirely of "good" objects.

Example A.24 (Subdivision changes the complex but not its cohomology) *Give S^1 a cellular decomposition with two vertices and two edges. As we saw, its cellular cochain complex is $C^\bullet_{\mathrm{reg}} = \left[k^2 \xrightarrow{D} k^2\right]$, where $D(a,b) = (b-a, a-b)$. The kernel of D is the diagonal copy of k and its image is $\{(x,-x) : x \in k\}$, so its cokernel is one-dimensional. Therefore*

$$H^0\big(C^\bullet_{\mathrm{reg}}\big) \cong k, \quad H^1\big(C^\bullet_{\mathrm{reg}}\big) \cong k.$$

Let $C^\bullet_{\mathrm{min}} = \left[k \xrightarrow{0} k\right]$ be the minimal complex of Example A.19. Define

$$p^0(a,b) = a, \quad p^1(x,y) = x + y$$

Since $p^1(D(a,b)) = (b-a) + (a-b) = 0$, the maps define a morphism of complexes $p : C^\bullet_{\mathrm{reg}} \to C^\bullet_{\mathrm{min}}$. It induces an isomorphism from the diagonal kernel to k in degree 0 and an isomorphism $k^2/\operatorname{im}(D) \to k$ in degree 1. Hence p is a quasi-isomorphism.

However, the two complexes are not isomorphic because their terms have different dimensions.

Quasi-isomorphisms determine when a morphism preserves all homology objects. But how does the homology of a third complex depend on the homology of two others in a short exact sequence?

Definition A.25 (long exact sequence) *A **long exact sequence** (LES) is an exact sequence, infinite in one or both directions: $\cdots \to X_{n+1} \to X_n \to X_{n-1} \to \cdots$ meaning it is exact at every interior term.*

Vanishing or explicit knowledge of neighbouring terms determines an unknown term, like in equations. The Snake Lemma constructs the connecting morphisms that join the separate degrees.

Lemma A.26 (Snake Lemma) *Given a commutative diagram with exact rows in an abelian category:*

$$\begin{array}{ccccccccc} & & & \overset{i}{\longrightarrow} & & \overset{p}{\longrightarrow} & & & \\ & & A & & B & & C & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle h} & & \\ 0 & \longrightarrow & A' & \xrightarrow[i']{} & B' & \xrightarrow[p']{} & C' & & \end{array}$$

there is an exact sequence:

$$\operatorname{Ker} f \to \operatorname{Ker} g \to \operatorname{Ker} h \xrightarrow{\partial} \operatorname{Coker} f \to \operatorname{Coker} g \to \operatorname{Coker} h,$$

*where ∂ is the **connecting homomorphism**.*

Remark *The differential d is an internal operator within a single chain complex ($C_n \xrightarrow{d_n} C_{n-1}$). The map ∂ is an external bridge between the homologies of two completely different complexes ($H_n(C) \xrightarrow{\partial_n} H_{n-1}(A)$).*

We recall the construction of ∂. Let $c \in \operatorname{Ker} h$. Choose $b \in B$ with $p(b) = c$. Commutativity gives $p'(g(b)) = h(p(b)) = h(c) = 0$, so exactness of the lower row produces a unique $a' \in A'$ such that $i'(a') = g(b)$. If b is replaced by another lift $b + i(a)$, then a' is replaced by $a' + f(a)$. Hence the class of a' in $\operatorname{Coker} f$ is independent of the chosen lift. Define $\partial(c) := [a'] \in \operatorname{Coker} f$. The same argument verifies that the construction is compatible with representatives and is natural in morphisms of diagrams; see [17].

By repeatedly applying the Snake Lemma degree-by-degree to a short exact sequence of chain complexes, we get the following theorem/

Theorem A.27 (Long Exact Sequence in Homology) *A short exact sequence of chain complexes* $0 \to A_\bullet \to B_\bullet \to C_\bullet \to 0$ *induces a natural long exact sequence in homology:*

$$\cdots \to H_n(A) \to H_n(B) \to H_n(C) \xrightarrow{\partial} H_{n-1}(A) \to H_{n-1}(B) \to \cdots$$

The construction of the connecting morphism and the proof of naturality are given in [11, Theorem 6.13].

The connecting map ∂ lowers the degree by one. If a cycle in C_n fails to lift to a cycle in B_n (i.e - have no source in the pre-image) its appears as a homology class in $H_{n-1}(A)$.

Exactness lets us determine unknown terms from the adjacent maps. For example, $0 \to H_n \to 0$ implies $H_n = 0$, while $0 \to A \to H_n \to 0$ implies $H_n \cong A$.

A.6 Projective and Injective Objects

As we just learned, SES's are the natural grammar of modules (and hence of abelian categories), but not every functor respects them. For left/right-exact functors, the missing exactness can be related to lifting through surjections or extending across injections. For instance, an element of $\mathrm{Hom}(P, Z)$ is a morphism $f : P \to Z$. If $h : Y \to Z$ is surjective, saying that f lies in the image of $\mathrm{Hom}(P, Y) \to \mathrm{Hom}(P, Z)$ means that there is some $g : P \to Y$ such that $f = h \circ g$. In simple terms, every map to the quotient can be lifted to its source.

This lifting property motivates the following definition and our strategy - to replace objects by "good enough" objects for which the relevant functor behaves better.

Definition A.28 (Projective Object) *In an abelian category* $\mathcal{A}$*, an object* P *is* ***projective*** *if* $\mathrm{Hom}_{\mathcal{A}}(P, -)$ *is exact.*

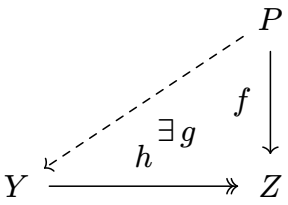
Dually,

Definition A.29 (Injective Object) *I is* ***injective*** *if the contravariant functor* $\mathrm{Hom}_{\mathcal{A}}(-, I)$ *is exact.*

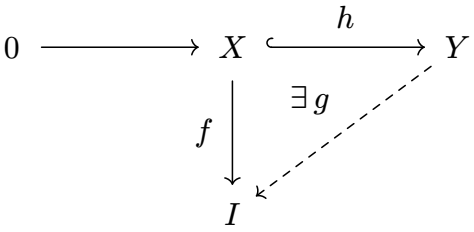
Those definitions unpack into:

Proposition A.30 (Lifting and extension) *The following are equivalent characterisations.*

- *P is projective $\Leftrightarrow$ for every epimorphism $h : Y \twoheadrightarrow Z$ and every $f : P \to Z$, there exists (not necessarily unique) $g : P \to Y$ with $f = h \circ g$. This is the* ***lifting property****:*



- *I is injective $\Leftrightarrow$ for every monomorphism $h : X \hookrightarrow Y$ and every $f : X \to I$, there exists $g : Y \to I$ with $f = g \circ h$. This is the* ***extension property****:*



Moreover P is projective $\Leftrightarrow$ every short exact sequence $0 \to A \to B \to P \to 0$ splits; dually every $0 \to I \to A \to B \to 0$ splits.

What these objects are, module-theoretically?

Theorem A.31 *In R-* **Mod***:*

(1) ***Free modules are projective****, and more precisely P is projective $\Leftrightarrow$ P is a direct summand of a free module.*

(2) Consequently R- **Mod** ***has enough projectives****: every module M admits an epimorphism $F \twoheadrightarrow M$ from a free (hence projective) module.*

The classification of projective modules as the direct summands of free modules, and the consequences drawn below, are treated in [11, Chapter 3].

Why free means projective. Let F be free on a basis $\{x_i\}$, let $h : Y \twoheadrightarrow Z$ be an epimorphism, and let $f : F \to Z$. For each basis element x_i, surjectivity of h allows us to choose $y_i \in Y$ with $h(y_i) = f(x_i)$. Since F is free, the assignment $x_i \mapsto y_i$ extends uniquely to an R-linear map $g : F \to Y$. On every basis element,

$$(h \circ g)(x_i) = h(y_i) = f(x_i),$$

and hence $h \circ g = f$ on all of F. Thus F has the lifting property and is projective.

Why projective implies a summand of a free module. Every module P is a quotient of some free module F. This means there exists a short exact sequence

$0 \to \mathrm{Ker}(h) \to F \xrightarrow{h} P \to 0$. Because P is projective, this sequence must split by the Splitting Lemma (Lemma A.7). A split exact sequence implies $F \cong P \oplus \mathrm{Ker}(h)$, hence P is a direct summand of F.

Why R- **Mod** *has enough projectives.* Since every R-module M can be generated by some set of elements, it is canonically a quotient of the free module generated by that set.

Example A.32 *Over a field, every module (vector space) is free, hence projective; and dually also injective.*

Over $\mathbb{Z}$, the projective modules are exactly the free abelian groups; e.g. $\mathbb{Z}$ is projective but $\mathbb{Z}/2$ is not (there is no lift of $\mathrm{id} : \mathbb{Z}/2 \to \mathbb{Z}/2$ *along $\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2$, since $\mathbb{Z}$ has no element of order 2).*

Injectives are subtler, but again explicit in module land.

Theorem A.33 (Baer's Criterion) *A module M over R is injective if and only if every R-module homomorphism from an ideal $I \subseteq R$ into M extends to a homomorphism from R into M.*

The non-trivial direction, is proved in [11, Theorem 3.30].

When R is a Principal Ideal Domain (PID), this algebraic condition simplifies: M is injective if and only if M is divisible; this is [11, Corollary 3.35]. Furthermore, R- **Mod** has enough injectives, see [10, Proposition 2.3.10].

A.6.1 Resolutions

A projective resolution replaces an arbitrary object by a sequence of projective objects without changing its degree-zero homology. In a module category we may choose the projectives to be free, which gives the construction a concrete interpretation in terms of generators and successive relations.

Definition A.34 (Projective Resolution) *A **projective resolution** of an object A is an exact sequence:*

$$\cdots \to P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} A \to 0$$

where every P_n is a projective object.

When the resolution is chosen to be free, P_0 has a basis of generators mapping onto A. The kernel of $P_0 \to A$ consists of the relations among these generators. Choose generators for this kernel and let them form a basis of P_1; the map $P_1 \to P_0$ sends each basis vector to the relation it represents. Its kernel consists of

relations among those relations, and the same construction produces P_2 and the subsequent terms.

Because the sequence is exact, the complex $P_\bullet = (\ldots \to P_2 \to P_1 \to P_0 \to 0)$ has zero homology everywhere except at degree 0, where $H_0(P) = P_0/\operatorname{im} d_1 \cong A$. In the language of Section A.5, the complex $P_\bullet$ is quasi-isomorphic to A.

In order to be able to succeed in the process, the category required to fullfile one condition:

Definition A.35 (Enough Projectives / Injectives) *An abelian category has* ***enough projectives*** *if every object M is a quotient of a projective object (i.e., there exists an epimorphism $P \twoheadrightarrow M$ with P projective).*

Dually, it has ***enough injectives*** *if every object can be embedded into an injective object (a monomorphism $M \hookrightarrow I$).*

A resolution requires making arbitrary choices. There are infinitely many ways to choose generators (P_0), and infinitely many ways to map relations (P_1). If we use these resolutions to compute invariants, won't our answers depend on our arbitrary choices?

The Fundamental Lemma of Homological Algebra guarantees that they do not.

Theorem A.36 (The Comparison Theorem) *Let $P_\bullet \to A$ be a projective resolution of A, and let $Q_\bullet \to B$ be* ***any*** *resolution of B (exact sequence). Then, any morphism $f : A \to B$ can be lifted to a chain map $f_\bullet : P_\bullet \to Q_\bullet$ between the resolutions:*

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & P_1 & \xrightarrow{d_1} & P_0 & \xrightarrow{\varepsilon_A} & A & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_0} & & \downarrow{\scriptstyle f} & & \\
\cdots & \longrightarrow & Q_1 & \xrightarrow{d_1'} & Q_0 & \xrightarrow{\varepsilon_B} & B & \longrightarrow & 0
\end{array}$$

Furthermore, any two such lifts $f_\bullet, g_\bullet$ are ***chain homotopic*** *($f_\bullet \simeq g_\bullet$).*

This is [10, Comparison Theorem 2.26]; an equivalent formulation, with the inductive construction of the lift written out degree by degree, is [11, Theorem 6.16].

If we take $A = B$ and $f = \mathrm{id}_A$, the Comparison Theorem tells us that any two projective resolutions of A are chain equivalent. If $P_\bullet \to A$ and $Q_\bullet \to A$

are projective resolutions, the Comparison Theorem applied to id_A produces chain maps

$$P_\bullet \to Q_\bullet \quad \text{and} \quad Q_\bullet \to P_\bullet$$

lifting the identity of A. Their composites also lift id_A, so uniqueness up to chain homotopy implies that the composites are homotopic to the corresponding identity maps. Hence $P_\bullet$ and $Q_\bullet$ are chain-homotopy equivalent and we can pick any projective resolution we want.

Dually, we have

Definition A.37 (Injective Resolution) *An **injective resolution** of an object A is an exact sequence:*

$$0 \to A \xrightarrow{\eta} I^0 \xrightarrow{d^0} I^1 \xrightarrow{d^1} I^2 \to \cdots$$

where every I^n is an injective object.

The theorems regarding existence and uniqueness up to chain homotopy apply symmetrically to injective resolutions [10, Section 2.3].

A.7 Derived Functors

We now combine these ingredients to define derived functors. Starting with a left/right exact functor F, we choose a nice projective or injective resolution, apply F, and take homology.

The resulting functors measure the failure of F to be exact. For a short exact sequence, they fit together through connecting morphisms to form a Long Exact Sequence. This is the basic construction of derived functors; its axiomatic formulation in terms of δ-functors is developed in [10, Section 2.1].

A.7.1 The Construction of Derived Functors

Suppose $F : \mathcal{A} \to \mathcal{B}$ is a right-exact additive functor (such as $- \otimes M$, Definition A.9), and the category $\mathcal{A}$ has enough projectives (Definition A.35). We want to measure its defect on an object A. We construct the left derived functors of F in four explicit steps; the same recipe, spelled out step by step, appears in [11, p. 344]:

Step 1: Resolve. Construct a projective resolution for A:

$$\cdots \to P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} A \to 0$$

Step 2: Remove the augmentation. Replace A by 0, obtaining

$$P_\bullet : \cdots \to P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \to 0.$$

Then $H_0(P_\bullet) \cong A$ and $H_{n(P_\bullet)} = 0$ for $n > 0$. The object A is no longer a term of this complex. The augmentation $\varepsilon : P_0 \to A$ extends to a chain map $\varepsilon : P^\bullet \to A[0]$, which is a quasi-isomorphism. Thus $P^\bullet$ is a projective replacement of $A[0]$.

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & P_2 & \xrightarrow{d_2} & P_1 & \xrightarrow{d_1} & P_0 & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle 0} & & \downarrow{\scriptstyle 0} & & \downarrow{\scriptstyle \varepsilon} & & \downarrow{\scriptstyle 0} \\
\cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & A & \longrightarrow & 0
\end{array}$$

Step 3: Apply the Functor. Apply the additive functor F to the truncated complex:

$$\cdots \to F(P_2) \xrightarrow{F(d_2)} F(P_1) \xrightarrow{F(d_1)} F(P_0) \to 0$$

Because F is only right-exact, applying it to $P_\bullet$ generally disrupts the exactness at higher degrees.

Step 4: Compute Homology. To quantify the deviation from exactness, we take the homology of this new complex.

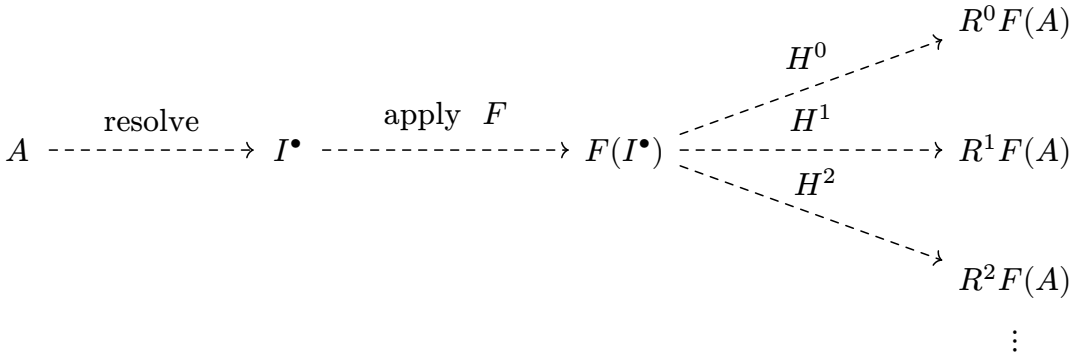
Definition A.38 (Classical Left Derived Functor) *Let $F : \mathcal{A} \to \mathcal{B}$ be right exact and additive, and suppose that $\mathcal{A}$ has enough projectives. For $A \in \mathcal{A}$, choose a projective resolution $P_\bullet \to A$. Define*

$$L_n F(A) \coloneqq H_n(F(P_\bullet))$$

and, dually -

Definition A.39 (Classical Right Derived Functor) *Let $F : \mathcal{A} \to \mathcal{B}$ be left exact and additive, and suppose that $\mathcal{A}$ has enough injectives. For $A \in \mathcal{A}$, choose an injective resolution $A \to I^\bullet$. Define*

$$R^n F(A) \coloneqq H^n(F(I^\bullet))$$



By the Comparison Theorem (Theorem A.36), any two projective resolutions of A are chain homotopy equivalent. Since F is an additive functor, it preserves chain homotopies. Chain homotopic complexes yield isomorphic homology groups. Therefore, $L_nF(A)$ is independent of the chosen projective resolution.

The construction for left-exact covariant functors uses injective resolutions and cohomological grading. In particular, $\mathrm{Hom}_R(M, -)$ is covariant in its second variable, so applying it does not reverse the arrows.

The Comparison Theorem (Theorem A.36) gives independence of the chosen injective resolution up to canonical isomorphism; see also [10, Section 2.5].

To get a Long Exact Sequence after applying F, we need a short exact sequence of resolutions

$$0 \to P_A^\bullet \to P_B^\bullet \to P_C^\bullet \to 0$$

Unfortanatley, resolving A, B, and C separately does not guarantee this. The Horseshoe Lemma constructs compatible resolutions for which the sequence is exact.

Lemma A.40 (The Horseshoe Lemma) *Given a short exact sequence* $0 \to A \to B \to C \to 0$ *and projective resolutions* $P_\bullet^A$ *for* A *and* $P_\bullet^C$ *for* C*, there exists a projective resolution* $P_\bullet^B$ *for* B *such that the sequence of complexes:*

$$0 \to P_\bullet^A \to P_\bullet^B \to P_\bullet^C \to 0$$

is exact. Furthermore, because P_n^C *is projective for every* n*, this sequence of complexes splits at every degree:* $P_n^B \cong P_n^A \oplus P_n^C$*.*

$$
\begin{array}{ccccccccc}
 & & \vdots & & \vdots & & \vdots & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \to & P_2^A & \longrightarrow & P_2^B & \longrightarrow & P_2^C & \to & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \to & P_1^A & \longrightarrow & P_1^B & \longrightarrow & P_1^C & \to & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \to & P_0^A & \longrightarrow & P_0^B & \longrightarrow & P_0^C & \to & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}
$$

Each column is a projective resolution, and each horizontal row is short exact. Since P_n^C is projective, the rows in the resolution split degree by degree. An additive functor F preserves these split sequences, giving a short exact sequence of complexes $0 \to F(P_\bullet^A) \to F(P_\bullet^B) \to F(P_\bullet^C) \to 0$. We apply the Snake Lemma (Lemma A.26) to this sequence of complexes to construct the Long Exact Sequence (Definition A.25). The resulting morphisms connects the homology groups of the three complexes into

$$\cdots \to H_{n(F(P_\bullet^A))} \to H_{n(F(P_\bullet^B))} \to H_{n(F(P_\bullet^C))} \xrightarrow{\partial} H_{n-1}\big(F(P_\bullet^A)\big) \to \cdots$$

. By definition these homology groups are the derived functors $L_nF(A)$, $L_nF(B)$, and $L_nF(C)$, which gives the long exact sequence of derived functors below. The construction of the resolution grid is [10, Horseshoe Lemma 2.2.8].

Theorem A.41 (Long Exact Sequence of Left Derived Functors) *A short exact sequence* $0 \to A \to B \to C \to 0$ *induces a natural long exact sequence of derived functors:*

$$\begin{aligned}\cdots \to L_2F(C) \xrightarrow{\partial} L_1F(A) \to L_1F(B) \to L_1F(C) \\ \xrightarrow{\partial} F(A) \to F(B) \to F(C) \to 0\end{aligned}$$

And, of course, a dual version exist almost verbatim. This, together with the universality statement below, is [10, Theorem 2.4.6 and Theorem 2.4.7].

Notice the termination of the sequence:

$$F(A) \to F(B) \to F(C) \to 0$$

The map $F(A) \to F(B)$ remains non-injective.

A.8 Tor, Ext

Tensor product and Hom provide the two basic examples of the derived-functor construction. The derived functors Ext and Tor formally measure their failures of exactness. Both are treated at length in [10, Chapter 3] and [11, Chapter 7].

Using the recepie from Section A.7.1, we construct the follows.

Definition A.42 (The Tor Functor) *Let $A \in$ **Mod** -R and $B \in R$- **Mod**. For a projective resolution $P_\bullet \to A$, define*

$$\mathrm{Tor}_n^R(A, B) \coloneqq H_n(P_\bullet \otimes_R B)$$

See [10, Section 3.1] for the basic computations, and [10, Theorem 2.7.2] for the fact that the two variables may be resolved interchangeably.

The name Tor originates from torsion. Apply the definition to the projective resolution $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$. After deleting the augmentation and tensoring the resulting resolution complex with $\mathbb{Z}/2\mathbb{Z}$, one obtains $0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z} \to 0$. Consequently, $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) = \mathrm{Ker}(0) \cong \mathbb{Z}/2\mathbb{Z}$. This is the kernel created in Section A.1 when multiplication by 2 became zero.

For a fixed left R-module A, the functor $\mathrm{Hom}_R(A, -)$ is left exact, so its right derived functors measure the failure of the final lifting map in a short exact sequence to remain surjective.

Definition A.43 (The Ext Functor) *For a fixed R-module A, define* $\mathrm{Ext}_R^n(A, -) \coloneqq R^n(\mathrm{Hom}_R(A, -))$.

To compute $\mathrm{Ext}_R^n(A, B)$, choose an injective resolution

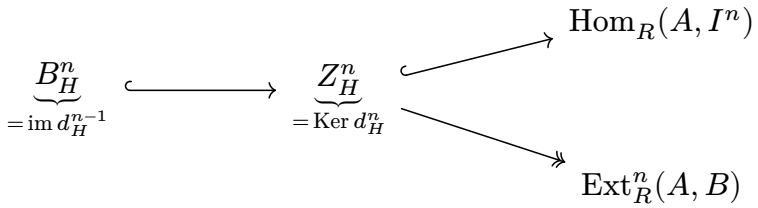
$$0 \to B \to I^0 \xrightarrow{d_I^0} I^1 \xrightarrow{d_I^1} I^2 \to \cdots$$

Applying $\mathrm{Hom}_R(A, -)$ term by term gives the cochain complex

$$\mathrm{Hom}_R(A, I^0) \xrightarrow{d_H^0} \mathrm{Hom}_R(A, I^1) \xrightarrow{d_H^1} \mathrm{Hom}_R(A, I^2) \to \cdots$$

, where $d_H^q(\varphi) \coloneqq d_I^q \circ \varphi$.

At degree n, the two differentials determine $B_H^n \subseteq Z_H^n \subseteq \mathrm{Hom}_R(A, I^n)$, and cohomology takes the quotient of the cocycles by the coboundaries.



Hence $\operatorname{Ext}_R^n(A,B) = Z_H^n/B_H^n = H^n(\operatorname{Hom}_R^\bullet(A, I^\bullet))$. This is the cohomology construction of Section A.5.1 applied to the Hom complex introduced in Section A.5.2. Agreement with the computation obtained from a projective resolution of the first variable is proved in [10, Theorem 2.7.6, p. 63].

The name Ext stands for **extensions**. A short exact sequence $0 \to B \xrightarrow{i} E \xrightarrow{q} A \to 0$ is called an extension of A by B.Two such extensions are equivalent if there is an isomorphism between their middle terms that makes the diagram of short exact sequences commute and induces the identity maps on A and B. The group $\operatorname{Ext}_R^1(A,B)$ classifies equivalence classes of extensions of A by B.

The Splitting Lemma (Lemma A.7) identifies the distinguished class among them. An extension represents zero precisely when it splits, equivalently when q has a section or i has a retraction. In that case $E \cong B \oplus A$ compatibly with the two structure maps. See [10, Section 3.4] and [11, Lemma 7.29].

The projective-resolution calculation is particularly simple for cyclic groups.

Example A.44 (Extensions of a cyclic group) *Let $n \geq 1$ and let B be an abelian group. The projective resolution $0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$ gives, after applying $\operatorname{Hom}_{\mathbb{Z}(-,B)}$, $0 \to B \xrightarrow{\cdot n} B \to 0$. Therefore*

$$\operatorname{Ext}_{\mathbb{Z}}^0(\mathbb{Z}/n\mathbb{Z}, B) \cong \operatorname{Ker}(n : B \to B), \quad \operatorname{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/n\mathbb{Z}, B) \cong B/nB.$$

For $n = 2$ and $B = \mathbb{Z}/2\mathbb{Z}$, multiplication by 2 is zero, so

$$\operatorname{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}.$$

The zero class is represented by

$$0 \to \mathbb{Z}/2\mathbb{Z} \to (\mathbb{Z}/2\mathbb{Z}) \oplus (\mathbb{Z}/2\mathbb{Z}) \to \mathbb{Z}/2\mathbb{Z} \to 0,$$

whereas the nonzero class is represented by the extension of Example A.8:

$$0 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0.$$

The preceding interpretation also recovers the projective splitting criterion of Proposition A.30. An R-module A is projective precisely when every extension

$0 \to B \to E \to A \to 0$ splits, for every R-module B. I.e, A is projective iff $\mathrm{Ext}^1_R(A, B) = 0$ for every B.

B Derived Categories

B.1 Motivation

Appendix A culminated in the definition of derived functors in their classical form. However, the output is a graded family of cohomology objects. The complex from which they were obtained, and the way in which the different degrees are organized within it, are not retained as part of the output object. The connecting morphisms in long exact sequences require a separate construction, and there is no single object that represents the action of the functor in every degree and on which further derived constructions can be performed.

To understand what information is retained in a complex, we begin with a morphism between complexes and attach another complex to it. The cohomology of this complex records the kernels and cokernels of the maps induced on cohomology and measures their failure to be isomorphisms. To turn cohomology-preserving morphisms into isomorphisms, we proceed in two stages. First, we seek a category in which homotopy-equivalent resolutions become isomorphic, so that we can retain the complex obtained from a resolution. However, the map from an object to its resolution is generally a quasi-isomorphism and is not necessarily a homotopy equivalence. For the resolution to represent the object itself, we therefore pass to a category in which quasi-isomorphisms become invertible. This passage will require an explicit description of the morphisms in the new category and a way to identify the morphisms that we seek to invert.

The resulting category allows us to replace the family $R^nF(A)_n$ by a single object $RF(A)$. Passing to a single object effectively gives us a new derived

functor, RF, that is defined without taking cohomology; taking its cohomology recovers the classical derived functors. Under suitable conditions, further derived constructions can be applied to these objects and functors can be composed. This category is not abelian, but one can isolate within it objects that form an abelian subcategory equivalent to the original category.

Throughout, $\mathcal{A}$ is an abelian category and $\mathrm{Ch}(\mathcal{A})$ denotes its category of cohomological complexes. Thus $d_A^n : A^n \to A^{n+1}$ and

$$H^n(A^\bullet) = \mathrm{Ker}(d_A^n)/\,\mathrm{im}\big(d_A^{n-1}\big).$$

Morphisms of complexes, cochain homotopies, acyclic complexes, and quasi-isomorphisms were defined in Section A.5.1. We write $\mathrm{Ch}^+(\mathcal{A})$, $\mathrm{Ch}^-(\mathcal{A})$, and $\mathrm{Ch}^{b(\mathcal{A})}$ for complexes that vanish in all sufficiently negative degrees, all sufficiently positive degrees, and all but finitely many degrees, respectively. We use the analogous notation for homotopy and derived categories. No boundedness hypothesis is imposed on finite direct sums, shifts, or cones; assumptions needed for resolutions or totalizations will be stated where they are used.

Finally, an object $X \in \mathcal{A}$ is often to be regarded as a complex with X in a single degree n and zero elsewhere. We write $X_{[n]}$ for that complex, the degree appearing as a bracketed subscript. This is only a shorthand for a shift of X placed in degree 0, but writing it as a shift forces the reader to negate an index in order to read off a position, and positions are what the truncation and t-structure arguments are about.

B.2 The Cone

To begin, let us eye the following morphism in **Ab** -

$$u : \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}, \quad u(x) = 2x$$

This morphism is neither a monomorphism nor an epimorphism. Its kernel, image, and cokernel are all isomorphic to $\mathbb{Z}/2\mathbb{Z}$, but this information is divided between two short exact sequences:

$$0 \to \mathrm{Ker}(u) \to \mathbb{Z}/4\mathbb{Z} \to \mathrm{im}(u) \to 0,$$

$$0 \to \mathrm{im}(u) \to \mathbb{Z}/4\mathbb{Z} \to \mathrm{coker}(u) \to 0.$$

The mapping cone will replace these two pieces of information by the cohomology of one complex attached directly to u.

An immediate observation is that although we can analyze every morphism in **Ab**, with two short exact sequences for that. Degreewise kernels and cokernels of a morphism of complexes are not invariant under homotopy: replacing f by a homotopic g can change $\operatorname{Ker}(f^n)$ and $\operatorname{coker}(f^n)$ altogether, so neither can descend to the homotopy category, and neither computes anything a derived construction is allowed to depend on. The cone, by contrast, will be shown in Proposition B.10 to depend on f only through its homotopy class.

The total complex (Definition A.20) construction of Appendix A already explains how several graded directions can be combined into one complex. For the cone the source complex must first be shifted by one degree.

Definition B.1 (Shift of a complex) *Let $A^\bullet \in \mathrm{Ch}(\mathcal{A})$ and $r \in \mathbb{Z}$. The shift $A[r]$ is the complex defined by*

$$A[r]^n = A^{n+r}, \quad d^n_{A[r]} = (-1)^r d^{n+r}_A.$$

For a morphism $f : A^\bullet \to B^\bullet$, define $f[r]^n = f^{n+r}$.

Proposition B.2 (Cohomology of a shift) *For every $A^\bullet \in \mathrm{Ch}(\mathcal{A})$ and every $n, r \in \mathbb{Z}$, there is a natural isomorphism*

$$H^n(A[r]) \cong H^{n+r}(A).$$

Proof. Multiplying a differential by $(-1)^r$ changes neither its kernel nor its image. □

In particular $-[1]$ moves an object concentrated in degree 0 to degree -1: the shift lowers the degree in which a complex sits by one, in the direction opposite to the one the notation may suggest.

Let $f : A^\bullet \to B^\bullet$ be a morphism of complexes. In degree n, the cone places B^n beside A^{n+1}. The morphism f^{n+1} is then inserted as an off-diagonal component of the new differential:

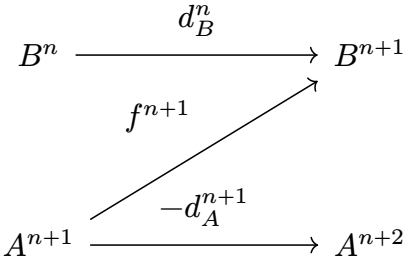


Figure B.1: The components of the cone differential from degree n to degree $n+1$.

Return to $f : \mathbb{Z}/4\mathbb{Z} \xrightarrow{2} \mathbb{Z}/4\mathbb{Z}$, with both groups regarded as complexes concentrated in degree 0. The shifted complex $\mathbb{Z}/4\mathbb{Z}[1]$ is concentrated in degree -1, so the graded object $\mathbb{Z}/4\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}[1]$ has one copy of $\mathbb{Z}/4\mathbb{Z}$ in degree -1 and another in degree 0.

Definition B.3 (Mapping cone) *The mapping cone of $f : A^\bullet \to B^\bullet$ is the complex* $\mathrm{Cone}(f)$ *defined by*

$$\mathrm{Cone}(f)^n := B^n \oplus A^{n+1},$$

with differential

$$d^n_{\mathrm{Cone}(f)} := \begin{pmatrix} d^n_B & f^{n+1} \\ 0 & -d^{n+1}_A \end{pmatrix}.$$

Equivalently,

$$d^n_{\mathrm{Cone}(f)}(b, a) = \big(d^n_B(b) + f^{n+1}(a), -d^{n+1}_A(a)\big).$$

The canonical maps are

$$i_f : B^\bullet \to \mathrm{Cone}(f), \quad i_f(b) = (b, 0),$$

$$p_f : \mathrm{Cone}(f) \to A[1], \quad p_f(b, a) = a.$$

The matrix formula displays why the sign and the off-diagonal entry are necessary.

$$d^{n+1}_{\mathrm{Cone}(f)} \circ d^n_{\mathrm{Cone}(f)} = \begin{pmatrix} d^2_B & d_B f - f d_A \\ 0 & d^2_A \end{pmatrix} = 0.$$

Thus the condition that f commute with the differentials is what makes the cone differential square to zero.

For the $f : \mathbb{Z}/4\mathbb{Z} \xrightarrow{2} \mathbb{Z}/4\mathbb{Z}$, the source and target have zero differentials, so the cone is the two-term complex

$$\cdots \to 0 \to \mathbb{Z}/4\mathbb{Z} \xrightarrow{2} \mathbb{Z}/4\mathbb{Z} \to 0 \to \cdots,$$

concentrated in degrees -1 and 0. The original morphism has become the differential of one complex.

B.3 The cone triangle

In order to establish a full replacement for short exact sequences, we need to show that we could use those cones in order to reproduce long exact sequences.

Note that the cone diagram

$$A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{i_f} \underbrace{\mathrm{Cone}(f)}_{B^\bullet \oplus A[1]} \xrightarrow{p_f} A[1]$$

form a degreewise-split short exact sequence of complexes:

$$0 \to B^\bullet \xrightarrow{i_f} \mathrm{Cone}(f) \xrightarrow{p_f} A[1] \to 0.$$

Applying the long exact cohomology sequence (Theorem A.27) to our short exact sequence, and using the identification $H^n(A[1]) \cong H^{n+1}(A)$, gives one sequence containing all cohomological defects of f.

Theorem B.4 (Long exact cohomology sequence of a cone) *For every morphism $f : A^\bullet \to B^\bullet$, there is a natural long exact sequence*

$$\cdots \to H^n(A) \xrightarrow{H^n(f)} H^n(B) \to H^n(\mathrm{Cone}(f)) \to H^{n+1}(A) \xrightarrow{H^{n+1}(f)} H^{n+1}(B) \to \cdots.$$

See [2, Lemma A.9], [5, Prop. 1.1.21], and [1, Prop. I.3.6].

The final map $p_f : \mathrm{Cone}(f) \to A[1]$ explains the shift in the fourth vertex of the later triangle. Since its target is shifted, the induced cohomology maps raise degree by one. The connecting morphisms are therefore represented by one degree-zero morphism of complexes.

For multiplication by 2, the relevant part of the long exact sequence is

$$0 \to H^{-1}(\mathrm{Cone}(2)) \to \mathbb{Z}/4\mathbb{Z} \xrightarrow{2} \mathbb{Z}/4\mathbb{Z} \to H^0(\mathrm{Cone}(2)) \to 0.$$

Hence

$$H^{-1}(\mathrm{Cone}(2)) \cong \mathrm{Ker}(2) \cong \mathbb{Z}/2\mathbb{Z},$$

$$H^0(\mathrm{Cone}(2)) \cong \mathrm{coker}(2) \cong \mathbb{Z}/2\mathbb{Z}.$$

The kernel and cokernel have not disappeared; the cone has reorganized them as adjacent cohomology objects of one complex.

The long exact sequence yields the following statement.

Proposition B.5 (Cohomology of the cone) *For every morphism $f : A^\bullet \to B^\bullet$ and every $n \in \mathbb{Z}$, there is a natural short exact sequence*

$$0 \to \operatorname{coker}(H^n(f)) \to H^n(\operatorname{Cone}(f)) \to \operatorname{Ker}\big(H^{n+1}(f)\big) \to 0.$$

The proposition converts the two failures of $H^n(f)$ into the cohomology of one object. In particular:

Corollary B.6 (Quasi-isomorphisms and acyclic cones) *A morphism $f : A^\bullet \to B^\bullet$ is a quasi-isomorphism if and only if* $\operatorname{Cone}(f)$ *is acyclic.*

Proof. If every $H^n(f)$ is an isomorphism, both outer terms in Proposition B.5 vanish. Conversely, if the cone is acyclic, the same short exact sequence shows that every $H^n(f)$ is both monic and epic. □

A short exact sequence of complexes also fits naturally into the cone construction. If

$$0 \to A^\bullet \xrightarrow{i} B^\bullet \xrightarrow{q} C^\bullet \to 0$$

is exact, then

$$\varphi : \operatorname{Cone}(i) \to C^\bullet, \quad \varphi(b, a) = q(b)$$

is a morphism of complexes and a quasi-isomorphism. Thus $\operatorname{Cone}(i)$ is a canonical complex attached to the injection, while $C^\bullet$ is a quasi-isomorphic replacement for it.Thus the cone triangle $A^\bullet \to B^\bullet \to \operatorname{Cone}(i) \to A^{\bullet[1]}$ has a third term quasi-isomorphic to $C^\bullet$.

B.4 Homotopy Category

The cone construction answers the first guiding question. We now address the non-uniqueness of resolutions. Given a morphism $u : A \to B$ and injective resolutions of A and B, the Comparison Theorem produces a morphism between the resolutions extending u, but that lift is unique only up to cochain homotopy. Literal chain maps therefore distinguish choices that derived constructions must regard as equivalent.

Recall that $f, g : A^\bullet \to B^\bullet$ are cochain-homotopic if there is a family $h^n : A^n \to B^{n-1}$ such that

$$f - g = d_B \circ h + h \circ d_A$$

We write $f \sim g$. A morphism homotopic to zero is called **null-homotopic**.

Homotopies add, so the relation $\sim$ is compatible with addition. It is also compatible with composition: if $f - g = d_B h + h d_A$, then for cochain maps $u : A' \to A$ and $v : B \to B'$ we have $vfu - vgu = d_{B'}(vhu) + (vhu)d_{A'}$. Composition therefore descends to homotopy classes.

Definition B.7 (Homotopy category) *The homotopy category* $\mathrm{K}(\mathcal{A})$ *has the same objects as* $\mathrm{Ch}(\mathcal{A})$*. Its morphisms are homotopy classes of morphisms of complexes*

$$\mathrm{Hom}_{\mathrm{K}(\mathcal{A})}(A^\bullet, B^\bullet) \coloneqq \mathrm{Hom}_{\mathrm{Ch}(\mathcal{A})}(A^\bullet, B^\bullet)/\sim$$

The class of f *is denoted* $[f]$*, as usual.*

The passage $\mathrm{Ch}(\mathcal{A}) \to \mathrm{K}(\mathcal{A})$ changes only the morphisms. Keeps the complexes unchanged and replaces cochain maps by their homotopy classes. Addition and shifts descend to the quotient, as does cohomology, since homotopic maps induce the same maps on cohomology by Lemma A.17 and its cochain version in Section A.5.1.

Proposition B.8 (Homotopy equivalences and contractible complexes) *A morphism of complexes is a homotopy equivalence if and only if its homotopy class is an isomorphism in* $\mathrm{K}(\mathcal{A})$*. Consequently, a complex* $C^\bullet$ *is contractible if and only if it is isomorphic to the zero object in* $\mathrm{K}(\mathcal{A})$*.*

Proof. By Definition B.7, $[f]$ is invertible when there is g with $gf \sim \mathrm{id}$ and $fg \sim \mathrm{id}$, which is the definition of a homotopy equivalence. Applying this to $M^\bullet \to 0$: the composites are $0 \sim \mathrm{id}_{M^\bullet}$ and id_0, and the first says precisely that $\mathrm{id}_{M^\bullet}$ is null-homotopic, that is, that $M^\bullet$ is contractible. □

For example, the two-term complex $\cdots \to 0 \to M \xrightarrow{\mathrm{id}} M \to 0 \to \cdots$ is generally not the zero complex in $\mathrm{Ch}(\mathcal{A})$, but it is contractible and hence isomorphic to zero in $\mathrm{K}(\mathcal{A})$. The homotopy category therefore cleanly removes parts of a complex that can be contracted without changing its cohomology.

More fundamentally, the homotopy category removes the non-uniqueness that appeared in the construction of resolutions.

Corollary B.9 (Uniqueness of injective resolutions in the homotopy category) *Let* $A \to I^\bullet$, $A \to J^\bullet$ *be injective resolutions. There is a unique isomorphism*

$$I^\bullet \to J^\bullet$$

in $\mathrm{K}^+(\mathcal{A})$ *represented by a comparison morphism extending* id_A*.*

Proof. The Comparison Theorem gives maps $I^\bullet \to J^\bullet$ and $J^\bullet \to I^\bullet$ extending id_A. Their composites are homotopic to the relevant identity maps, and any two comparison morphisms extending id_A are homotopic. □

The cone was defined for a morphism of complexes, but $\mathrm{K}(\mathcal{A})$ has only homotopy classes. To formulate cone diagrams intrinsically in $\mathrm{K}(\mathcal{A})$, we must verify that the cone does not depend on the chosen chain-level representative.

Proposition B.10 (Homotopic morphisms have isomorphic cones) *If $f, g : A^\bullet \to B^\bullet$ are homotopic, then* $\mathrm{Cone}(f)$ *and* $\mathrm{Cone}(g)$ *are isomorphic as complexes.*

Compare [5, Lemma 1.1.20(iii)].

Proof. Choose h with $f - g = d_B h + h d_A$. The block matrix

$$\Phi_h^n := \begin{pmatrix} \mathrm{id}_{B^n} & h^{n+1} \\ 0 & \mathrm{id}_{A^{n+1}} \end{pmatrix}$$

defines a chain isomorphism $\mathrm{Cone}(f) \to \mathrm{Cone}(g)$, with inverse Φ_{-h}. □

The resulting isomorphism depends on the chosen homotopy. Consequently, the cone of a morphism in $\mathrm{K}(\mathcal{A})$ is defined only up to noncanonical isomorphism.

For a strictly commutative square of chain maps

$$b \circ f = f' \circ a$$

in $\mathrm{Ch}(\mathcal{A})$, the block map

$$\begin{pmatrix} b & 0 \\ 0 & a \end{pmatrix} : \mathrm{Cone}(f) \to \mathrm{Cone}(f')$$

is canonical. A square that commutes in $\mathrm{K}(\mathcal{A})$ supplies only a homotopy between bf and $f'a$. Choosing such a homotopy gives the required off-diagonal entry in a map between the cones, and different choices can give different maps.

Consequently, a morphism in $\mathrm{K}(\mathcal{A})$ has a cone determined up to isomorphism, while a functorial choice of cone would require additional data. see [5, Remark 1.2.7] and [1, Lemma I.4.2].

The homotopy category solves the correct uniqueness problem, but it does not yet invert every quasi-isomorphism.

Lemma B.11 (Split exact sequences and contractible complexes) *Let*

$$0 \to A \xrightarrow{i} B \xrightarrow{p} C \to 0$$

be a short exact sequence in $\mathcal{A}$, and let

$$E^\bullet := \left[A \overset{i}{\to} B \overset{p}{\to} C\right]$$

be the associated complex, concentrated in degrees $-1, 0, 1$*. Then* $E^\bullet$ *is contractible if and only if the short exact sequence splits.*

This is the splitting lemma in complex form; see [10, §1.4] and [15].

Proof. If $E^\bullet$ is contractible, a contracting homotopy gives maps $r : B \to A$ and $s : C \to B$ with $ri = \mathrm{id}_A$ and $ps = \mathrm{id}_C$, so the sequence splits. Conversely, a splitting gives maps satisfying $ir + sp = \mathrm{id}_B$; these maps form a contracting homotopy of $E^\bullet$. □

Example B.12 (An acyclic complex that is not contractible) *Consider the extension of Example A.8*

$$0 \to \mathbb{Z}/2\mathbb{Z} \overset{\iota}{\to} \mathbb{Z}/4\mathbb{Z} \overset{\pi}{\to} \mathbb{Z}/2\mathbb{Z} \to 0$$

The associated complex

$$E^\bullet = \left[\mathbb{Z}/2\mathbb{Z} \overset{\iota}{\to} \mathbb{Z}/4\mathbb{Z} \overset{\pi}{\to} \mathbb{Z}/2\mathbb{Z}\right]$$

is acyclic because the sequence is exact. It is not contractible by Lemma B.11, since the short exact sequence does not split. Hence $E^\bullet$ *is nonzero in* $\mathrm{K}(\mathbf{Ab})$ *even though* $E^\bullet \to 0$ *is a quasi-isomorphism. Compare [2, Appendix A.18], where the same two maps appear as an injective resolution over* $\mathbb{Z}/4\mathbb{Z}$*.*

This is the exact gap between $\mathrm{K}(\mathcal{A})$ and $D(\mathcal{A})$. The homotopy category kills contractible complexes but not all acyclic complexes.

We have achieved three things. First, passing from $\mathrm{Ch}(\mathcal{A})$ to $\mathrm{K}(\mathcal{A})$ makes morphisms between resolutions unique whenever they were previously unique up to homotopy. Second, homotopy equivalences become isomorphisms. Third, the cone of a morphism in $\mathrm{K}(\mathcal{A})$ is well defined (up to noncanonical isomorphism).

This is enough to formulate cone triangles intrinsically in the homotopy category, as we wished. It is not yet enough to identify quasi-isomorphic complexes: the acyclic noncontractible complex above is nonzero in $\mathrm{K}(\mathcal{A})$. Thus we still have two further steps. We first organize cone triangles into the triangulated structure of $\mathrm{K}(\mathcal{A})$; we then invert quasi-isomorphisms to obtain the derived category.

B.5 Triangulated Homotopy Category

Cone triangles in $\mathrm{K}(\mathcal{A})$ already carry the long exact cohomology sequence proved in Section B.3. Later we will use the same form of exactness in settings where an object is presented only inside a triangulated category. We therefore isolate the properties of cone triangles that can be formulated entirely in terms of objects, morphisms, shifts, and commutative diagrams.

We first specify the relevant diagrams and then verify four operations on them - Normalization, rotation, completion of a commutative square, and compatibility with composition. These become the axioms of a triangulated category.

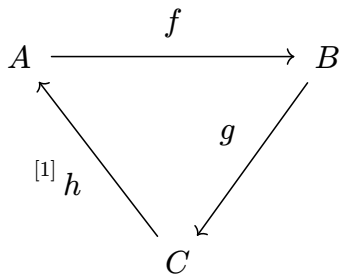


Figure B.2: Geometric notation for $A \to B \to C \to A[1]$. The label [1] indicates the shifted target of the returning morphism.

Definition B.13 (Triangle and morphism of triangles) *A **triangle** in* $\mathrm{K}(\mathcal{A})$ *is a diagram*

$$A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet \xrightarrow{h} A^\bullet[1].$$

*A **morphism** from this triangle to*

$$A'^\bullet \xrightarrow{f'} B'^\bullet \xrightarrow{g'} C'^\bullet \xrightarrow{h'} A'^\bullet[1]$$

is a triple (a, b, c) *such that*

$$bf = f'a, \quad cg = g'b, \quad a[1]h = h'c$$

in $\mathrm{K}(\mathcal{A})$.

Definition B.14 (Cone triangle and distinguished triangle) *The **cone triangle** of* $f : A^\bullet \to B^\bullet$ *is*

$$A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{i_f} \mathrm{Cone}(f) \xrightarrow{p_f} A^\bullet[1].$$

A triangle in $\mathrm{K}(\mathcal{A})$ *is **distinguished** if it is isomorphic, as a triangle, to a cone triangle.*

The closure under isomorphism is essential. The cone of a homotopy class is determined only up to noncanonical isomorphism, as we saw. Distinguishedness must therefore depend on the diagram in $\mathrm{K}(\mathcal{A})$, not on a particular chain-level presentation.

Proposition B.15 (Cohomology of a distinguished triangle) *Every distinguished triangle*

$$A^\bullet \to B^\bullet \to C^\bullet \to A^\bullet[1]$$

induces a natural long exact sequence

$$\cdots \to H^n(A) \to H^n(B) \to H^n(C) \to H^{n+1}(A) \to \cdots.$$

Proof. The result was proved for cone triangles in Theorem B.4. Cohomology descends to $\mathrm{K}(\mathcal{A})$, and isomorphic triangles induce isomorphic long exact sequences. □

In particular, consecutive morphisms in a distinguished triangle compose to zero. The converse fails, and it fails already in the smallest case.

Example B.16 (Vanishing composites do not suffice) *In* $\mathrm{K}(\mathbf{Ab})$ *take*

$$\mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z}[1],$$

all three objects concentrated in degree 0 *and all three morphisms zero. Consecutive composites vanish. If the triangle were distinguished, Proposition B.15 would force*

$$0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z} \to 0$$

to be exact, which already fails at the first term. Vanishing consecutive composites are therefore necessary for distinguishedness. The additional condition is that applying cohomology produce the exact sequence of Theorem B.4 in every degree.

B.5.1 Identity and normalization

Proposition B.17 (Normalization) *The cone of* id_A *is contractible, while*

$$\mathrm{Cone}(0 : A \to B) = B \oplus A[1].$$

Hence the identity triangle is distinguished, and every morphism begins a distinguished triangle.

Proof. Both statements follow directly from the cone differential. □

B.5.2 Rotation

The left rotation of

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} A[1]$$

is

$$B \xrightarrow{g} C \xrightarrow{h} A[1] \xrightarrow{-f[1]} B[1].$$

Proposition B.18 (Rotation preserves distinguished triangles) *A triangle is distinguished if and only if its left rotation is distinguished.*

Proof sketch. For a cone triangle, $\mathrm{Cone}(i_f)$ is homotopy equivalent to $A[1]$ after removing a contractible summand. Under this equivalence, its cone triangle becomes the left rotation. See [1, Lemma I.4.2]. □

The long exact cohomology sequence of the rotated triangle is the same sequence with its starting point and indices shifted.

B.5.3 Completion of a square

Suppose $a : A \to A'$ and $b : B \to B'$ satisfy $bf = f'a$ in $\mathrm{K}(\mathcal{A})$. Choosing a homotopy

$$bf - f'a = d_{B'}s + sd_A$$

defines a map

$$c^n = \begin{pmatrix} b^n & s^{n+1} \\ 0 & a^{n+1} \end{pmatrix} : \mathrm{Cone}(f)^n \to \mathrm{Cone}(f')^n.$$

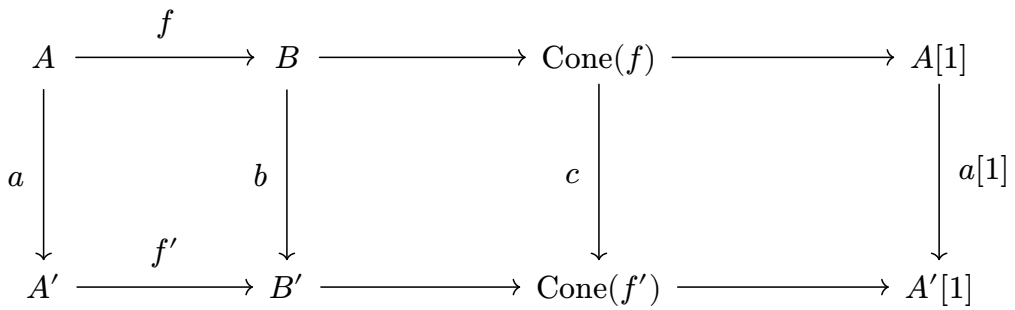


Figure B.3: A commuting square in $\mathrm{K}(\mathcal{A})$ completed to a morphism of cone triangles.

Proposition B.19 (Completion to a morphism of triangles) *Every commutative square between the first morphisms of two distinguished triangles extends to a morphism of triangles.*

Proof sketch. Replace both triangles by cone triangles. The chosen homotopy s gives the displayed map c, which completes the diagram. □

The completion need not be unique: different choices of s may produce different maps c. For conditions implying uniqueness, see [7, 1.1.9–1.1.10].

B.5.4 Composition and the octahedron

Let $A \xrightarrow{f} B \xrightarrow{g} C$ be composable. Define

$$\alpha(b, a) = (g(b), a), \quad \beta(c, a) = (c, f(a)), \quad \gamma(c, b) = (b, 0).$$

These maps compare the cones of f, g, and gf.

Proposition B.20 (The cone of a composition) *The sequence*

$$\mathrm{Cone}(f) \xrightarrow{\alpha} \mathrm{Cone}(gf) \xrightarrow{\beta} \mathrm{Cone}(g) \xrightarrow{\gamma} \mathrm{Cone}(f)[1]$$

is a distinguished triangle in $\mathrm{K}(\mathcal{A})$.

Proof sketch. The cone of α splits, up to homotopy, as $\mathrm{Cone}(g)$ together with a contractible summand. Under this equivalence, the remaining maps become β and γ; see [1, Proposition I.4.4] and [2, Appendix A.10–A.11]. □

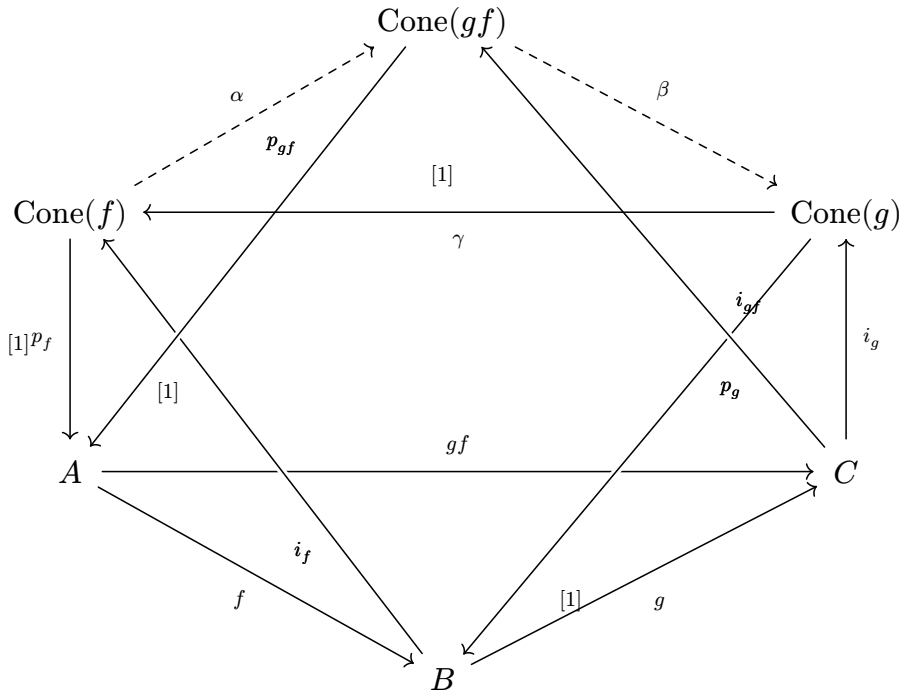


Figure B.4: The octahedral diagram associated with the composable morphisms $A \xrightarrow{f} B \xrightarrow{g} C$. The lower face records f, g, and gf, while the upper face is the distinguished triangle $\mathrm{Cone}(f) \xrightarrow{\alpha} \mathrm{Cone}(gf) \xrightarrow{\beta} \mathrm{Cone}(g) \xrightarrow{\gamma} \mathrm{Cone}(f)[1]$. A label [1] indicates that the endpoint represented at that vertex is shifted.

The distinguished triangle of Proposition B.20 implies that if two of $\mathrm{Cone}(f)$, $\mathrm{Cone}(g)$, and $\mathrm{Cone}(gf)$ are acyclic, then so is the third. By Corollary B.6, if any two of f, g, and gf are quasi-isomorphisms, then the remaining morphism is also a quasi-isomorphism.

Definition B.21 (Triangulated category) *A triangulated category consists of an additive category $\mathcal{T}$, an additive autoequivalence[3] $-[1]$ with quasi-inverse $-[-1]$, and a class of distinguished triangles satisfying:*

- ***(TR1)*** *Distinguished triangles are closed under isomorphism; $A \xrightarrow{\mathrm{id}_A} A \to 0 \to A[1]$ is distinguished; and every morphism begins a distinguished triangle.*
- ***(TR2)*** *A triangle is distinguished if and only if its left rotation is distinguished.*
- ***(TR3)*** *Every commutative square between the first morphisms of distinguished triangles extends to a morphism of triangles.*
- ***(TR4)*** *For composable morphisms $A \xrightarrow{f} B \xrightarrow{g} C$, the distinguished triangles[4] of f, g, and $g \circ f$ can be completed to an octahedral diagram[5].*

These are Verdier's axioms. See [1, Definition I.5.1], [8, Definition B.3.6] and [7, 1.1.6–1.1.7].

Theorem B.22 (The homotopy category is triangulated) *With the shift and distinguished triangles defined above, $\mathrm{K}(\mathcal{A})$ is a triangulated category. The bounded subcategories $\mathrm{K}^+(\mathcal{A})$, $\mathrm{K}^-(\mathcal{A})$, and $\mathrm{K}^{b(\mathcal{A})}$ are triangulated subcategories.*

Proof sketch. Normalization gives TR1, rotation gives TR2, completion of squares gives TR3, and the octahedral construction gives TR4. Shifts and mapping cones preserve bounded-below, bounded-above, and bounded complexes. The same constructions therefore restrict to each of the three bounded variants. □

For every cohomological functor, a distinguished triangle induces a long exact sequence. Rotation reindexes that sequence, TR3 makes it natural with respect to commutative squares, and TR4 relates the sequences associated with a composition. A third vertex exists for every morphism, although its choice and the completion of a square are not canonical.

---

[3] An autoequivalence of a category is an equivalence from the category to itself.

[4] $A \xrightarrow{f} B \xrightarrow{i_f} C_f \xrightarrow{p_f} A[1]$, $B \xrightarrow{g} C \xrightarrow{i_g} C_g \xrightarrow{p_g} B[1]$, $A \xrightarrow{gf} C \xrightarrow{i_{gf}} C_{gf} \xrightarrow{p_{gf}} A[1]$.

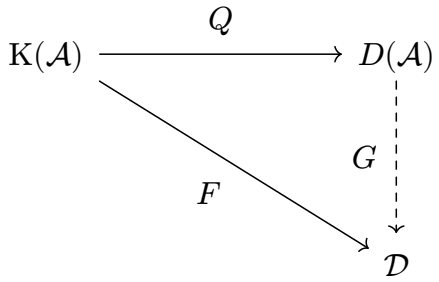
[5] I.e, there exist morphisms $u : C_f \to C_{gf}$ and $v : C_{gf} \to C_g$ such that (id_A, g, u) and (f, id_C, v) are morphisms of triangles, and $C_f \xrightarrow{u} C_{gf} \xrightarrow{v} C_g \xrightarrow{i_f[1] \circ p_g} C_f[1]$ is distinguished

The acyclic noncontractible complex of Example B.12 remains nonzero in $\mathrm{K}(\mathcal{A})$. Localization is what we need to make it vanish.

B.6 The derived category

We now invert the quasi-isomorphisms. The operation is the categorical analogue of inverting a set of elements in a ring - one adds formal inverses and asks that nothing else be identified. This operation called "localization", and it is a universal property in the category of categories.

Theorem B.23 (Existence of the localization) *Let $\mathcal{C}$ be a category and S a class of morphisms in $\mathcal{C}$. There is a category $\mathcal{C}[S^{-1}]$, possibly after passing to a larger universe, and a functor $Q : \mathcal{C} \to \mathcal{C}[S^{-1}]$ with the following universal property: $Q(s)$ is an isomorphism for every $s \in S$, and every functor $F : \mathcal{C} \to \mathcal{D}$ such that $F(s)$ is an isomorphism for all $s \in S$ factors as $F \cong G \circ Q$ for a functor G that is unique up to natural isomorphism. The pair $(\mathcal{C}[S^{-1}], Q)$ is determined by this property up to equivalence.*



See [1, §I.6] and [8, Def. B.4.1].

We use Theorem B.23 as stated and do not construct $\mathcal{C}[S^{-1}]$. The construction is delicate — the naive one produces morphism classes that need not be sets — and for the class of quasi-isomorphisms in $\mathrm{K}(\mathcal{A})$ the difficulty is resolved by a calculus of fractions, which is what makes the roof description of the next subsection valid. [2, Appendix A.23] gives the roof description directly, including the composition rule; [5, Def. 1.3.1] states the universal property and refers to Verdier; the construction itself is carried out in [1, §I.6–I.7]. None of the 'calculus of fractions' machinery is needed below, and we shall not introduce it.

Definition B.24 (Derived category) *Let S be the class of quasi-isomorphisms in $\mathrm{K}(\mathcal{A})$. The **derived category** of $\mathcal{A}$ is*

$$D(\mathcal{A}) := \mathrm{K}(\mathcal{A})[S^{-1}].$$

Its bounded-below, bounded-above, and bounded versions are denoted $D^+(\mathcal{A})$, $D^-(\mathcal{A})$, *and* $D^b(\mathcal{A})$.

The objects remain complexes, since the localization changes only the morphisms. Every quasi-isomorphism becomes invertible, so a resolution becomes isomorphic to the object it resolves. The graded family $(H^n(K))_n$ does not generally determine K in $D(\mathcal{A})$, since the morphisms joining its adjacent cohomological pieces also affect its isomorphism class.

By Corollary B.6, a morphism belongs to S precisely when its cone is acyclic. Thus the morphisms to be inverted are detected entirely by the triangulated structure already constructed.

However, a derived morphism may exist only after replacing its source by a quasi-isomorphic complex.

Definition B.25 (Roof) *A* ***roof*** *from* $A^\bullet$ *to* $B^\bullet$ *is a diagram in* $\mathrm{K}(\mathcal{A})$

$$\begin{array}{ccccc} s\ (\text{qis}) & & X^\bullet & & f \\ & \swarrow & & \searrow & \\ A^\bullet & & & & B^\bullet \end{array}$$

where s *is a quasi-isomorphism.*

It represents the formal composite $Q(f) \circ Q(s)^{-1} : A^\bullet \to B^\bullet$ in $D(\mathcal{A})$.

The intermediate complex $X^\bullet$ is a replacement of $A^\bullet$ by a quasi-isomorphic model from which an actual morphism to $B^\bullet$ is available.

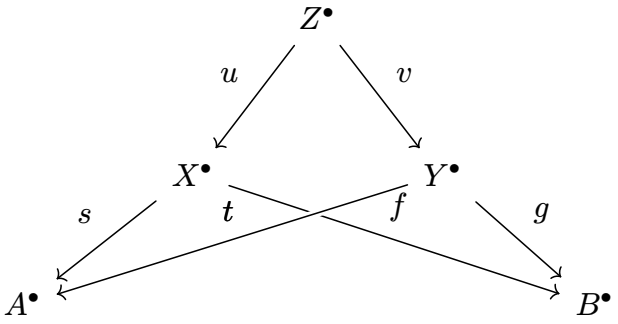
Proposition B.26 (Morphisms represented by roofs) *Every morphism* $\alpha : A^\bullet \to B^\bullet$ *in* $D(\mathcal{A})$ *can be represented by a roof*

$$A^\bullet \xleftarrow{s} X^\bullet \xrightarrow{f} B^\bullet$$

with s *a quasi-isomorphism.*

This is the fraction description of the localization at quasi-isomorphisms; see [2, Appendix A.23] and [1, §I.6–I.7]. A morphism already represented by a map $A^\bullet \to B^\bullet$ corresponds to the special roof with $X^\bullet = A^\bullet$ and $s = \mathrm{id}_A$.

Different roofs may represent the same morphism. Their equality is determined by passing to a common refinement.



Definition B.27 (Equivalent roofs) *Two roofs $A^\bullet \xleftarrow{s} X^\bullet \xrightarrow{f} B^\bullet$ and $A^\bullet \xleftarrow{t} Y^\bullet \xrightarrow{g} B^\bullet$ are **equivalent** if there exist a complex $Z^\bullet$ and morphisms $u : Z^\bullet \to X^\bullet$, $v : Z^\bullet \to Y^\bullet$ such that*

$$s \circ u = t \circ v, \quad f \circ u = g \circ v,$$

and the common morphism $s \circ u = t \circ v$ is a quasi-isomorphism.

The requirement that $s \circ u = t \circ v$ be a quasi-isomorphism excludes the trivial refinement $Z^\bullet = 0$ which would identify every morphism with zero. The calculus of fractions for quasi-isomorphisms implies that the resulting relation is reflexive, symmetric, and transitive. Its equivalence classes are the morphisms of $D(\mathcal{A})$.

Composition requires the same calculus of fractions. Given roofs $A^\bullet \xleftarrow{s} X^\bullet \xrightarrow{f} B^\bullet$ and $B^\bullet \xleftarrow{t} Y^\bullet \xrightarrow{g} C^\bullet$, the arrows f and t cannot be concatenated. They must first be replaced by a commutative roof over their common target.

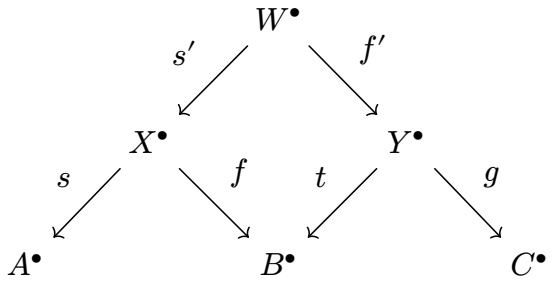


Figure B.7: Composition of roofs. The upper roof $(W^\bullet, s', f')$ satisfies $fs' = tf'$ with s' a quasi-isomorphism; the composite is then $(W^\bullet, ss', gf')$.

Producing that upper roof for every pair is precisely the content of the calculus of fractions referred to after Theorem B.23; it is [1, Def. I.6.1, conditions (S 1)–(S 4)]. We take it for granted. The injective model (Section B.7) will remove the need for it entirely, by replacing roofs with ordinary homotopy classes of chain maps.

Before drawing any consequence we record that the exactness formalism of Section B.5 survives the passage.

Definition B.28 (Distinguished triangles in the derived category) *A triangle in $D(\mathcal{A})$ is distinguished if it is isomorphic to the image of a distinguished triangle in $\mathrm{K}(\mathcal{A})$ under the localization functor Q.*

Theorem B.29 (The derived category is triangulated) *The shift and distinguished triangles above make $D(\mathcal{A})$ a triangulated category, and the localization functor $Q : \mathrm{K}(\mathcal{A}) \to D(\mathcal{A})$ is triangulated.*

The triangulated structure is obtained by localizing $\mathrm{K}(\mathcal{A})$ along the triangulated subcategory of acyclic complexes; this is the Verdier localization described in Theorem B.32 below. See [1, §I.7] for the construction.

Cohomology has a simpler consequence of the universal property. Each functor

$$H^n : \mathrm{K}(\mathcal{A}) \to \mathcal{A}$$

sends quasi-isomorphisms to isomorphisms, so it factors through $D(\mathcal{A})$. We continue to denote the induced functor by H^n. Consequently, every distinguished triangle in $D(\mathcal{A})$ gives the long exact cohomology sequence of Proposition B.15.

Proposition B.30 (Acyclic complexes vanish in the derived category) *For a complex $A^\bullet$, the following are equivalent:*

- *$A^\bullet$ is acyclic;*
- *$0 \to A^\bullet$ is a quasi-isomorphism;*
- *$A^\bullet \cong 0$ in $D(\mathcal{A})$.*

Proof. The first two are the same statement, since $H^n(0) = 0$. If $0 \to A^\bullet$ is a quasi-isomorphism it becomes invertible in $D(\mathcal{A})$, giving the third. Conversely, if $A^\bullet \cong 0$ in $D(\mathcal{A})$ then $H^n(A^\bullet) \cong H^n(0) = 0$ for all n, because H^n is a functor on $D(\mathcal{A})$. □

Thus the complex $E^\bullet$ of Example B.12, nonzero in $\mathrm{K}(\mathbf{Ab})$, becomes zero in $D(\mathbf{Ab})$. This was the gap of Section B.4 and now we closed it.

Proposition B.31 (Cohomology detects isomorphisms) *A morphism $f : A^\bullet \to B^\bullet$ in $D(\mathcal{A})$ is an isomorphism if and only if $H^n(f)$ is an isomorphism for every $n \in \mathbb{Z}$.*

Proof. By Theorem B.29 we may complete f to a distinguished triangle $A^\bullet \to B^\bullet \to C^\bullet \to A^\bullet[1]$, and the associated long exact sequence shows that every $H^n(f)$ is an isomorphism precisely when $C^\bullet$ is acyclic, hence precisely when $C^\bullet \cong 0$ by Proposition B.30. It remains to note that a morphism in a distinguished triangle is an isomorphism when the third vertex vanishes; this follows

by applying $\mathrm{Hom}_{D(\mathcal{A})}(W, -)$ to the triangle and using the five lemma, as in [1, Cor. I.5.5]. □

Theorem B.32 (Two descriptions of the derived category) *Let* $\mathrm{K}_{\mathrm{ac}(\mathcal{A})} \subseteq \mathrm{K}(\mathcal{A})$ *be the full subcategory of acyclic complexes. It is a triangulated subcategory, and there is an equivalence of triangulated categories*

$$D(\mathcal{A}) \equiv \mathrm{K}(\mathcal{A}) / \mathrm{K}_{\mathrm{ac}(\mathcal{A})}$$

Proof. By Corollary B.6 a morphism of $\mathrm{K}(\mathcal{A})$ is a quasi-isomorphism if and only if its cone is acyclic. Inverting the quasi-isomorphisms and killing the acyclic complexes are therefore two descriptions of one operation: a morphism becomes invertible when its third vertex becomes zero. For the verification that $\mathrm{K}_{\mathrm{ac}(\mathcal{A})}$ is triangulated and that the quotient carries the stated structure, see [1, Def. I.7.1, with Props. I.6.7 and I.6.9], where the quotient description is taken as the definition. □

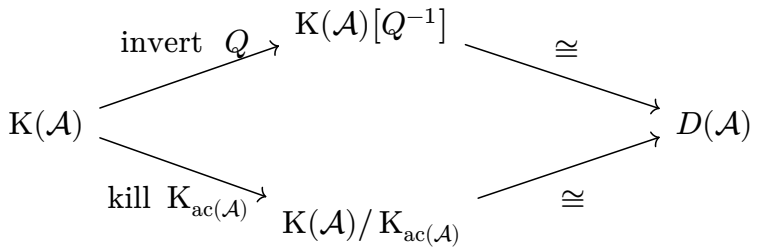


Figure B.8: Inverting the quasi-isomorphisms and killing the acyclic complexes give the same category, because a morphism is a quasi-isomorphism when its cone is acyclic.

The long exact cohomology sequence of a cone was first obtained from a short exact sequence of complexes, using the Snake Lemma in the abelian setting of Appendix A. It then descends to $\mathrm{K}(\mathcal{A})$, because homotopic maps induce the same maps on cohomology, and to $D(\mathcal{A})$, because quasi-isomorphisms induce cohomology isomorphisms. Thus the later categories retain these exact sequences without themselves being abelian. It remains to describe how their morphisms and derived functors can be computed using injective replacements.

B.7 Injective model

The localization $D(\mathcal{A}) = \mathrm{K}(\mathcal{A})[Q^{-1}]$ gives the correct category, but its intrinsic description is not convenient for calculations. A morphism in $D(\mathcal{A})$ is represented by a roof

$$K^\bullet \xleftarrow{s} X^\bullet \xrightarrow{a} L^\bullet,$$

with s a quasi-isomorphism. Roofs are necessary for the universal definition of localization, but comparing roofs and composing them requires common refinements. For bounded-below complexes, injective replacements remove this difficulty: after replacing the target by a bounded-below complex of injective objects, every derived morphism is represented by an ordinary chain map, uniquely up to homotopy.

Theorem B.33 (Injective replacement theorem) *Assume that $\mathcal{A}$ has enough injectives.*

- *Every bounded-below complex $K^\bullet \in \mathrm{Ch}^+(\mathcal{A})$ admits a quasi-isomorphism $K^\bullet \to I_K^\bullet$, where $I_K^\bullet$ is bounded below and every I_K^n is injective in $\mathcal{A}$.*
- *For every bounded-below complex $A^\bullet$ and every bounded-below complex of injectives $I^\bullet$, localization induces a natural isomorphism*

$$\mathrm{Hom}_{\mathrm{K}^+(\mathcal{A})}(A^\bullet, I^\bullet) \cong \mathrm{Hom}_{D^+(\mathcal{A})}(A^\bullet, I^\bullet).$$

- *Consequently, the localization functor induces an equivalence of triangulated categories*

$$\mathrm{K}^+(\mathrm{Inj}\,\mathcal{A}) \equiv D^+(\mathcal{A}),$$

where $\mathrm{K}^+(\mathrm{Inj}\,\mathcal{A})$ denotes the homotopy category of bounded-below complexes whose terms are injective.

We use the general injective replacement theorem without reproducing its full proof. The two-term construction below illustrates how the differential of a complex is lifted to resolutions and how the resulting data are assembled into one complex. For the general theorem, see [2, Appendix A.16–A.23], [5, Props. 1.3.9–1.3.10], and [1, Props. I.7.7 and I.7.10].

Remark (Where the bounded-below hypothesis comes from) *The role of the lower bound is visible for a two-term complex $K^\bullet = \left[A \xrightarrow{u} B\right]$ in degrees 0 and 1. Choose injective resolutions $A \to I^\bullet$ and $B \to J^\bullet$. By the dual Comparison Theorem (Theorem A.36) u extends to a cochain map $f^\bullet : I^\bullet \to J^\bullet$, and totalizing the resulting two-column double complex gives*

$$T^n = I^n \oplus J^{n-1}, \quad d_T^n = \begin{pmatrix} d_I^n & 0 \\ f^n & -d_J^{n-1} \end{pmatrix}$$

The resolution maps induce a cochain map $K^\bullet \to T^\bullet$. In the degreewise split short exact sequence $0 \to J^\bullet[-1] \to T^\bullet \to I^\bullet \to 0$, the connecting morphism $H^0(I^\bullet) \to H^0(J^\bullet)$ identifies with u. Hence $H^0(T^\bullet) \cong \mathrm{Ker}(u)$, $H^1(T^\bullet) \cong \mathrm{coker}(u)$, and all

other cohomology vanishes. Thus $K^\bullet \to T^\bullet$ is a quasi-isomorphism, and every T^n is injective.

For a general bounded-below complex, choose a degree m below which $K^\bullet$ vanishes. Starting in degree m, one constructs an injective replacement upward, at each stage embedding a suitable cokernel into an injective object. The lower bound gives this induction a starting point; for a complex unbounded below, this particular construction has none.

The lower bound enters a second time in the computation of derived morphisms. Every cochain map from an acyclic complex to a bounded-below complex of injectives is null-homotopic. The homotopy is constructed upward, again using injectivity at each stage. This is why morphisms into the replacement may be computed up to homotopy, as in Theorem B.33.

For an unbounded complex of injectives, this null-homotopy property need not hold: injectivity of the individual terms is insufficient. A complex satisfying it is called K-injective, and K-injective replacements provide the corresponding method in the unbounded setting when they exist. The bounded-below theorem suffices for every derived-functor and derived-Hom construction in this chapter.

The three categories of this chapter, and the shortcut between the two ends, may be displayed at once.

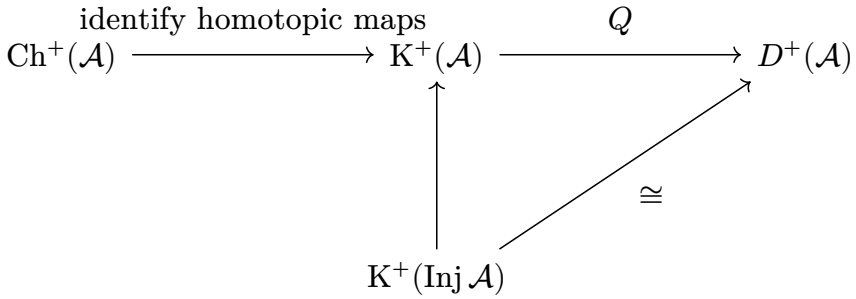


Figure B.9: The long route and the shortcut. Objects are unchanged along the top row; only the morphisms change. The injective model identifies the bottom vertex with the right-hand one, so a derived morphism may be computed as a homotopy class of chain maps.

To see how the theorem removes roofs, choose a replacement $j_L : L^\bullet \to I_L^\bullet$. A roof from $K^\bullet$ to $L^\bullet$ determines a unique homotopy class of maps $g : K^\bullet \to I_L^\bullet$ satisfying $gs \sim j_L a$:

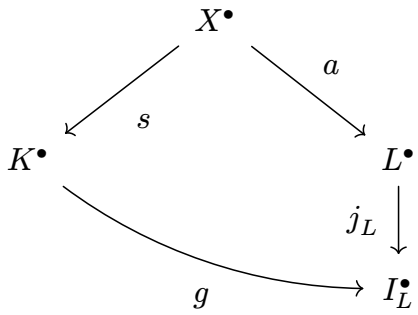


Figure B.10: A roof becomes an ordinary map after the target is replaced by a complex of injectives.

Thus, for $K, L \in D^+(\mathcal{A})$ and an injective replacement $L \to I_L^\bullet$,

$$\mathrm{Hom}_{D^+(\mathcal{A})}(K, L) \cong \mathrm{Hom}_{\mathrm{K}^+(\mathcal{A})}(K^\bullet, I_L^\bullet).$$

The left-hand side is intrinsic; the right-hand side is the form used in calculations.

B.8 Derived functors

In Appendix A, the classical right derived functors of a left-exact additive functor $F : \mathcal{A} \to \mathcal{B}$ were defined term-by-term $R^n F(A) \coloneqq H^n(F(I^\bullet))$, The derived category retains the entire complex $F(I^\bullet)$ rather than only its individual cohomology objects, and allows the input itself to be a complex.

The same distinction between an implementation and its interface applies here. An injective resolution is an implementation of the derived functor: one chooses $A \to I^\bullet$, applies F, and computes. The interface is the resulting object $RF(A)$, characterised up to canonical isomorphism and independent of the chosen resolution. Later arguments may use that interface without reopening the particular resolution that produced it, as the triangle axioms of Definition B.21 are used without reopening the cone construction behind them.

An additive functor acts termwise on complexes:

$$F(A^\bullet)^n \coloneqq F(A^n), \quad d^n_{F(A)} \coloneqq F(d^n_A).$$

The functor also acts termwise on cochain maps. Additivity carries a homotopy identity $f - g = dh + hd$ to $F(f) - F(g) = F(d)F(h) + F(h)F(d)$, so homotopic maps remain homotopic. It also preserves shifts and finite biproducts, and the cone differential gives a natural identification $F(\mathrm{Cone}(f)) \approx \mathrm{Cone}(F(f))$. Thus termwise application induces a triangulated functor $F : \mathrm{K}^+(\mathcal{A}) \to \mathrm{K}^+(\mathcal{B})$.

In general, however, it does not descend directly to the derived categories. A non-exact functor need not preserve quasi-isomorphisms, so applying F to an arbitrary representative of a derived object is not well defined. The injective model (Section B.7) resolves this problem: first replace the complex by an injective representative, and only then apply F.

Example B.34 (A functor that destroys a quasi-isomorphism) *Let $\mathcal{A} = \mathbf{Ab}$ and let $F = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, -)$, which is additive and left exact. Consider the complex $P^\bullet := \left[\mathbb{Z} \xrightarrow{2} \mathbb{Z}\right]$ in degrees -1 and 0. This is the projective resolution of $\mathbb{Z}/2\mathbb{Z}$ used in Example A.44, now regarded as an object of $\mathrm{Ch}(\mathbf{Ab})$. Since multiplication by 2 is injective with cokernel $\mathbb{Z}/2\mathbb{Z}$, the reduction map $P^\bullet \to \mathbb{Z}/2\mathbb{Z}_{[0]}$ is a quasi-isomorphism. Applying F termwise gives $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) = 0$ in both degrees, so $F(P^\bullet) = 0$, while $F\big(\mathbb{Z}/2\mathbb{Z}_{[0]}\big) = \mathbb{Z}/2\mathbb{Z}_{[0]}$. The induced map*

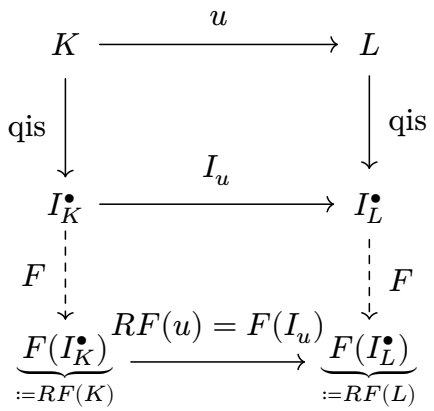
$$0 \to \mathbb{Z}/2\mathbb{Z}_{[0]}$$

is not a quasi-isomorphism. Two representatives of one object of $D^+(\mathbf{Ab})$ therefore have non-isomorphic images under F, and F cannot descend to $D^+(\mathbf{Ab})$ as it stands.

The injective replacement theorem solves this problem by restricting the calculation to representatives on which quasi-isomorphisms have already become homotopy equivalences.

Definition B.35 (Right derived functor) *Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor, and assume that $\mathcal{A}$ has enough injectives. For $K \in D^+(\mathcal{A})$, choose an injective replacement $K^\bullet \to I_K^\bullet$. The **right derived functor** of F is defined by $RF(K) := F(I_K^\bullet)$ as an object of $D^+(\mathcal{B})$.*

A morphism $u : K \to L$ in $D^+(\mathcal{A})$ need not be represented by a chain map between the initially chosen complexes. Under Theorem B.33, it is represented by a homotopy class of actual chain maps $I_u : I_K^\bullet \to I_L^\bullet$ between injective replacements. This middle horizontal arrow is the chain-level morphism to which F is applied.



Theorem B.36 (Basic properties of the right derived functor) *The object $RF(K)$ is independent of the chosen injective replacement up to a natural isomorphism. The construction on objects and on the chain maps I_u defines a functor*

$$RF : D^+(\mathcal{A}) \to D^+(\mathcal{B}),$$

and this functor is triangulated.

Proof. The equivalence $\mathrm{K}^+(\mathrm{Inj}\,\mathcal{A}) \equiv D^+(\mathcal{A})$ identifies different injective replacements in the homotopy category. Since an additive functor preserves homotopies, shifts, finite direct sums, and cone differentials, termwise application of F is triangulated on injective representatives. Transport across the equivalence proves all three assertions. □

The definition applies to an additive functor without assuming left exactness. Left exactness is needed when the derived object is compared with the classical right derived functors (Definition A.39).

Proposition B.37 (Recovery of the classical derived functors) *Let $F : \mathcal{A} \to \mathcal{B}$ be additive and left exact, and let $A \in \mathcal{A}$ be regarded as a complex concentrated in degree 0. Then, for every $n \geq 0$,*

$$H^n(RF(A)) \cong R^n F(A).$$

Moreover, $H^n(RF(A)) = 0$ for $n < 0$, and $H^0(RF(A)) \cong F(A)$.

Proof. Choose an injective resolution $0 \to A \to I^0 \to I^1 \to ...$, with $I^\bullet$ concentrated in nonnegative degrees. It is an injective replacement of $A[0]$, so $RF(A)$ is represented by $F(I^\bullet)$. By the classical definition, $H^{n(F(I^\bullet))} = R^n F(A)$ for $n \geq 0$, while the absence of negative terms gives $H^{n(RF(A))} = 0$ for $n < 0$. Finally, left exactness gives the exact sequence $0 \to F(A) \to F(I^0) \to F(I^1)$, so $H^0(F(I^\bullet)) = \ker(F(I^0) \to F(I^1)) \cong F(A)$. □

Derived functors compose under an additional acyclicity hypothesis. Let $\mathcal{B}$ have enough injectives, let $F : \mathcal{A} \to \mathcal{B}$, $G : \mathcal{B} \to \mathcal{C}$ be left-exact additive functors, and assume that $F(I)$ is G-acyclic for every injective object $I \in \mathcal{A}$. Then the canonical comparison morphism is an isomorphism $R(G \circ F) \cong RG \circ RF$. The individual derived functors are obtained by applying cohomology after this comparison. See [5, Props. 1.3.18–1.3.19].

Derived functors also preserve the triangle formalism. If

$$A \to B \to C \to A[1]$$

is a distinguished triangle, then so is

$$RF(A) \to RF(B) \to RF(C) \to RF(A)[1].$$

Applying cohomology produces the long exact sequence of the classical derived functors when A, B, C are objects of the original abelian category.

Later chapters will apply the same derived-functor construction to sheaf functors such as Γ, f_*, and $f_!$. Their derived versions will be introduced only after the underlying sheaf functors have been defined. The next section treats derived Hom directly in an abstract abelian category.

Remark (Boundedness) *The construction above naturally defines $RF : D^+(\mathcal{A}) \to D^+(\mathcal{B})$. It does not automatically restrict to $D^b(\mathcal{A}) \to D^b(\mathcal{B})$: the higher cohomology objects of $RF(K)$ may remain nonzero in arbitrarily high degrees. A bounded restriction requires a separate finite-cohomological-dimension or boundedness theorem for the particular functor. Such hypotheses will be stated when the sheaf operations are introduced; the relevant finiteness is the cohomological dimension of [5, §3.1].*

B.8.1 Derived Hom

The Hom complex was constructed in Appendix A (Definition A.22). Its differential compares a degree-shifted family of maps with the differentials of the source and target complexes. The cocycle condition is therefore the chain-map condition into a shift, while a coboundary gives a null-homotopic map. The following lemma states this interpretation in the homotopy category.

Lemma B.38 (The Hom complex computes homotopy classes) *For all $K^\bullet, L^\bullet \in \mathrm{Ch}(\mathcal{A})$ and every $n \in \mathbb{Z}$, $H^n(\mathrm{Hom}^\bullet_{\mathcal{A}}(K^\bullet, L^\bullet)) \cong \mathrm{Hom}_{\mathrm{K}(\mathcal{A})}(K^\bullet, L^\bullet[n])$.*

Proof. A degree-n cocycle is a family $\varphi = (\varphi^p)_p$ satisfying $d_L^{p+n} \circ \varphi^p = (-1)^n \varphi^{p+1} \circ d_K^p$. This is the condition that φ define a morphism of complexes

$K^\bullet \to L^\bullet[n]$. A degree-$n$ coboundary is a null-homotopic such morphism. Passing from cocycles to cohomology therefore passes from morphisms into the shift to their homotopy classes. See [2, Lemma A.15] and [5, Lemma 1.4.1]. $\square$

The passage from homotopy classes to derived morphisms still requires inversion of quasi-isomorphisms. To obtain a construction intrinsic to the derived category, we replace the second variable by the injective model.

Definition B.39 (Derived Hom) *Assume that $\mathcal{A}$ has enough injectives. Let $K \in D^b(\mathcal{A})$ and $L \in D^+(\mathcal{A})$. Choose a bounded representative $K^\bullet$ of K and an injective replacement $L \to I_L^\bullet$. Define* $\mathrm{RHom}_{\mathcal{A}}(K, L) \coloneqq \mathrm{Hom}^\bullet_{\mathcal{A}}(K^\bullet, I_L^\bullet)$ *as an object of $D(\mathbf{Ab})$.*

Note that the second variable must be bounded below because that is where Theorem B.33 applies. The first variable is restricted so that each group

$$\mathrm{Hom}^n_{\mathcal{A}}(K^\bullet, I_L^\bullet) = \prod_p \mathrm{Hom}_{\mathcal{A}}\left(K^p, I_L^{p+n}\right)$$

has only finitely many nonzero factors; with $K^\bullet$ bounded and $I_L^\bullet$ bounded below, all but finitely many indices p give a zero factor, so the product is a finite direct sum and no convergence question arises. Bounded above would in fact suffice, and the natural domain of the bifunctor is $D^-(\mathcal{A})^{\mathrm{op}} \times D^+(\mathcal{A})$, as in [5, §1.4] and [1, Example I.10.11]. Since $D^b(\mathcal{A}) \subseteq D^-(\mathcal{A})$, the definition above is a restriction of that bifunctor and not a different construction.

Theorem B.40 (Derived Hom represents shifted derived morphisms) *The object* $\mathrm{RHom}_{\mathcal{A}}(K, L)$ *is independent, up to natural isomorphism, of the chosen representatives. For every $n \in \mathbb{Z}$, there is a natural isomorphism*

$$H^n\left(\mathrm{RHom}_{\mathcal{A}(K,L)}\right) \cong \mathrm{Hom}_{D(\mathcal{A})}(K, L[n]).$$

Proof. The Hom-complex interpretation (Lemma B.38) gives $H^n(\mathrm{Hom}^\bullet_{\mathcal{A}}(K^\bullet, I_L^\bullet)) \cong \mathrm{Hom}_{\mathrm{K}(\mathcal{A})}(K^\bullet, I_L^\bullet[n])$. The injective model (Theorem B.33) identifies the latter group with $\mathrm{Hom}_{D(\mathcal{A})}(K, L[n])$.

The same calculation proves independence of the representatives. If $q : K'^\bullet \to K^\bullet$ is a quasi-isomorphism between bounded representatives, precomposition induces

$$\mathrm{Hom}^\bullet_{\mathcal{A}}(K^\bullet, I_L^\bullet) \to \mathrm{Hom}^\bullet_{\mathcal{A}}(K'^\bullet, I_L^\bullet)$$

By the isomorphism, this map induces an isomorphism on every cohomology group and is therefore a quasi-isomorphism. Zigzags of quasi-isomorphisms give the same conclusion for (arbitrary) bounded representatives.

If $I_L^\bullet$ and $J_L^\bullet$ are injective replacements of L, Theorem B.33 identifies them with the same object of $D^+(\mathcal{A})$ and hence with each other in $\mathrm{K}^+(\operatorname{Inj}\mathcal{A})$. A representing homotopy equivalence induces a homotopy equivalence

$$\operatorname{Hom}_{\mathcal{A}}^\bullet(K^\bullet, I_L^\bullet) \to \operatorname{Hom}_{\mathcal{A}}^\bullet(K^\bullet, J_L^\bullet).$$

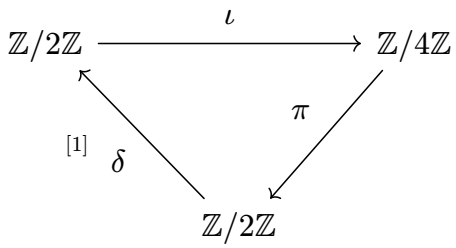
Thus $\operatorname{RHom}_{\mathcal{A}}(K, L)$ is independent of both choices. See [10, Chapter 10] and [1, Example I.10.11]. □

Corollary B.41 (Ext as morphisms in the derived category) *For objects $A, B \in \mathcal{A}$ and every $n \geq 0$, $\operatorname{Ext}_{\mathcal{A}}^n(A, B) \cong H^n(\operatorname{RHom}_{\mathcal{A}}(A, B)) \cong \operatorname{Hom}_{D(\mathcal{A})}(A, B[n])$.*

Proof. Choose an injective resolution $B \to I^\bullet$. The Ext calculation (Section A.8) gives $\operatorname{Ext}_{\mathcal{A}}^n(A, B) = \operatorname{Ker}(d_H^n)/\operatorname{im}(d_H^{n-1})$, where $d_H^q(\varphi) = d_I^q \circ \varphi$. An element of $\operatorname{Ker}(d_H^n)$ is therefore a map $\varphi : A \to I^n$ satisfying $d_I^n \circ \varphi = 0$. Since A is concentrated in degree 0, this is a morphism of complexes $A_{[0]} \to I^\bullet[n]$. Replacing φ by $\varphi + d_I^{n-1} \circ h$ changes this chain map by a null-homotopy, so passing to the quotient by $\operatorname{im}(d_H^{n-1})$ gives its homotopy class. The injective model (Theorem B.33) identifies these homotopy classes with $\operatorname{Hom}_{D(\mathcal{A})}(A, B[n])$. □

The nonzero class of Example A.44 determines the connecting morphism $\delta : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}[1]$ in the distinguished triangle associated with $0 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$. By Corollary B.41, this degree-1 morphism is precisely the corresponding element of

$$\operatorname{Ext}_{\mathbf{Ab}}^1(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}).$$



The label [1] on the returning edge indicates that its actual target is $\mathbb{Z}/2\mathbb{Z}[1]$. Thus δ is a degree-1 derived morphism, and the Ext interpretation (Corollary B.41) identifies it with the class of the extension. See [7, 1.1.8].

Example B.42 (A complete derived-Hom calculation) *In* $\mathbf{Ab}$*, the divisible group* $\mathbb{Q}/\mathbb{Z}$ *is injective, and* $0 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Q}/\mathbb{Z} \xrightarrow{2} \mathbb{Q}/\mathbb{Z} \to 0$ *is an injective resolution of* $\mathbb{Z}/2\mathbb{Z}$*. Applying* $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, -)$ *gives* $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$*. Multiplication by* 2 *induces the zero map on this group, so* $\mathrm{RHom}_{\mathbf{Ab}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})$ *is represented by* $\left[\mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z}\right]$ *in degrees* 0 *and* 1*. Therefore*

$$\mathrm{Ext}^0_{\mathbf{Ab}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}, \quad \mathrm{Ext}^1_{\mathbf{Ab}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z},$$

and $\mathrm{Ext}^n_{\mathbf{Ab}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) = 0$ *for* $n \geq 2$*.*

The object $\mathrm{RHom}_{\mathcal{A}}(K, L)$ is the external derived Hom: its value is a complex of abelian groups.

B.9 Canonical truncations

The derived category keeps all cohomological degrees available at once. We now need a controlled way to separate a derived object into a lower and an upper range of degrees. Deleting terms at a chosen boundary can change the cohomology there, because the incoming and outgoing differentials still contribute to its kernel and image. Canonical truncation uses the relevant kernel or cokernel at the boundary and therefore preserves the cohomology on the selected side of the cut.

Definition B.43 (Canonical truncations) *Let* $K^\bullet$ *be a cohomological complex in* $\mathcal{A}$*, and let* $n \in \mathbb{Z}$*. Define the* ***canonical truncation from above*** *by*

$$\left(\tau_{\leq n} K\right)^i := \begin{cases} K^i & i < n \\ \mathrm{Ker}(d_K^n) & i = n. \\ 0 & i > n \end{cases}$$

Its differential agrees with d_K *below degree* $n-1$*, and the final differential is the factorization* $K^{n-1} \to \mathrm{Ker}(d_K^n)$ *of* d_K^{n-1}*.*

Define the ***canonical truncation from below*** *by*

$$\left(\tau_{\geq n} K\right)^i := \begin{cases} 0 & i < n \\ \mathrm{coker}\left(d_K^{n-1}\right) & i = n. \\ K^i & i > n \end{cases}$$

Its first differential is the morphism $\mathrm{coker}\left(d_K^{n-1}\right) \to K^{n+1}$ *induced by* d_K^n*.*

Graphic, those cumbersome definitions are pretty simple:

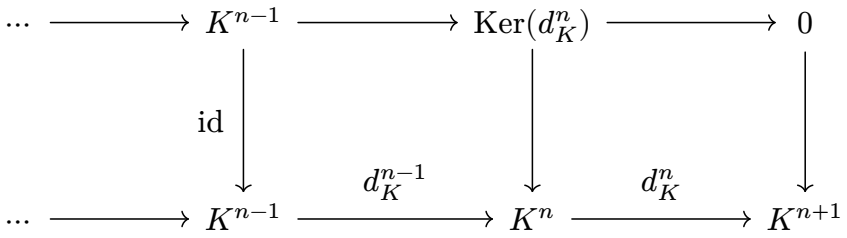


Figure B.12: The natural inclusion $\tau_{\leq n}K \to K$.

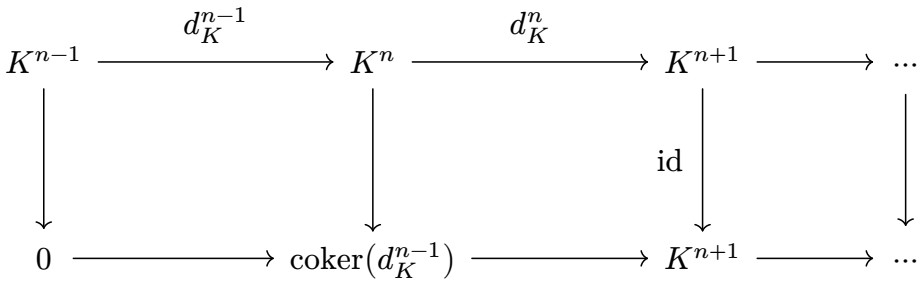


Figure B.13: The natural projection $K \to \tau_{\geq n}K$.

Proposition B.44 (Cohomology and the truncation triangle) *Canonical truncations preserve quasi-isomorphisms and therefore define functors on $D(\mathcal{A})$. Their cohomology is*

$$H^i(\tau_{\leq n}K) \cong \begin{cases} H^i(K) & i \leq n \\ 0 & i > n \end{cases},$$

$$H^i(\tau_{\geq n}K) \cong \begin{cases} 0 & i < n \\ H^i(K) & i \geq n \end{cases}.$$

Moreover, the natural morphisms extend to a canonical distinguished triangle

$$\tau_{\leq n}K \to K \to \tau_{\geq n+1}K \to \tau_{\leq n}K[1].$$

Proof sketch. The definitions change the complex only at the boundary degree, where the kernel and cokernel preserve the appropriate cohomology. The displayed formulas follow directly, and they imply that a quasi-isomorphism $K^\bullet \to L^\bullet$ induces quasi-isomorphisms on both truncations, so the functors descend to $D(\mathcal{A})$.

Finally, the cone of $\tau_{\leq n}K \to K$ is naturally quasi-isomorphic to $\tau_{\geq n+1}K$, which gives the distinguished triangle. See [5, Def. 1.1.14 with Remark 1.1.15], [2, §19.1], and [1, (I.3.10)–(I.3.11), Prop. I.3.7]. □

The content of the proposition is best seen degreewise. Each row below lists the cohomology of one of the three complexes in the triangle; the triangle says that the middle row is assembled from the other two, with the boundary degree n falling entirely to the left.

degree	$\cdots$	$n-1$	n	$n+1$	$\cdots$
$\tau_{\leq n}K$	$\cdots$	$H^{n-1}(K)$	$H^n(K)$	0	0
K	$\cdots$	$H^{n-1}(K)$	$H^n(K)$	$H^{n+1}(K)$	$\cdots$
$\tau_{\geq n+1}K$	0	0	0	$H^{n+1}(K)$	$\cdots$

Table B.1: Cohomology of the truncation triangle $\tau_{\leq n}K \to K \to \tau_{\geq n+1}K \to \tau_{\leq n}K[1]$. The two truncations partition the cohomology of K at the cut between degrees n and $n+1$; no degree is lost and none is counted twice.

Canonical truncation divides a derived object into a part with cohomology in low degrees and a part with cohomology in high degrees. The definition of a t-structure isolates the categorical properties that make such a division possible without referring to an underlying complex. This is needed because later the relevant notion of being concentrated in degree 0 will not be the standard one.

Definition B.45 (t-structure) *Let $\mathcal{T}$ be a triangulated category. A **t-structure** on $\mathcal{T}$ is a pair of full subcategories $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$, closed under isomorphisms, such that:*

- $\mathcal{T}^{\leq 0}[1] \subseteq \mathcal{T}^{\leq 0}$ *and* $\mathcal{T}^{\geq 0}[-1] \subseteq \mathcal{T}^{\geq 0}$*;*
- $\mathrm{Hom}_{\mathcal{T}}(A, B) = 0$ *whenever* $A \in \mathcal{T}^{\leq 0}$ *and* $B \in \mathcal{T}^{\geq 1}$*, where* $\mathcal{T}^{\geq 1} := \mathcal{T}^{\geq 0}[-1]$*;*
- *for every $K \in \mathcal{T}$, there is a distinguished triangle*

$$A \to K \to B \to A[1]$$

with $A \in \mathcal{T}^{\leq 0}$ and $B \in \mathcal{T}^{\geq 1}$.

For $n \in \mathbb{Z}$, set $\mathcal{T}^{\leq n} := \mathcal{T}^{\leq 0}[-n]$ and $\mathcal{T}^{\geq n} := \mathcal{T}^{\geq 0}[-n]$.

The axioms have this shape since the two truncation functors attached to a t-structure are adjoint to the inclusions, and the decomposition triangle is unique up to unique isomorphism - see [7, 1.3.3].

The first condition makes the two classes stable in the expected directions. The second says that the lower and upper parts do not admit morphisms across a positive degree gap. The third says that every object can actually be decomposed into such parts by a distinguished triangle.

We shall use the following notation throughout: $\mathcal{H}$ denotes the heart, defined below, and for the standard t-structure the truncation functors of Definition B.43 are the ones realizing the third axiom.

Theorem B.46 (The standard t-structure) *Define*

$$D^{\leq 0}(\mathcal{A}) \coloneqq \{K \in D(\mathcal{A}) \mid H^i(K) = 0 \ \text{ for every } \ i > 0\},$$

$$D^{\geq 0}(\mathcal{A}) \coloneqq \{K \in D(\mathcal{A}) \mid H^i(K) = 0 \ \text{ for every } \ i < 0\}.$$

Then $(D^{\leq 0}(\mathcal{A}), D^{\geq 0}(\mathcal{A}))$ *is a t-structure on* $D(\mathcal{A})$ *and on its bounded variants. It is called the* ***standard t-structure****.*

Proof. The shift conditions follow from Proposition B.2. The decomposition axiom is the canonical triangle

$$\tau_{\leq 0}K \to K \to \tau_{\geq 1}K \to \tau_{\leq 0}K[1]$$

of Proposition B.44, whose first term lies in $D^{\leq 0}(\mathcal{A})$ and whose third lies in $D^{\geq 1}(\mathcal{A})$, again by that proposition.

For orthogonality, let $A \in D^{\leq 0}(\mathcal{A})$, let $B \in D^{\geq 1}(\mathcal{A})$, and let $f : A \to B$ be a morphism in $D(\mathcal{A})$. Since $\tau_{\leq 0}$ is a functor on $D(\mathcal{A})$ and the morphisms $\varepsilon_X : \tau_{\leq 0}X \to X$ are natural, the square

$$\tau_{\leq 0}A \xrightarrow{\tau_{\leq 0}f} \tau_{\leq 0}B, \quad \varepsilon_A : \tau_{\leq 0}A \to A, \quad \varepsilon_B : \tau_{\leq 0}B \to B$$

commutes, that is, $f \circ \varepsilon_A = \varepsilon_B \circ \tau_{\leq 0}f$. Now ε_A is an isomorphism: it induces an isomorphism on H^i for $i \leq 0$ by Proposition B.44, and for $i > 0$ both sides vanish because $A \in D^{\leq 0}(\mathcal{A})$, so Proposition B.31 applies. On the other side, $B \in D^{\geq 1}(\mathcal{A})$ gives $H^{i(B)} = 0$ for $i \leq 0$, hence $H^{i(\tau_{\leq 0}B)} = 0$ for every i, so $\tau_{\leq 0}B$ is acyclic and therefore zero in $D(\mathcal{A})$ by Proposition B.30. Consequently $f \circ \varepsilon_A = 0$, and since ε_A is invertible, $f = 0$. □

see [5, Example 5.1.3] and [8, Example 8.1.2]. After cones, roofs, and localizations, the category finally acquires a heart. We are almost in Oz.

Definition B.47 (Heart of a t-structure) *The* ***heart*** *of a t-structure is the full subcategory*

$$\mathcal{H}^{\heartsuit} \coloneqq \mathcal{T}^{\leq 0} \cap \mathcal{T}^{\geq 0}.$$

An object of the heart is regarded as concentrated in degree 0 relative to the chosen t-structure. The meaning of "degree 0" therefore depends on the t-structure, not only on the triangulated category.

For the standard t-structure the intersection can be drawn. Each row records in which degrees an object of the indicated class is permitted to have nonzero cohomology.

degree	$\cdots$	-1	0	1	$\cdots$
$D^{\leq 0}(\mathcal{A})$	H^{-2}	H^{-1}	H^0	0	0
$D^{\geq 0}(\mathcal{A})$	0	0	H^0	H^1	H^2
$\mathcal{H}^\heartsuit$	0	0	H^0	0	0

Table B.2: The heart of the standard t-structure, as the overlap of the two classes. Only degree 0 survives both conditions, and Proposition B.49 identifies what survives with $\mathcal{A}$ itself. A different t-structure moves both boundaries at once, and the surviving column is then a different abelian category inside the same triangulated one.

Theorem B.48 (The heart is abelian) *The heart of every t-structure is an abelian category. Its short exact sequences are precisely the sequences*

$$0 \to A \to B \to C \to 0$$

whose two internal morphisms can be completed to a distinguished triangle

$$A \to B \to C \to A[1]$$

in the ambient triangulated category.

Given $f : X \to Y$ in $\mathcal{H}^\heartsuit$, complete it to a distinguished triangle with third vertex Z. Then $Z \in \mathcal{T}^{\leq 0} \cap \mathcal{T}^{\geq -1}$, and truncation gives

$$\operatorname{Ker}(f) \cong \tau_{\leq 0}(Z[-1]), \quad \operatorname{coker}(f) \cong \tau_{\geq 0} Z$$

Hence kernels and cokernels in the heart are obtained from cones by truncation. See [7, 1.2.2–1.2.5, Théorème 1.3.6], [1, Prop. X.1.11], and [8, Thm. 8.1.9].

Proposition B.49 (The standard heart) *The heart of the standard t-structure on $D(\mathcal{A})$ is equivalent to $\mathcal{A}$. The equivalence is induced by H^0, and its quasi-inverse sends an object $A \in \mathcal{A}$ to the complex concentrated in degree 0 with value A.*

Proof. An object K belongs to the standard heart when $H^i(K) = 0$ for every $i \neq 0$. For such a K, apply Proposition B.44 twice. The morphism $\tau_{\leq 0} K \to K$ is an isomorphism in every cohomological degree, hence an isomorphism in $D(\mathcal{A})$ by Proposition B.31, and likewise for $\tau_{\leq 0} K \to \tau_{\geq 0}\tau_{\leq 0} K$. The composite functor $\tau_{\geq 0}\tau_{\leq 0}$ therefore carries K to a complex concentrated in degree 0 with value $H^0(K)$, and the two natural morphisms give an isomorphism

$$K \cong H^0(K)_{[0]}$$

in $D(\mathcal{A})$. Conversely, $A_{[0]}$ has cohomology only in degree 0 for every $A \in \mathcal{A}$, so it lies in both $D^{\leq 0}(\mathcal{A})$ and $D^{\geq 0}(\mathcal{A})$. The two constructions are mutually inverse, and H^0 is fully faithful on the heart because it is inverse to a functor. $\square$

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